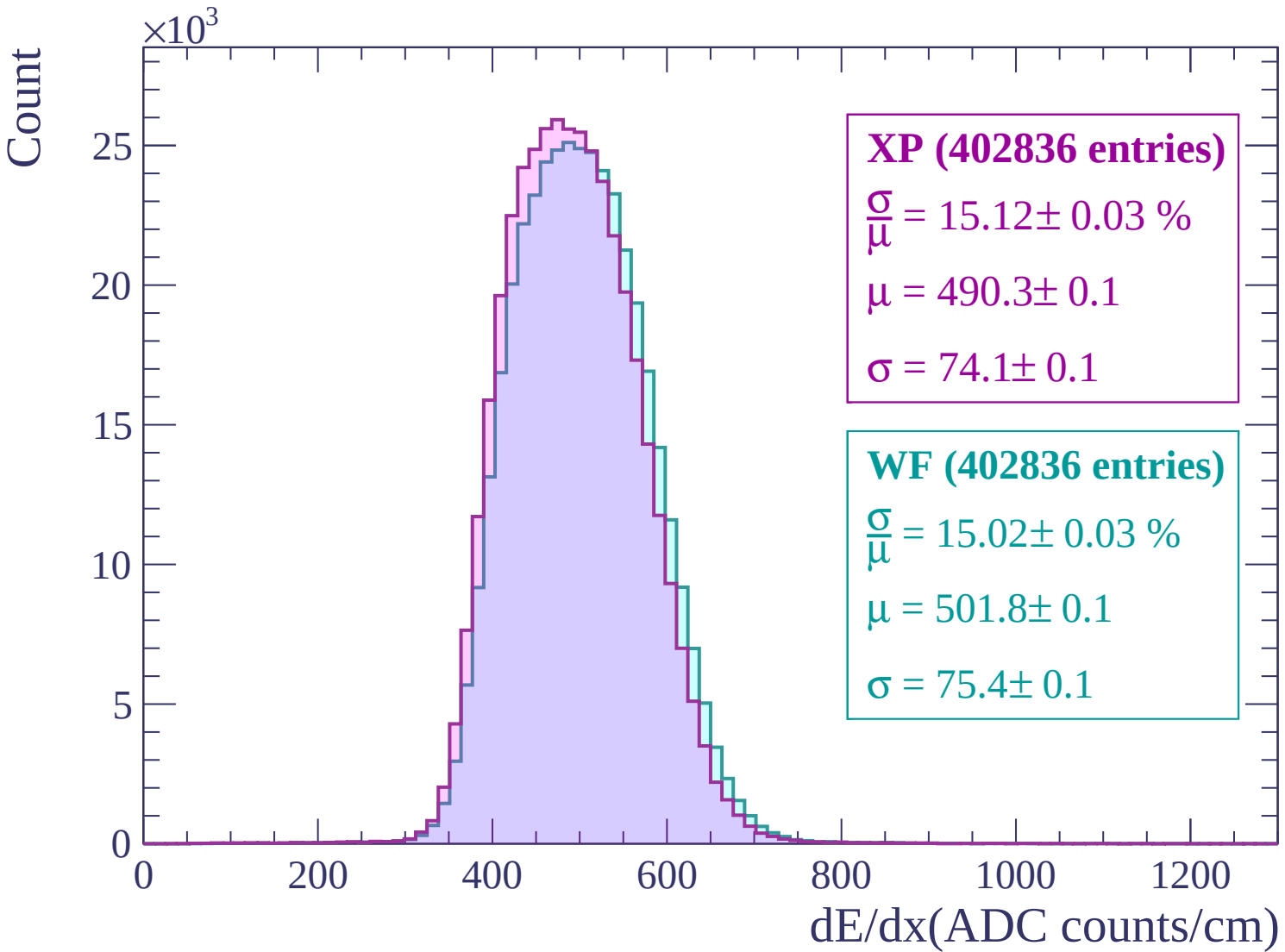
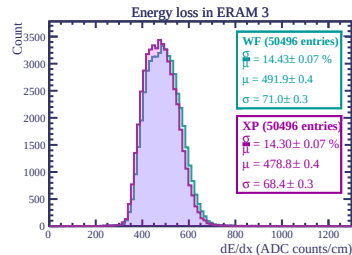
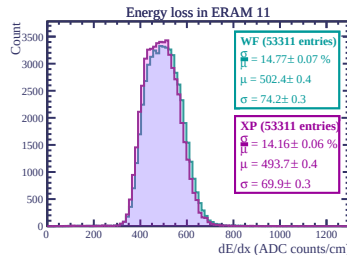
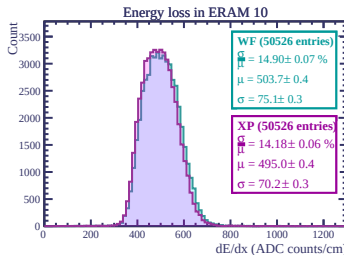
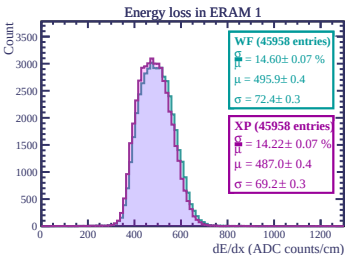
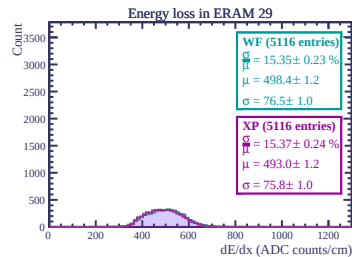
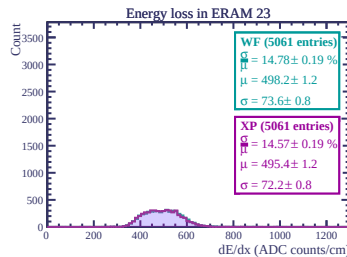
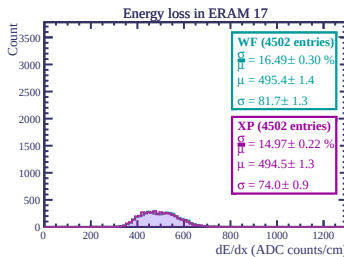
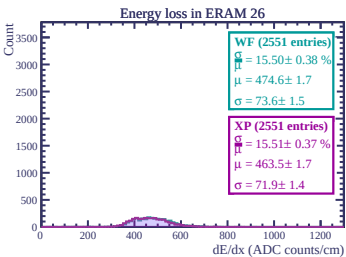
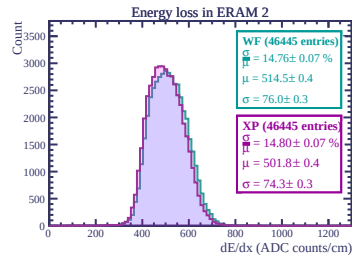
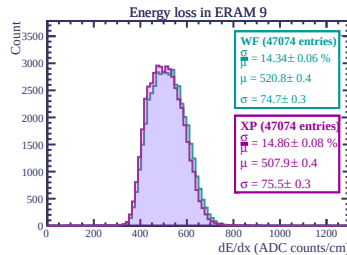
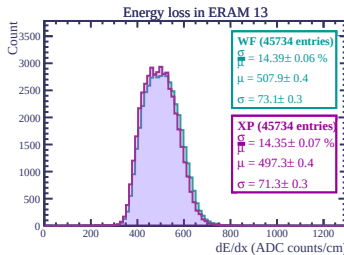
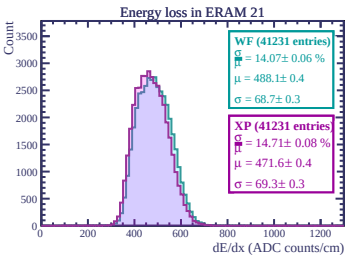
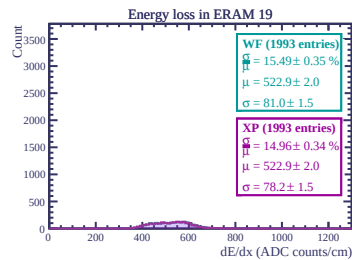
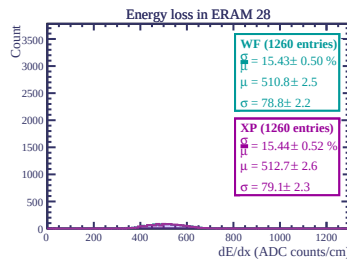
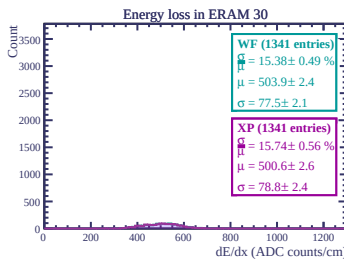
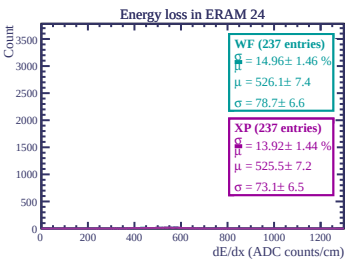
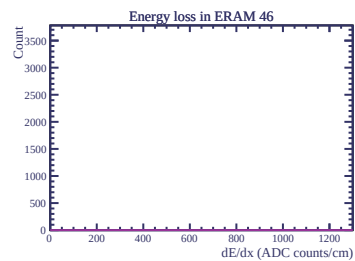
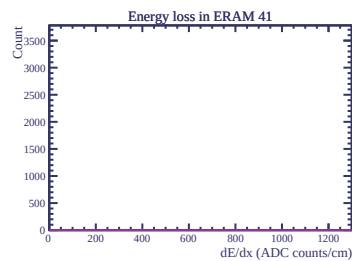
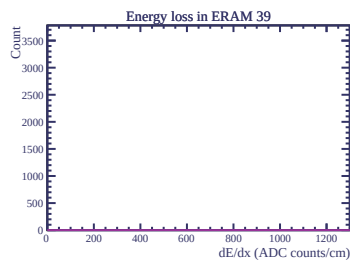
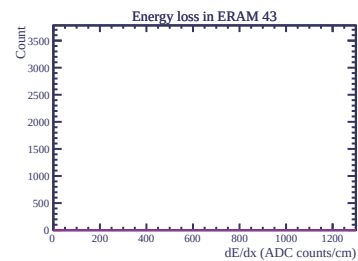
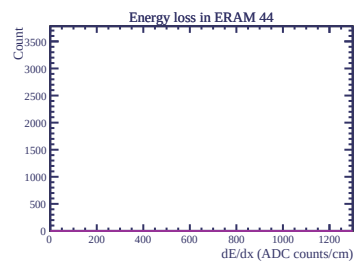
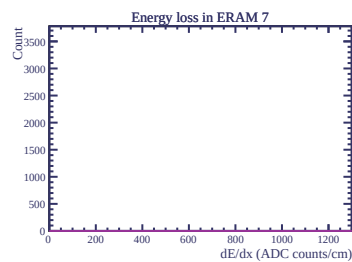
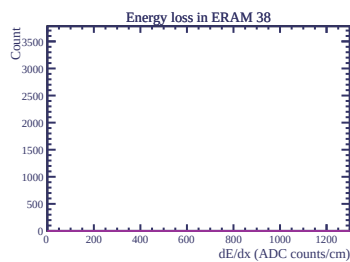
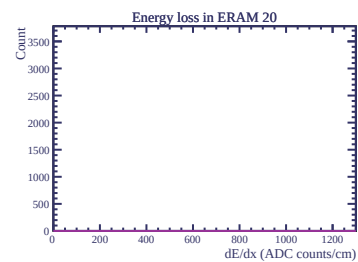
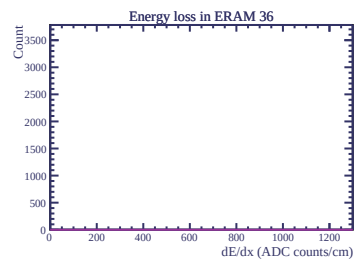
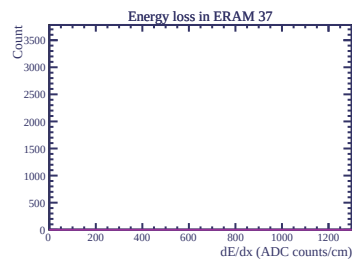
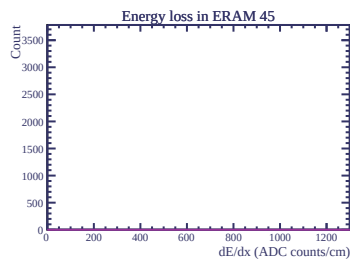
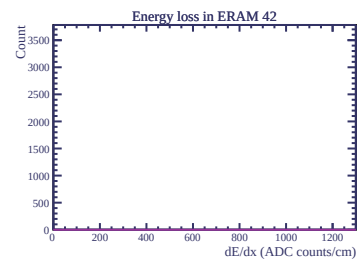
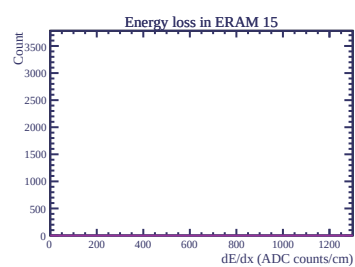
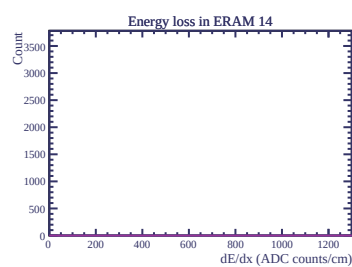
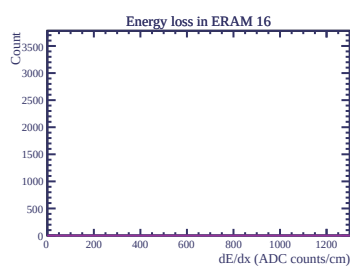
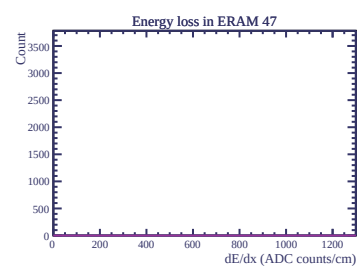
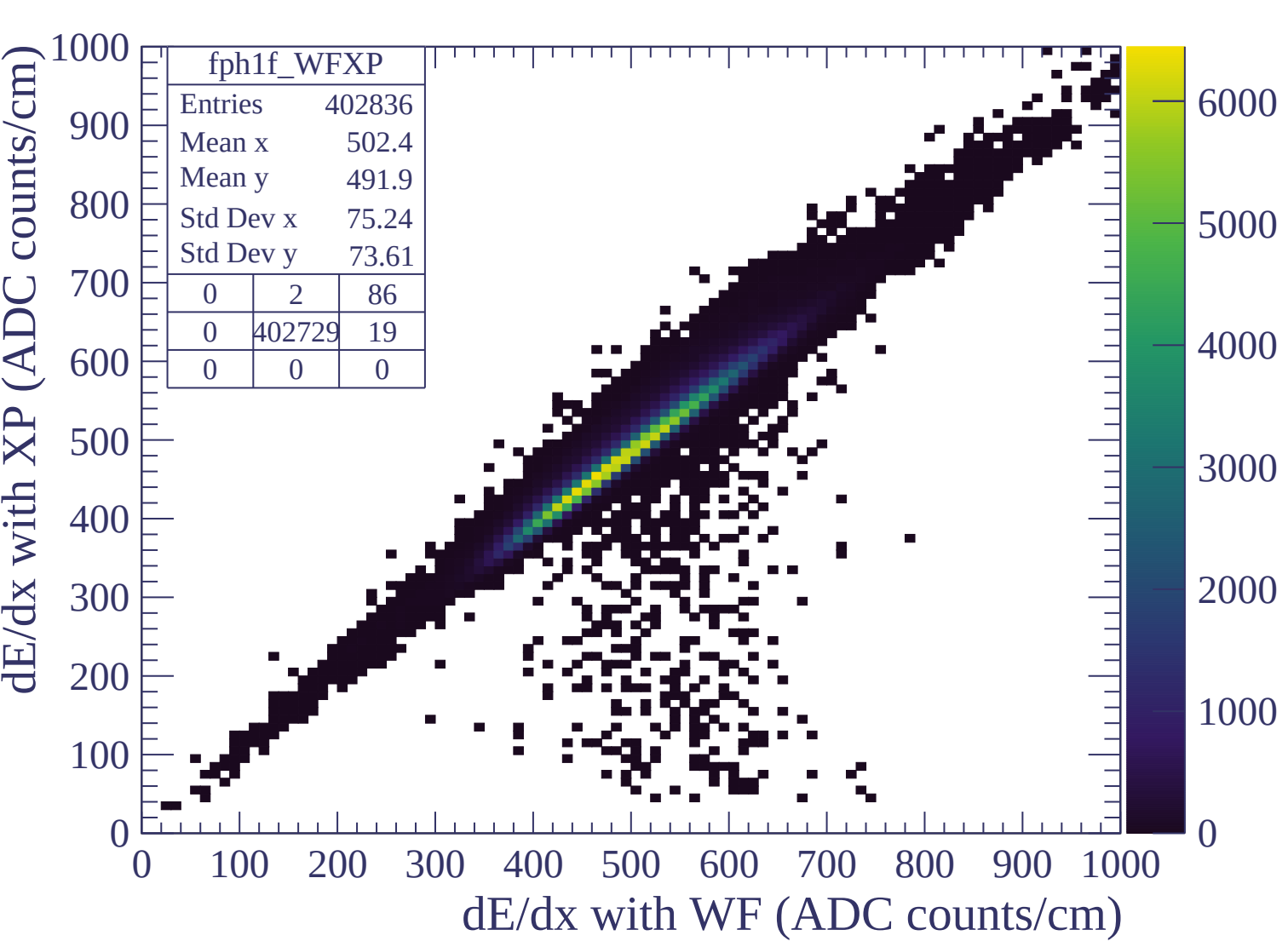
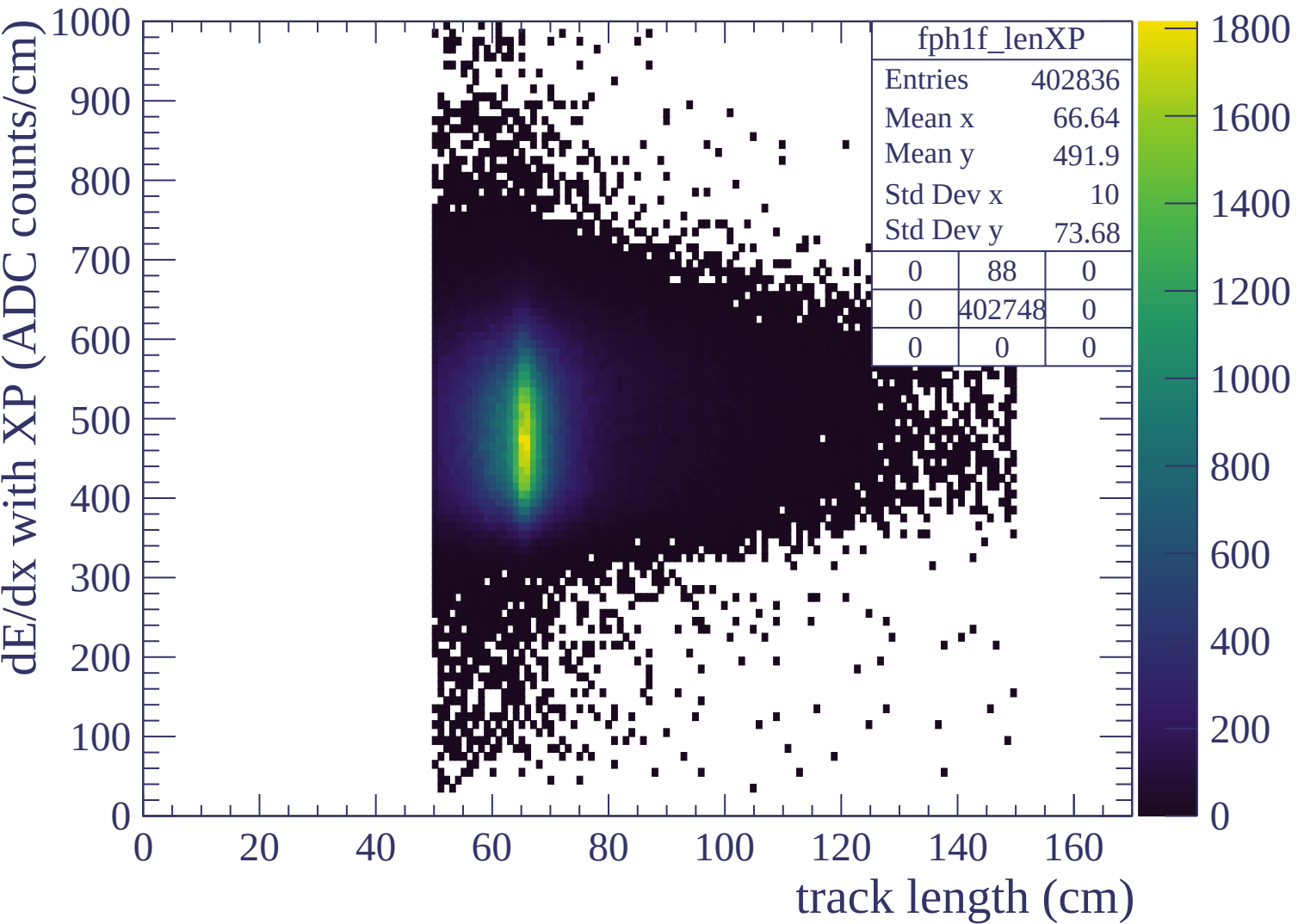


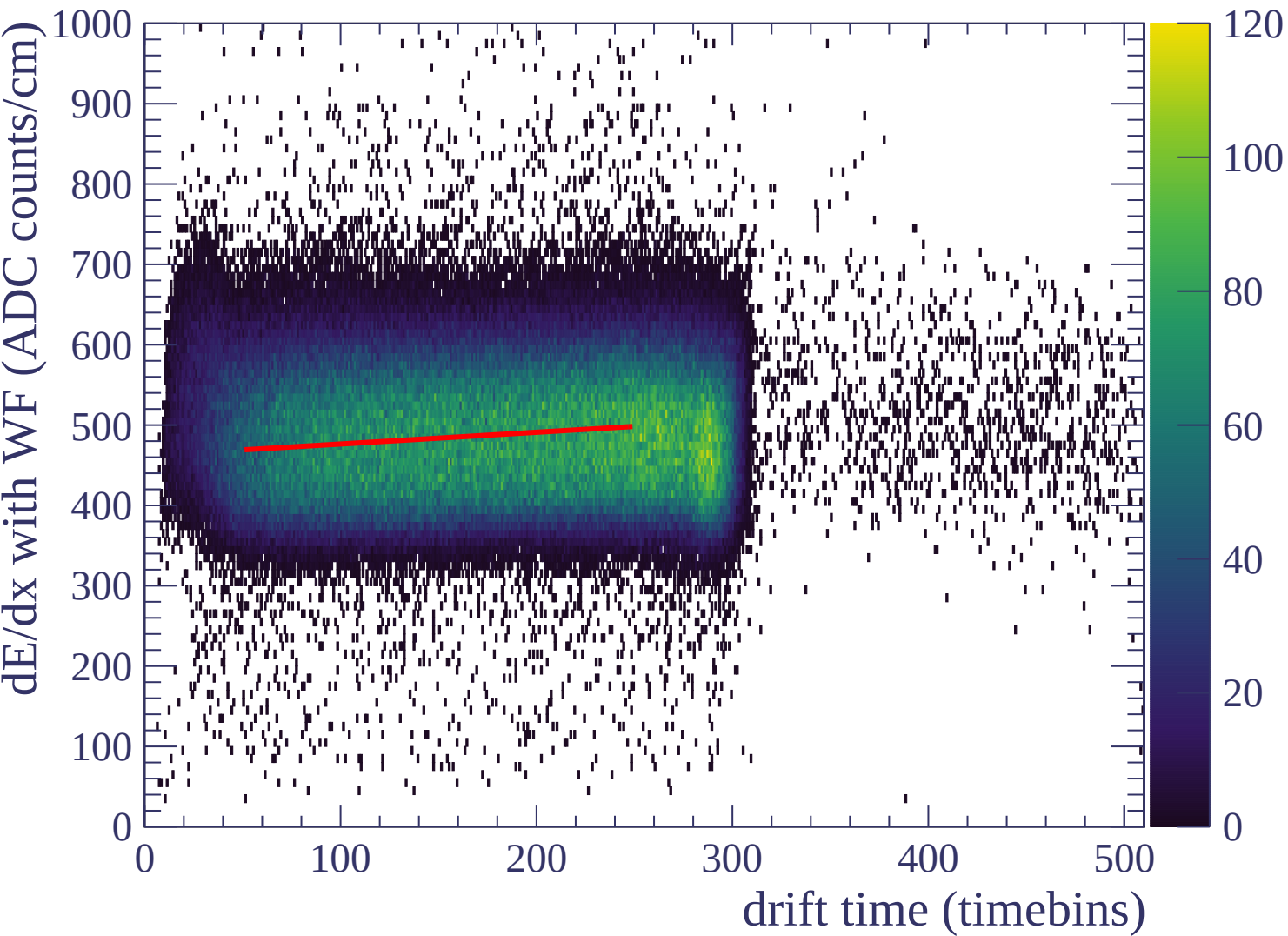


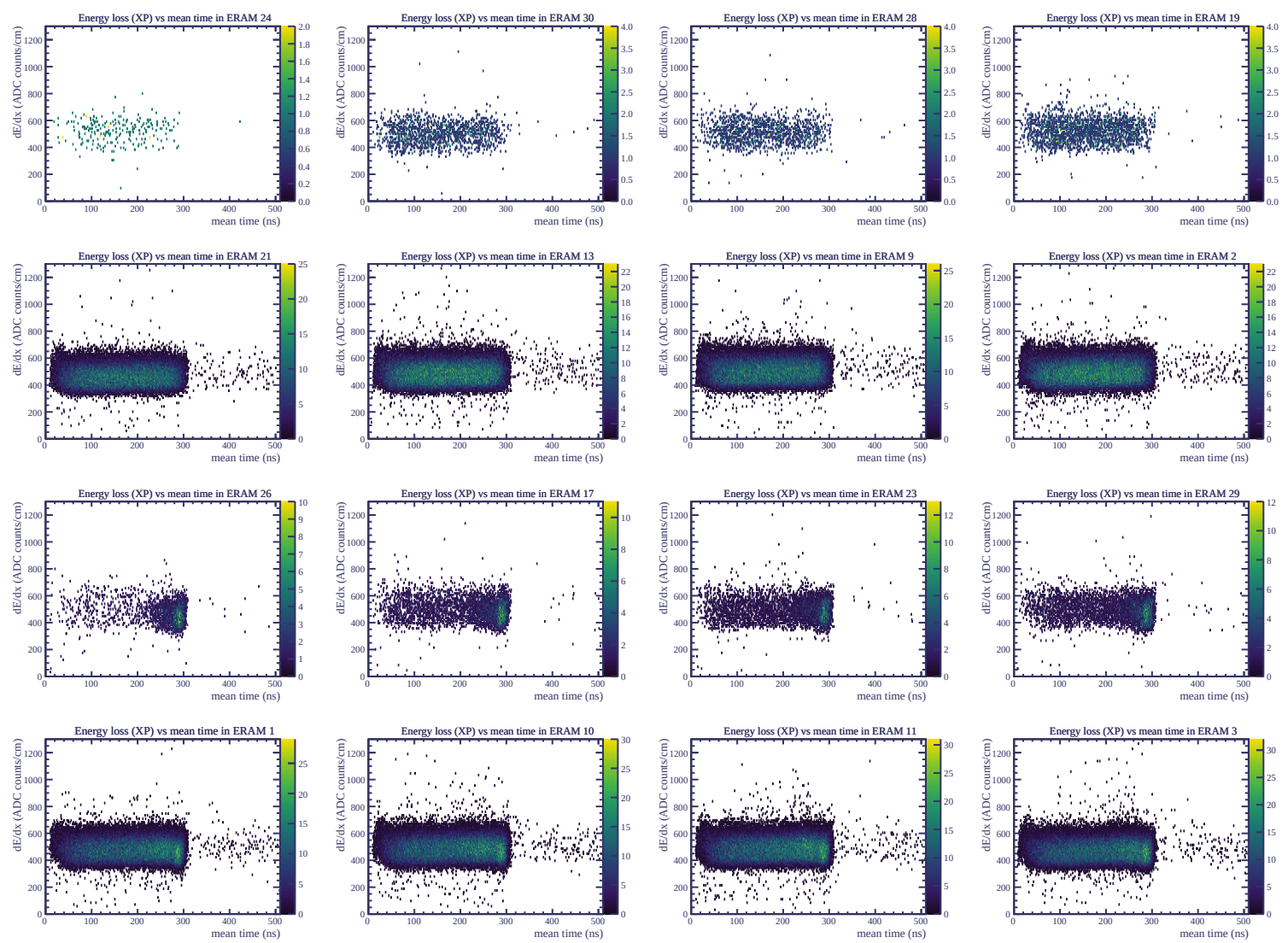


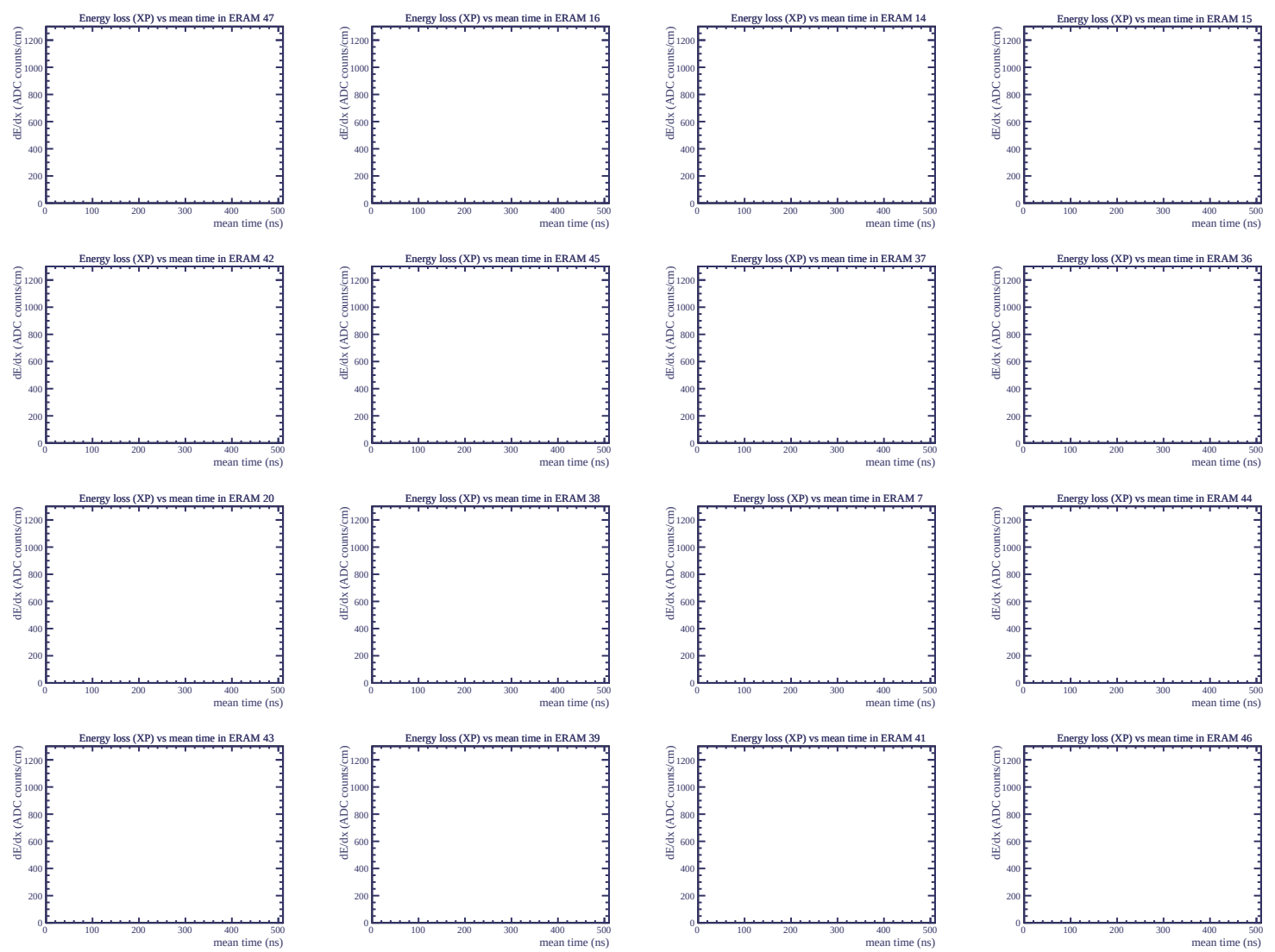




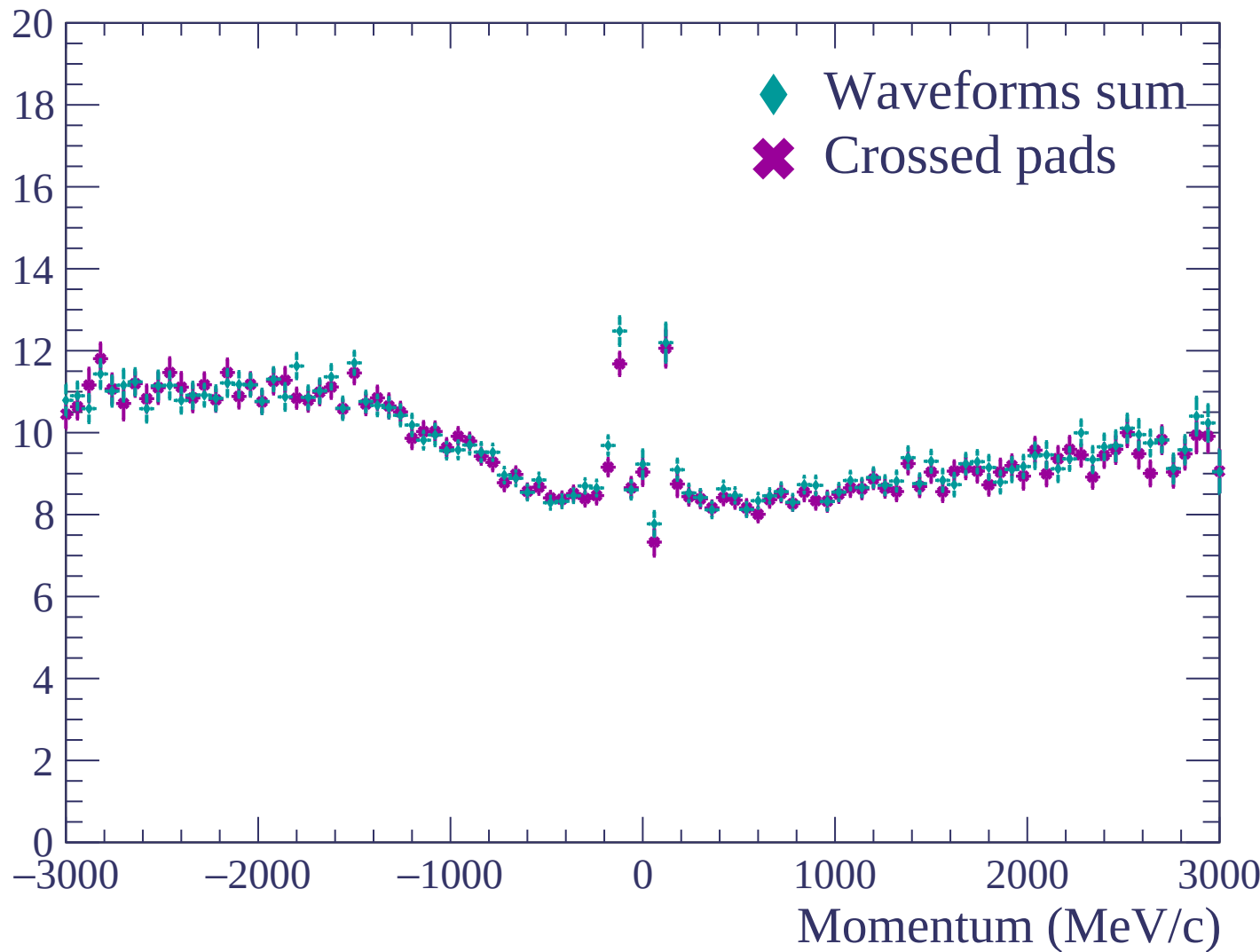


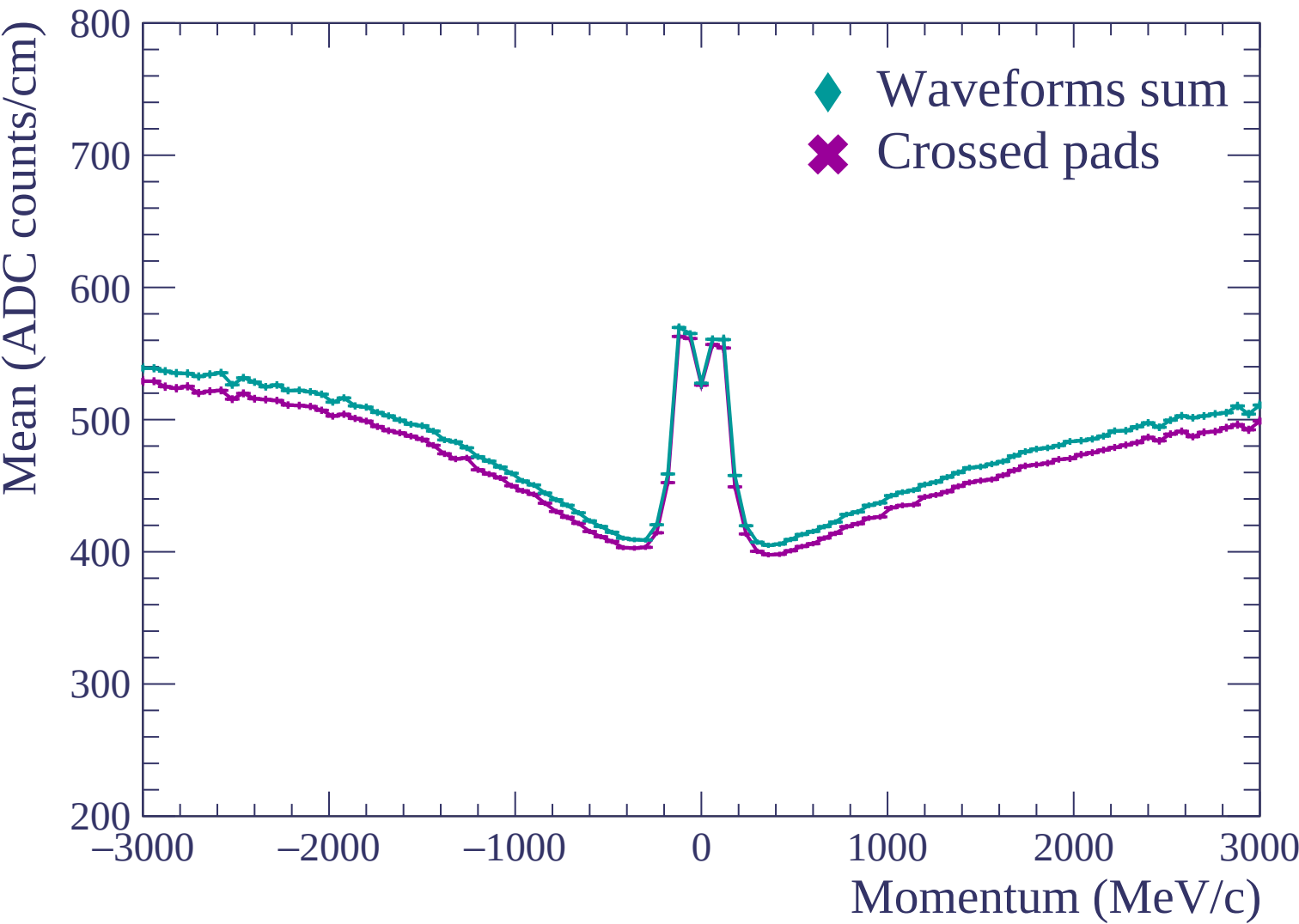


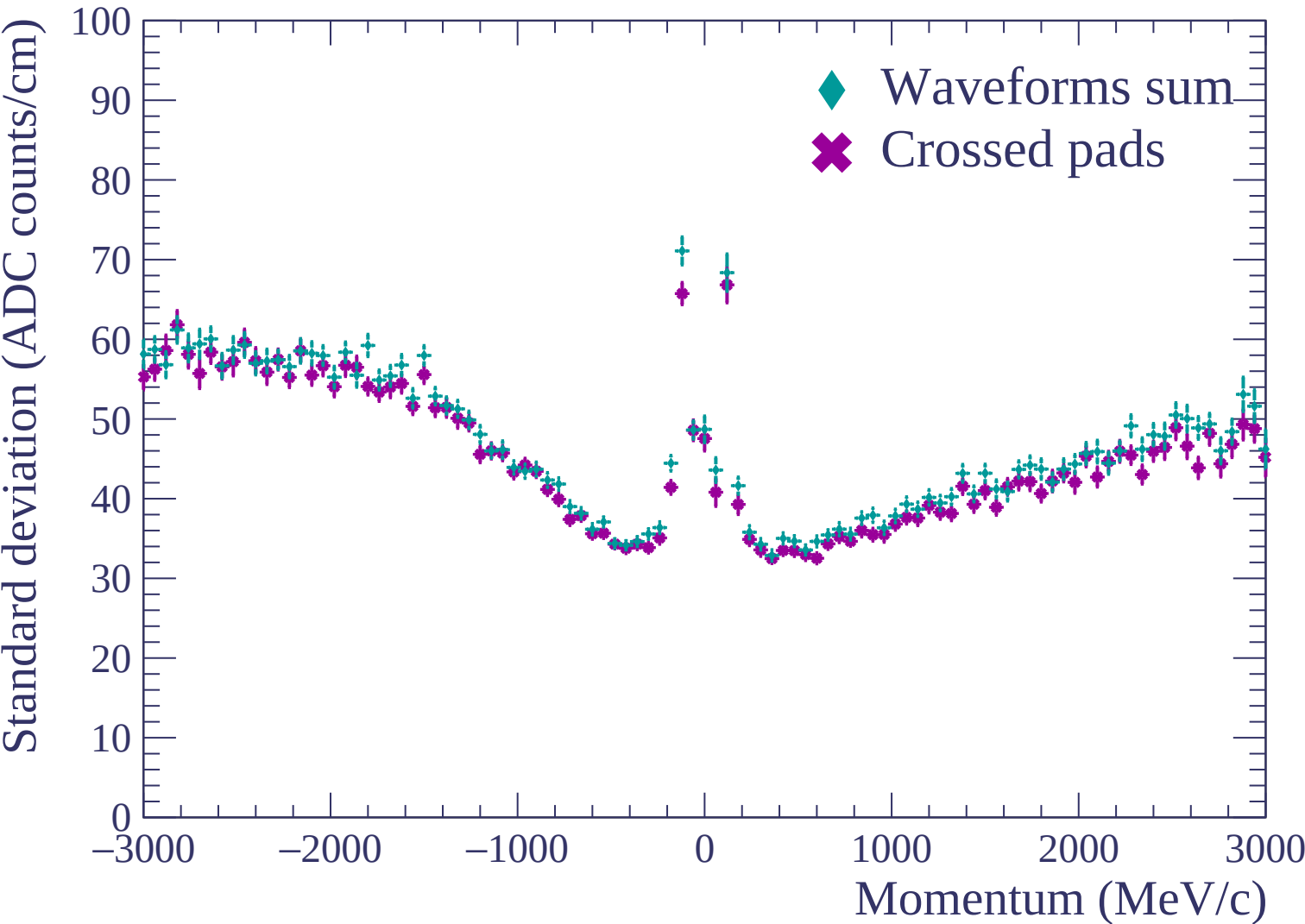


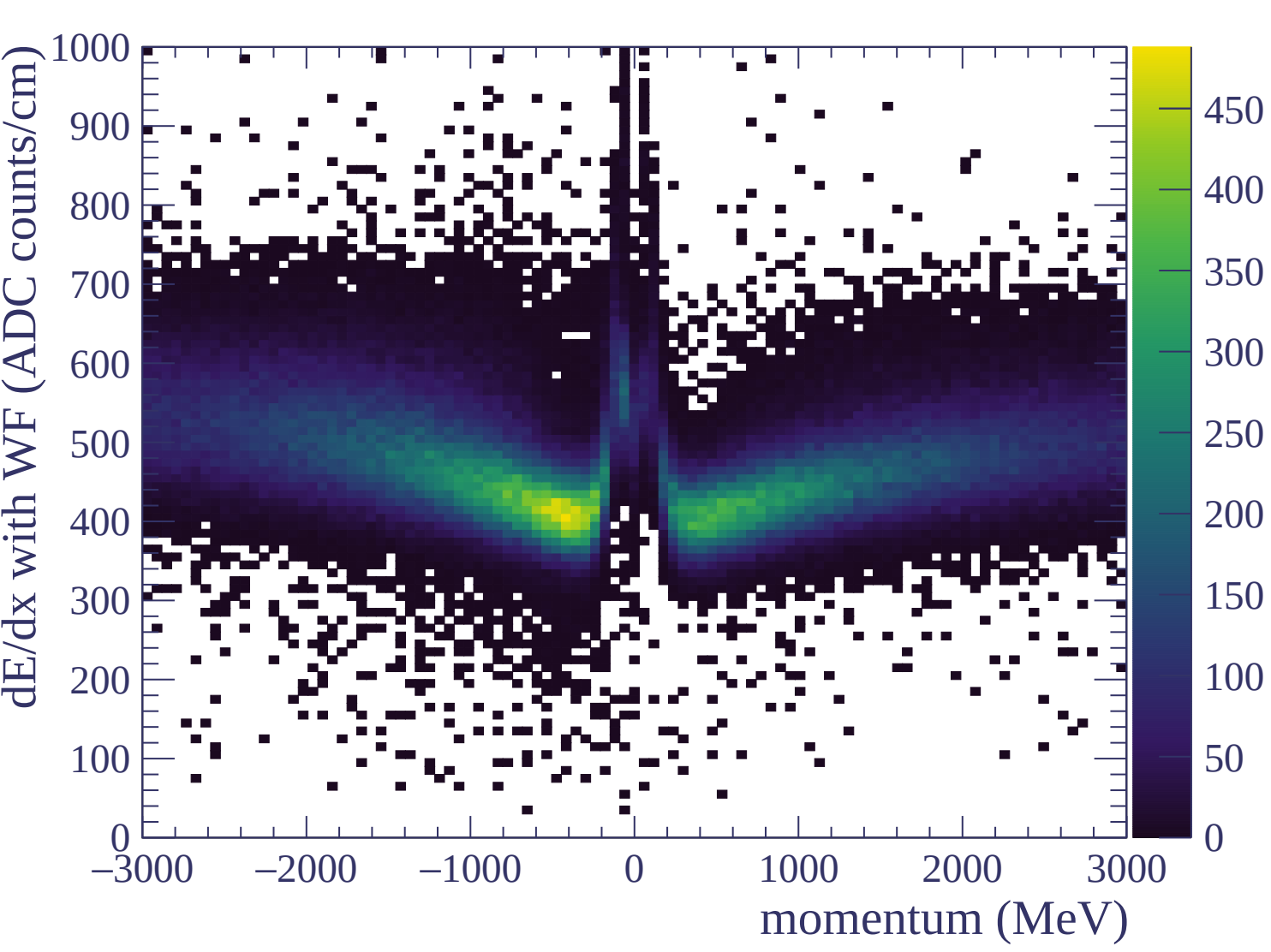


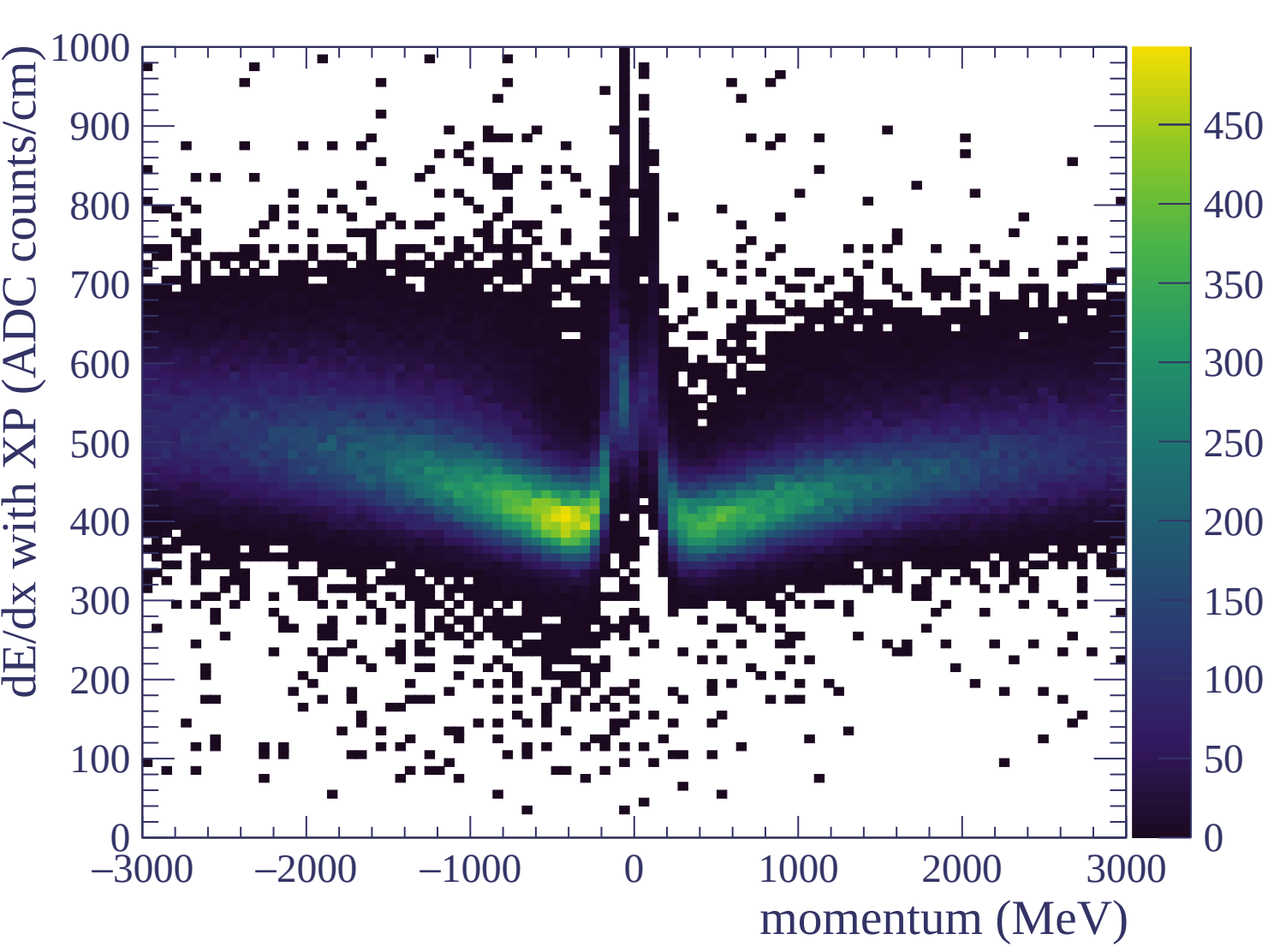
Resolution (%)

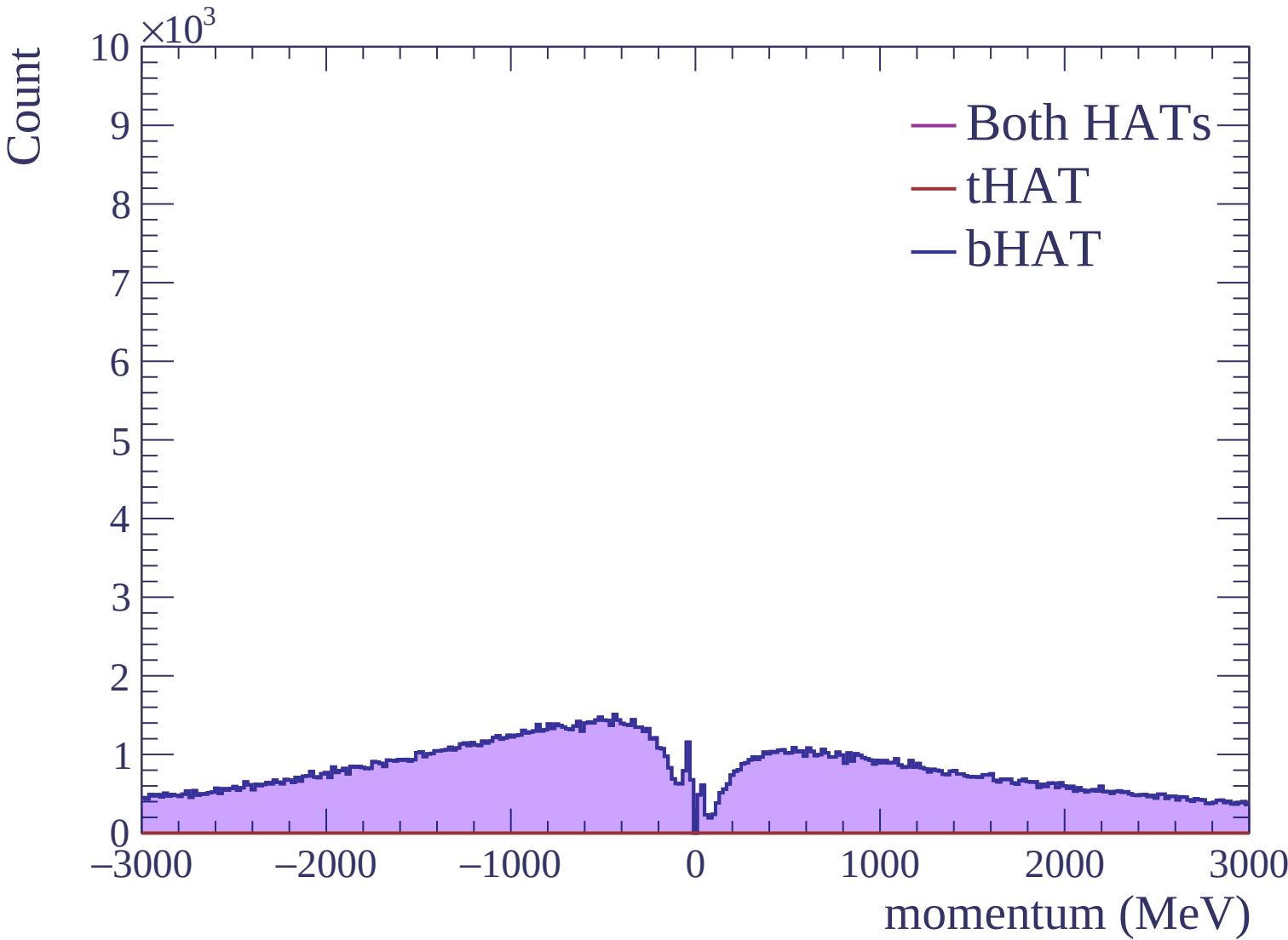






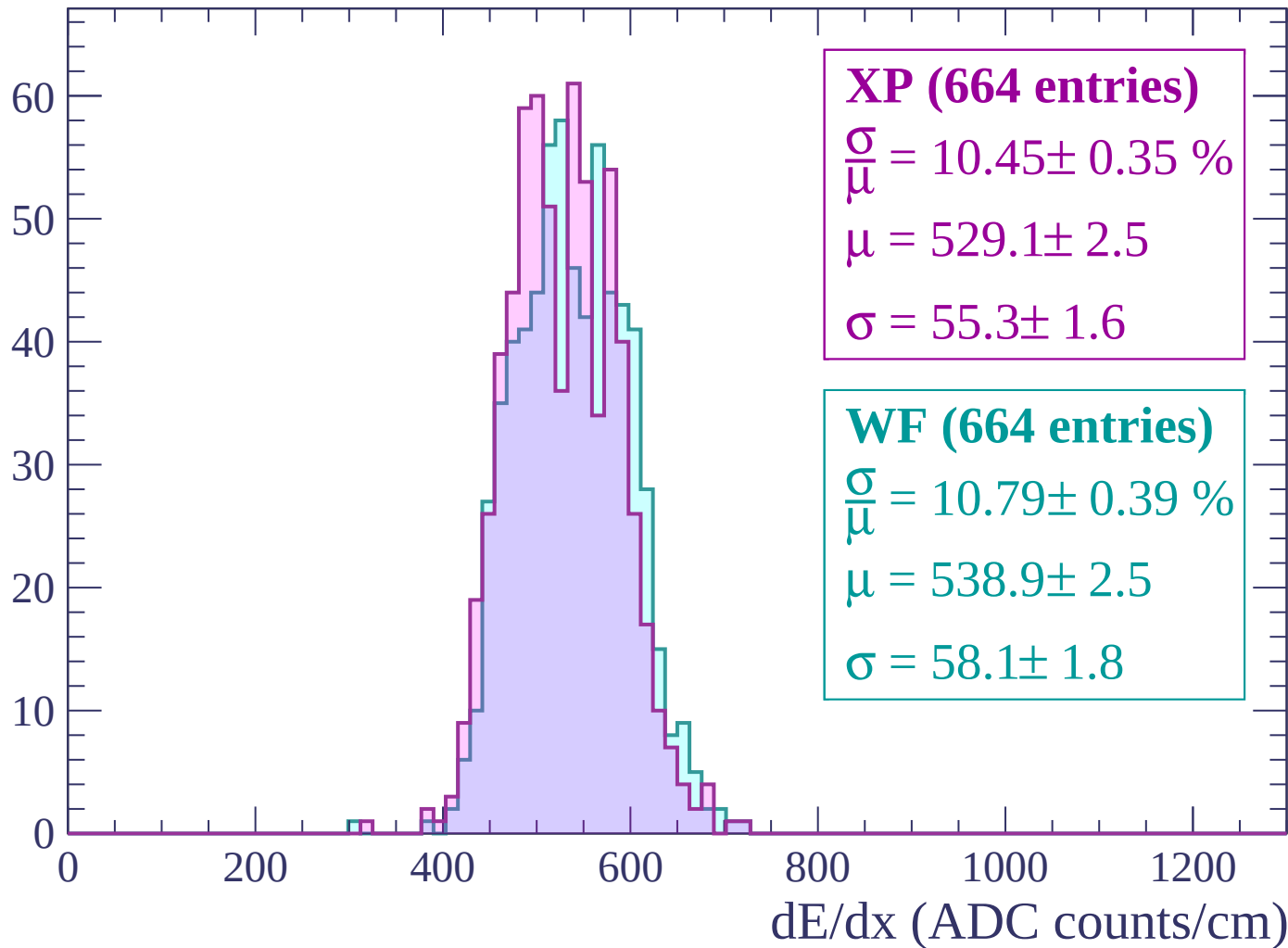






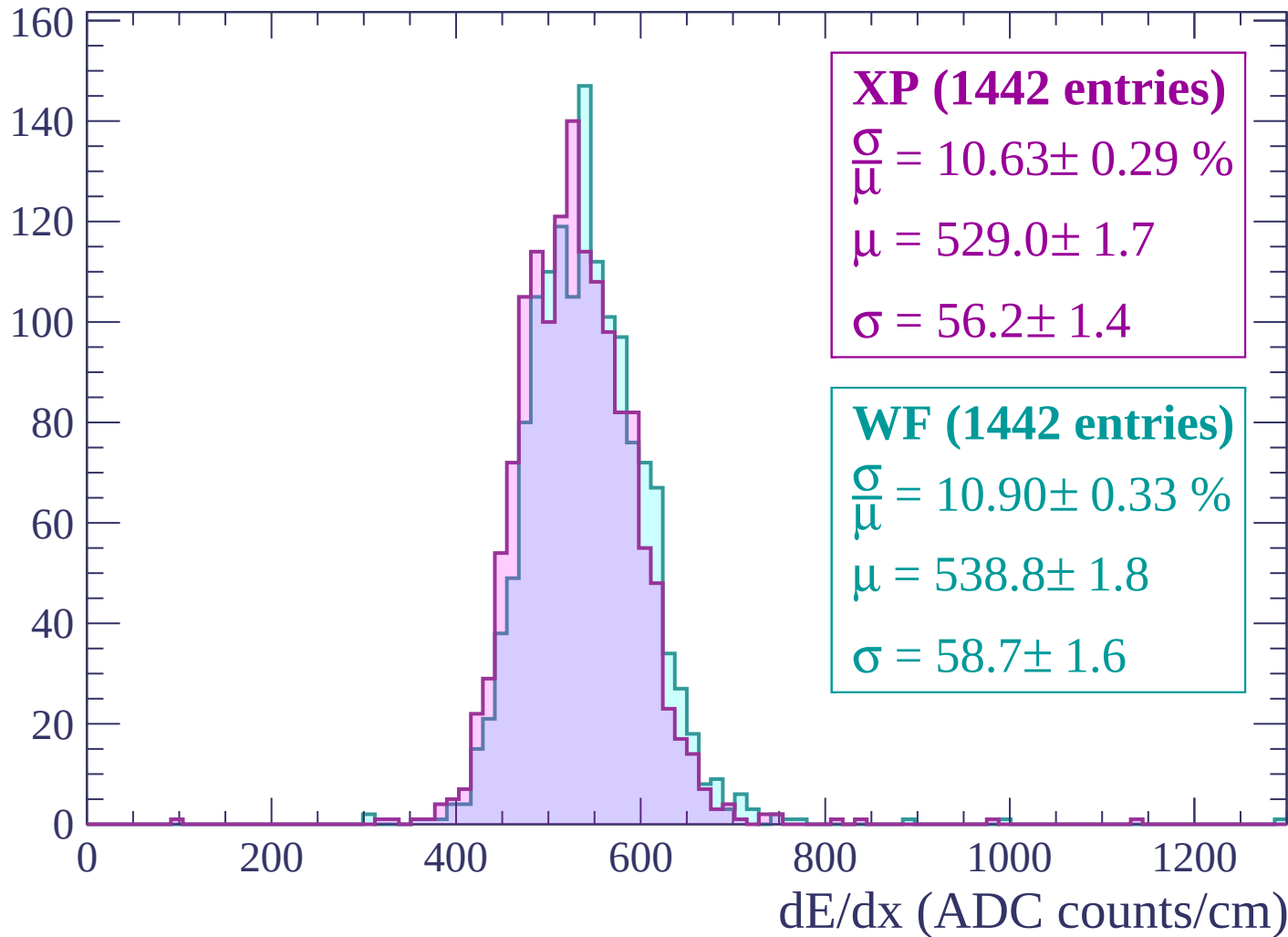
Energy loss | $-3000 < p < -2940$

Count



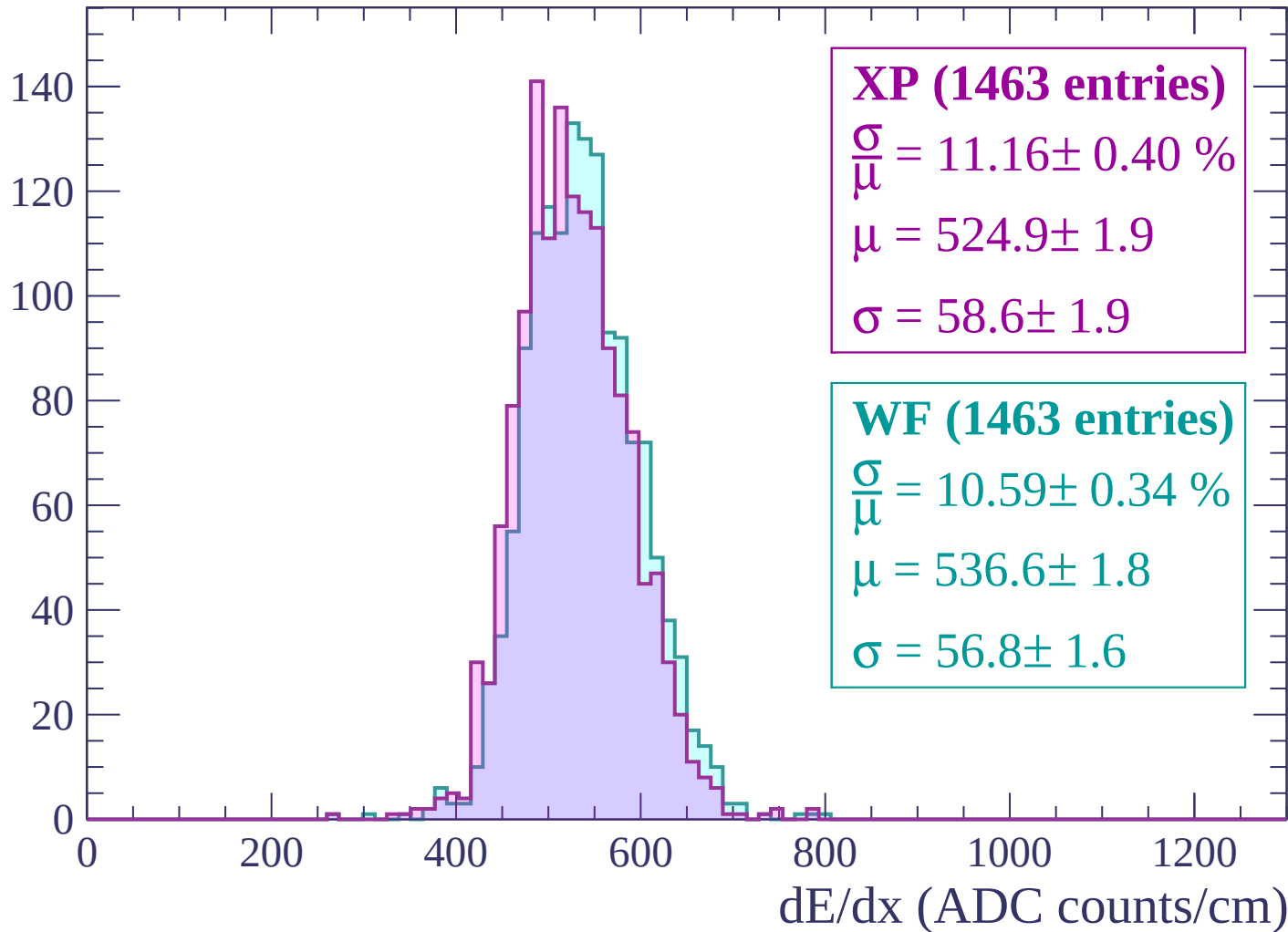
Energy loss | $-2940 < p < -2880$

Count



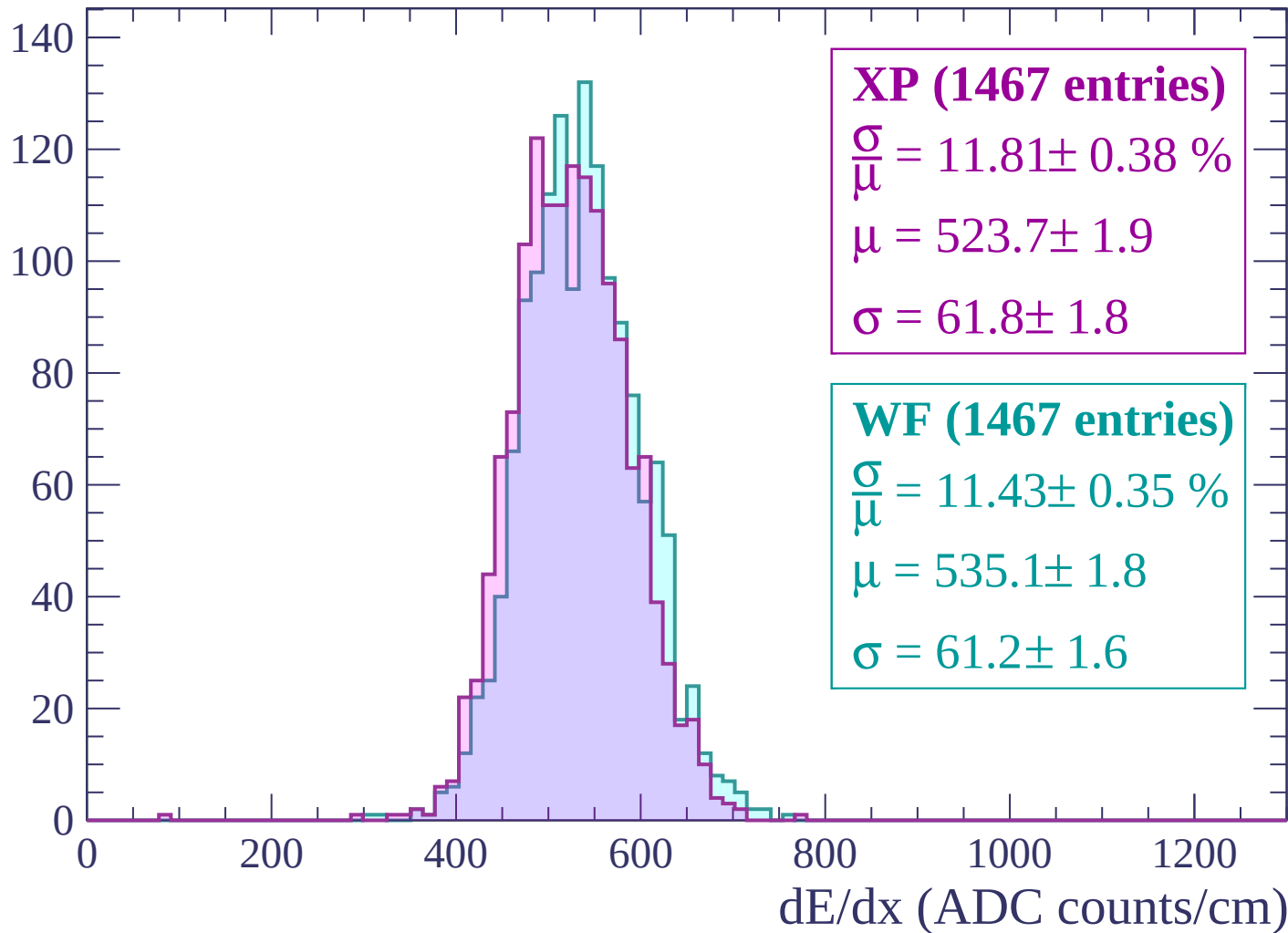
Energy loss | $-2880 < p < -2820$

Count



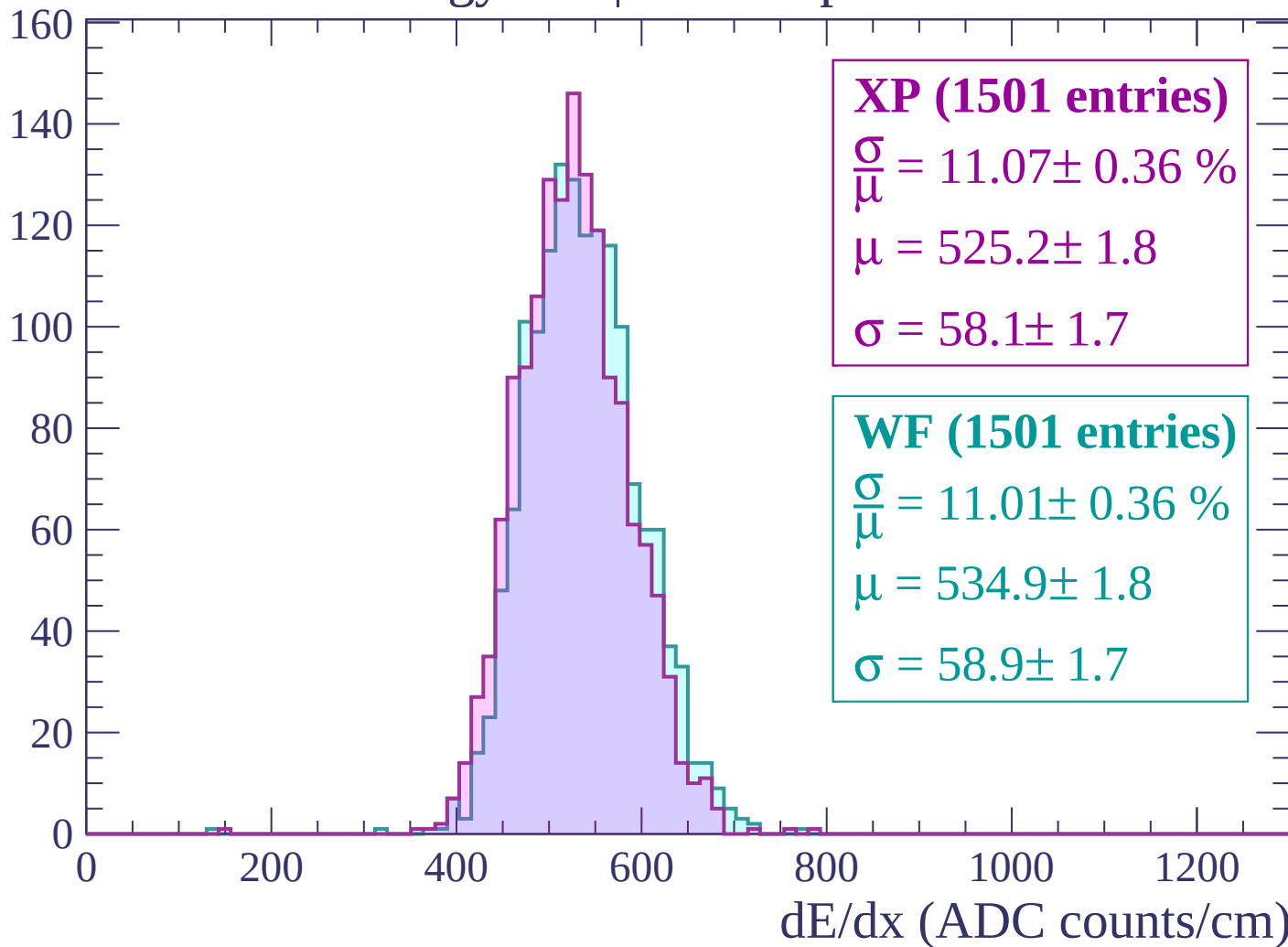
Energy loss | $-2820 < p < -2760$

Count



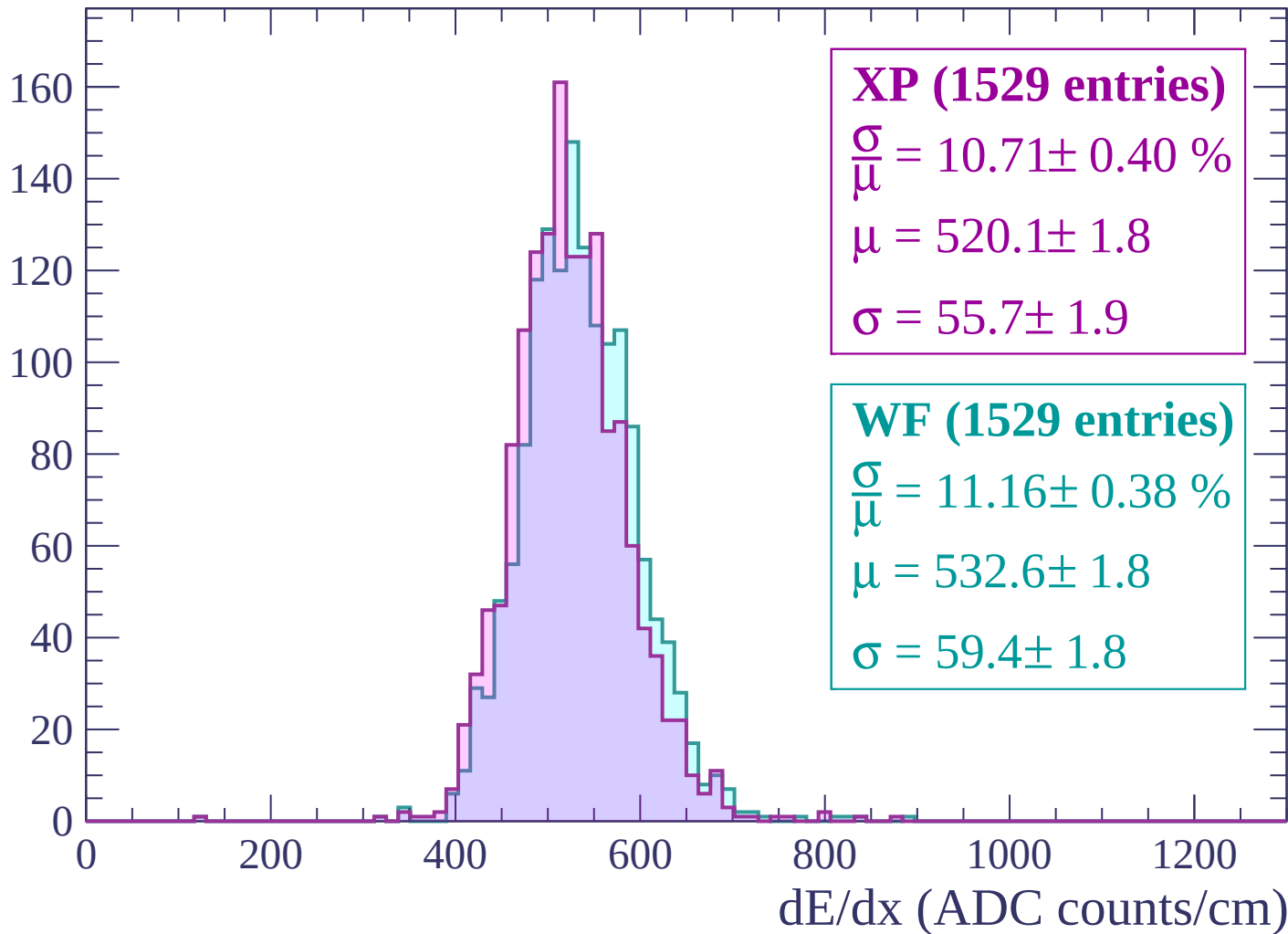
Energy loss | $-2760 < p < -2700$

Count



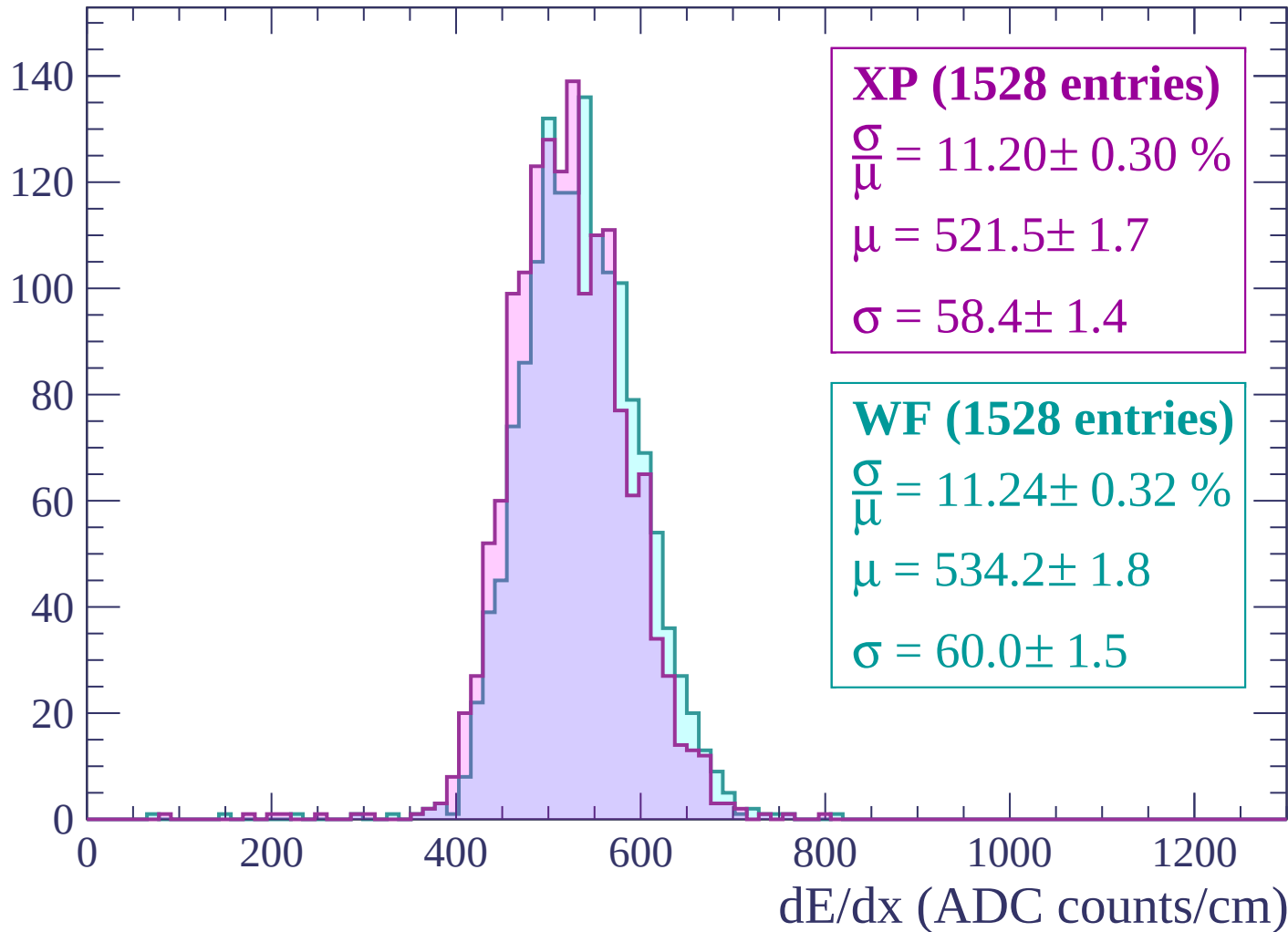
Energy loss | -2700 < p < -2640

Count



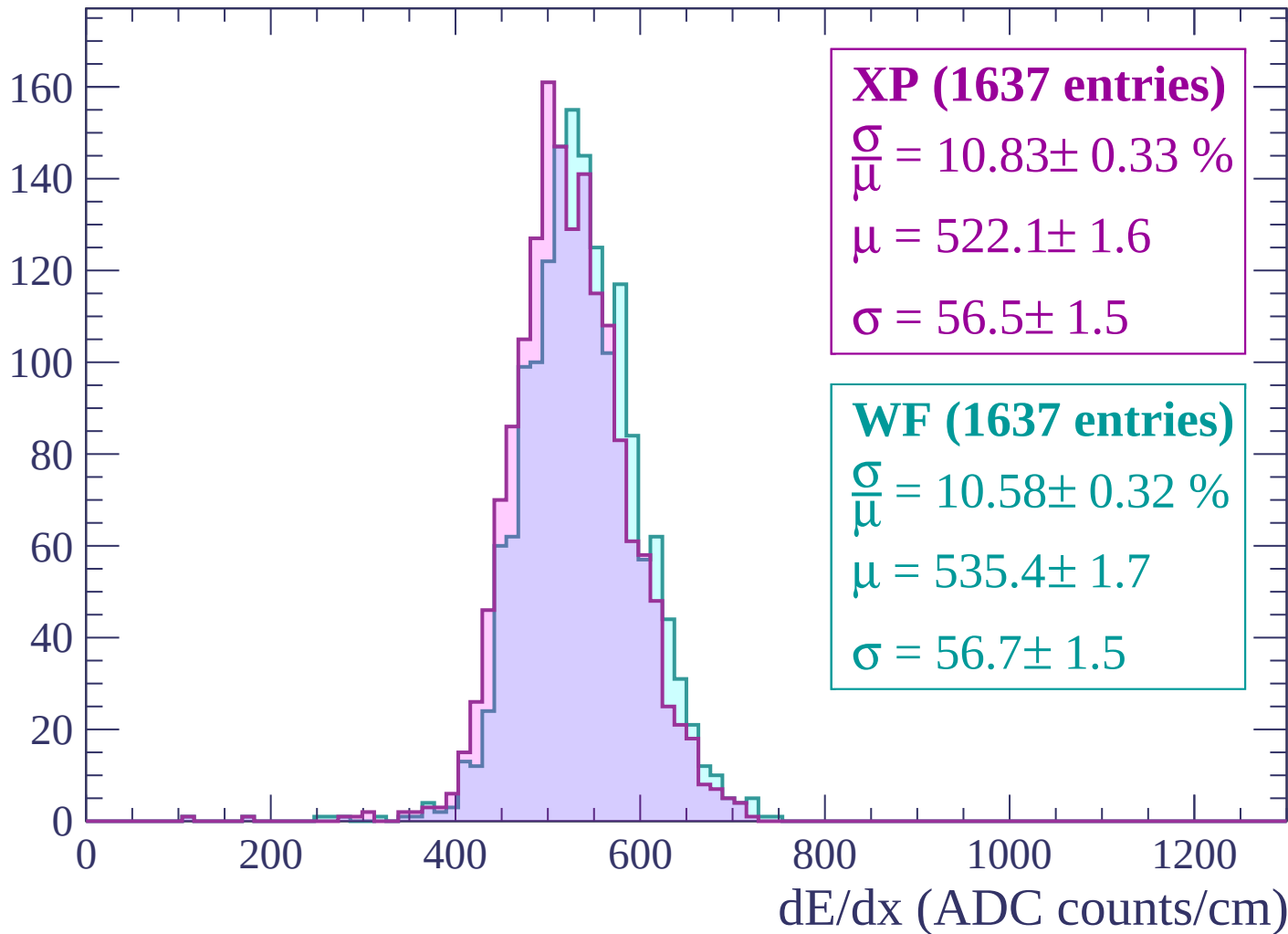
Energy loss | $-2640 < p < -2580$

Count

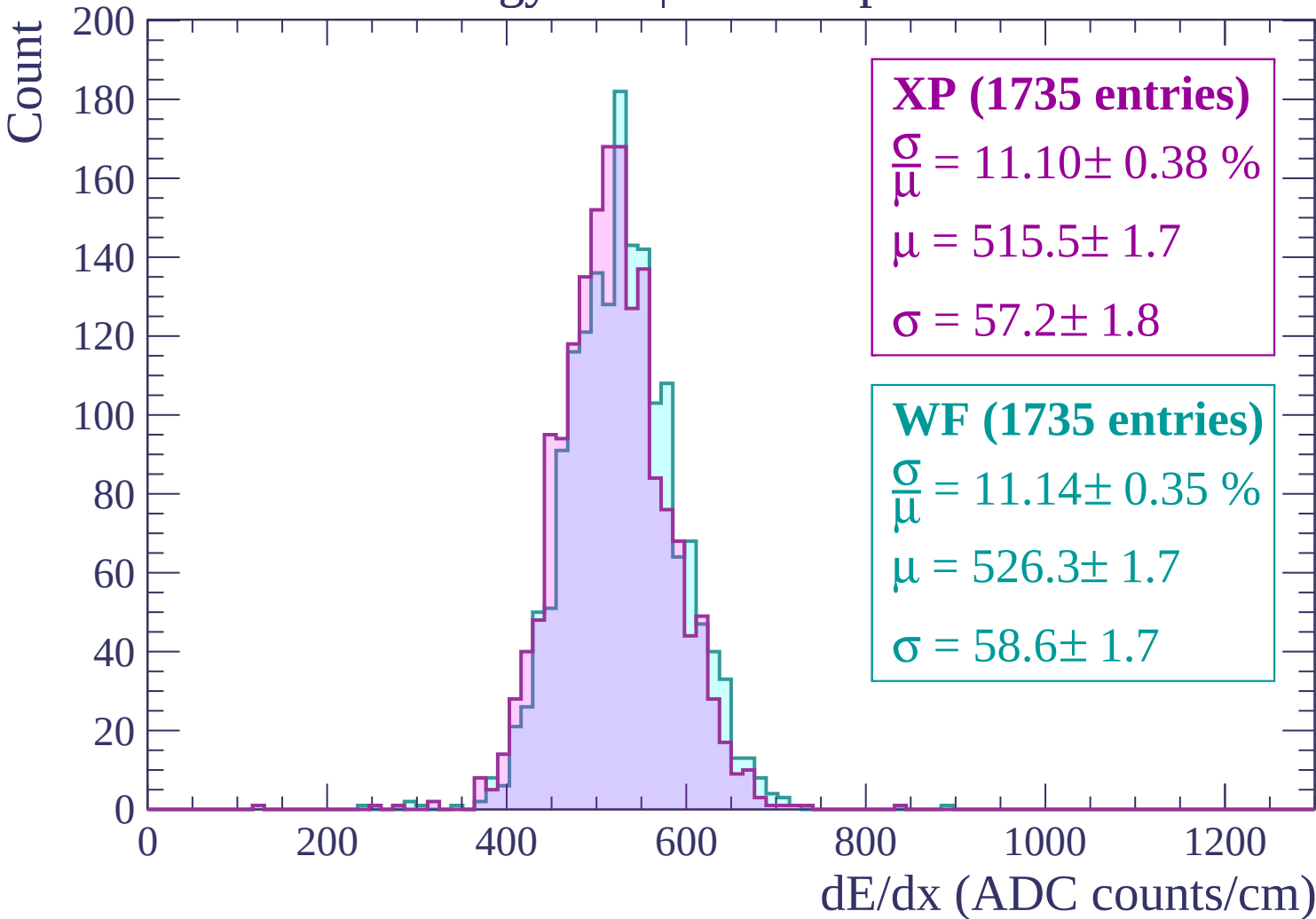


Energy loss | $-2580 < p < -2520$

Count

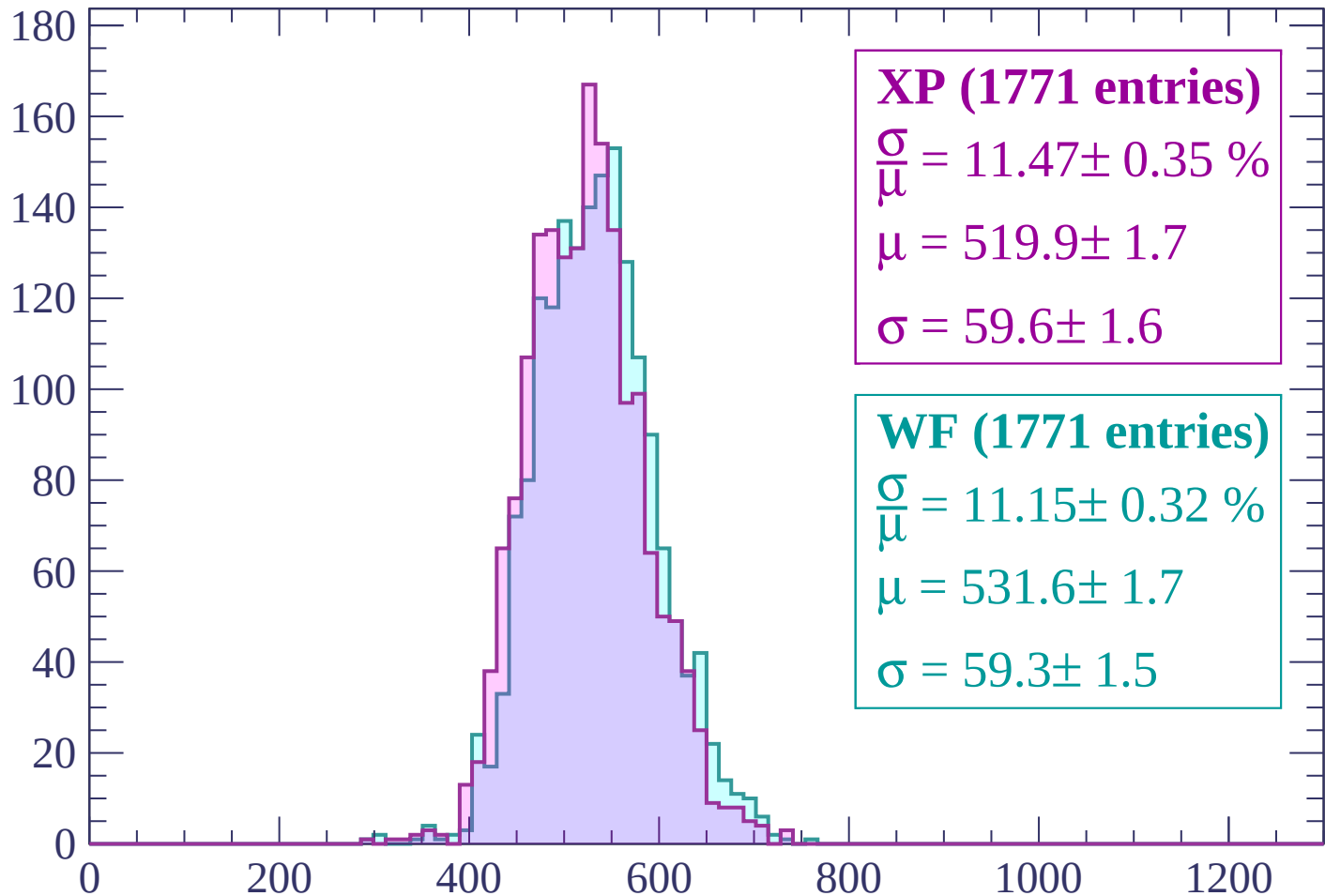


Energy loss | $-2520 < p < -2460$



Energy loss | $-2460 < p < -2400$

Count



XP (1771 entries)

$\frac{\sigma}{\mu} = 11.47 \pm 0.35 \%$

$\mu = 519.9 \pm 1.7$

$\sigma = 59.6 \pm 1.6$

WF (1771 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.32 \%$

$\mu = 531.6 \pm 1.7$

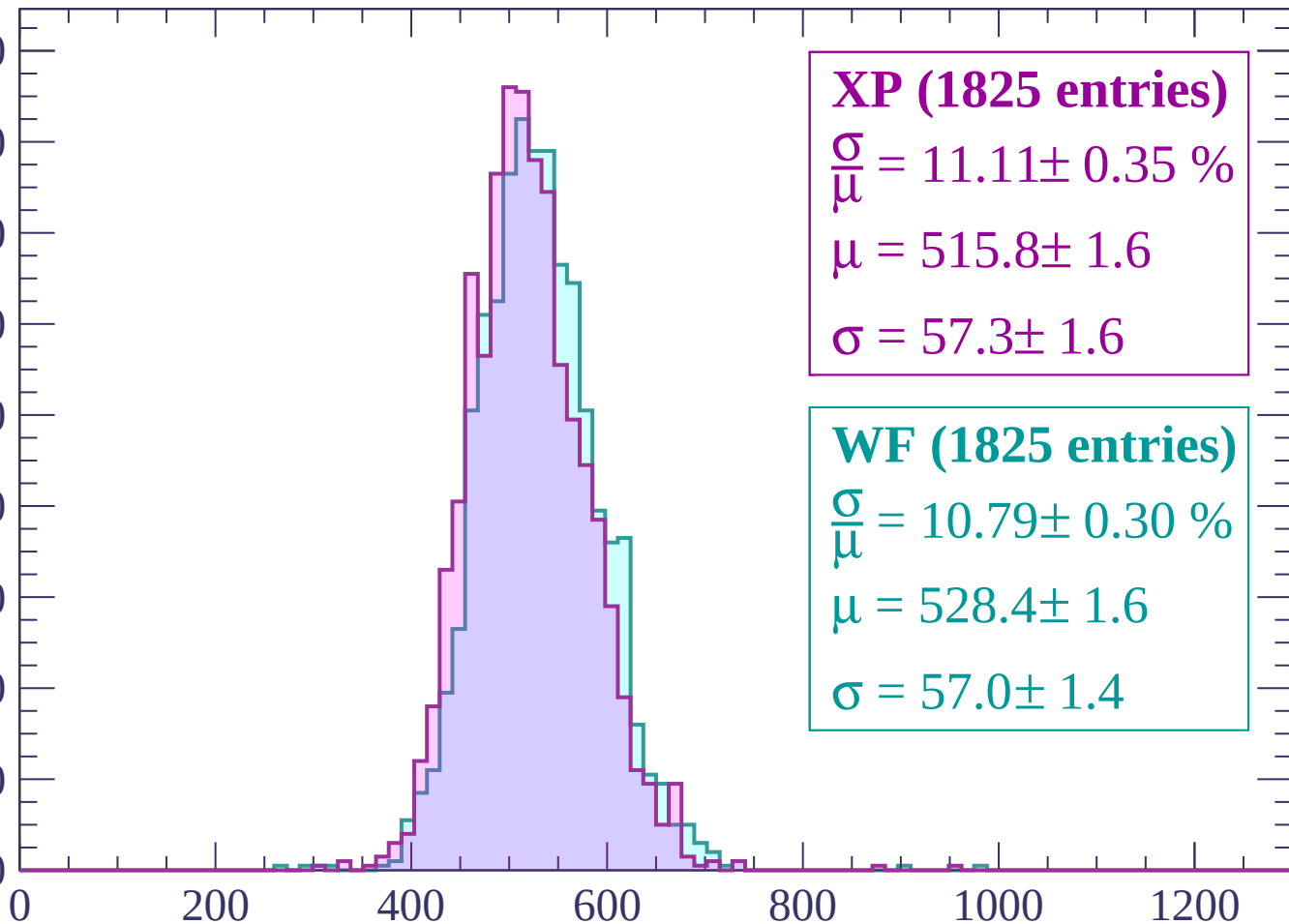
$\sigma = 59.3 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | $-2400 < p < -2340$

Count

180
160
140
120
100
80
60
40
20
0



XP (1825 entries)

$$\frac{\sigma}{\mu} = 11.11 \pm 0.35 \%$$

$$\mu = 515.8 \pm 1.6$$

$$\sigma = 57.3 \pm 1.6$$

WF (1825 entries)

$$\frac{\sigma}{\mu} = 10.79 \pm 0.30 \%$$

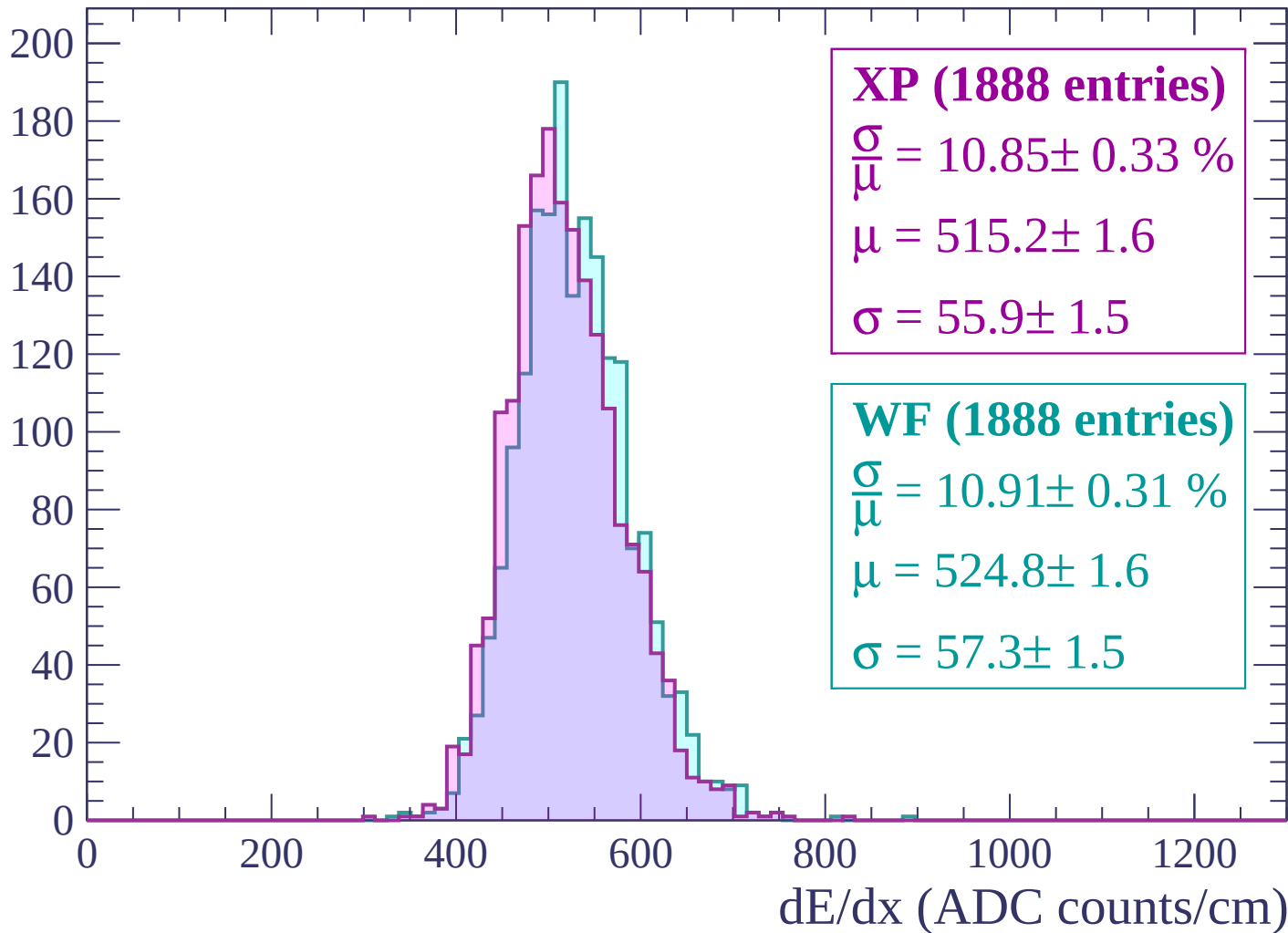
$$\mu = 528.4 \pm 1.6$$

$$\sigma = 57.0 \pm 1.4$$

dE/dx (ADC counts/cm)

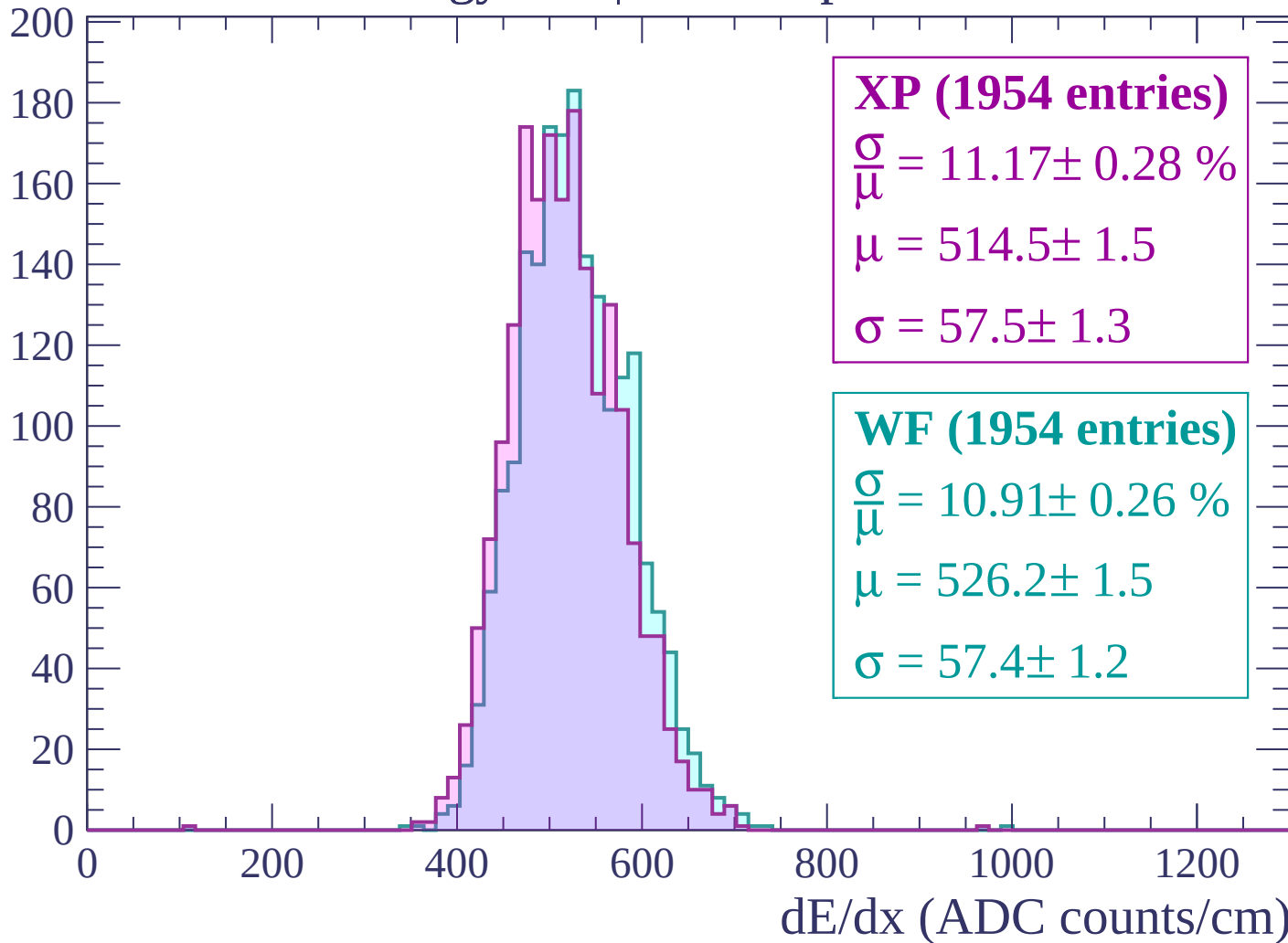
Energy loss | $-2340 < p < -2280$

Count



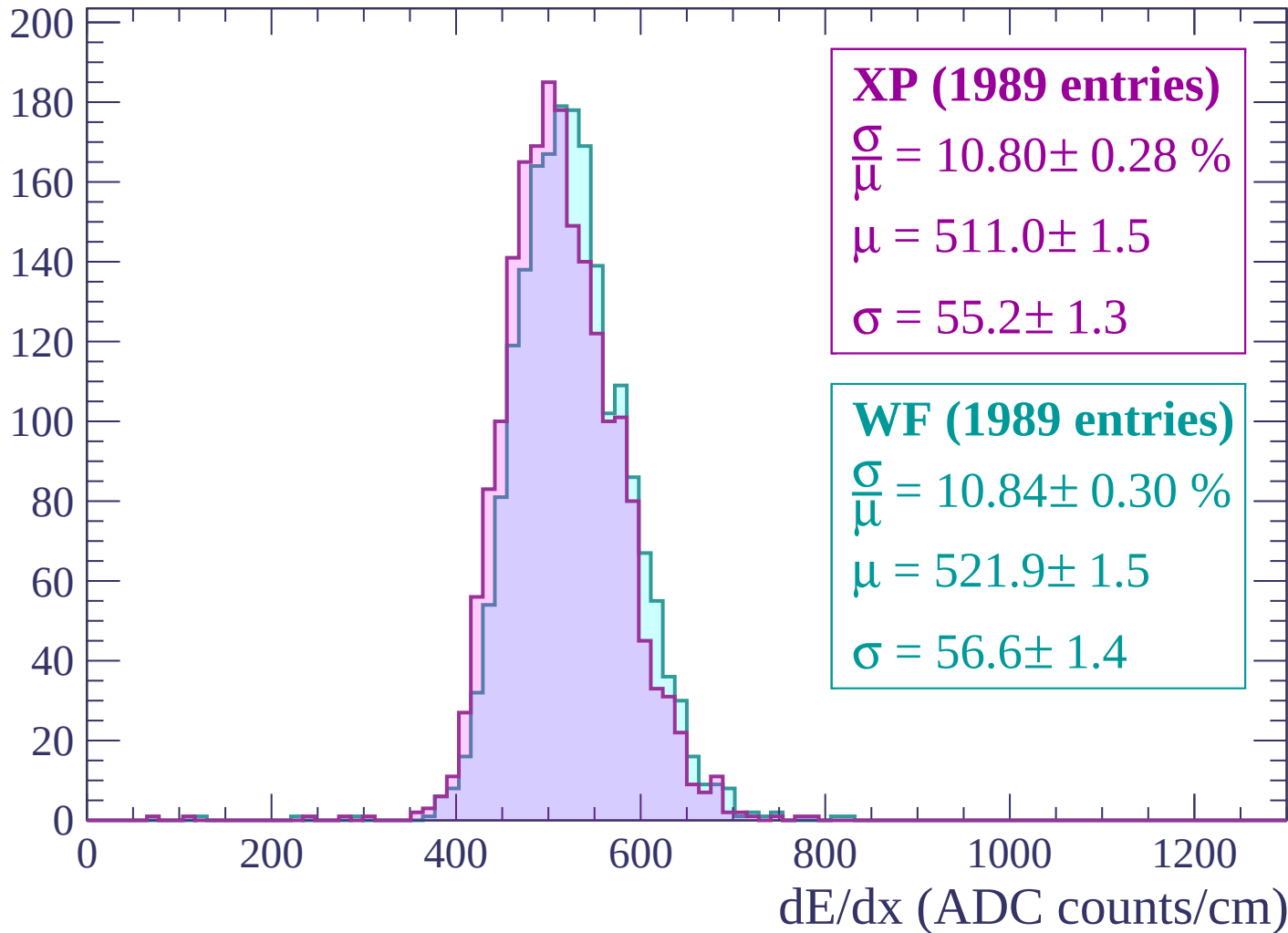
Energy loss | $-2280 < p < -2220$

Count



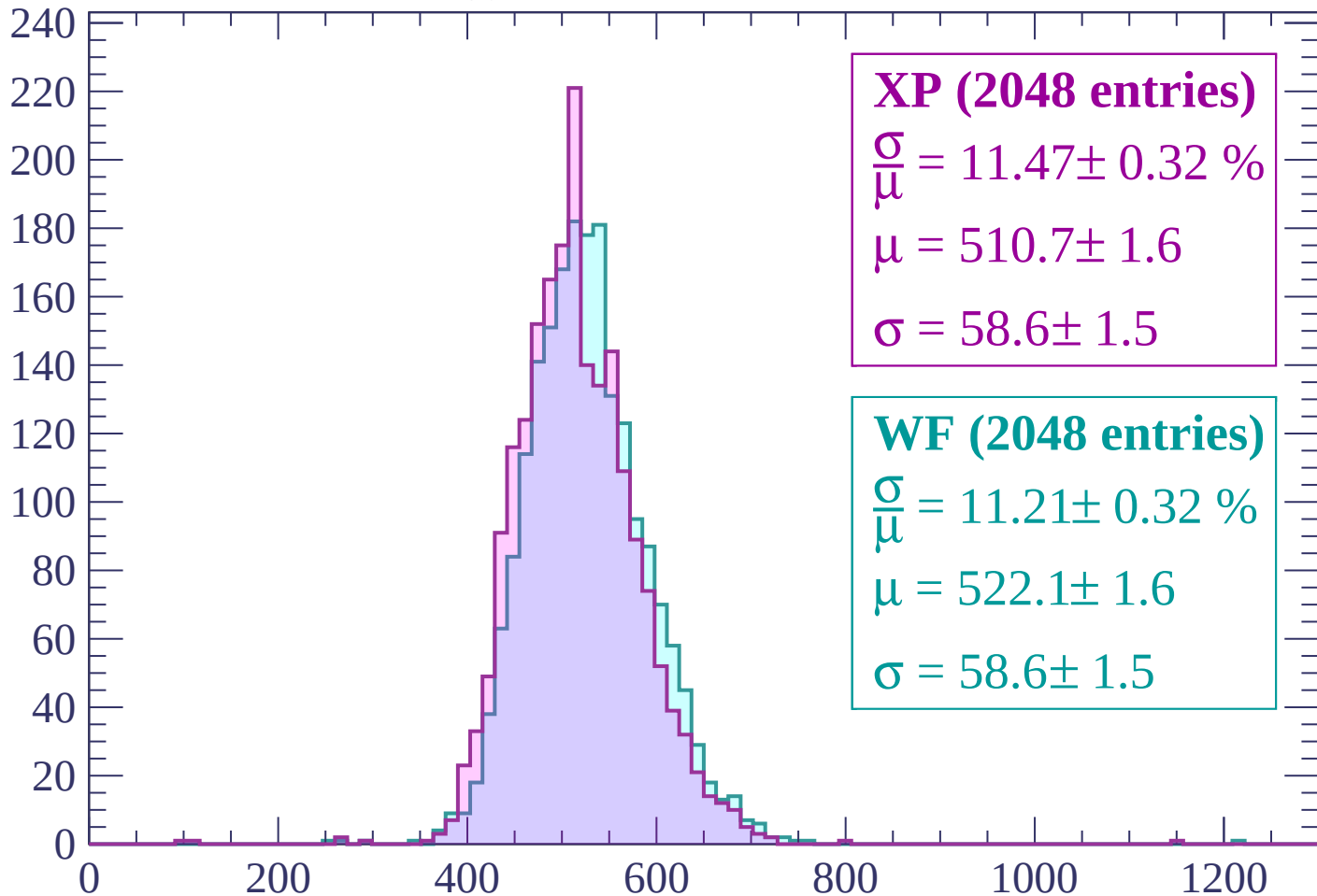
Energy loss | -2220 < p < -2160

Count



Energy loss | -2160 < p < -2100

Count



XP (2048 entries)

$$\frac{\sigma}{\mu} = 11.47 \pm 0.32 \%$$

$$\mu = 510.7 \pm 1.6$$

$$\sigma = 58.6 \pm 1.5$$

WF (2048 entries)

$$\frac{\sigma}{\mu} = 11.21 \pm 0.32 \%$$

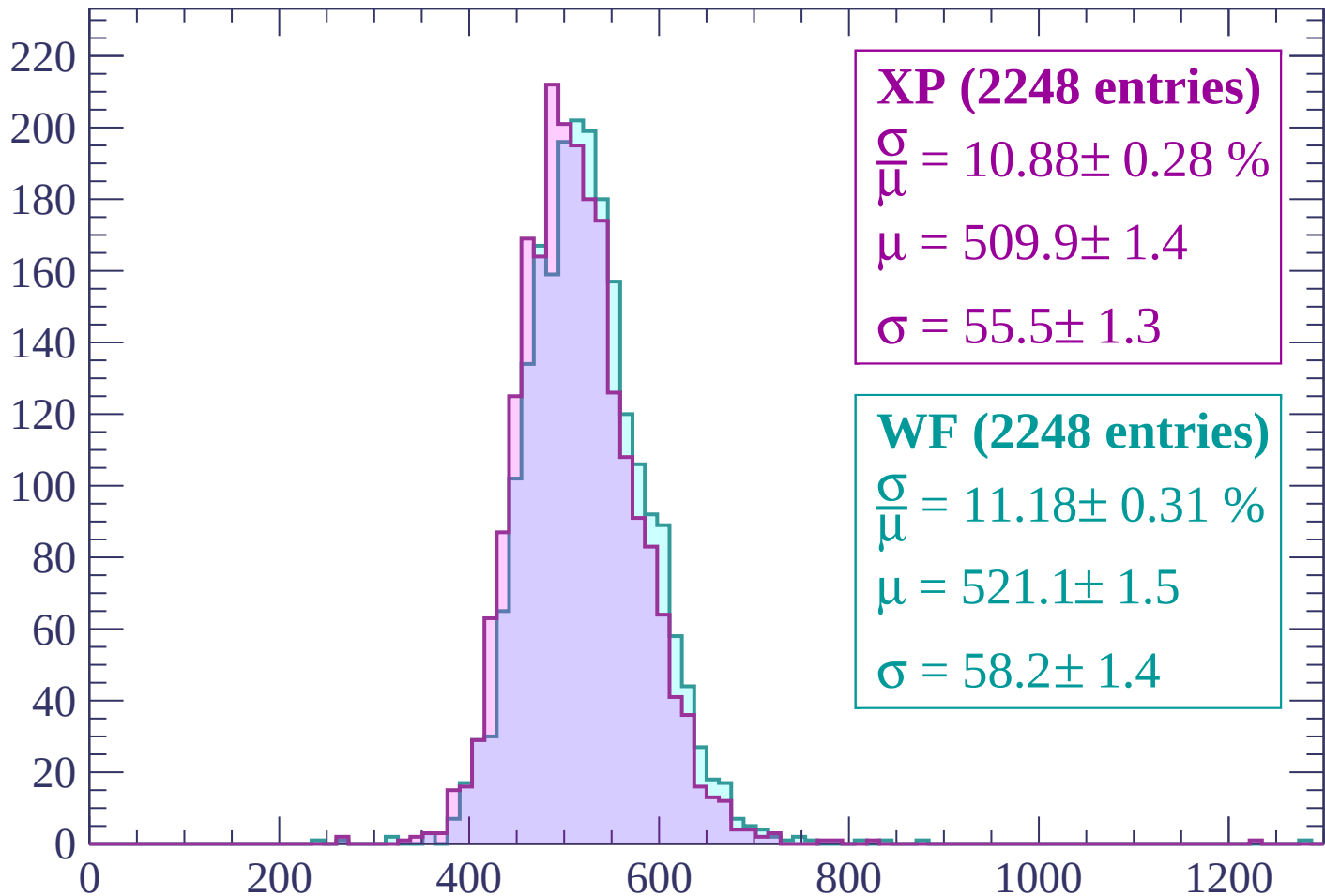
$$\mu = 522.1 \pm 1.6$$

$$\sigma = 58.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP (2248 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.28 \%$$

$$\mu = 509.9 \pm 1.4$$

$$\sigma = 55.5 \pm 1.3$$

WF (2248 entries)

$$\frac{\sigma}{\mu} = 11.18 \pm 0.31 \%$$

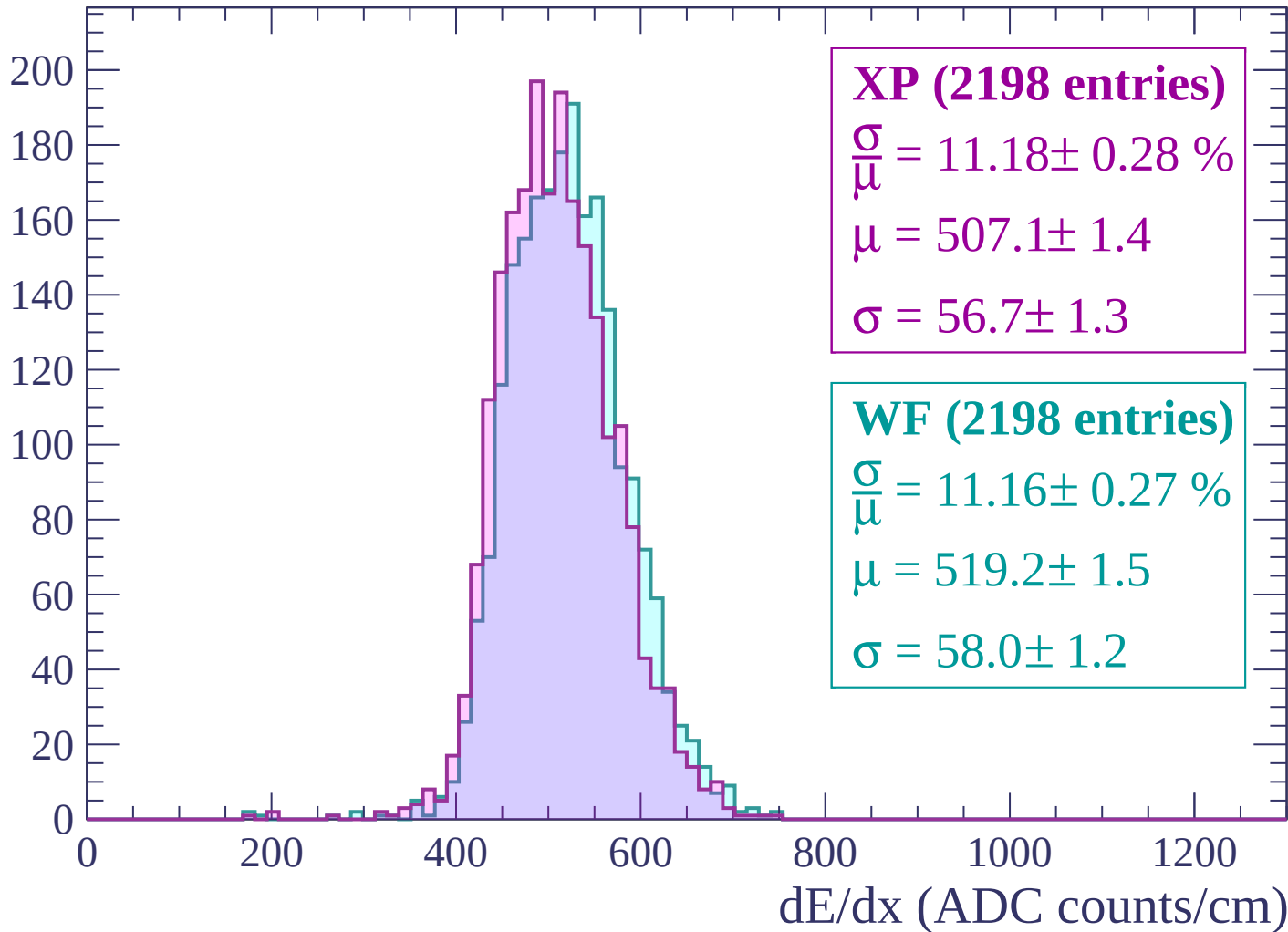
$$\mu = 521.1 \pm 1.5$$

$$\sigma = 58.2 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

XP (2336 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.28 \%$

$\mu = 502.7 \pm 1.4$

$\sigma = 54.0 \pm 1.3$

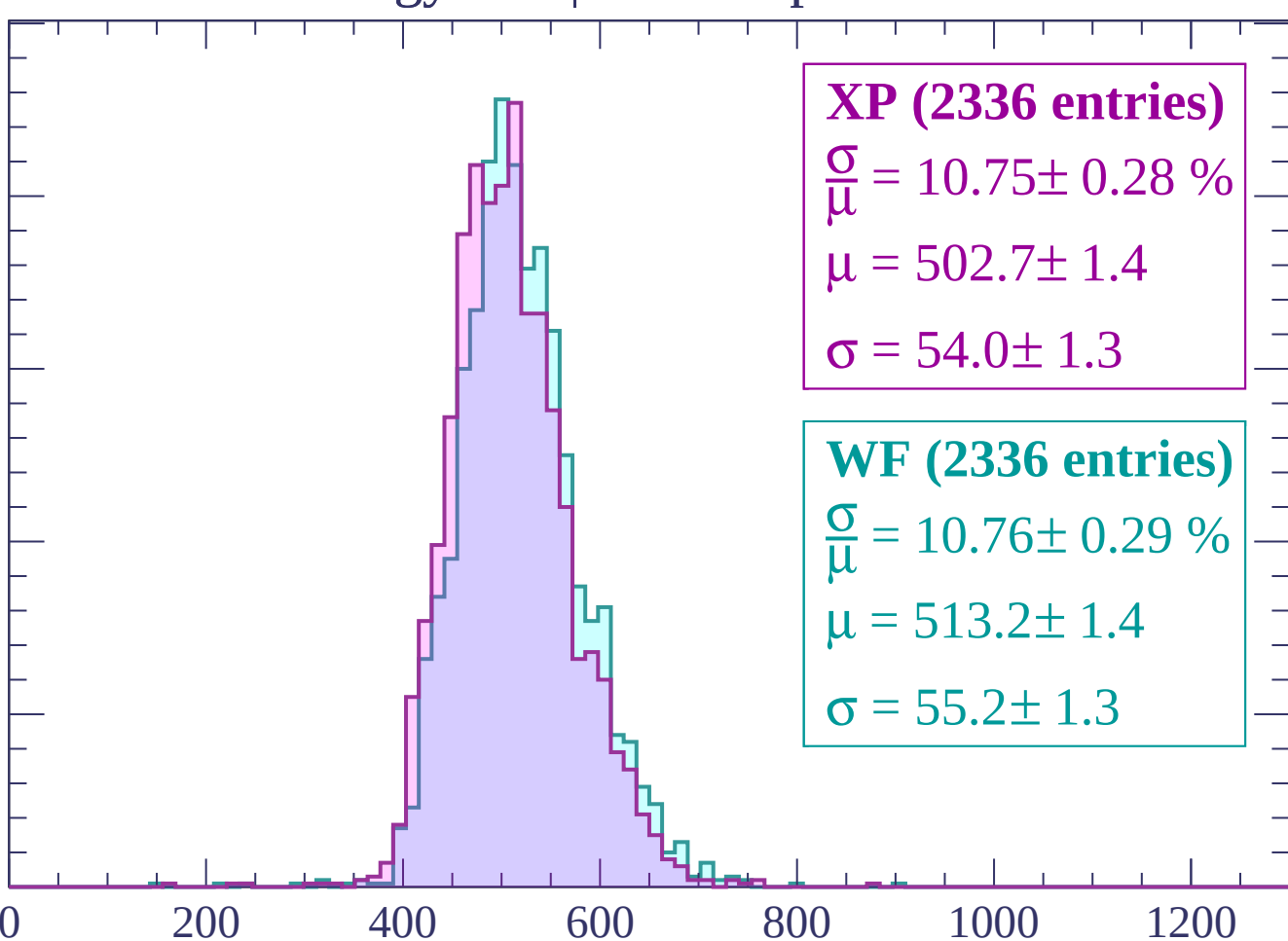
WF (2336 entries)

$\frac{\sigma}{\mu} = 10.76 \pm 0.29 \%$

$\mu = 513.2 \pm 1.4$

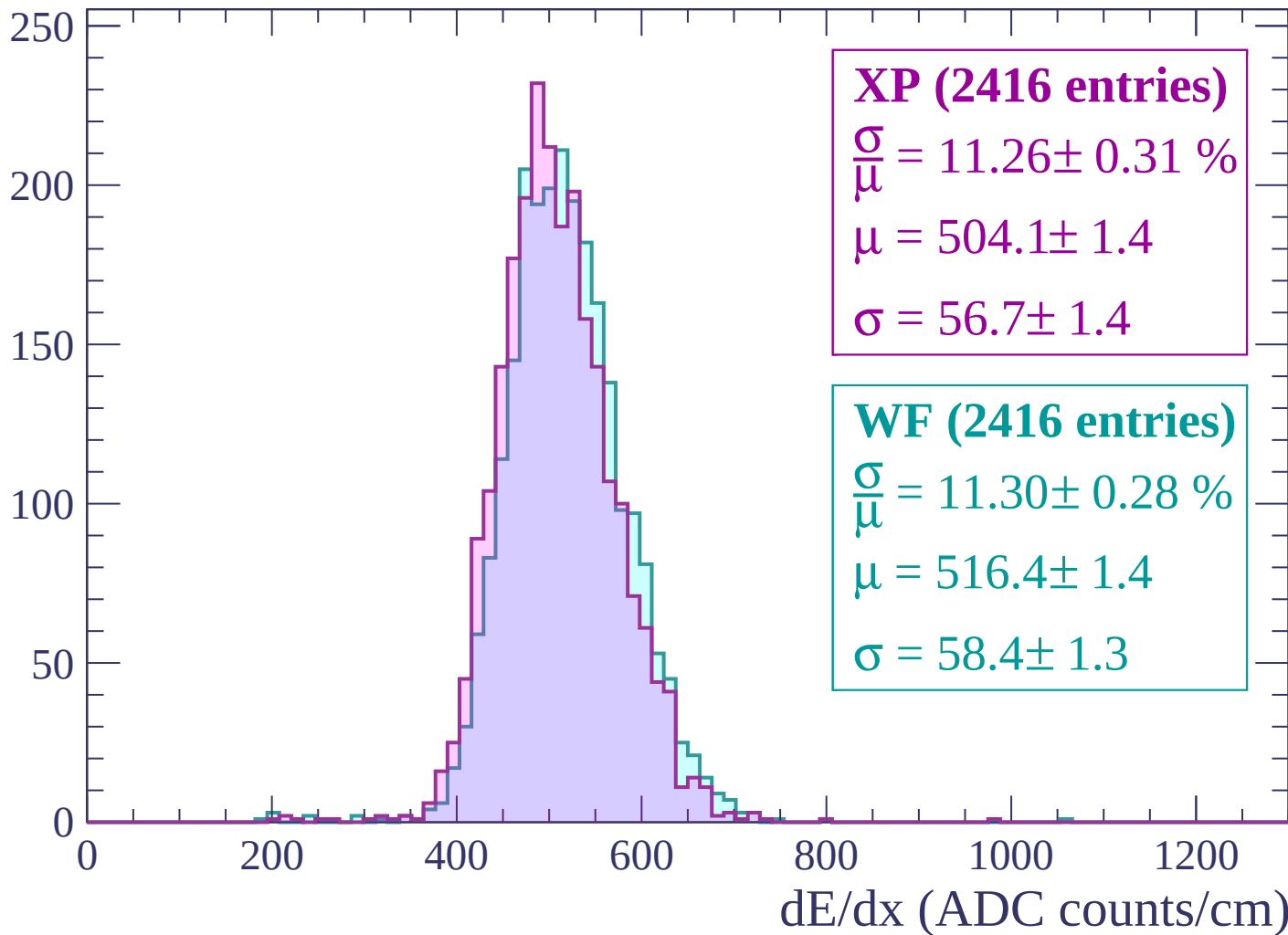
$\sigma = 55.2 \pm 1.3$

dE/dx (ADC counts/cm)



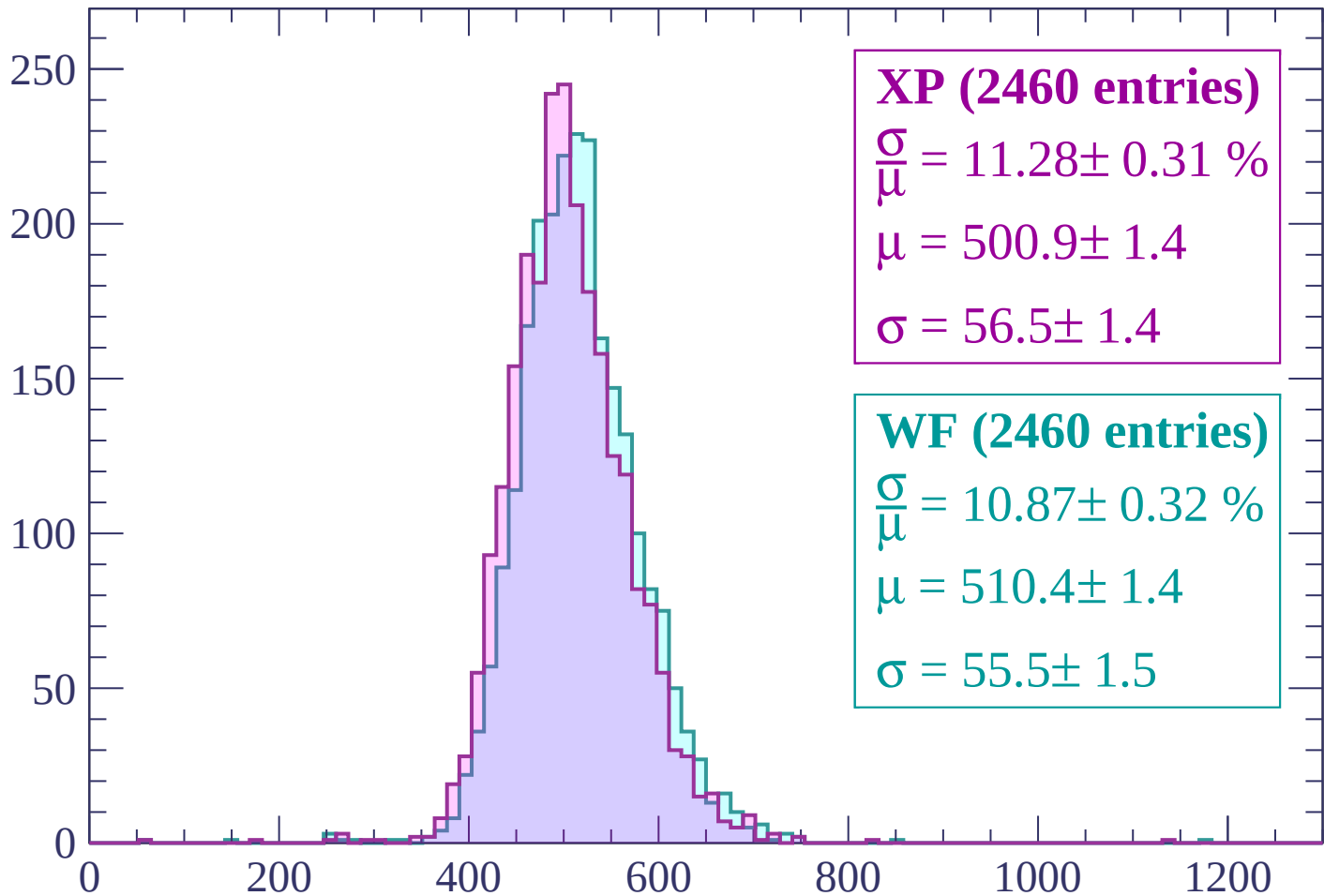
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP (2460 entries)

$\frac{\sigma}{\mu} = 11.28 \pm 0.31 \%$

$\mu = 500.9 \pm 1.4$

$\sigma = 56.5 \pm 1.4$

WF (2460 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.32 \%$

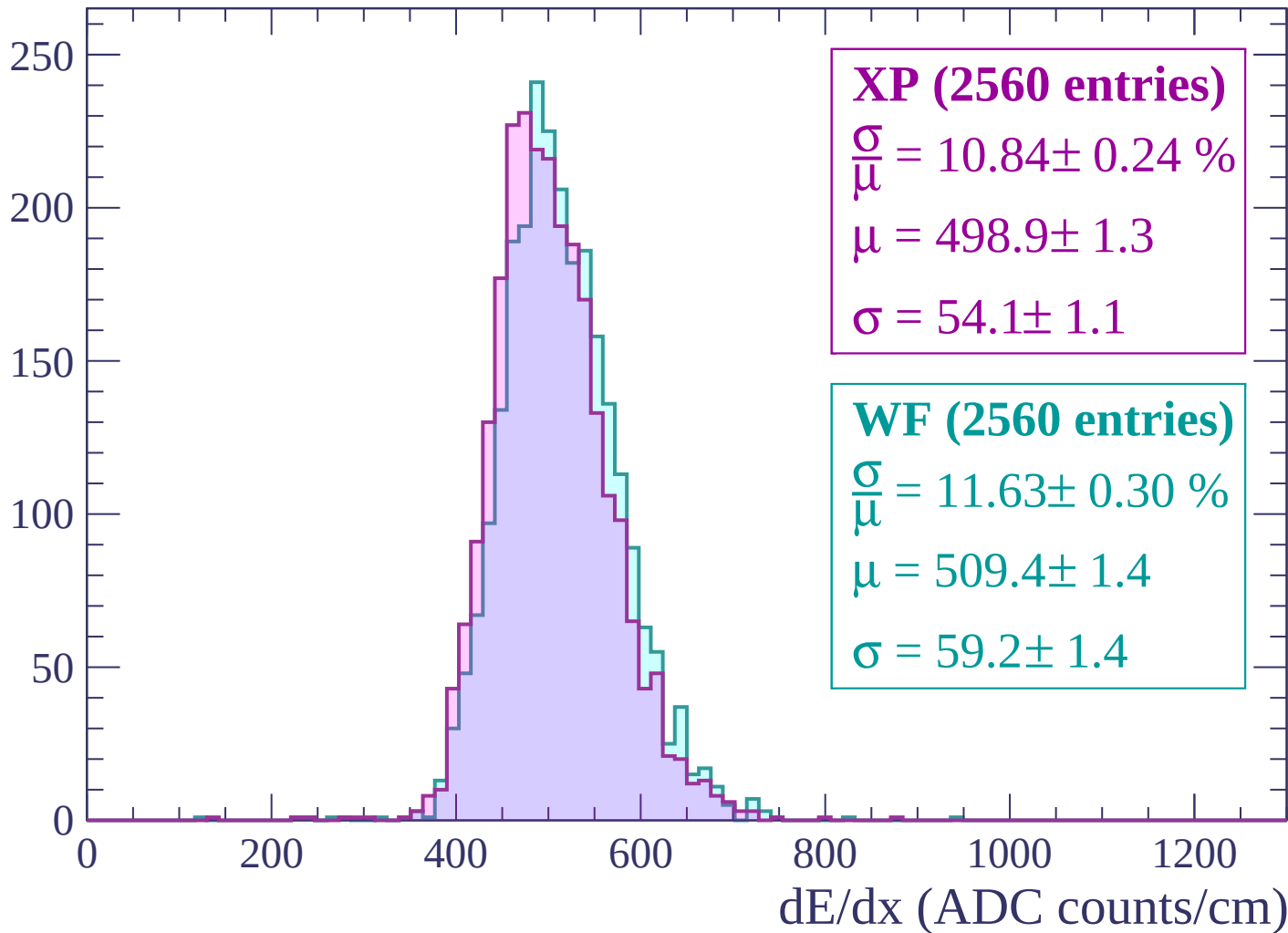
$\mu = 510.4 \pm 1.4$

$\sigma = 55.5 \pm 1.5$

dE/dx (ADC counts/cm)

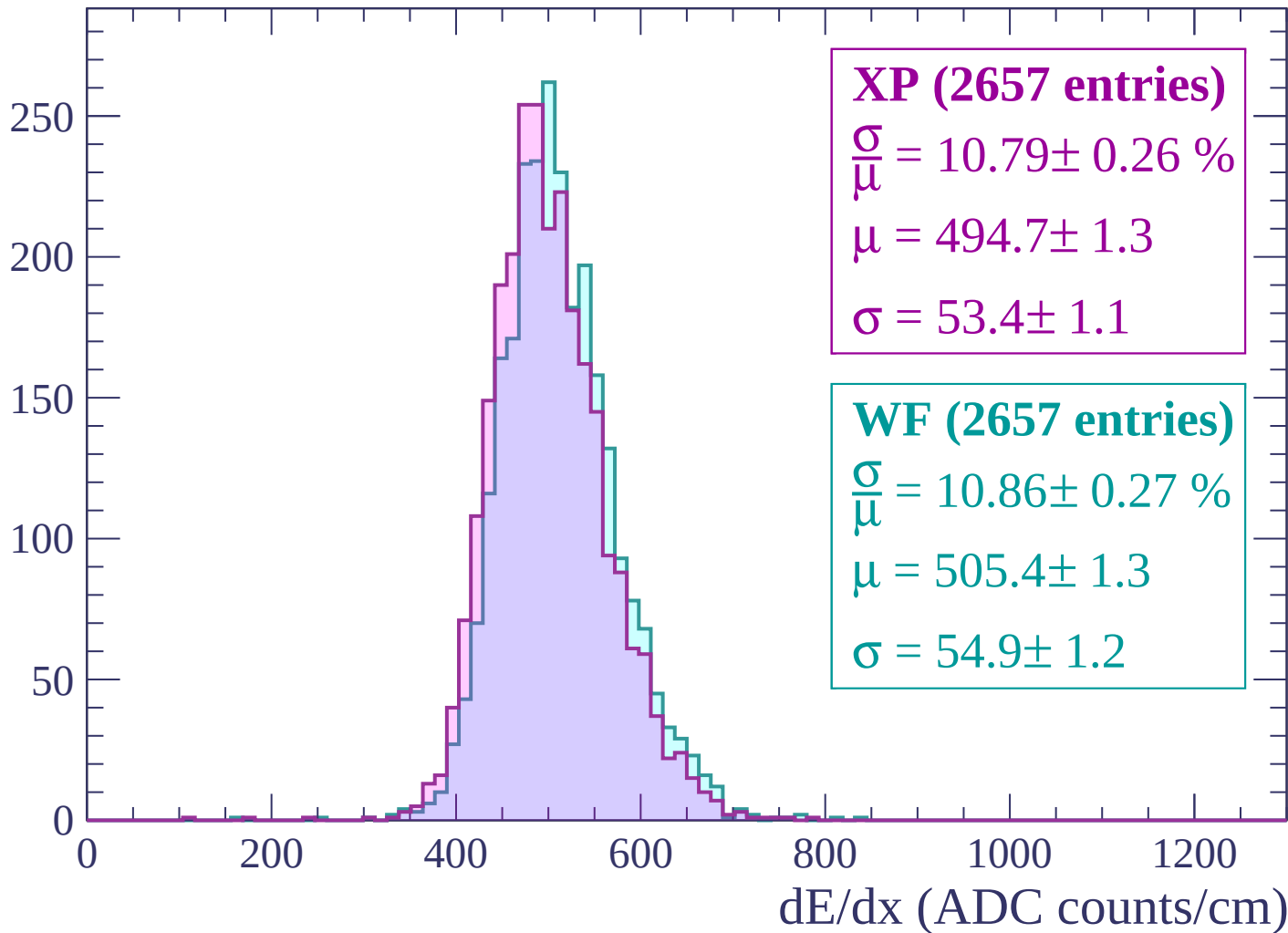
Energy loss | $-1800 < p < -1740$

Count



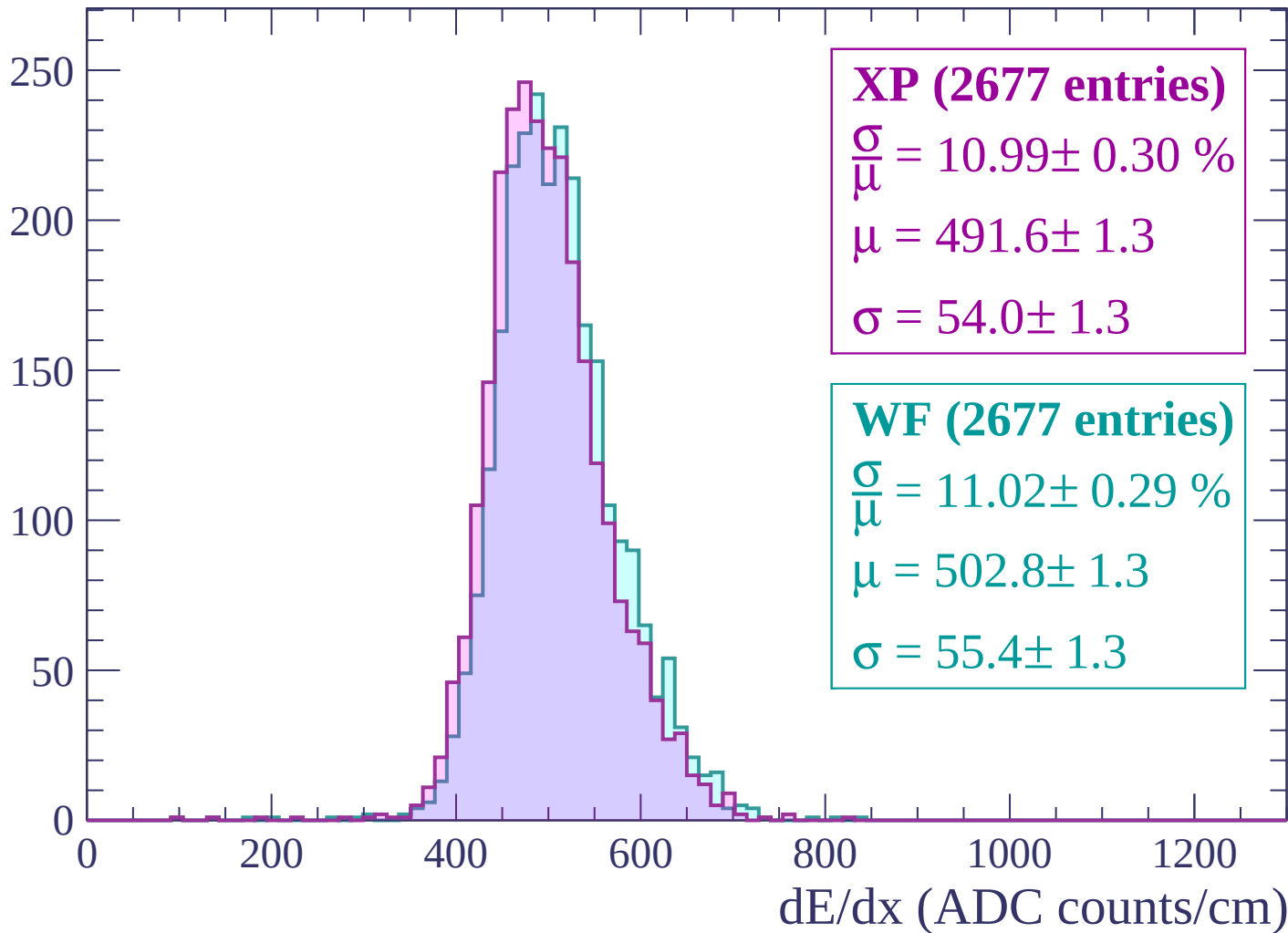
Energy loss | $-1740 < p < -1680$

Count



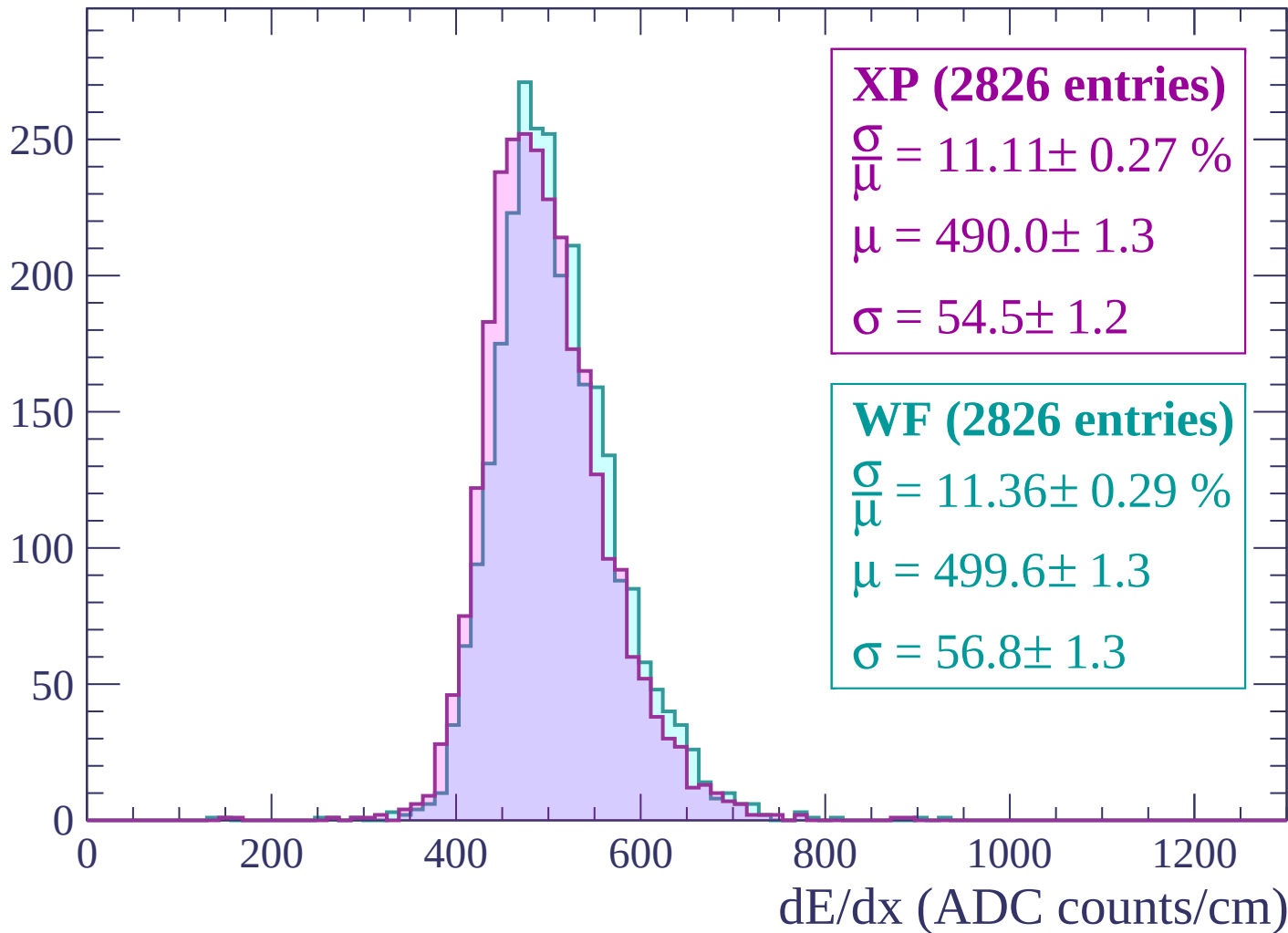
Energy loss | -1680 < p < -1620

Count



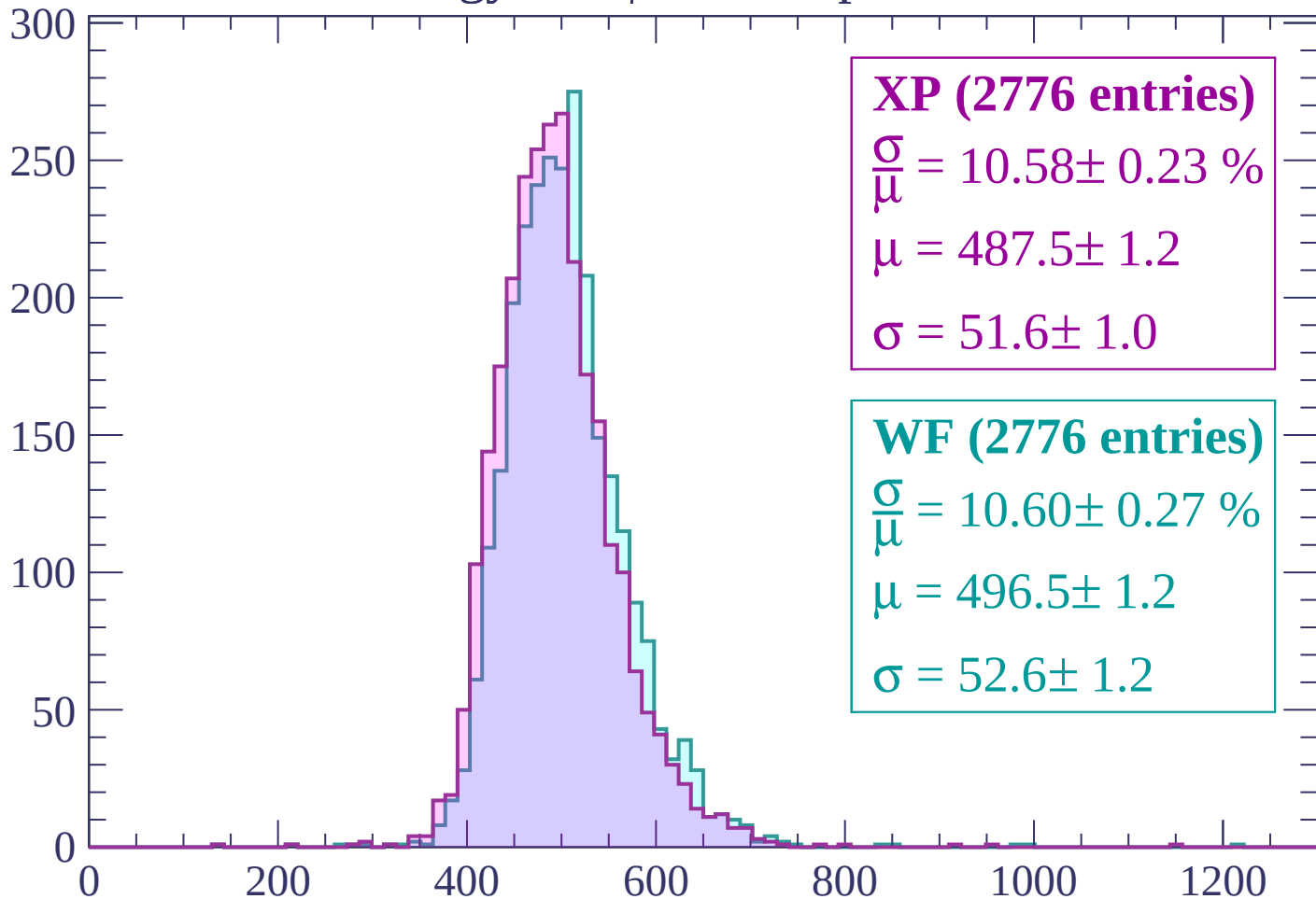
Energy loss | -1620 < p < -1560

Count



Energy loss | -1560 < p < -1500

Count



XP (2776 entries)

$$\frac{\sigma}{\mu} = 10.58 \pm 0.23 \%$$

$$\mu = 487.5 \pm 1.2$$

$$\sigma = 51.6 \pm 1.0$$

WF (2776 entries)

$$\frac{\sigma}{\mu} = 10.60 \pm 0.27 \%$$

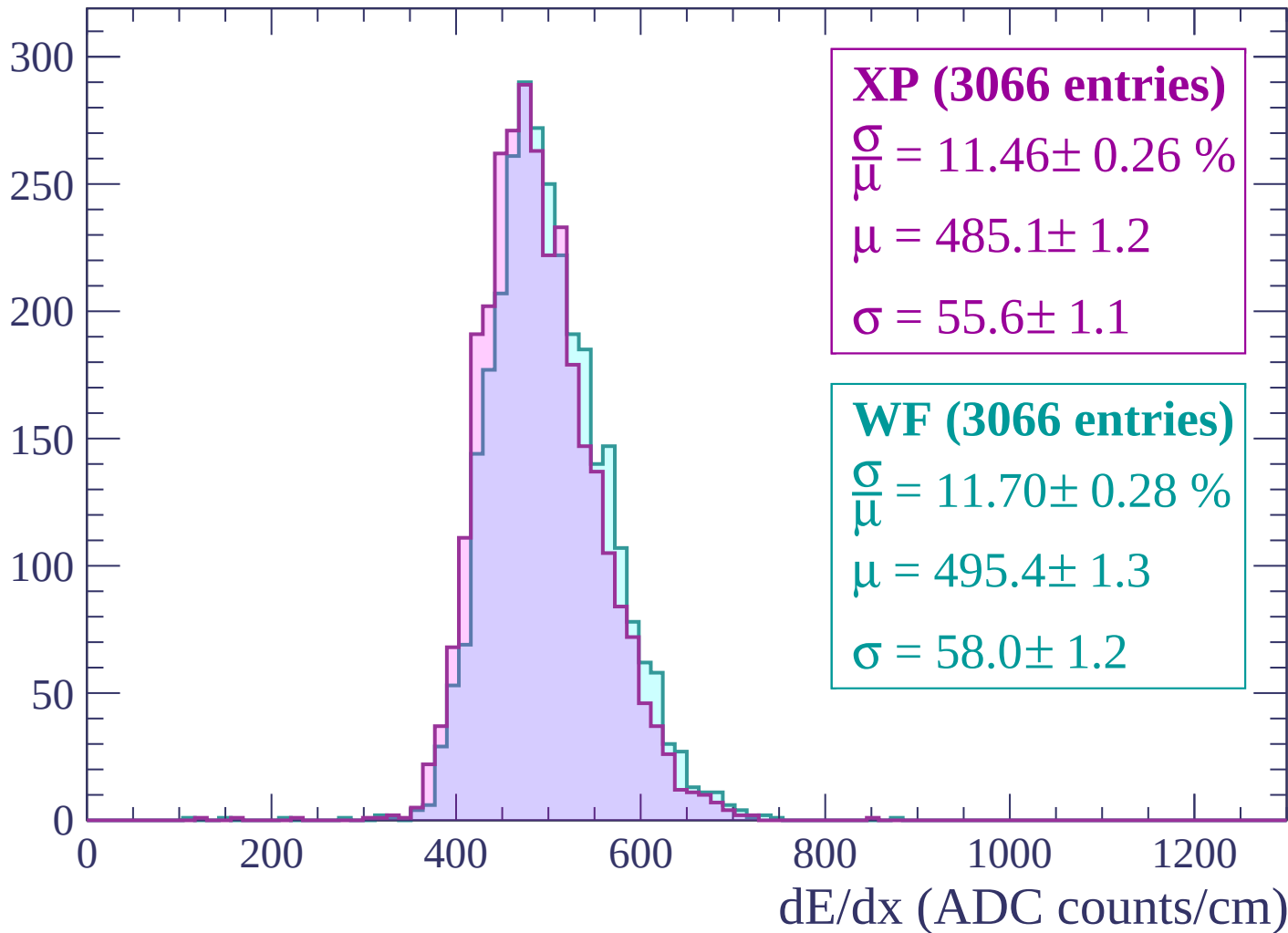
$$\mu = 496.5 \pm 1.2$$

$$\sigma = 52.6 \pm 1.2$$

dE/dx (ADC counts/cm)

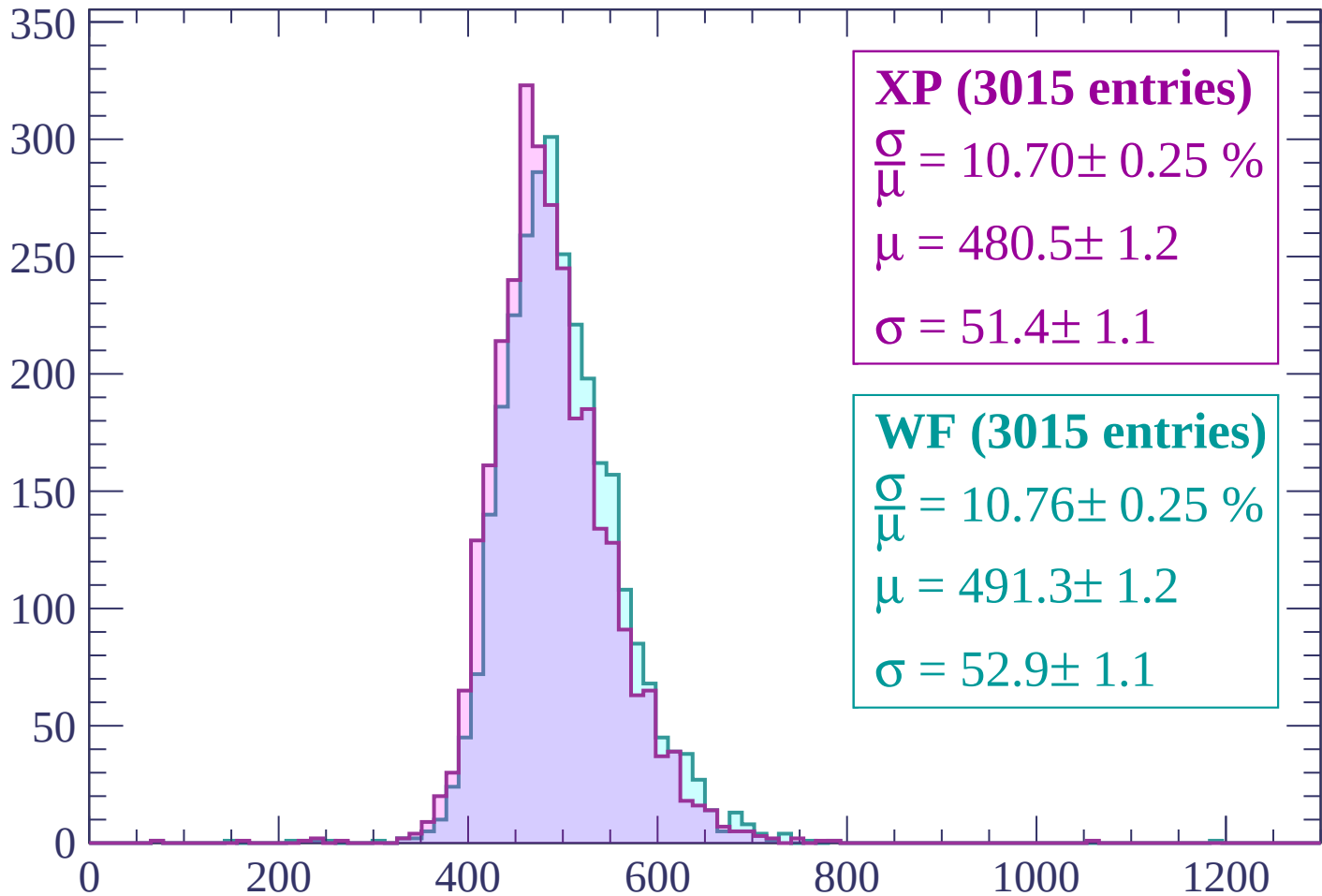
Energy loss | -1500 < p < -1440

Count



Energy loss | $-1440 < p < -1380$

Count



XP (3015 entries)

$$\frac{\sigma}{\mu} = 10.70 \pm 0.25 \%$$

$$\mu = 480.5 \pm 1.2$$

$$\sigma = 51.4 \pm 1.1$$

WF (3015 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.25 \%$$

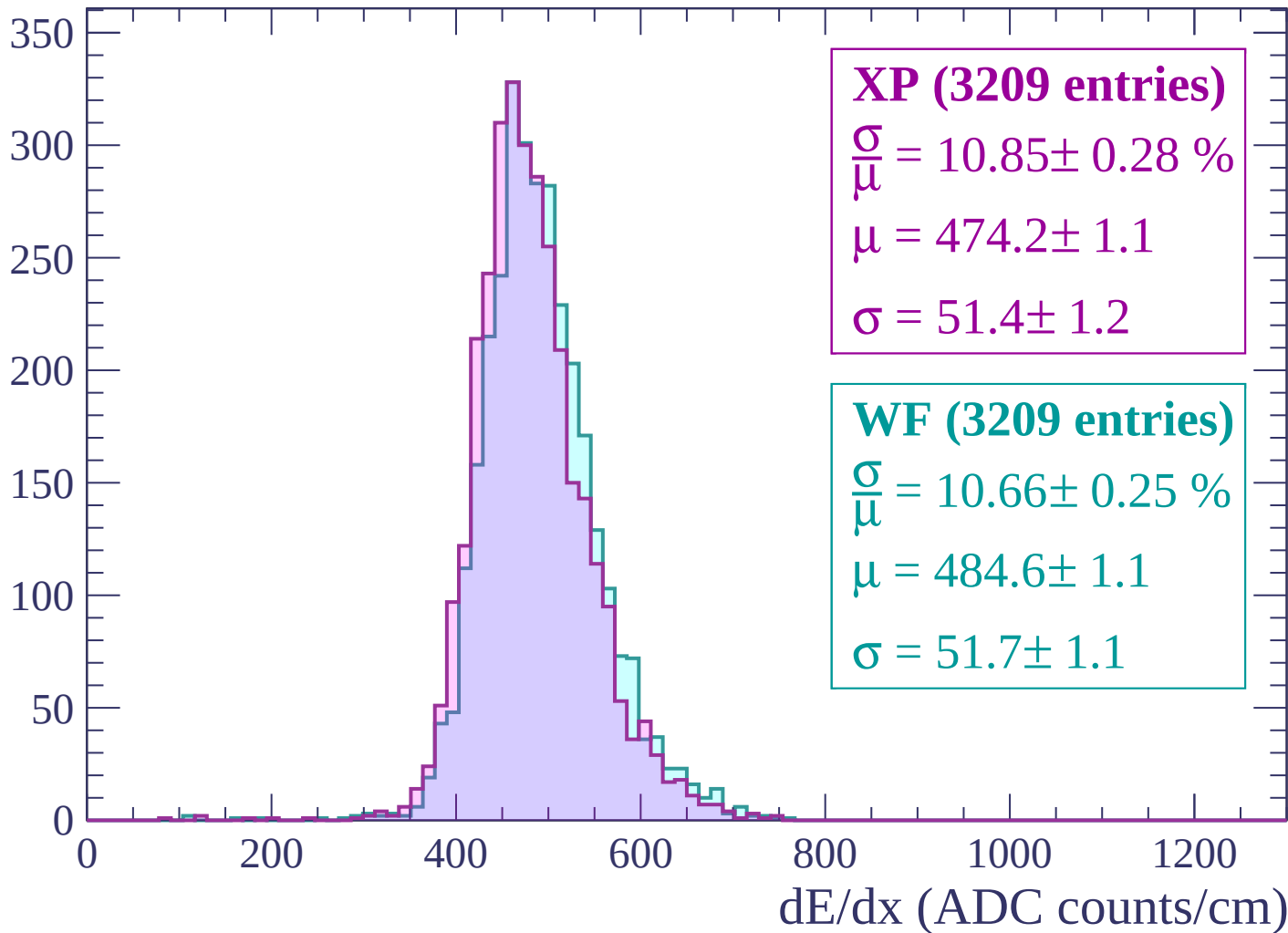
$$\mu = 491.3 \pm 1.2$$

$$\sigma = 52.9 \pm 1.1$$

dE/dx (ADC counts/cm)

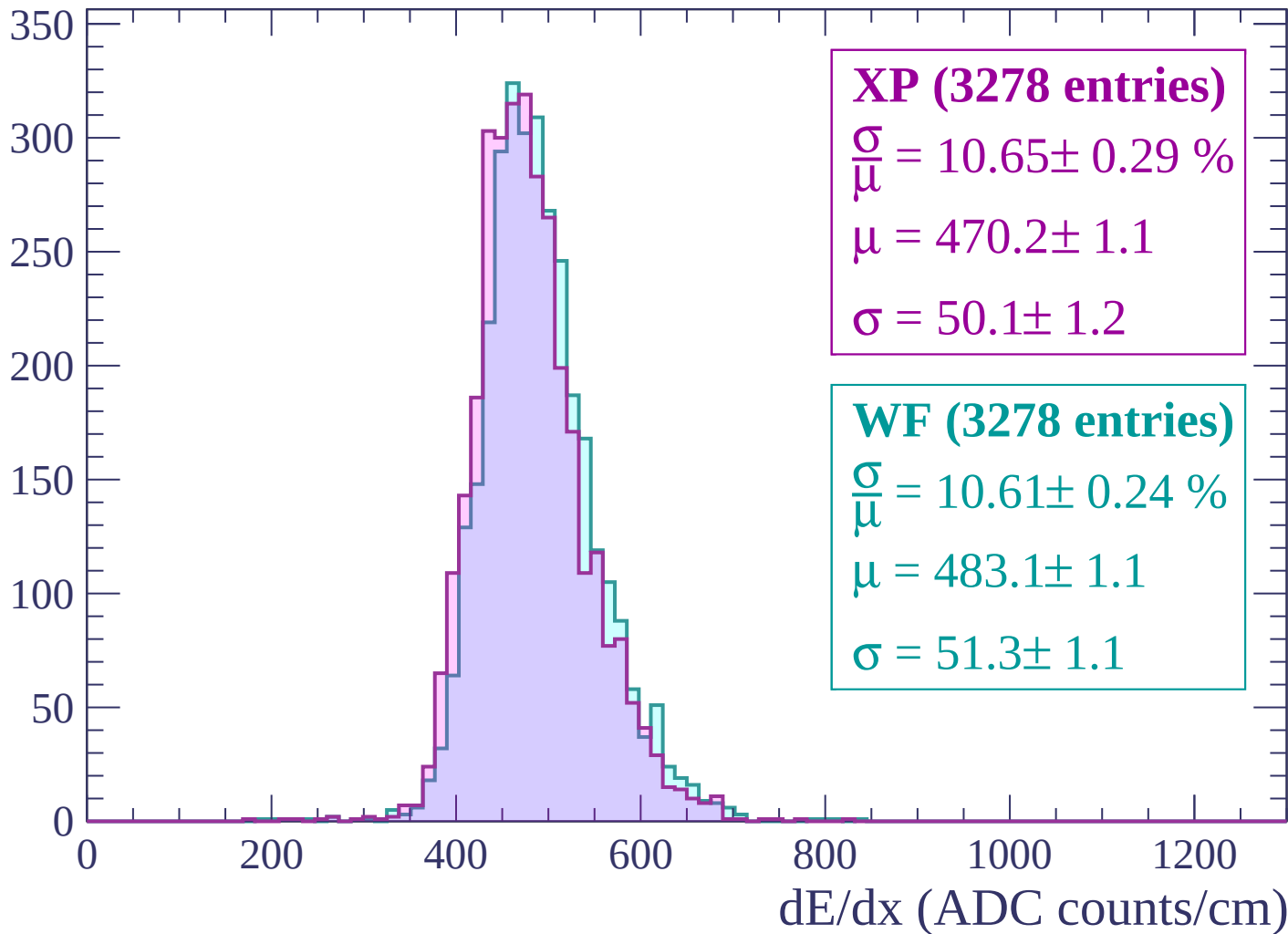
Energy loss | $-1380 < p < -1320$

Count



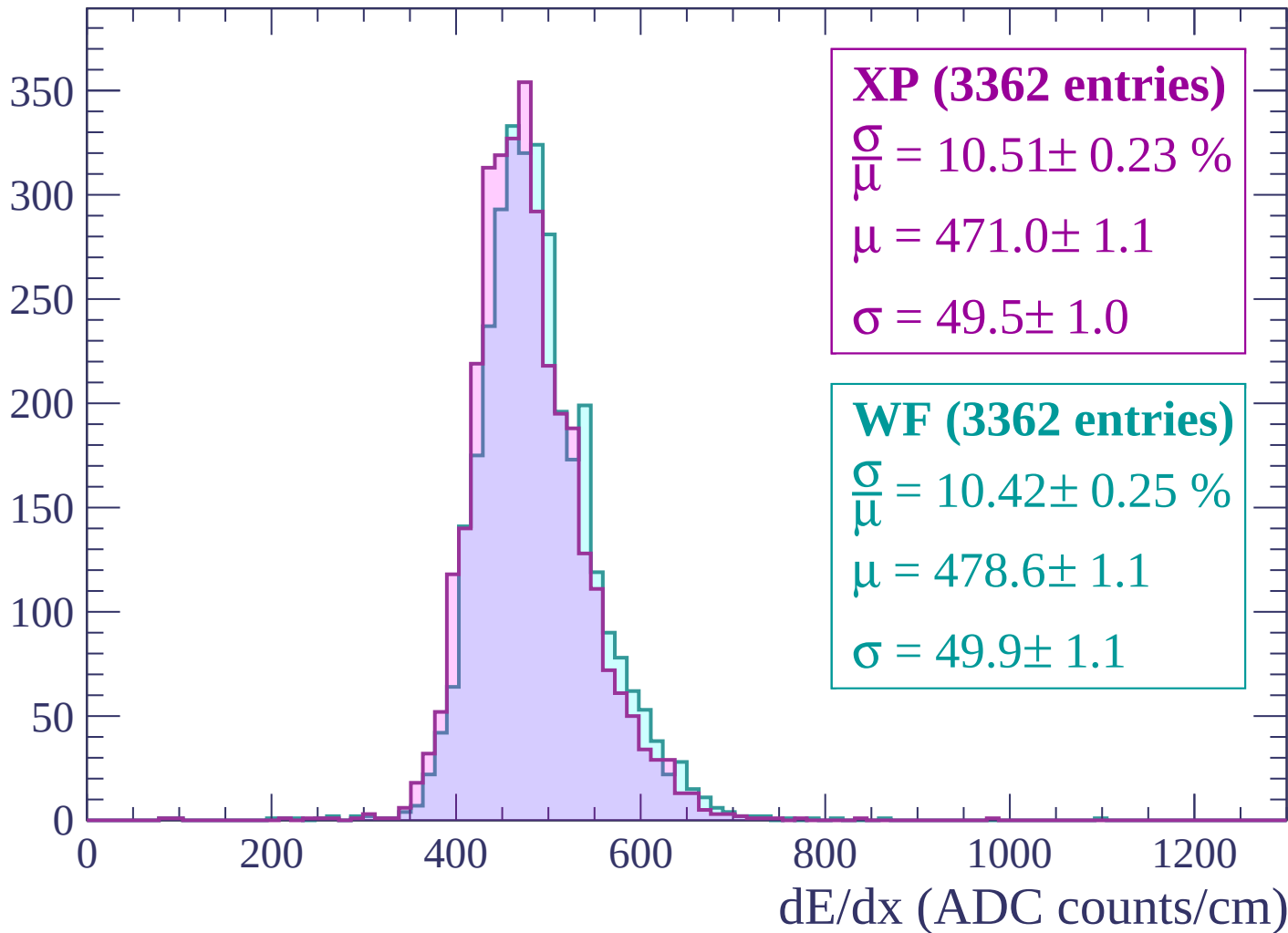
Energy loss | -1320 < p < -1260

Count



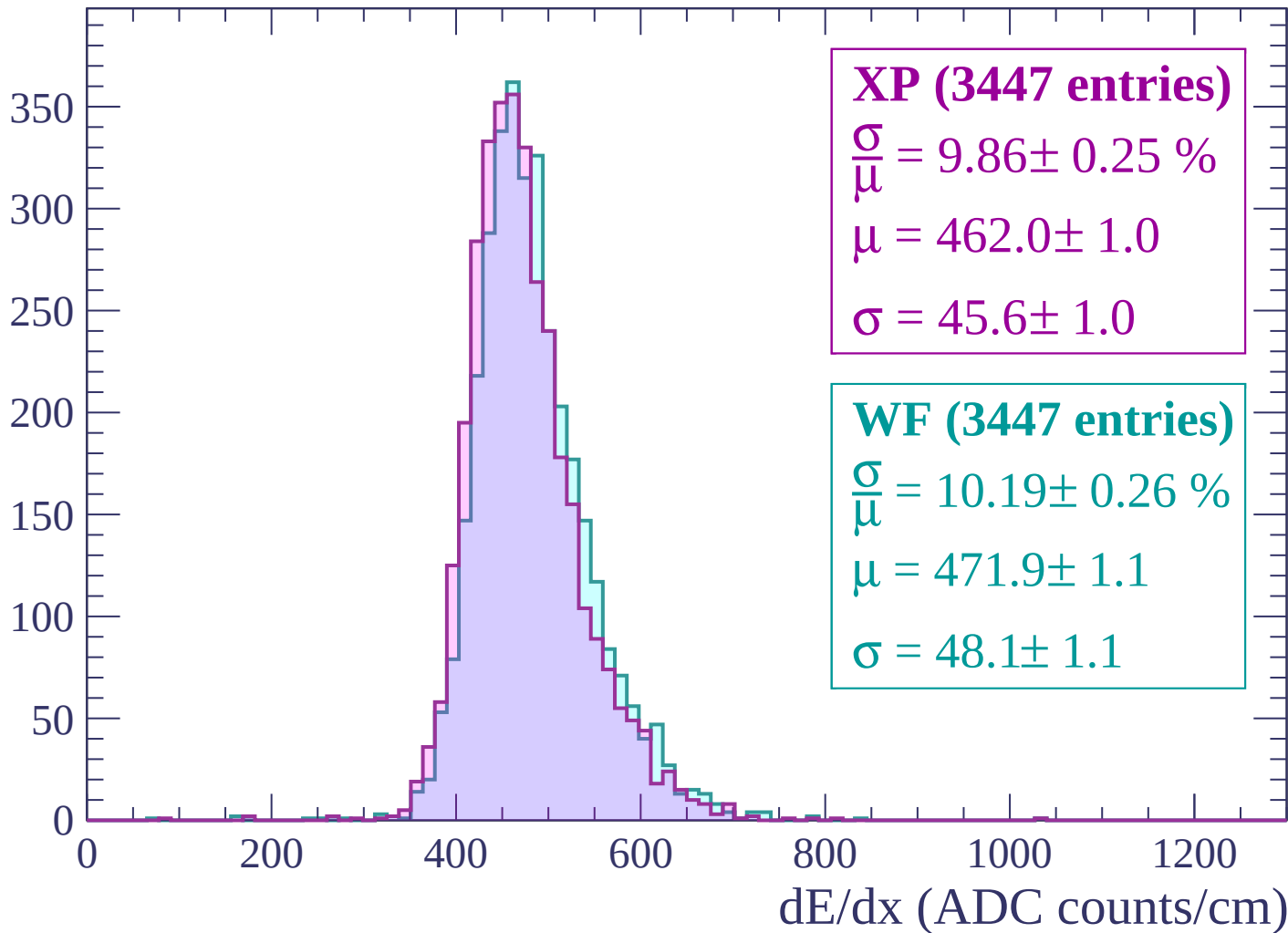
Energy loss | -1260 < p < -1200

Count



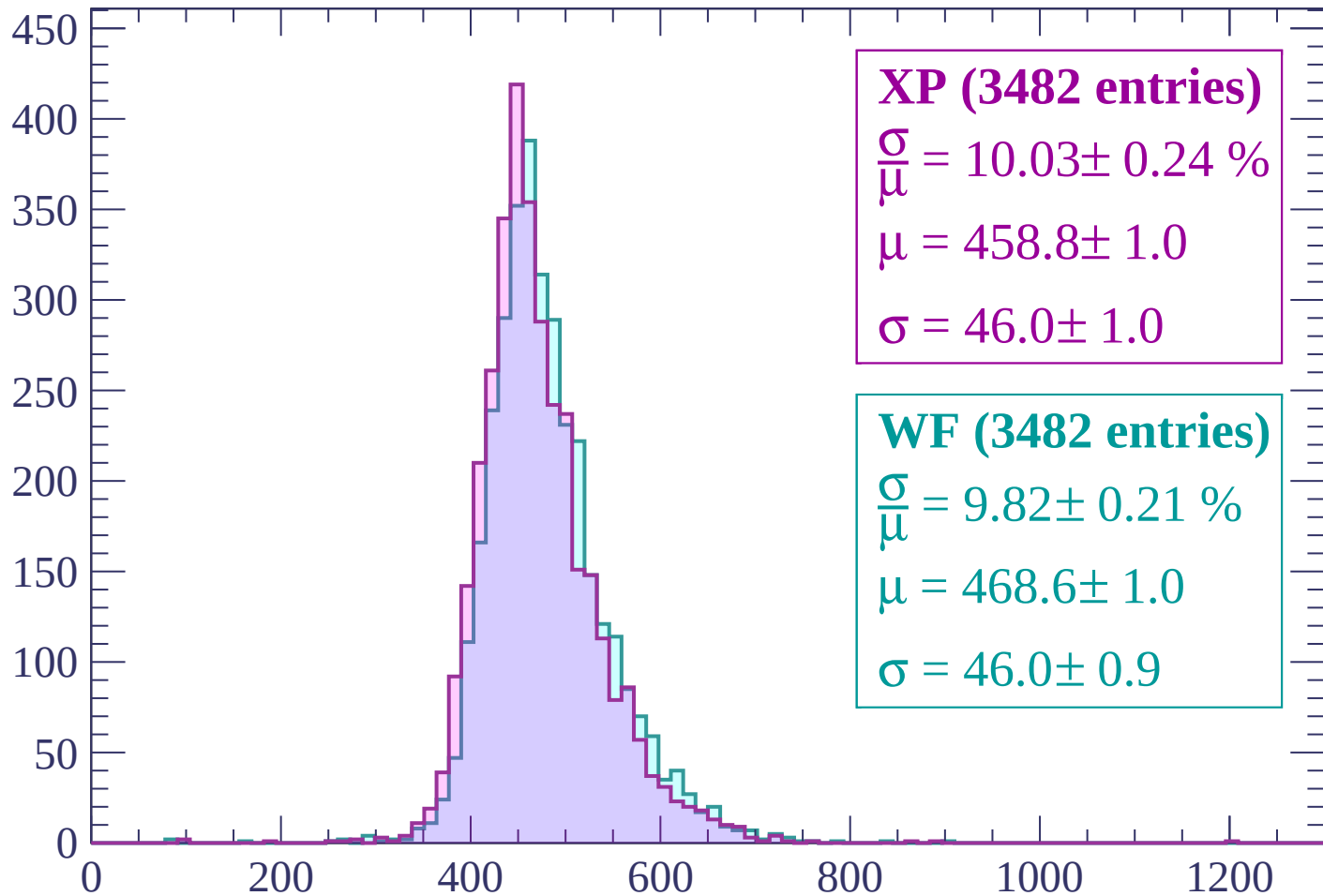
Energy loss | -1200 < p < -1140

Count



Energy loss | -1140 < p < -1080

Count



XP (3482 entries)

$\frac{\sigma}{\mu} = 10.03 \pm 0.24 \%$

$\mu = 458.8 \pm 1.0$

$\sigma = 46.0 \pm 1.0$

WF (3482 entries)

$\frac{\sigma}{\mu} = 9.82 \pm 0.21 \%$

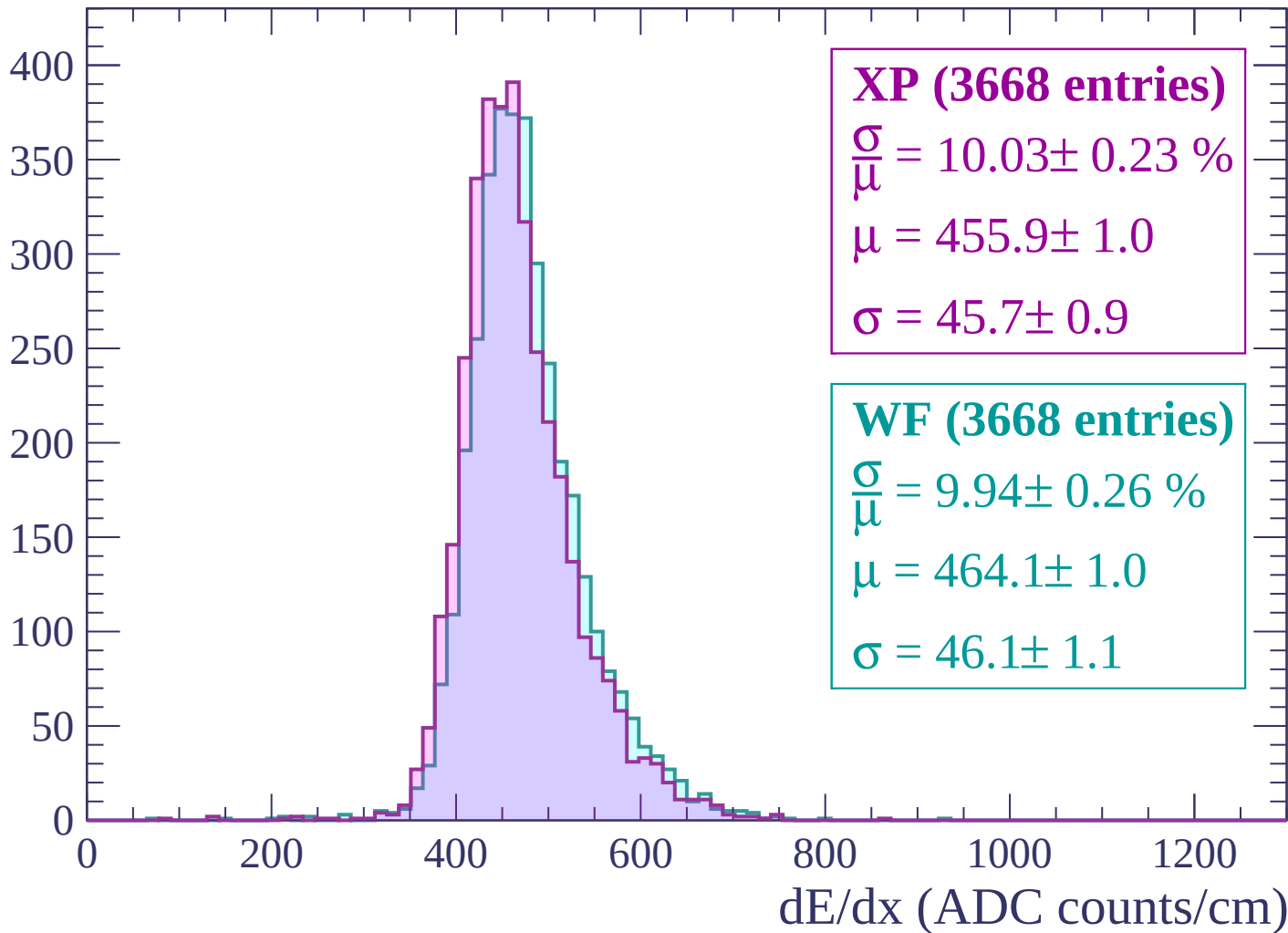
$\mu = 468.6 \pm 1.0$

$\sigma = 46.0 \pm 0.9$

dE/dx (ADC counts/cm)

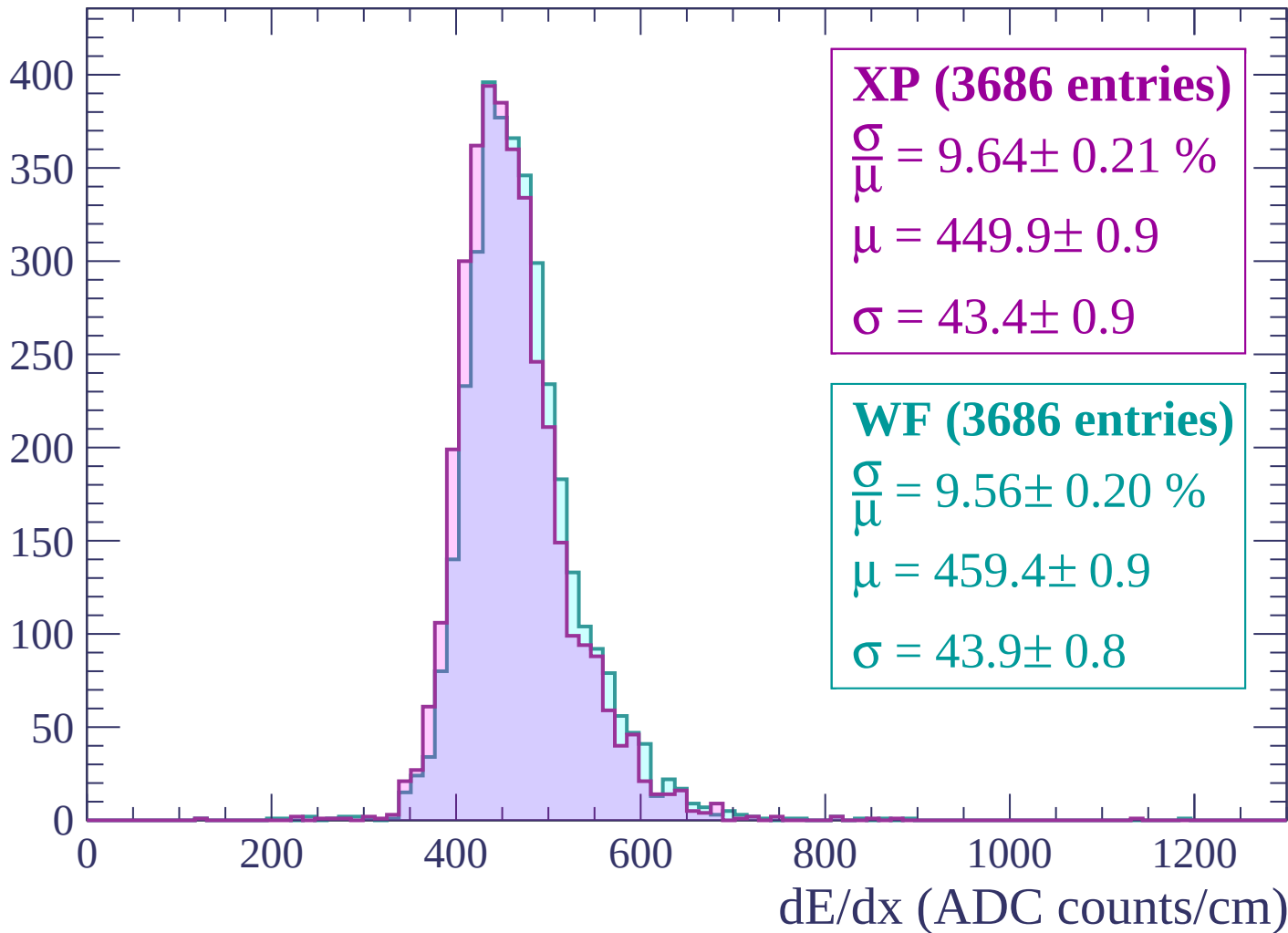
Energy loss | $-1080 < p < -1020$

Count



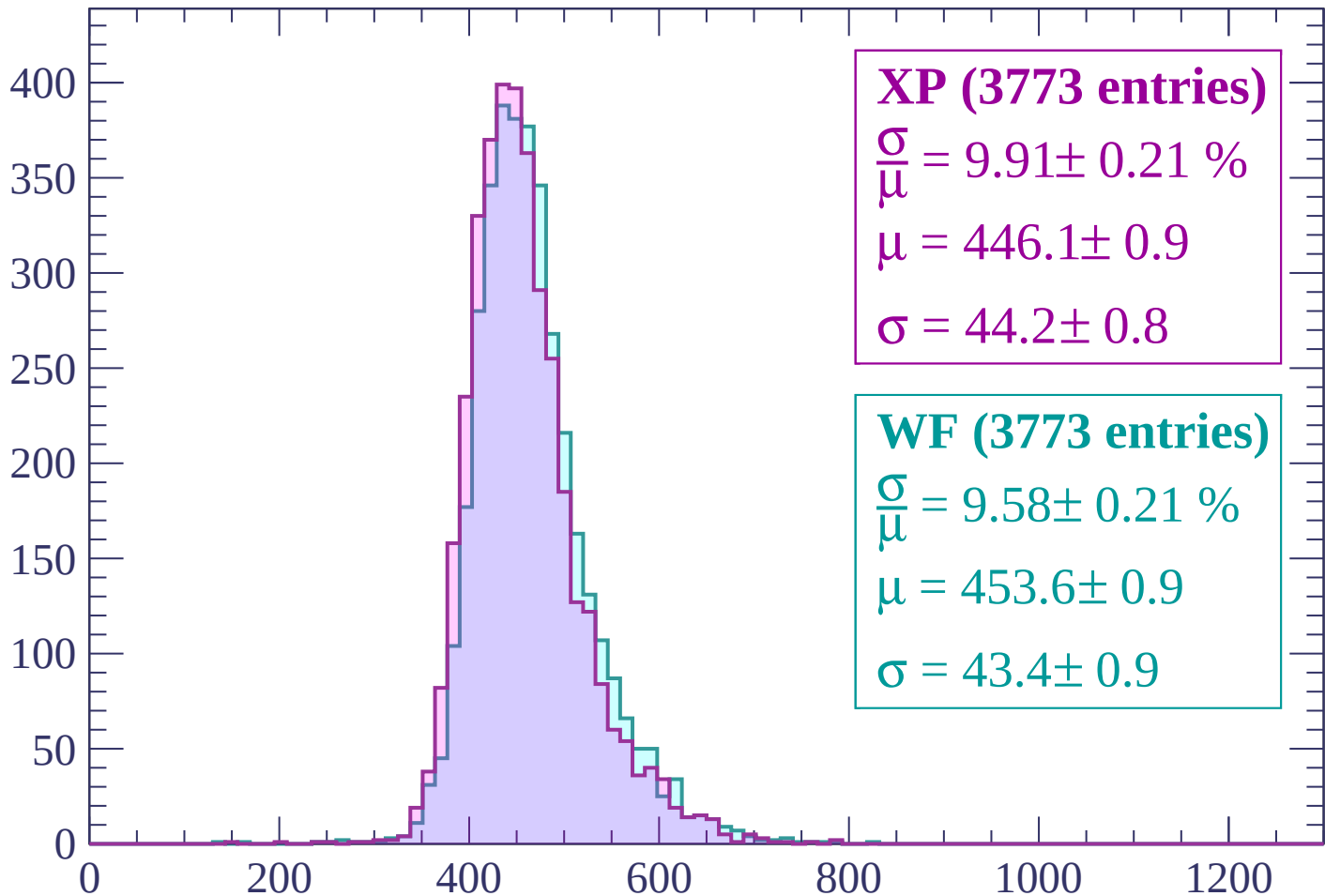
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (3773 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.21 \%$$

$$\mu = 446.1 \pm 0.9$$

$$\sigma = 44.2 \pm 0.8$$

WF (3773 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.21 \%$$

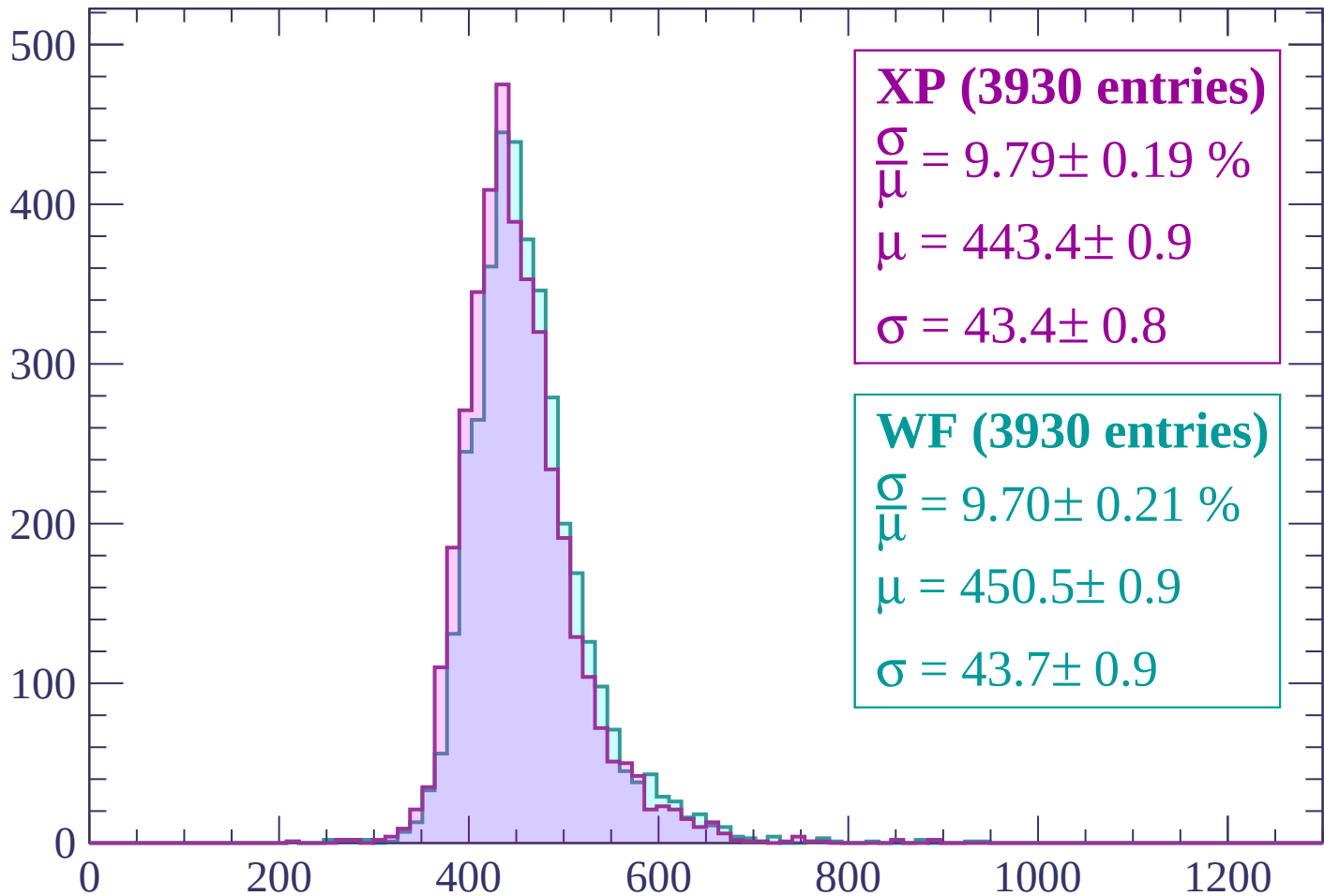
$$\mu = 453.6 \pm 0.9$$

$$\sigma = 43.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (3930 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.19 \%$$

$$\mu = 443.4 \pm 0.9$$

$$\sigma = 43.4 \pm 0.8$$

WF (3930 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.21 \%$$

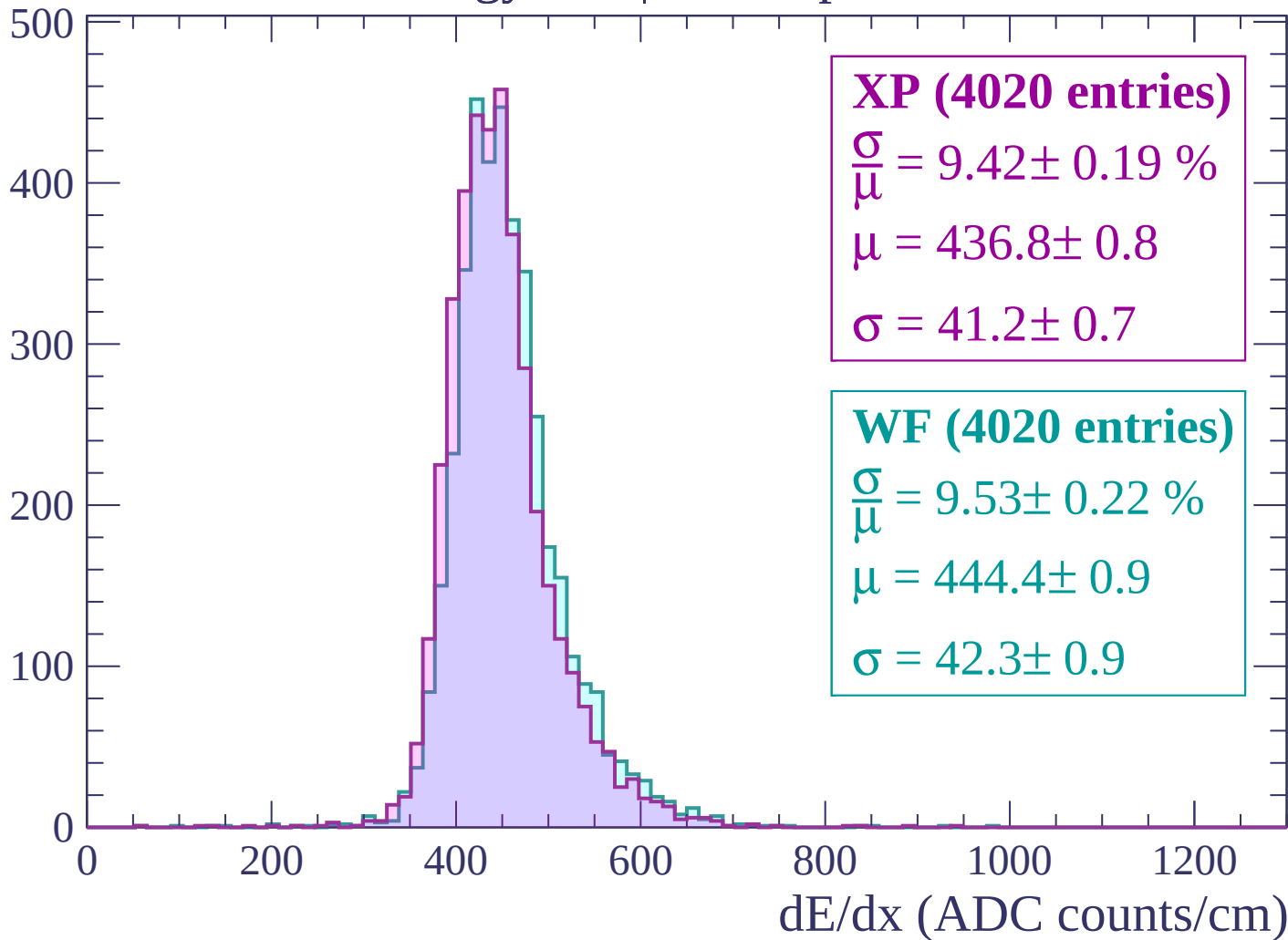
$$\mu = 450.5 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

dE/dx (ADC counts/cm)

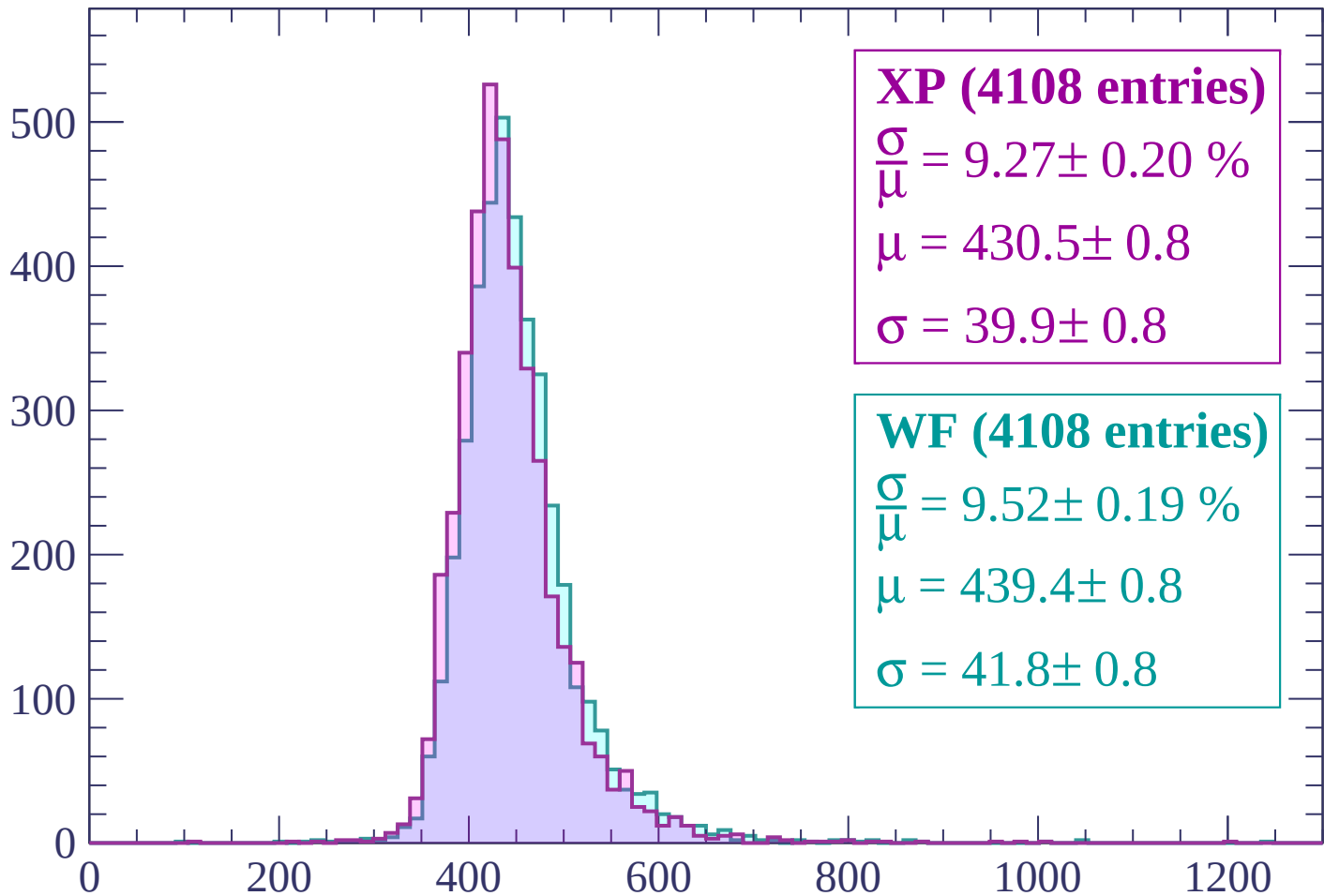
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP (4108 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.20 \%$$

$$\mu = 430.5 \pm 0.8$$

$$\sigma = 39.9 \pm 0.8$$

WF (4108 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.19 \%$$

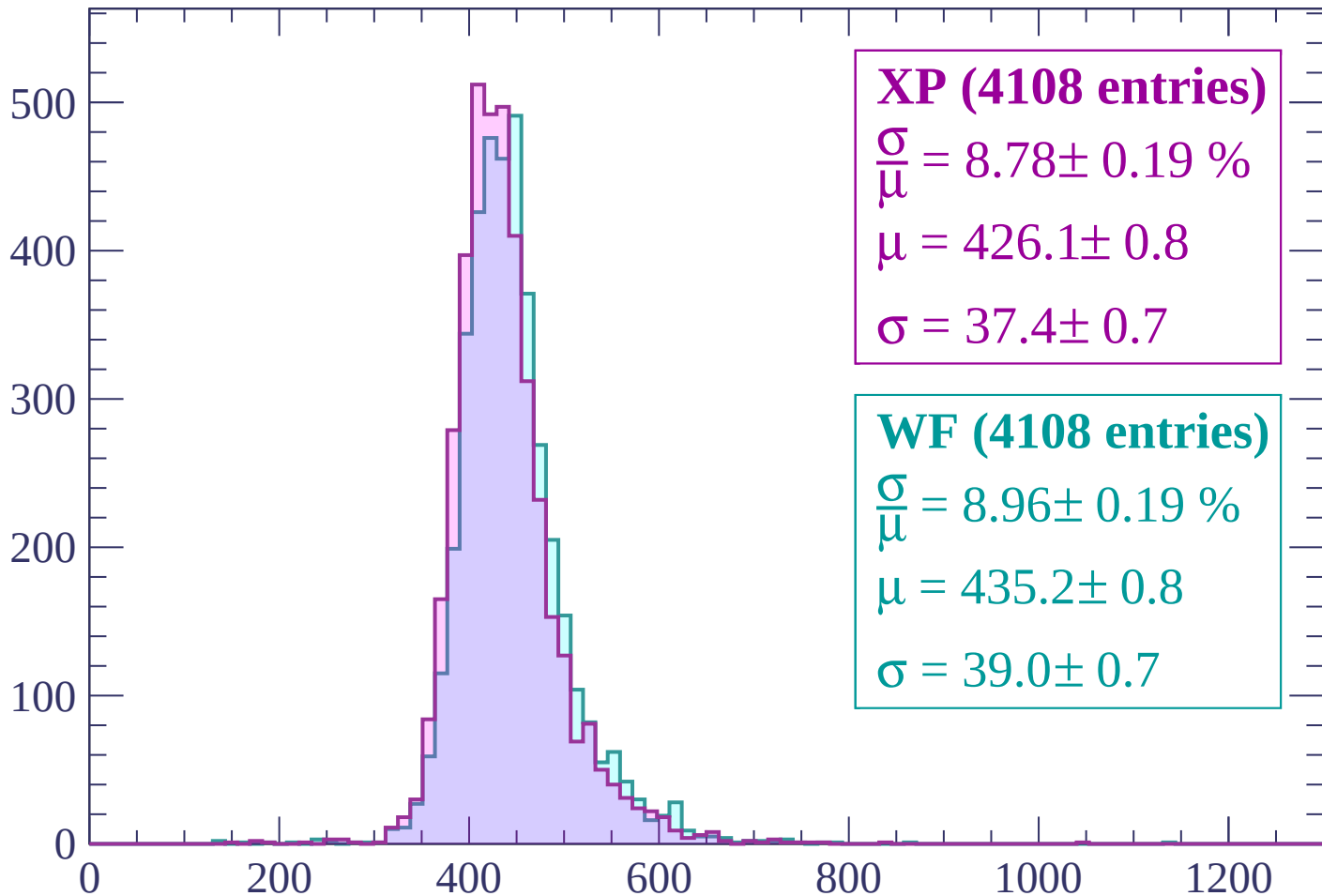
$$\mu = 439.4 \pm 0.8$$

$$\sigma = 41.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (4108 entries)

$$\frac{\sigma}{\mu} = 8.78 \pm 0.19 \%$$

$$\mu = 426.1 \pm 0.8$$

$$\sigma = 37.4 \pm 0.7$$

WF (4108 entries)

$$\frac{\sigma}{\mu} = 8.96 \pm 0.19 \%$$

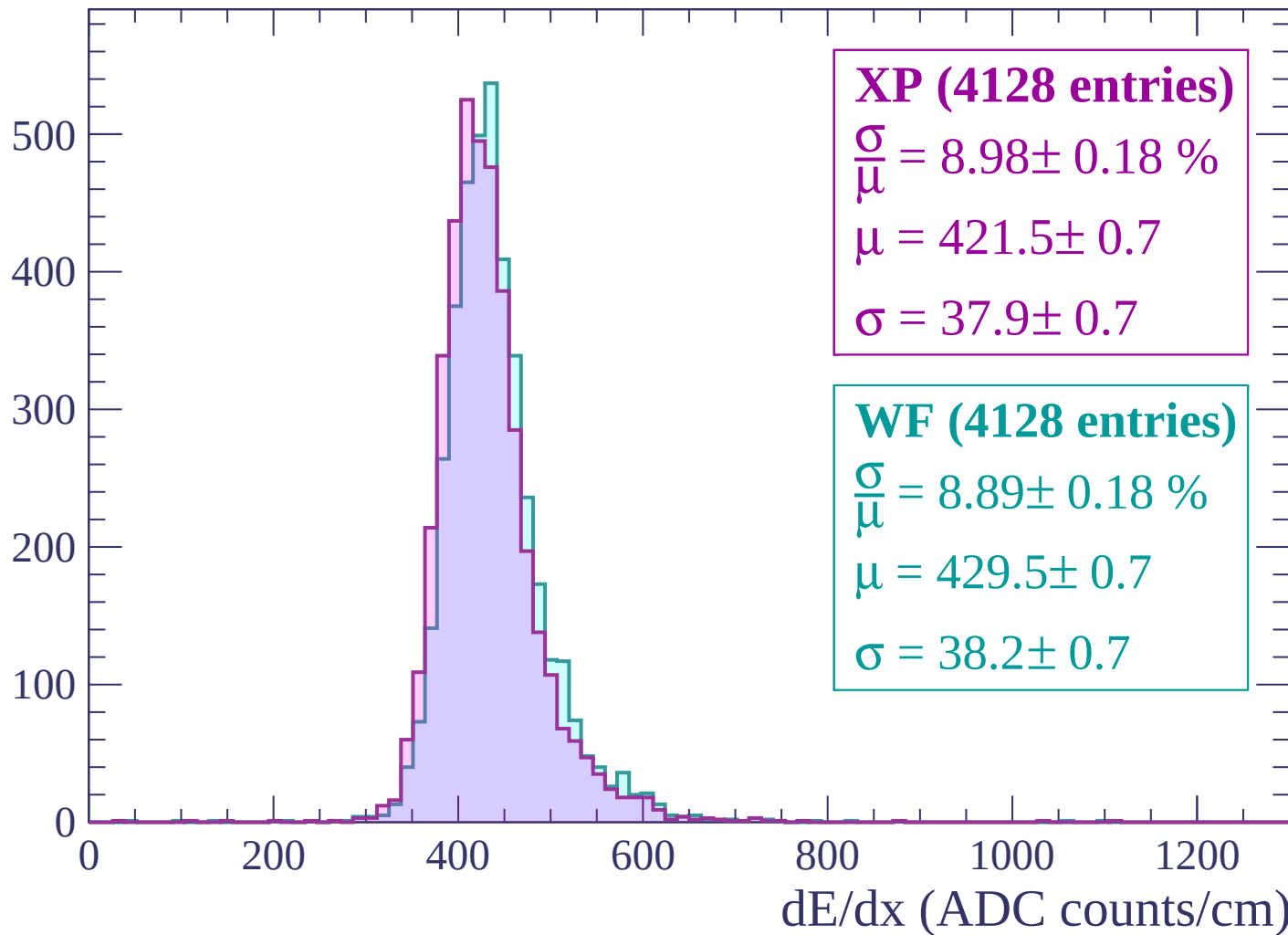
$$\mu = 435.2 \pm 0.8$$

$$\sigma = 39.0 \pm 0.7$$

dE/dx (ADC counts/cm)

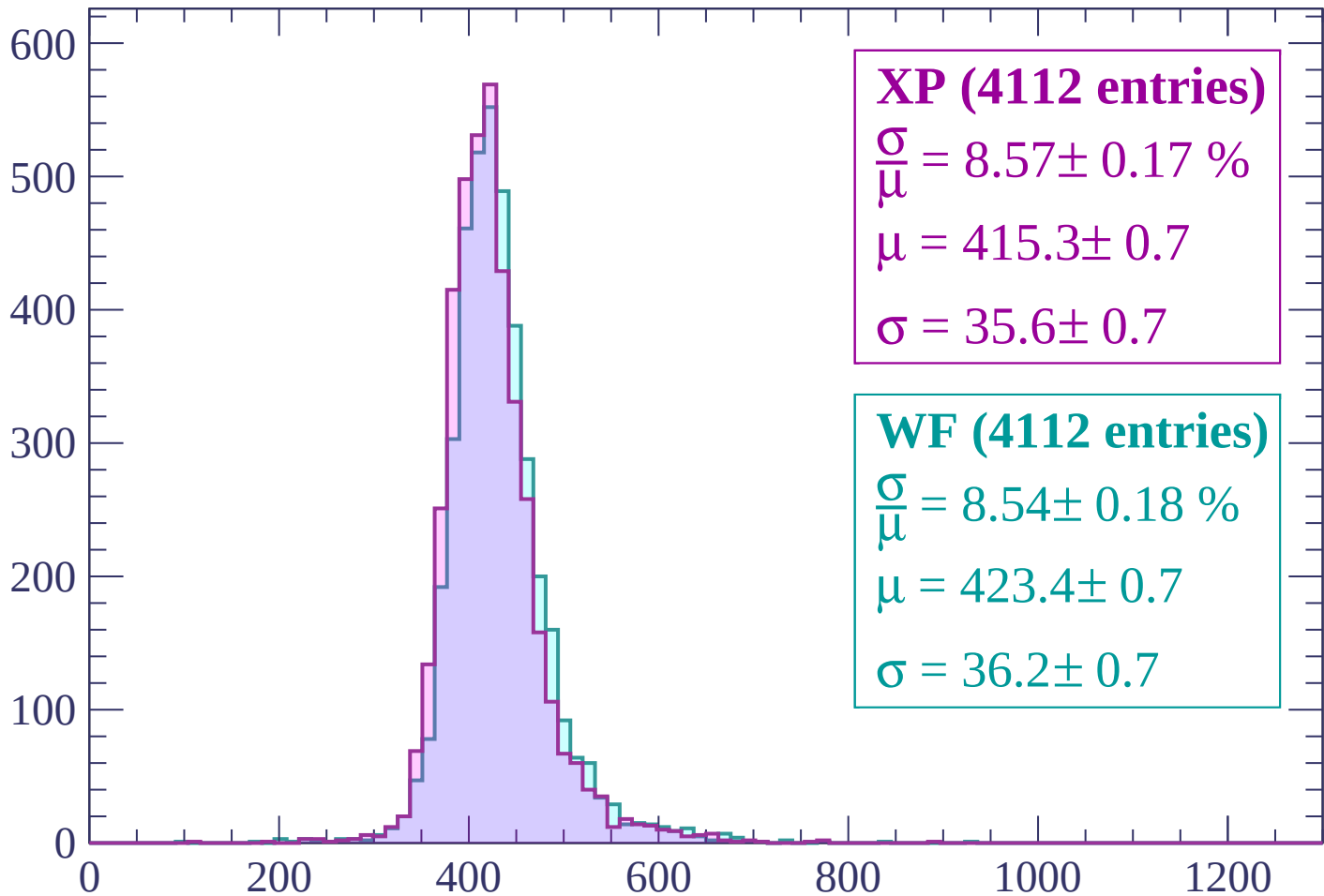
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP (4112 entries)

$\frac{\sigma}{\mu} = 8.57 \pm 0.17 \%$

$\mu = 415.3 \pm 0.7$

$\sigma = 35.6 \pm 0.7$

WF (4112 entries)

$\frac{\sigma}{\mu} = 8.54 \pm 0.18 \%$

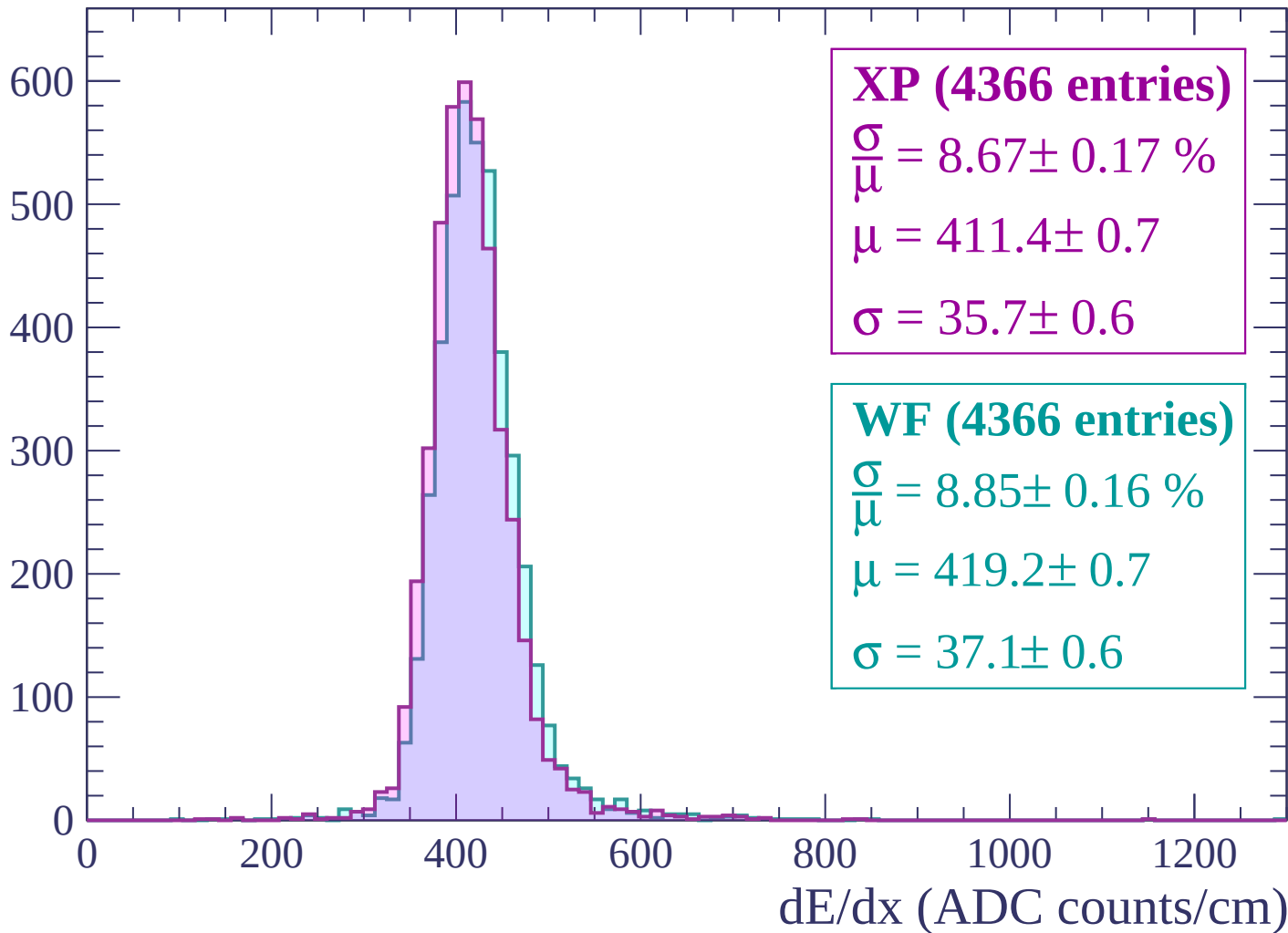
$\mu = 423.4 \pm 0.7$

$\sigma = 36.2 \pm 0.7$

dE/dx (ADC counts/cm)

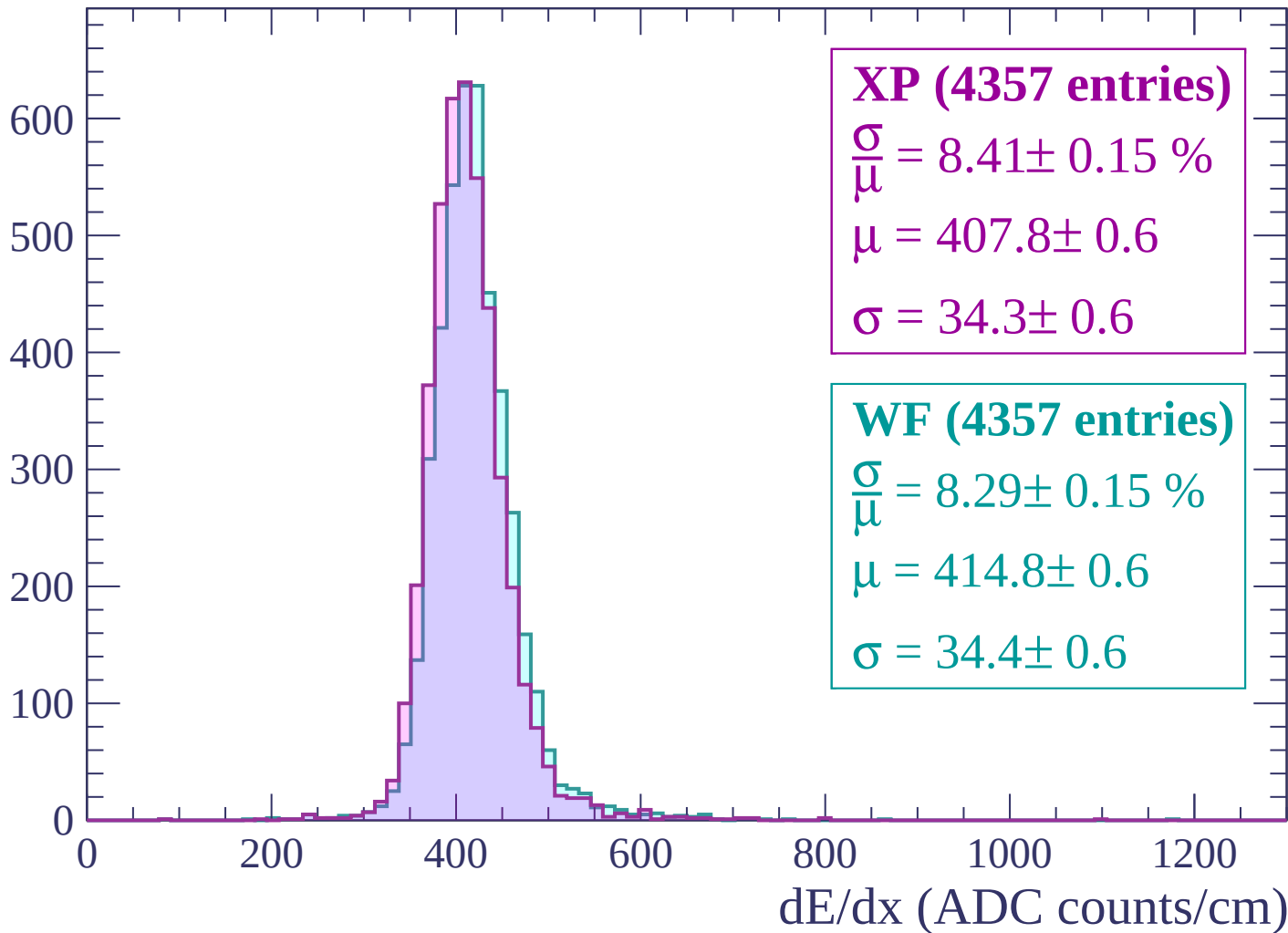
Energy loss | $-540 < p < -480$

Count



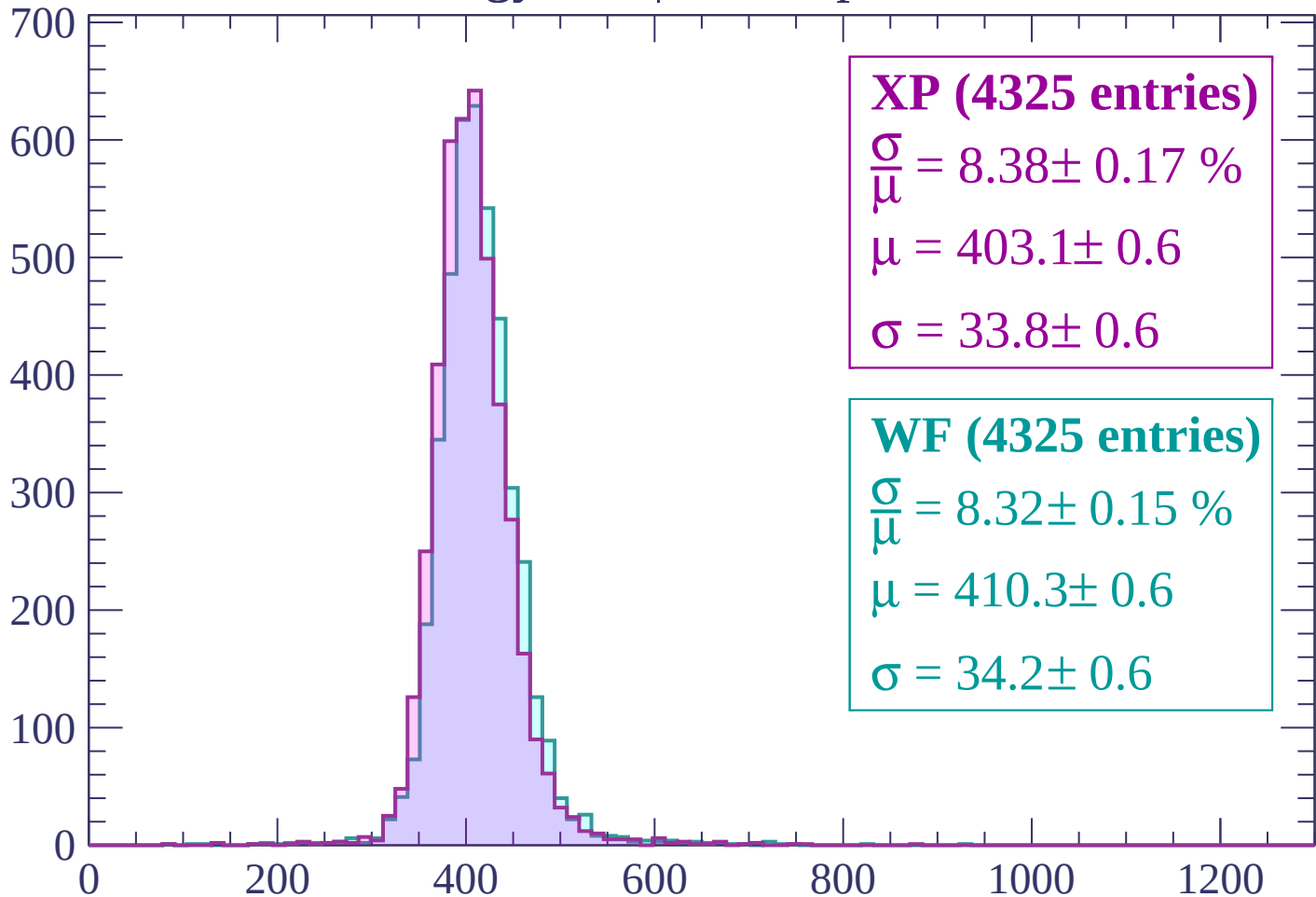
Energy loss | $-480 < p < -420$

Count



Energy loss | $-420 < p < -360$

Count



XP (4325 entries)

$$\frac{\sigma}{\mu} = 8.38 \pm 0.17 \%$$

$$\mu = 403.1 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

WF (4325 entries)

$$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$$

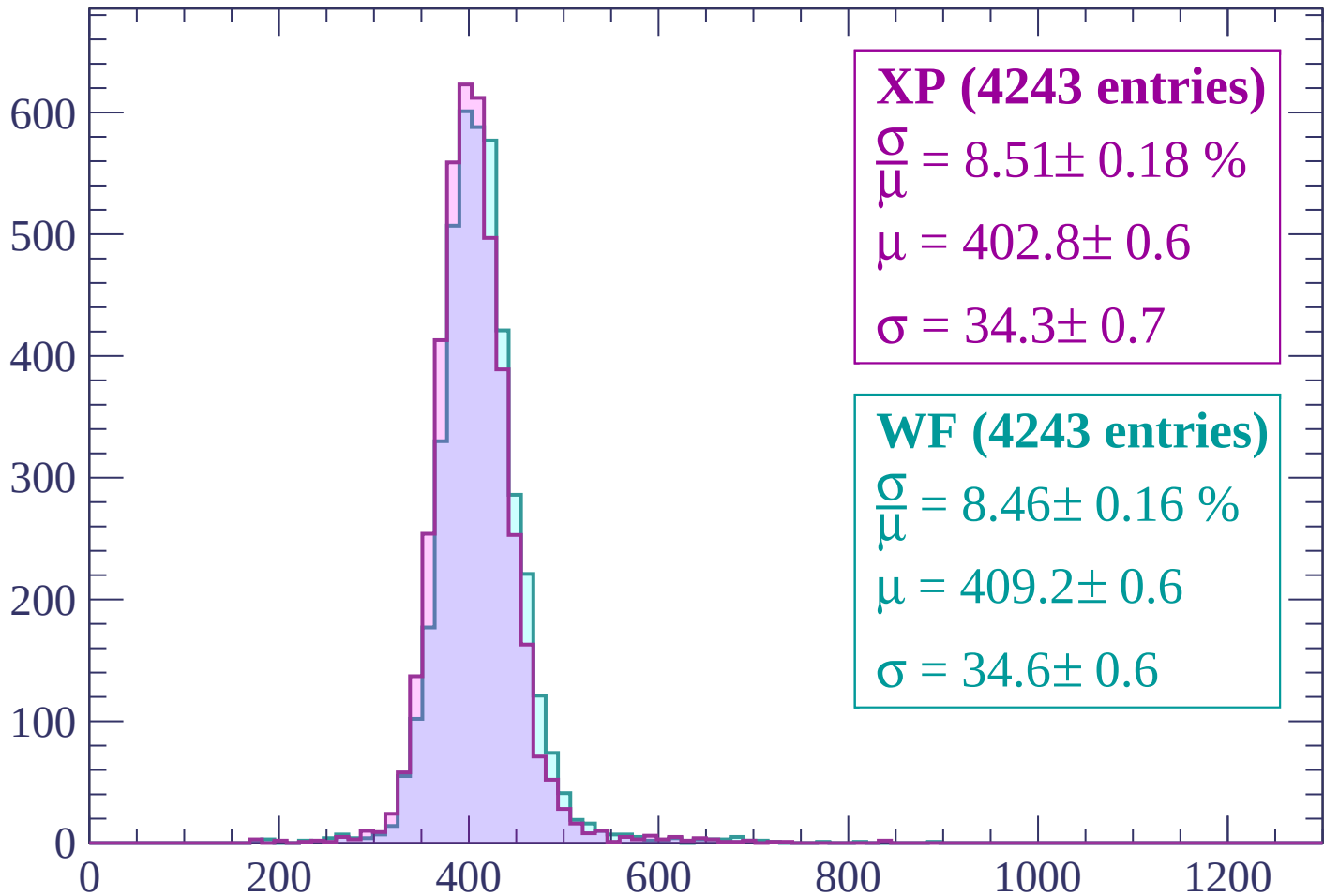
$$\mu = 410.3 \pm 0.6$$

$$\sigma = 34.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (4243 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.18 \%$$

$$\mu = 402.8 \pm 0.6$$

$$\sigma = 34.3 \pm 0.7$$

WF (4243 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.16 \%$$

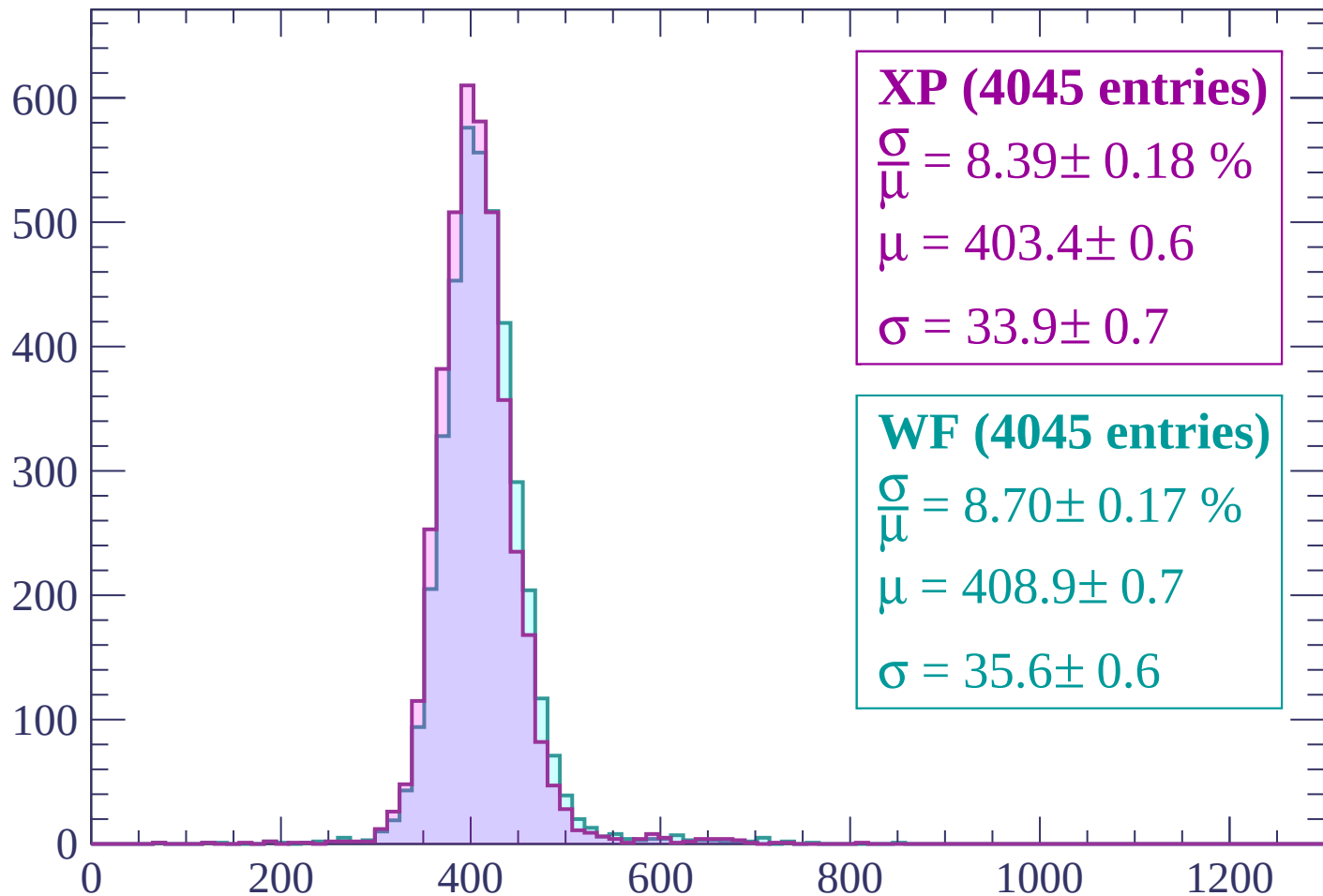
$$\mu = 409.2 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (4045 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.18 \%$$

$$\mu = 403.4 \pm 0.6$$

$$\sigma = 33.9 \pm 0.7$$

WF (4045 entries)

$$\frac{\sigma}{\mu} = 8.70 \pm 0.17 \%$$

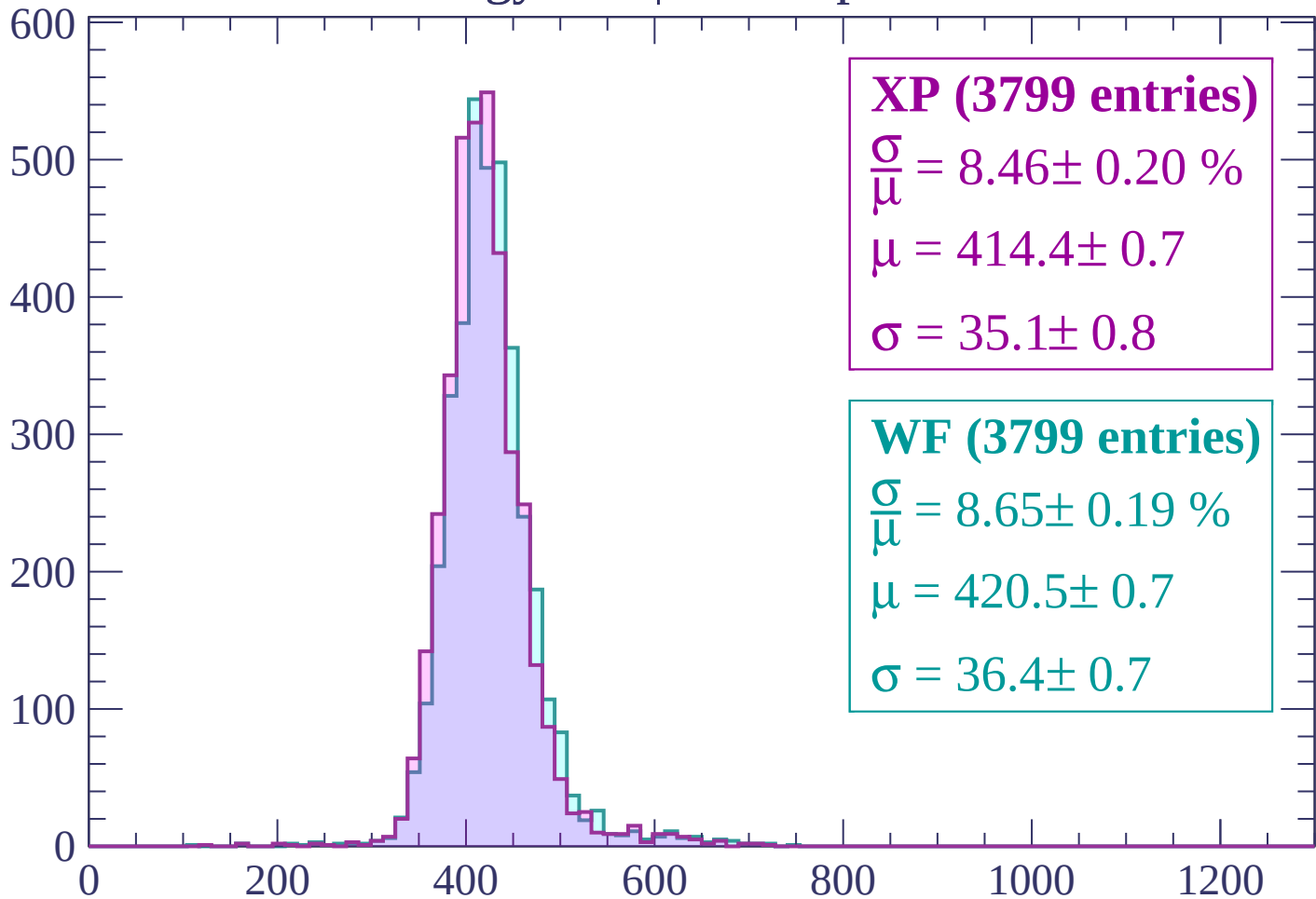
$$\mu = 408.9 \pm 0.7$$

$$\sigma = 35.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (3799 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.20 \%$$

$$\mu = 414.4 \pm 0.7$$

$$\sigma = 35.1 \pm 0.8$$

WF (3799 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.19 \%$$

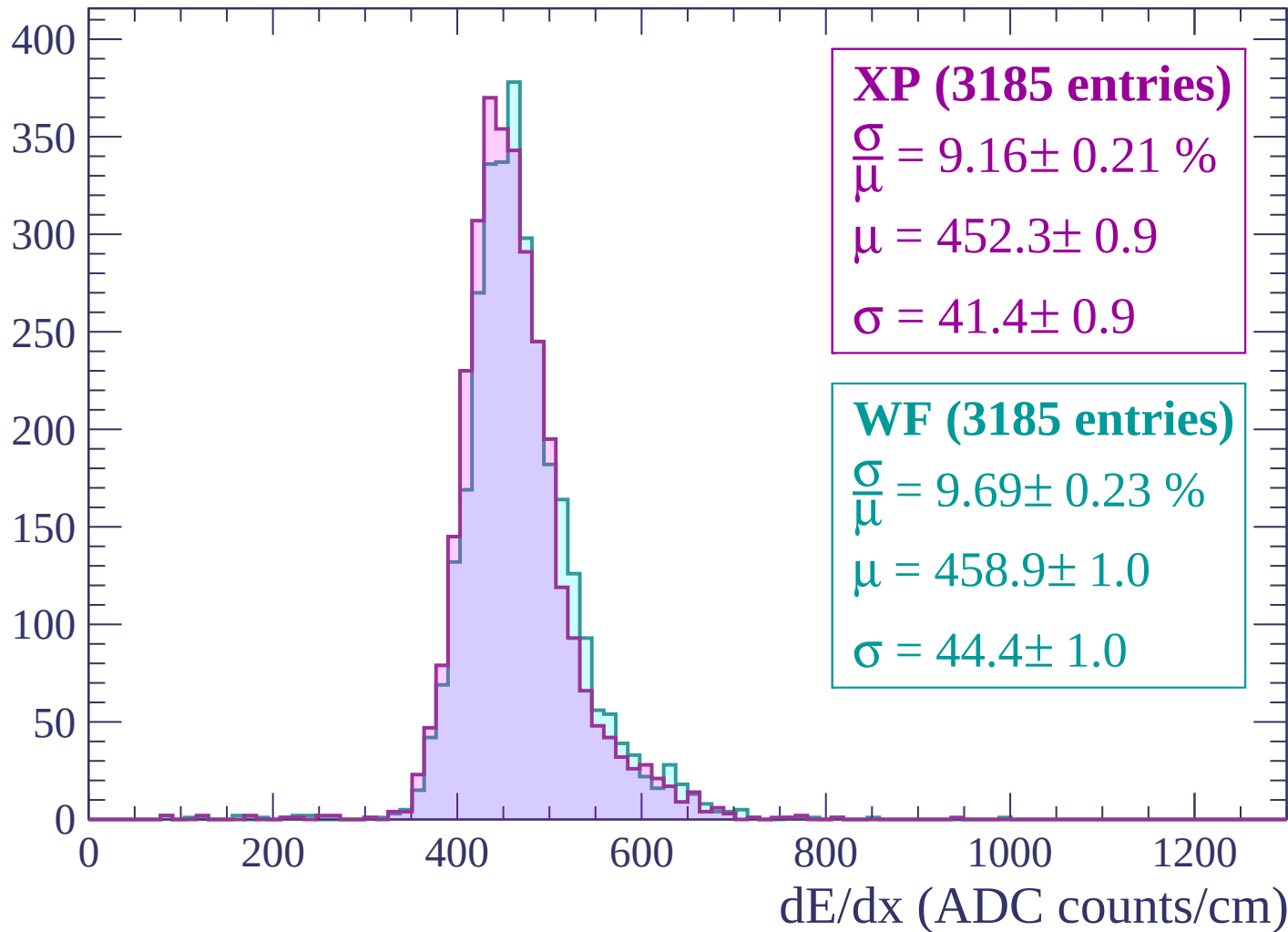
$$\mu = 420.5 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

dE/dx (ADC counts/cm)

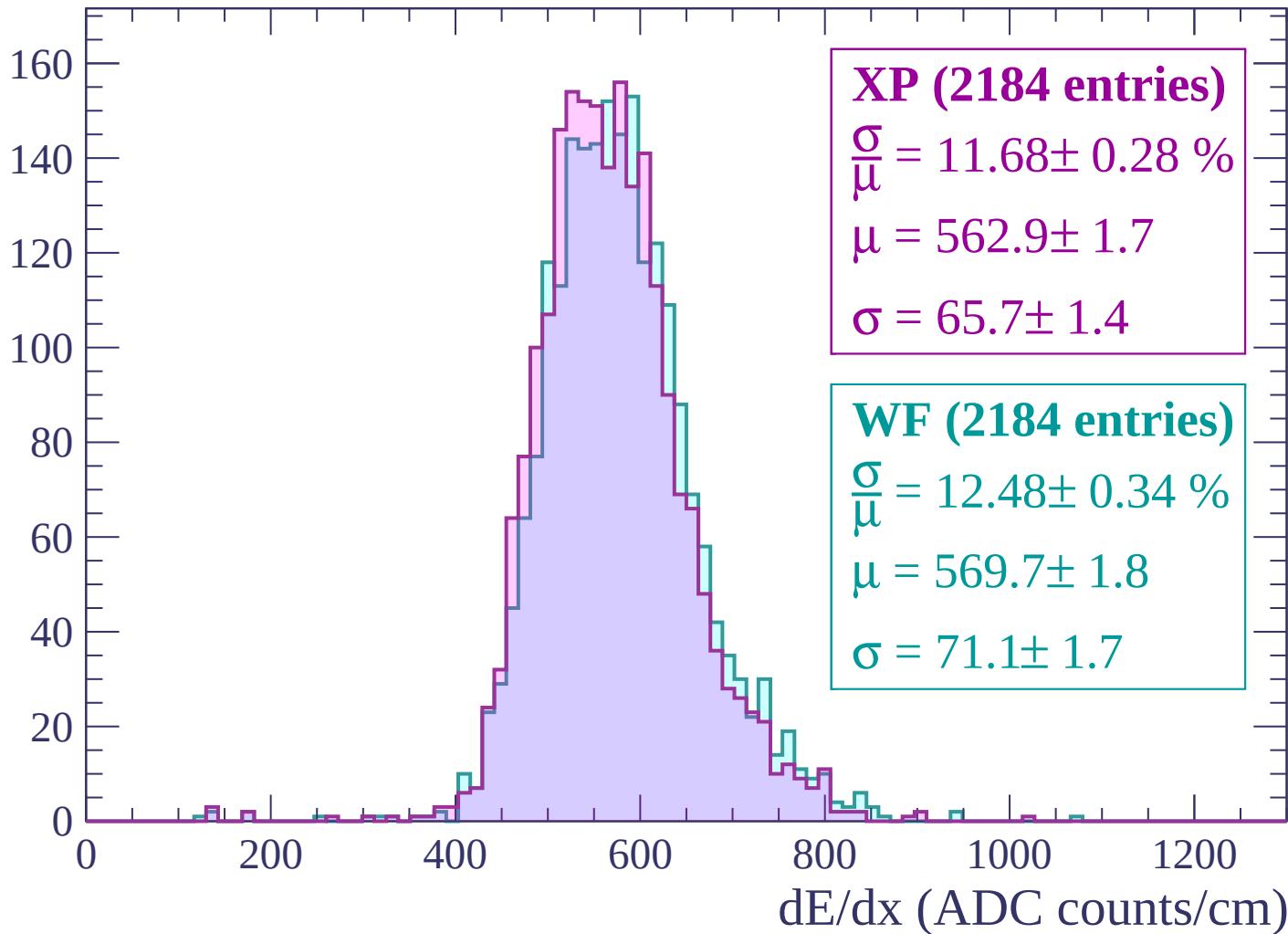
Energy loss | $-180 < p < -120$

Count



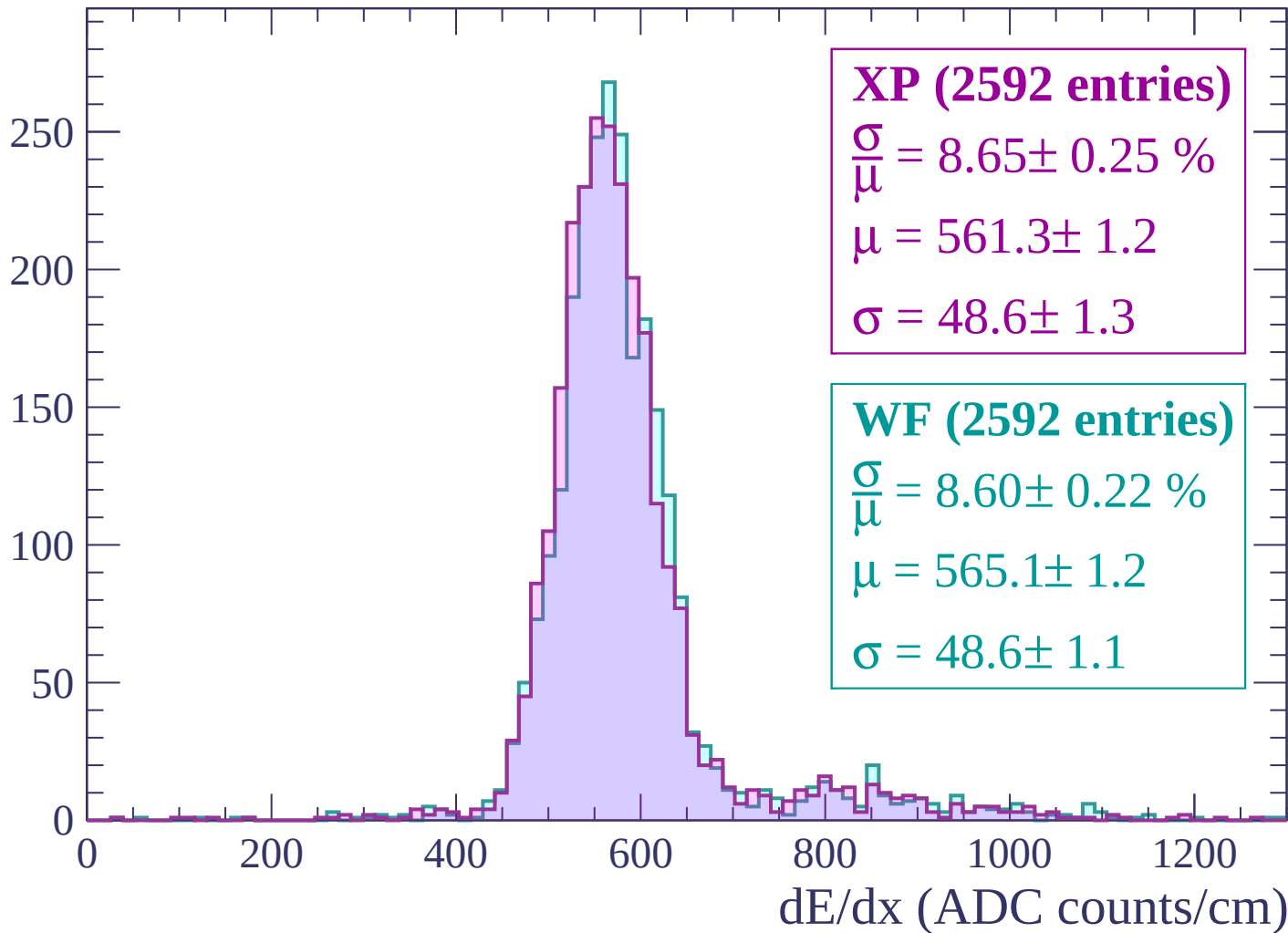
Energy loss | $-120 < p < -60$

Count



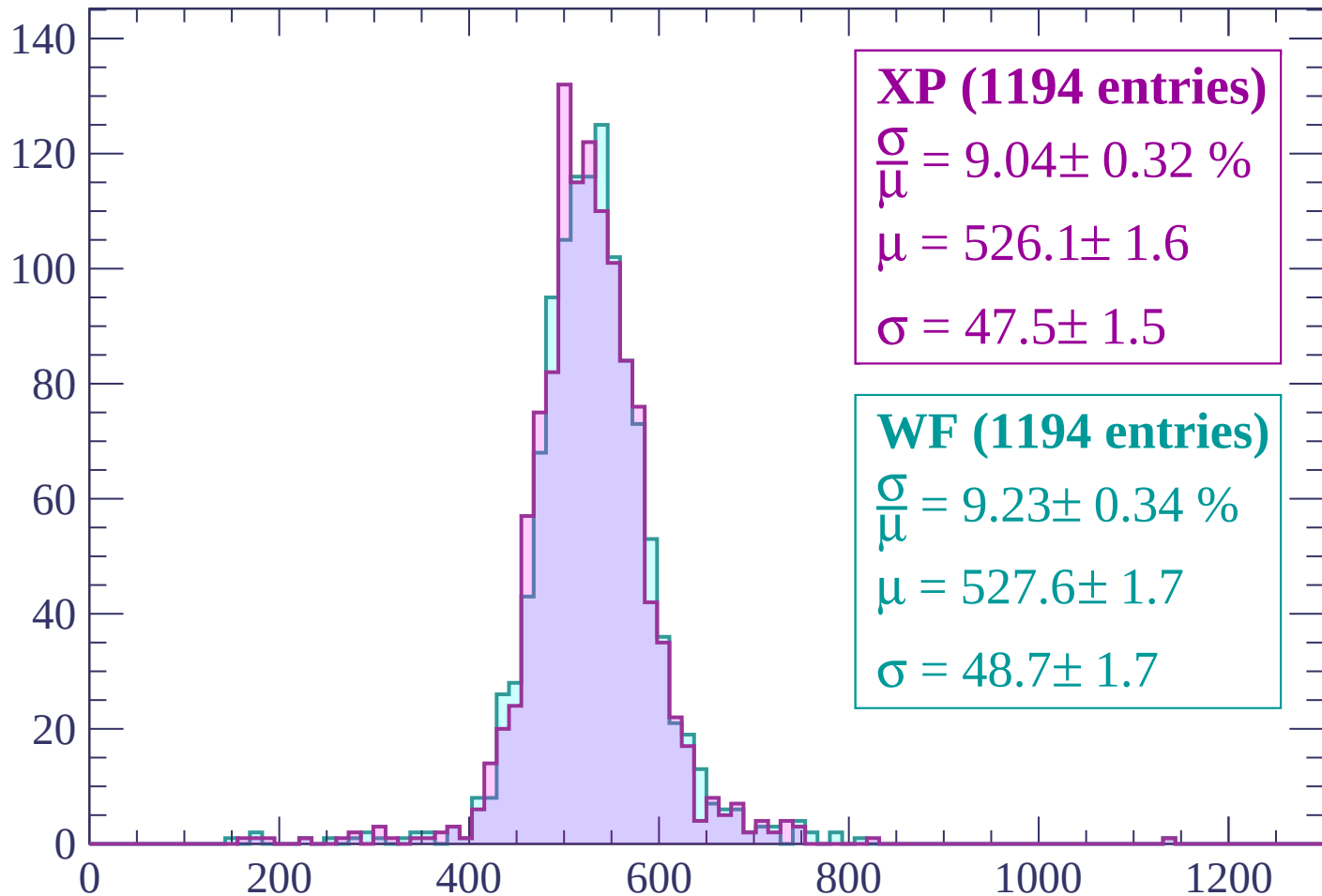
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP (1194 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.32 \%$$

$$\mu = 526.1 \pm 1.6$$

$$\sigma = 47.5 \pm 1.5$$

WF (1194 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.34 \%$$

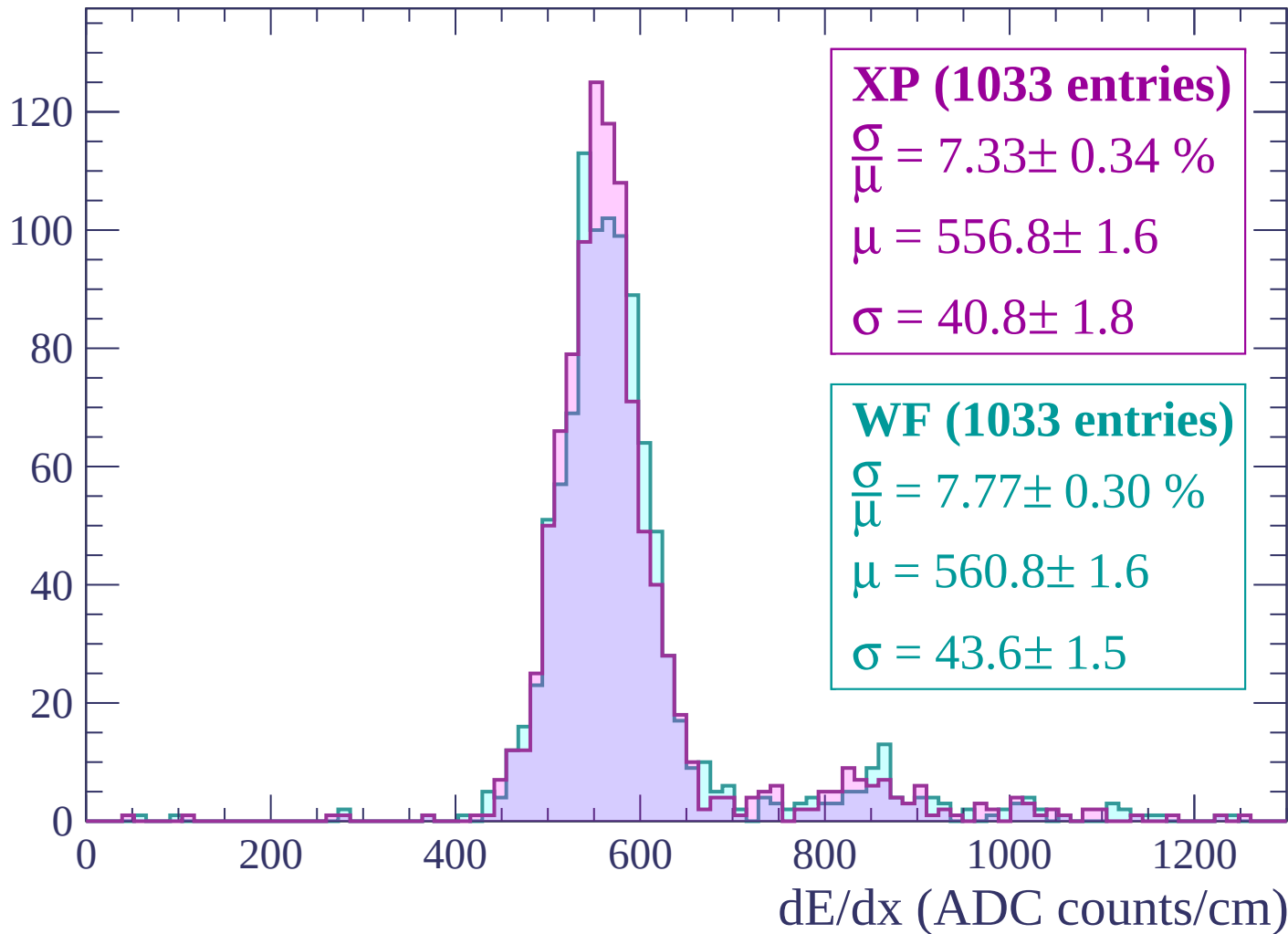
$$\mu = 527.6 \pm 1.7$$

$$\sigma = 48.7 \pm 1.7$$

dE/dx (ADC counts/cm)

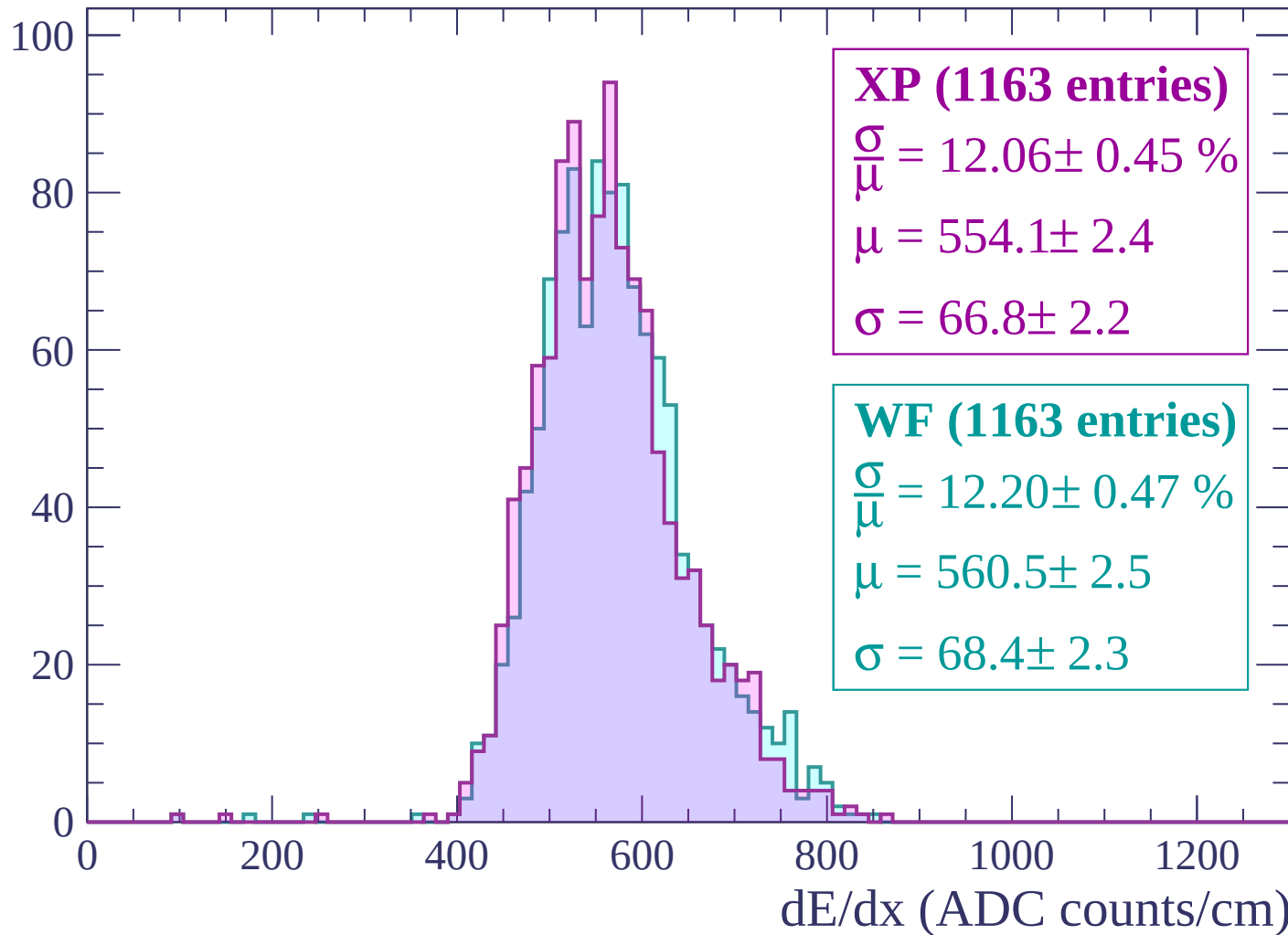
Energy loss | $60 < p < 120$

Count



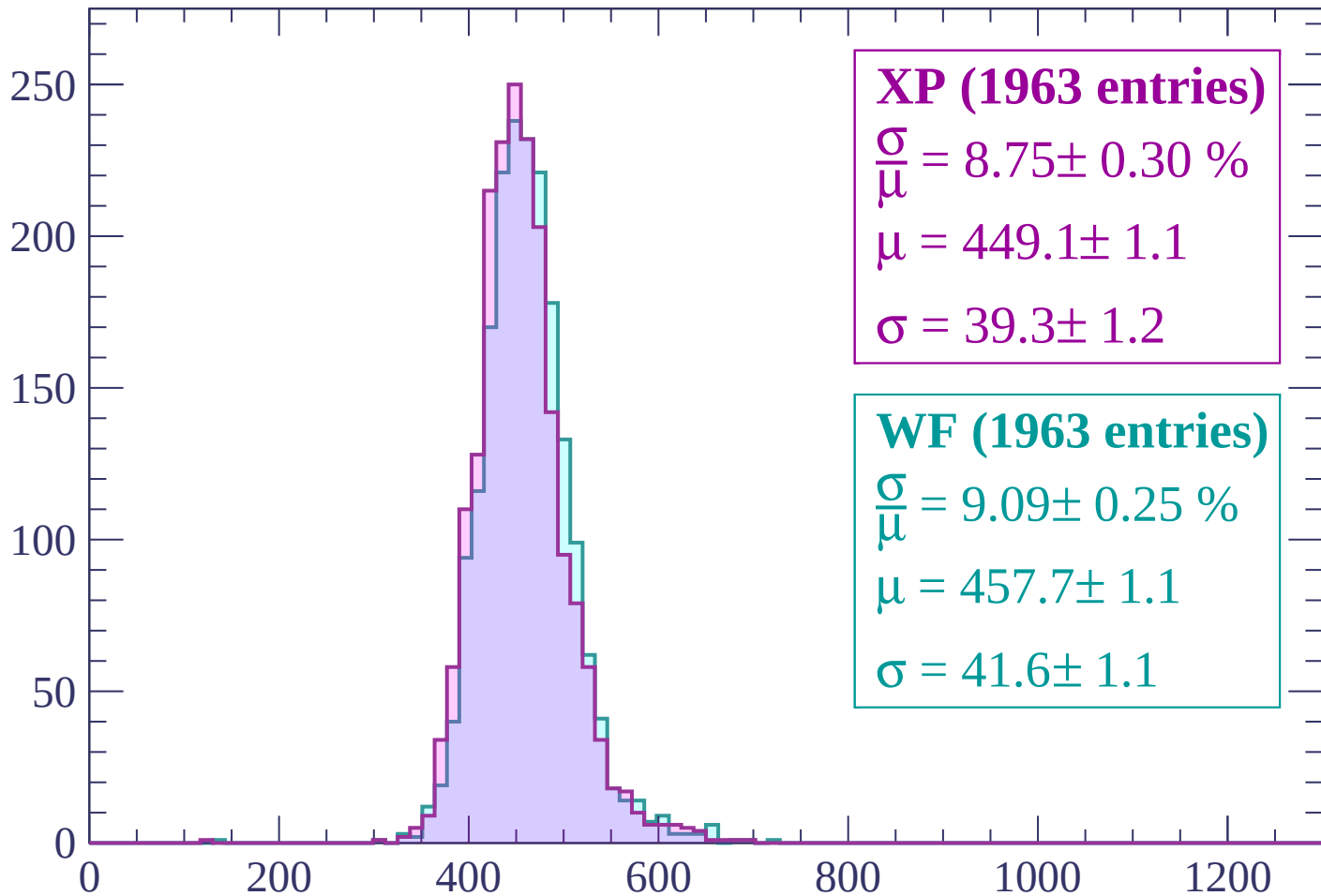
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (1963 entries)

$\frac{\sigma}{\mu} = 8.75 \pm 0.30 \%$

$\mu = 449.1 \pm 1.1$

$\sigma = 39.3 \pm 1.2$

WF (1963 entries)

$\frac{\sigma}{\mu} = 9.09 \pm 0.25 \%$

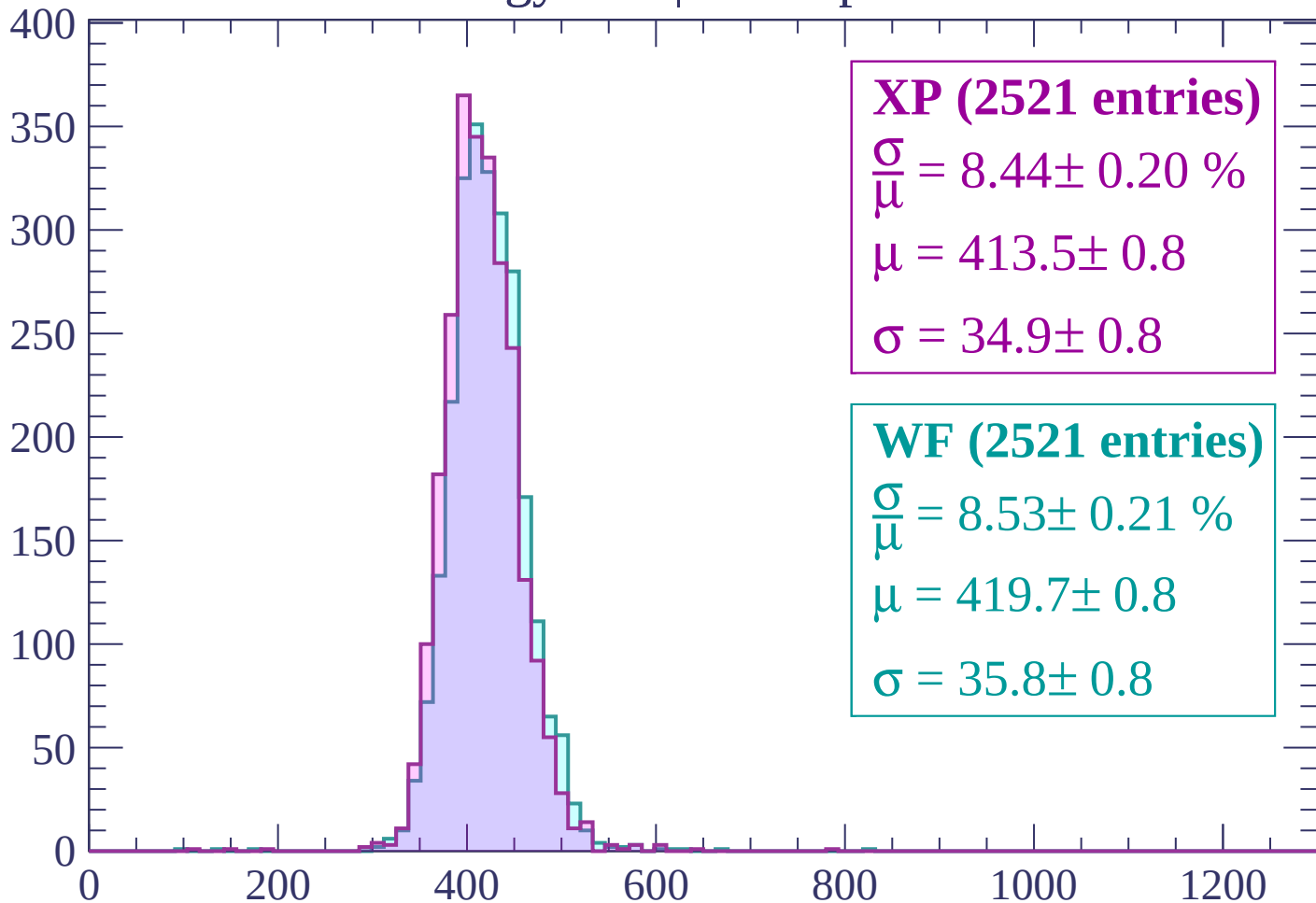
$\mu = 457.7 \pm 1.1$

$\sigma = 41.6 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (2521 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.20 \%$$

$$\mu = 413.5 \pm 0.8$$

$$\sigma = 34.9 \pm 0.8$$

WF (2521 entries)

$$\frac{\sigma}{\mu} = 8.53 \pm 0.21 \%$$

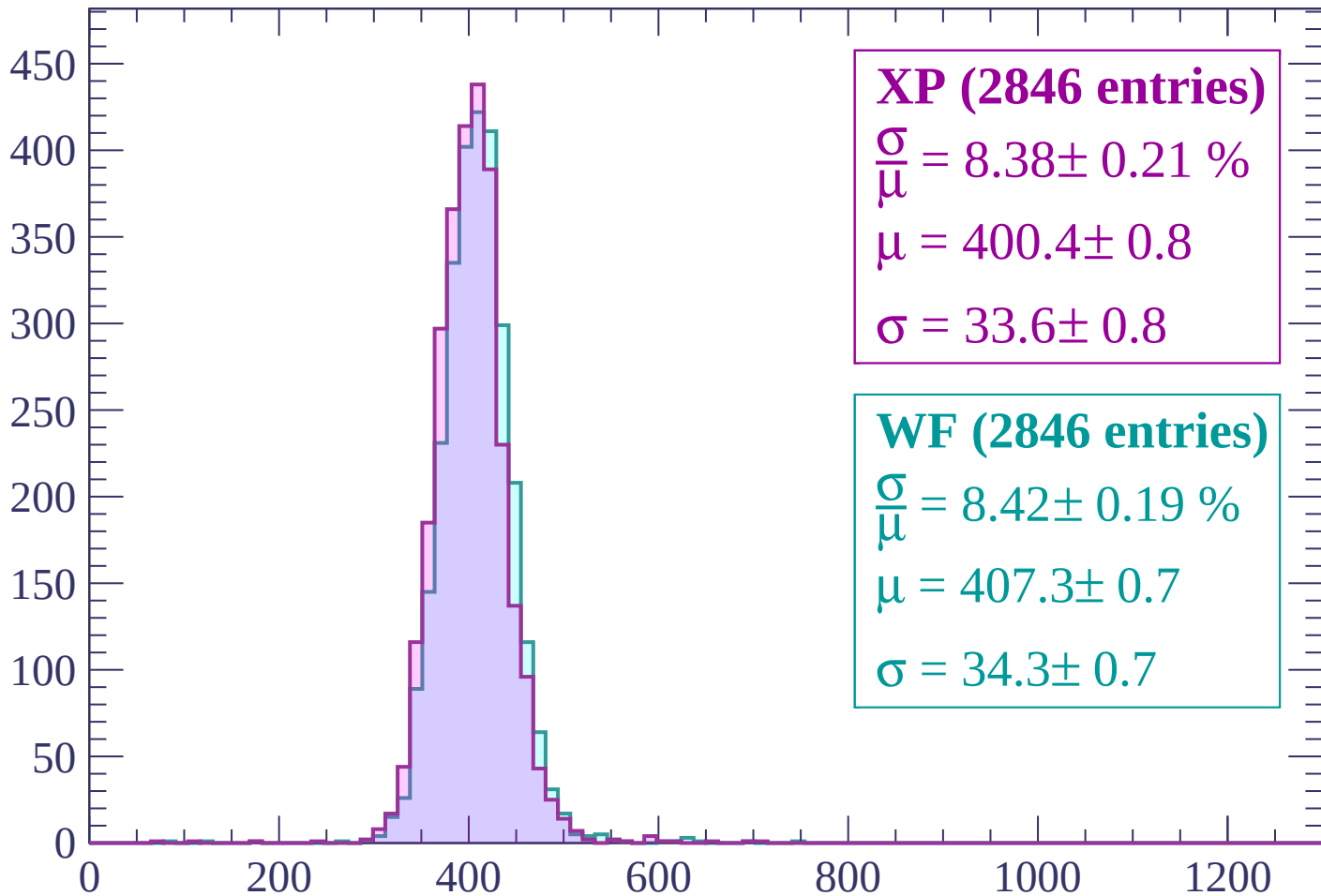
$$\mu = 419.7 \pm 0.8$$

$$\sigma = 35.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (2846 entries)

$$\frac{\sigma}{\mu} = 8.38 \pm 0.21 \%$$

$$\mu = 400.4 \pm 0.8$$

$$\sigma = 33.6 \pm 0.8$$

WF (2846 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.19 \%$$

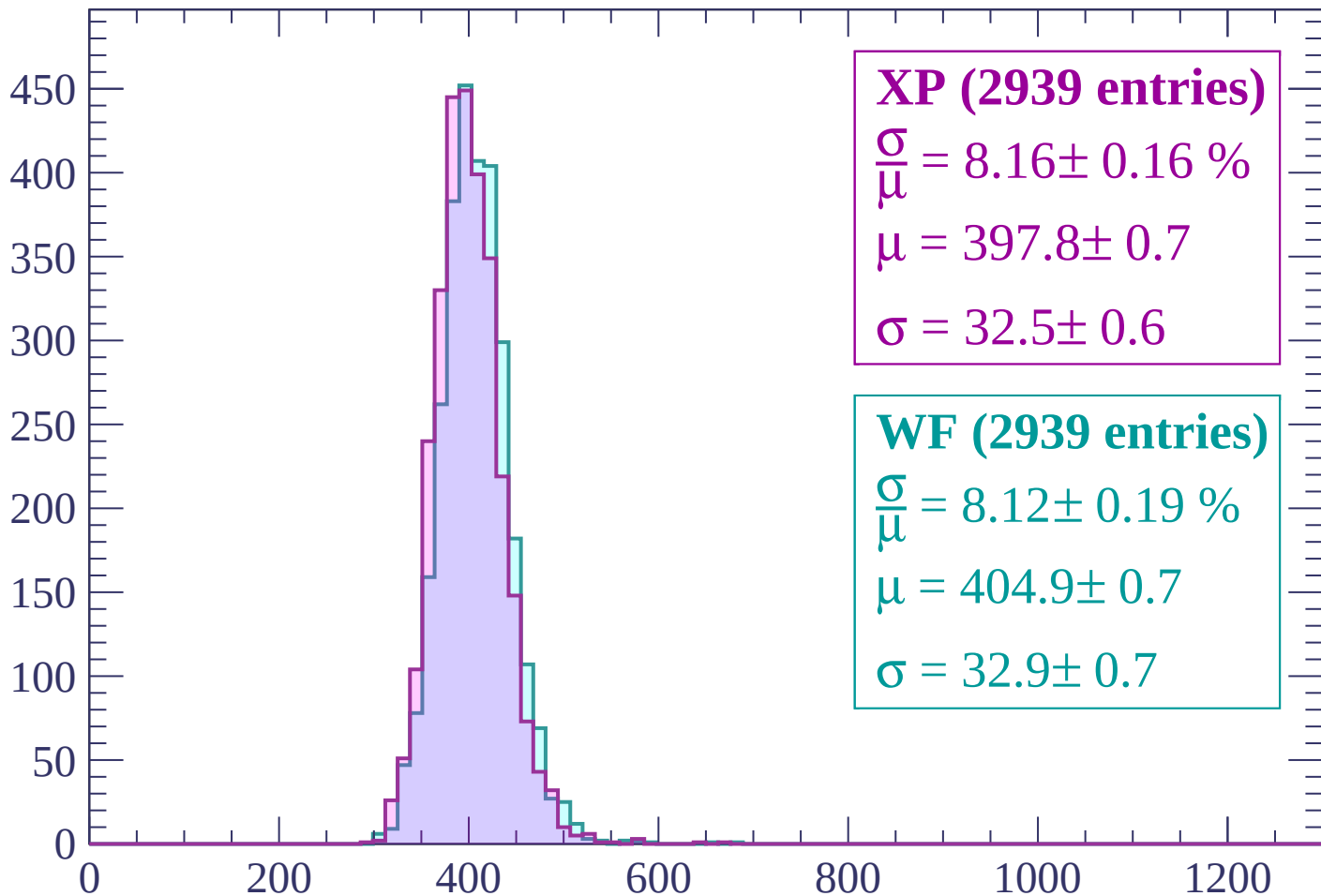
$$\mu = 407.3 \pm 0.7$$

$$\sigma = 34.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (2939 entries)

$$\frac{\sigma}{\mu} = 8.16 \pm 0.16 \%$$

$$\mu = 397.8 \pm 0.7$$

$$\sigma = 32.5 \pm 0.6$$

WF (2939 entries)

$$\frac{\sigma}{\mu} = 8.12 \pm 0.19 \%$$

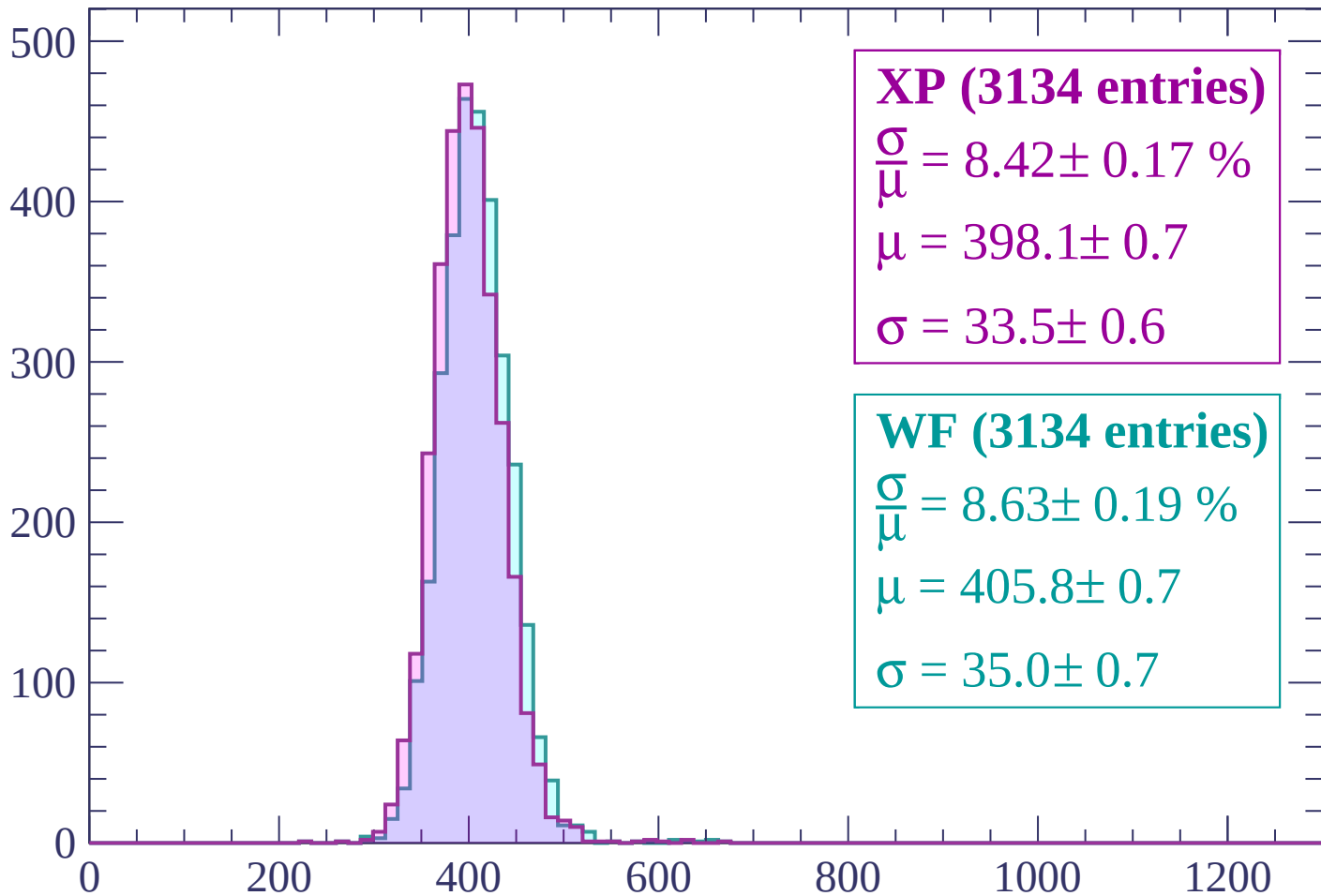
$$\mu = 404.9 \pm 0.7$$

$$\sigma = 32.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (3134 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.17 \%$$

$$\mu = 398.1 \pm 0.7$$

$$\sigma = 33.5 \pm 0.6$$

WF (3134 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.19 \%$$

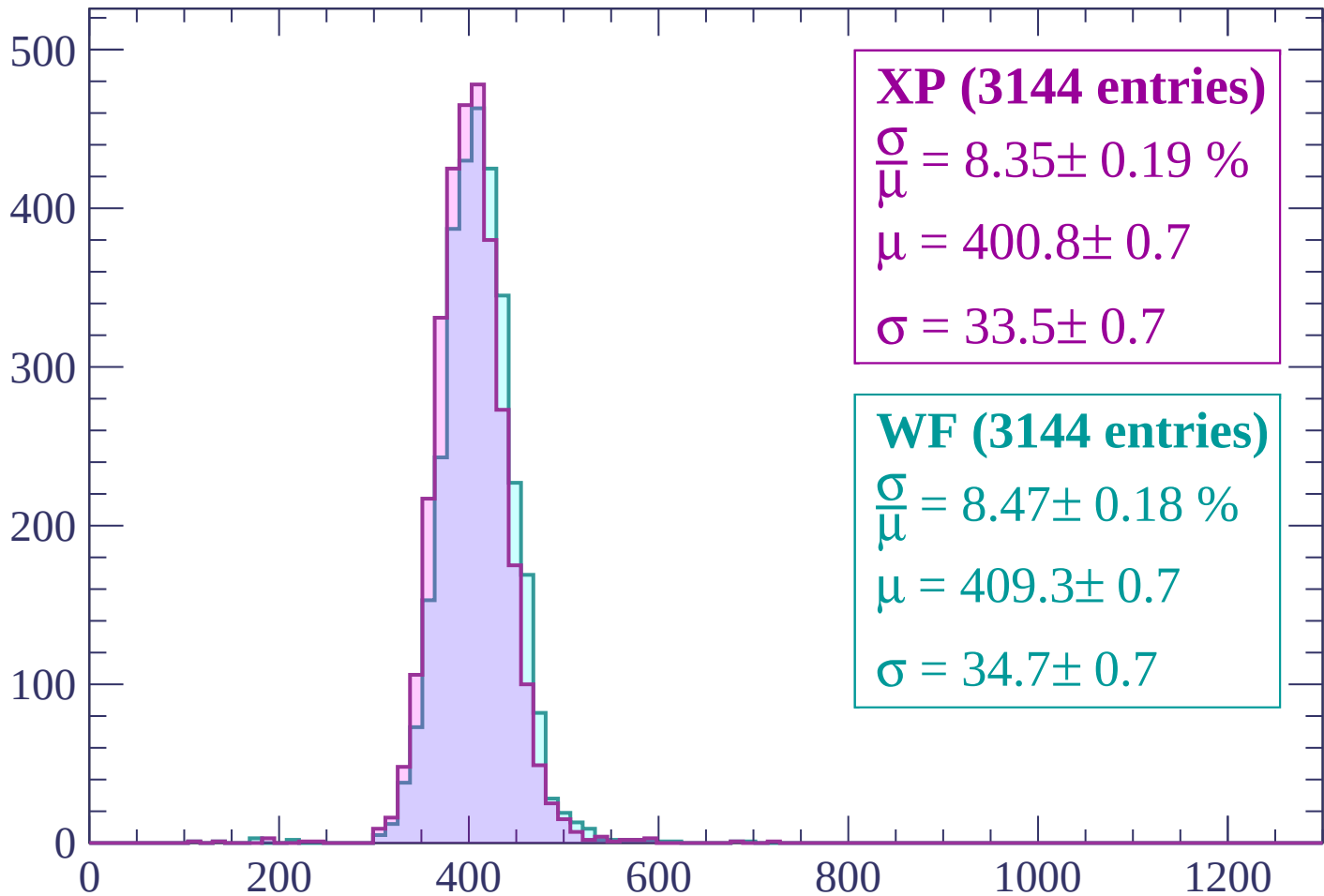
$$\mu = 405.8 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (3144 entries)

$\frac{\sigma}{\mu} = 8.35 \pm 0.19 \%$

$\mu = 400.8 \pm 0.7$

$\sigma = 33.5 \pm 0.7$

WF (3144 entries)

$\frac{\sigma}{\mu} = 8.47 \pm 0.18 \%$

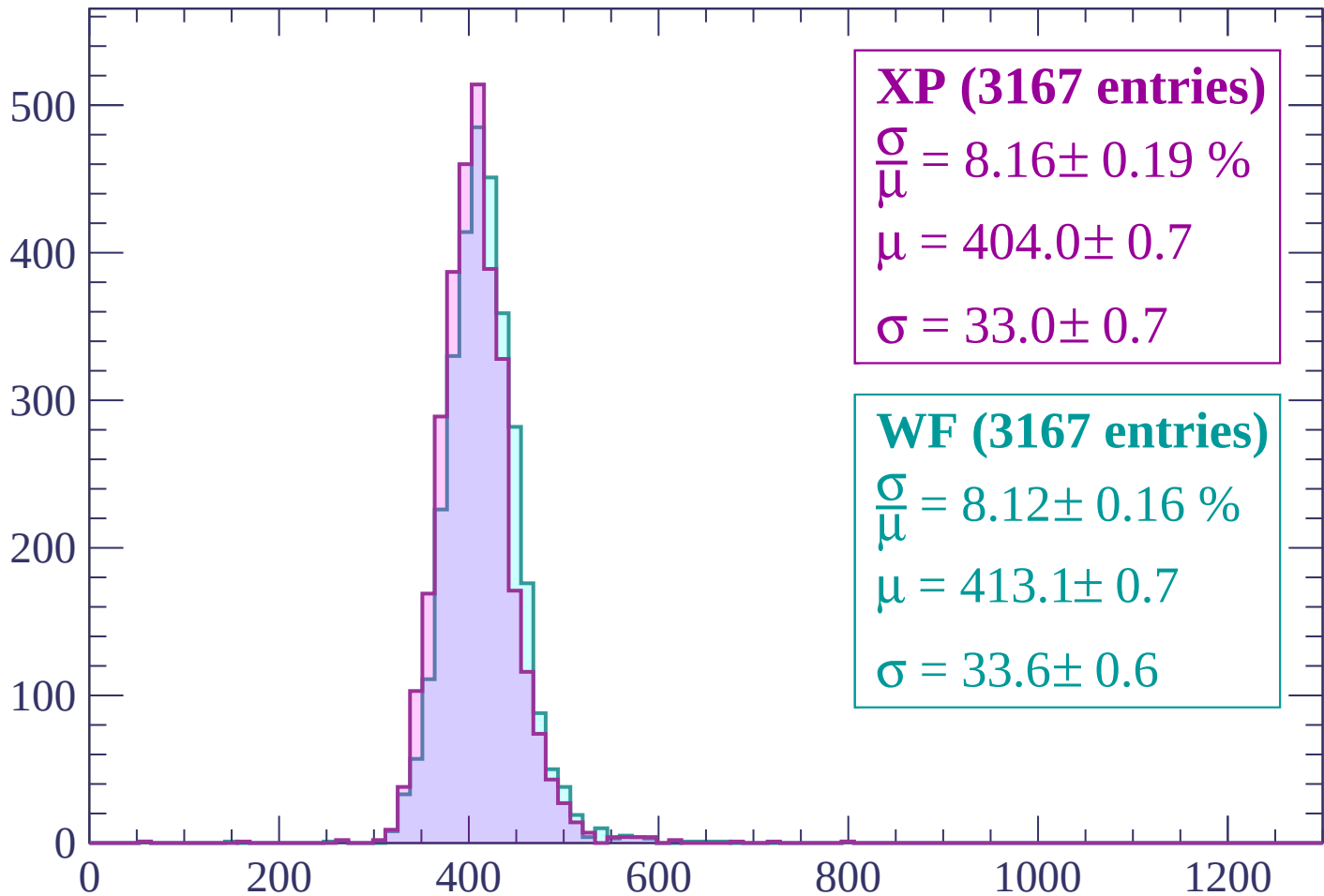
$\mu = 409.3 \pm 0.7$

$\sigma = 34.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (3167 entries)

$\frac{\sigma}{\mu} = 8.16 \pm 0.19 \%$

$\mu = 404.0 \pm 0.7$

$\sigma = 33.0 \pm 0.7$

WF (3167 entries)

$\frac{\sigma}{\mu} = 8.12 \pm 0.16 \%$

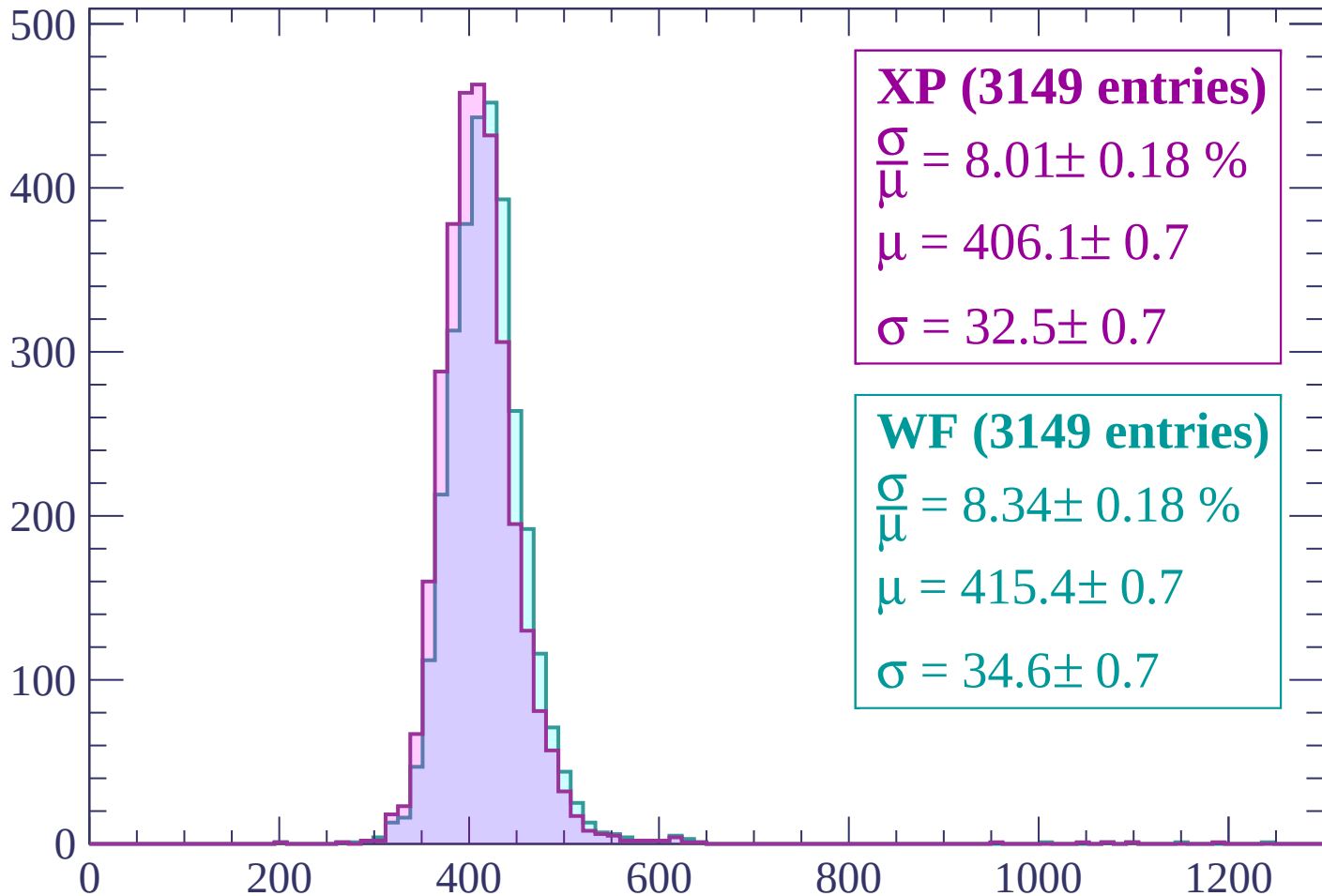
$\mu = 413.1 \pm 0.7$

$\sigma = 33.6 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (3149 entries)

$$\frac{\sigma}{\mu} = 8.01 \pm 0.18 \%$$

$$\mu = 406.1 \pm 0.7$$

$$\sigma = 32.5 \pm 0.7$$

WF (3149 entries)

$$\frac{\sigma}{\mu} = 8.34 \pm 0.18 \%$$

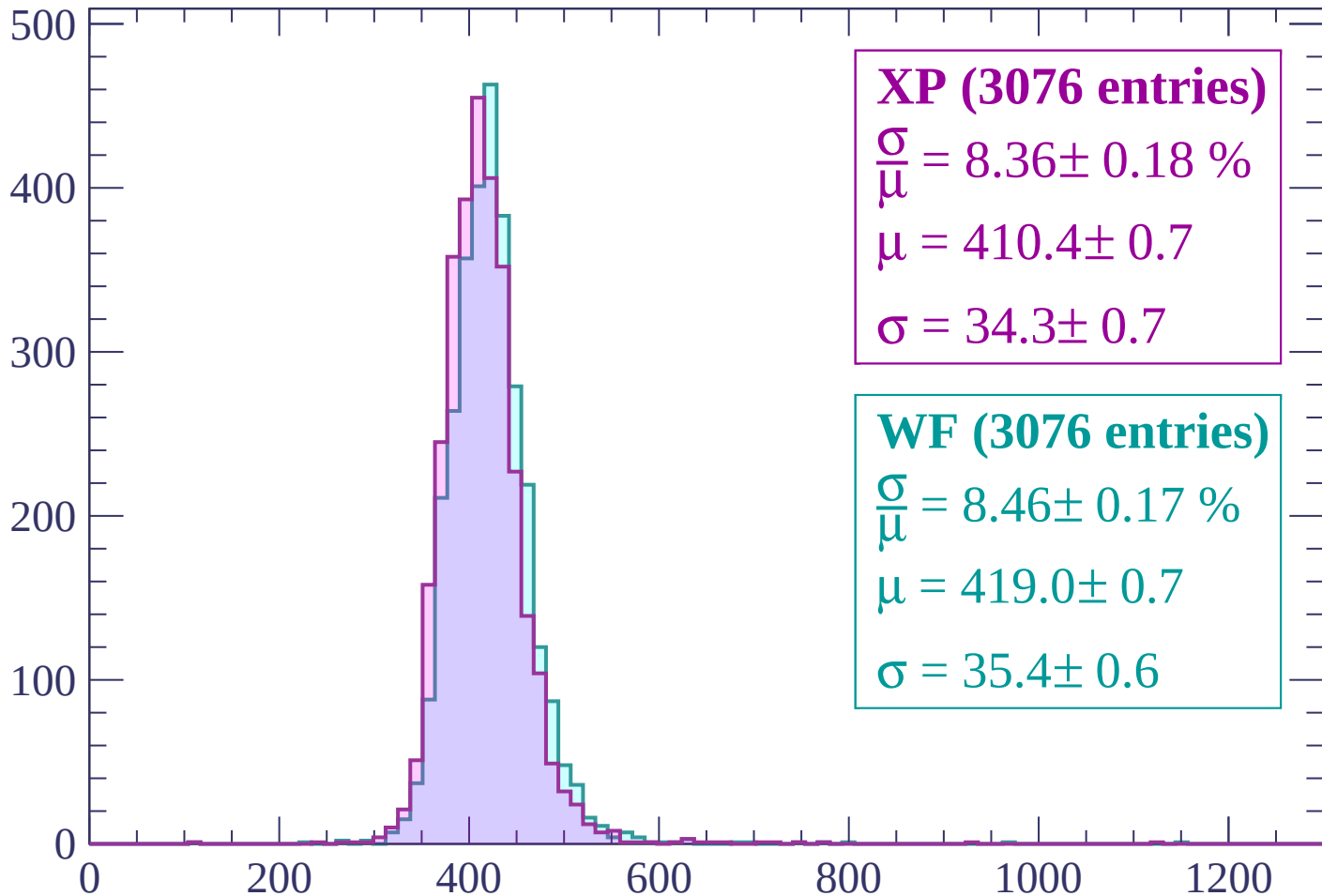
$$\mu = 415.4 \pm 0.7$$

$$\sigma = 34.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (3076 entries)

$$\frac{\sigma}{\mu} = 8.36 \pm 0.18 \%$$

$$\mu = 410.4 \pm 0.7$$

$$\sigma = 34.3 \pm 0.7$$

WF (3076 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

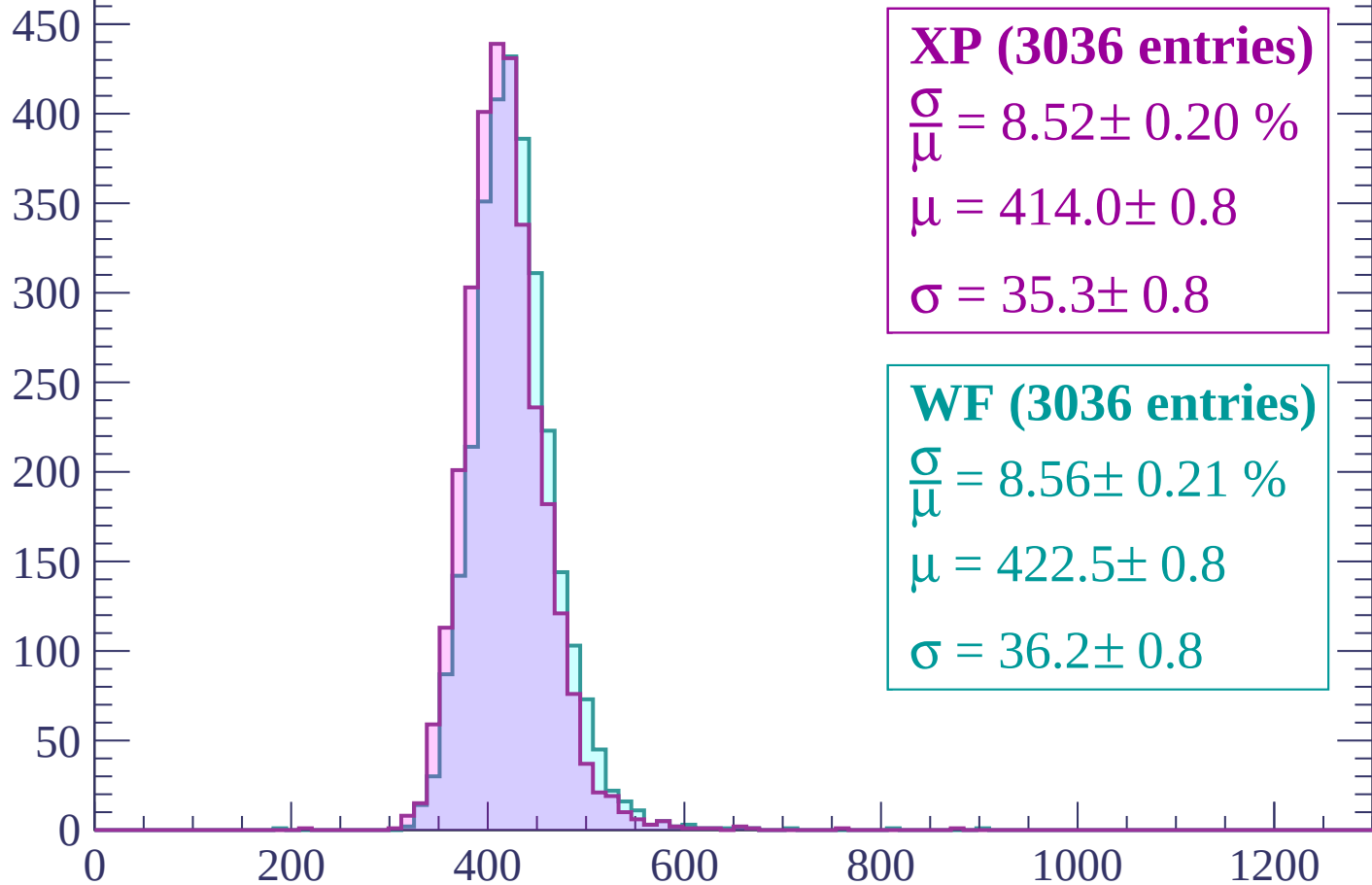
$$\mu = 419.0 \pm 0.7$$

$$\sigma = 35.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (3036 entries)

$$\frac{\sigma}{\mu} = 8.52 \pm 0.20 \%$$

$$\mu = 414.0 \pm 0.8$$

$$\sigma = 35.3 \pm 0.8$$

WF (3036 entries)

$$\frac{\sigma}{\mu} = 8.56 \pm 0.21 \%$$

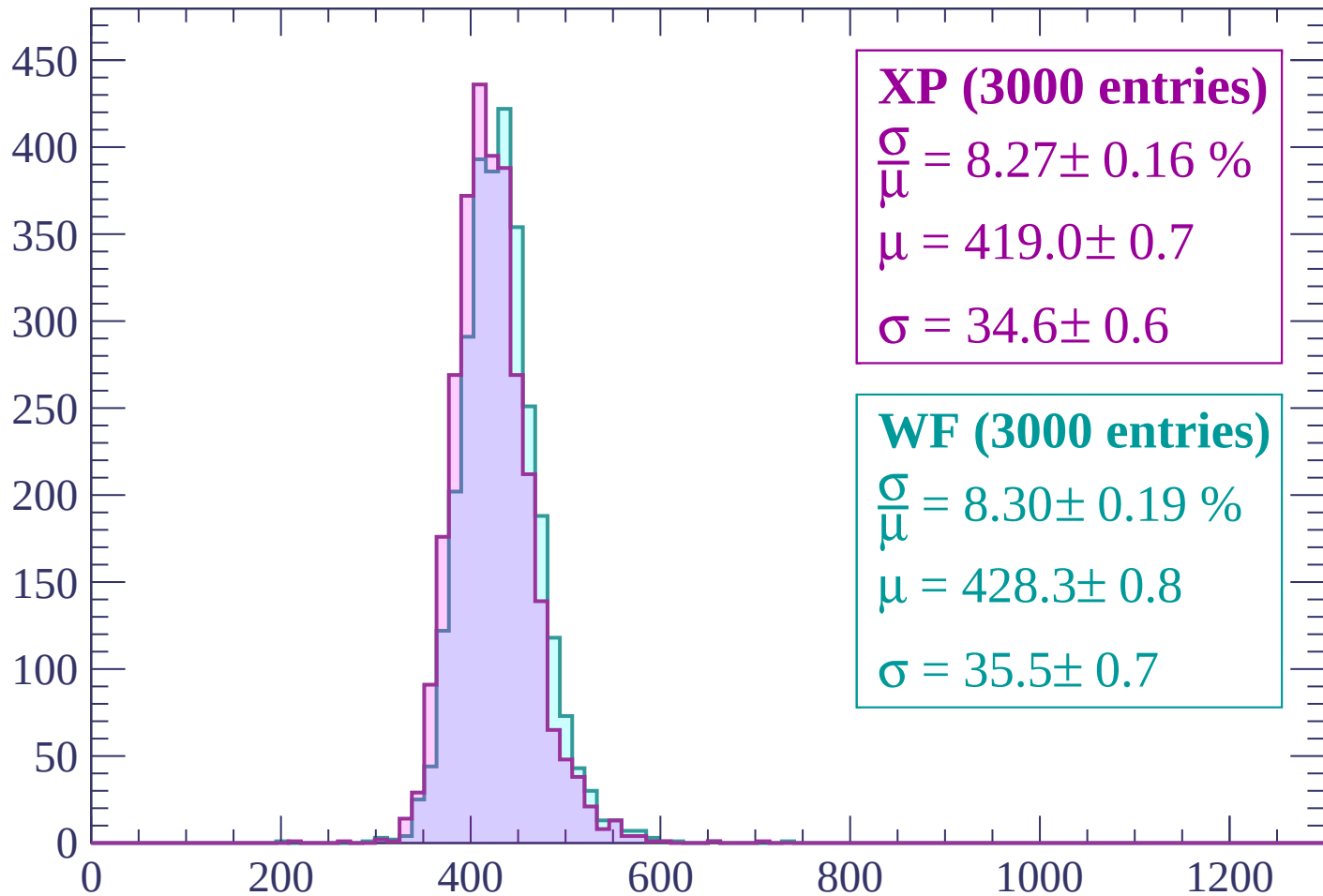
$$\mu = 422.5 \pm 0.8$$

$$\sigma = 36.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (3000 entries)

$$\frac{\sigma}{\mu} = 8.27 \pm 0.16 \%$$

$$\mu = 419.0 \pm 0.7$$

$$\sigma = 34.6 \pm 0.6$$

WF (3000 entries)

$$\frac{\sigma}{\mu} = 8.30 \pm 0.19 \%$$

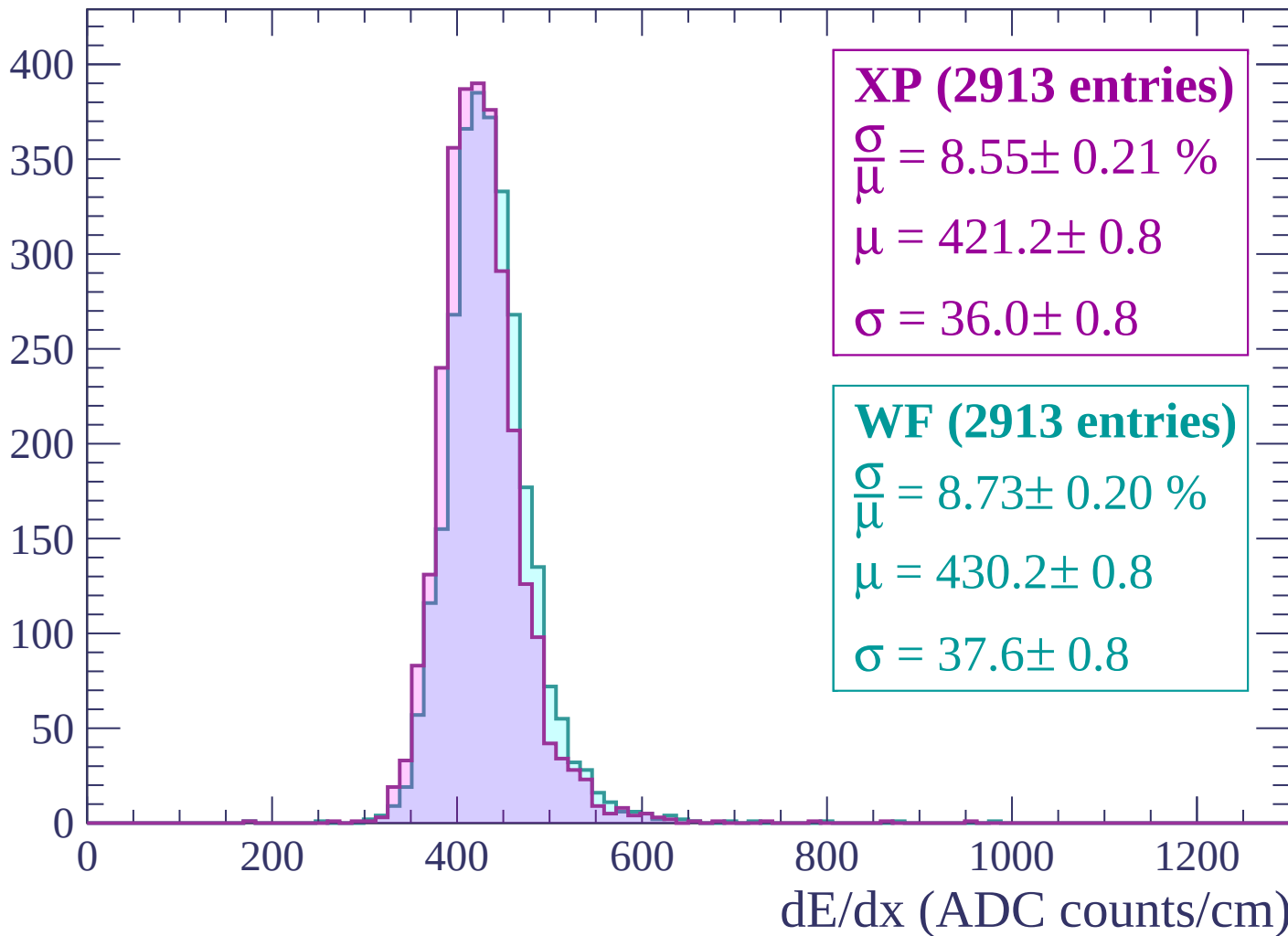
$$\mu = 428.3 \pm 0.8$$

$$\sigma = 35.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

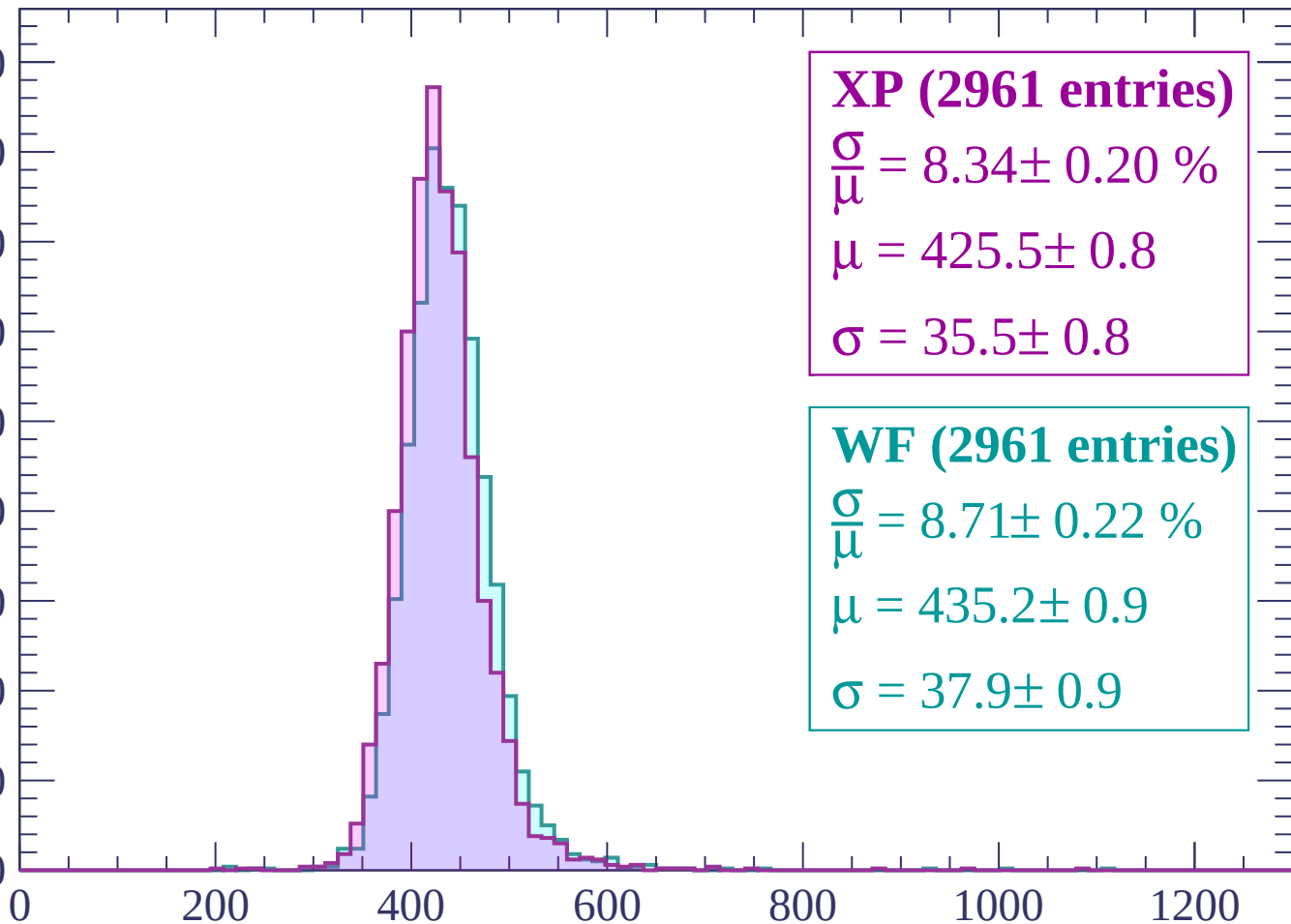
Count



Energy loss | $900 < p < 960$

Count

450
400
350
300
250
200
150
100
50
0



XP (2961 entries)

$\frac{\sigma}{\mu} = 8.34 \pm 0.20 \%$

$\mu = 425.5 \pm 0.8$

$\sigma = 35.5 \pm 0.8$

WF (2961 entries)

$\frac{\sigma}{\mu} = 8.71 \pm 0.22 \%$

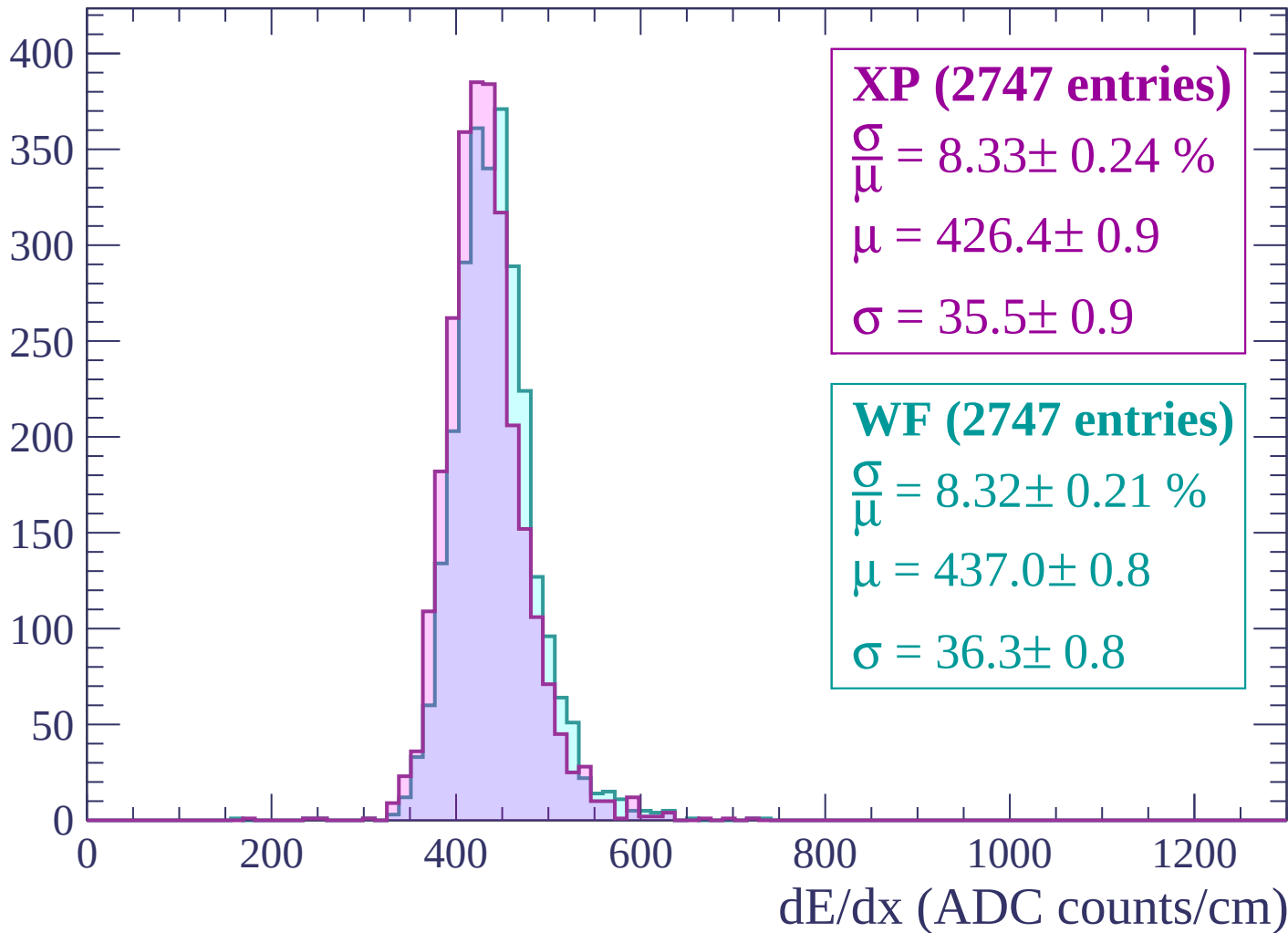
$\mu = 435.2 \pm 0.9$

$\sigma = 37.9 \pm 0.9$

dE/dx (ADC counts/cm)

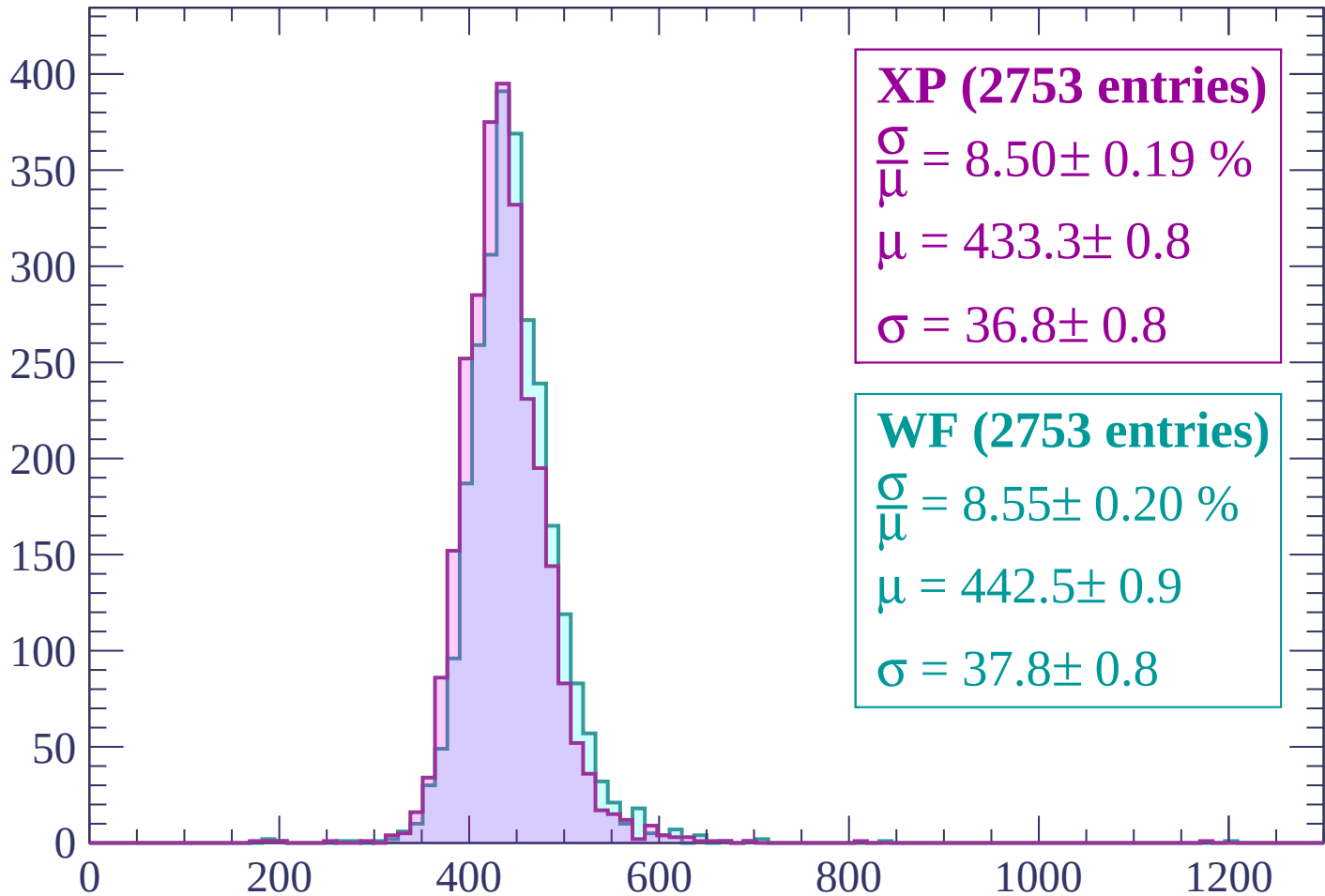
Energy loss | $960 < p < 1020$

Count



Energy loss | $1020 < p < 1080$

Count



XP (2753 entries)

$\frac{\sigma}{\mu} = 8.50 \pm 0.19 \%$

$\mu = 433.3 \pm 0.8$

$\sigma = 36.8 \pm 0.8$

WF (2753 entries)

$\frac{\sigma}{\mu} = 8.55 \pm 0.20 \%$

$\mu = 442.5 \pm 0.9$

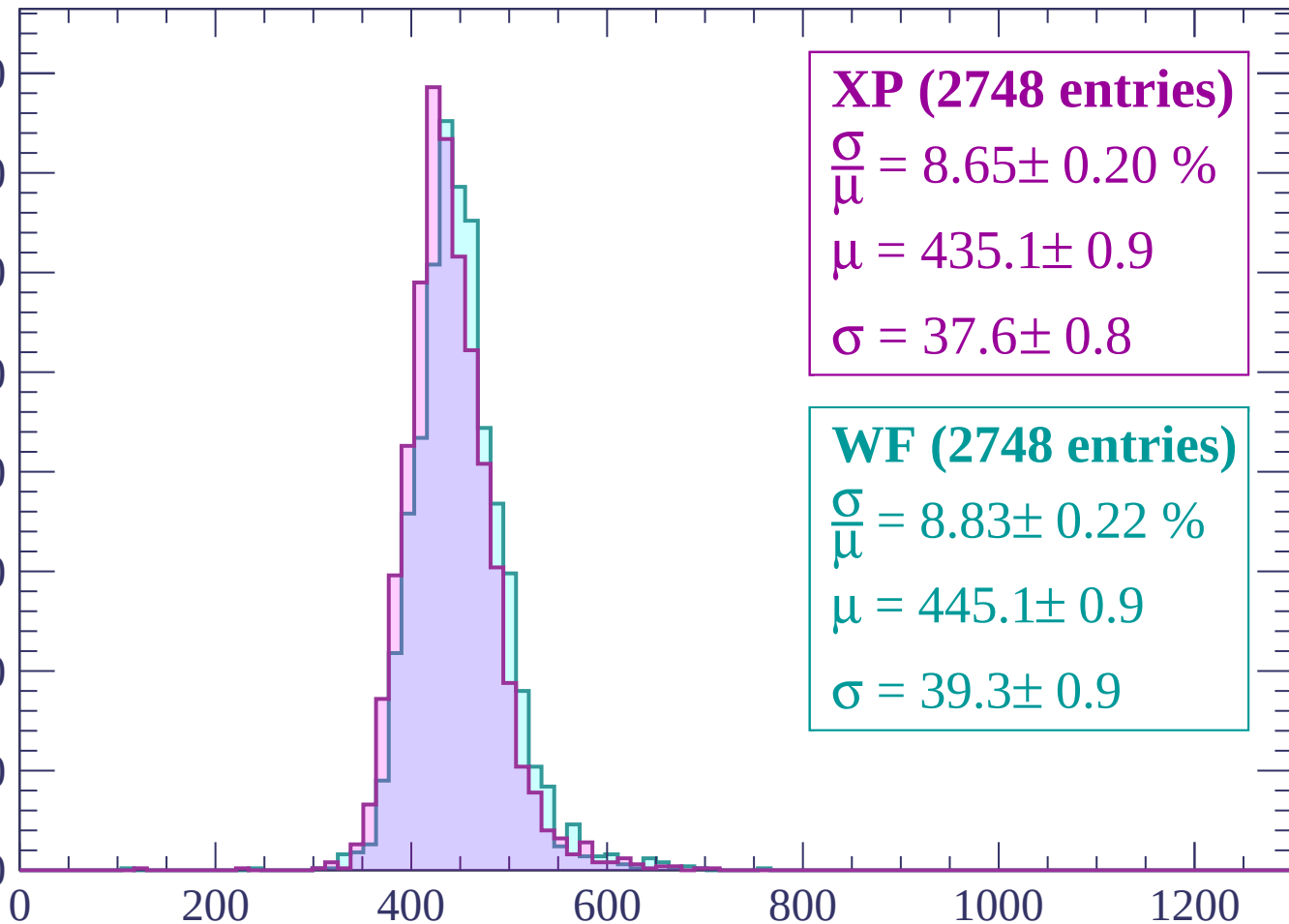
$\sigma = 37.8 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count

400
350
300
250
200
150
100
50
0



XP (2748 entries)

$\frac{\sigma}{\mu} = 8.65 \pm 0.20 \%$

$\mu = 435.1 \pm 0.9$

$\sigma = 37.6 \pm 0.8$

WF (2748 entries)

$\frac{\sigma}{\mu} = 8.83 \pm 0.22 \%$

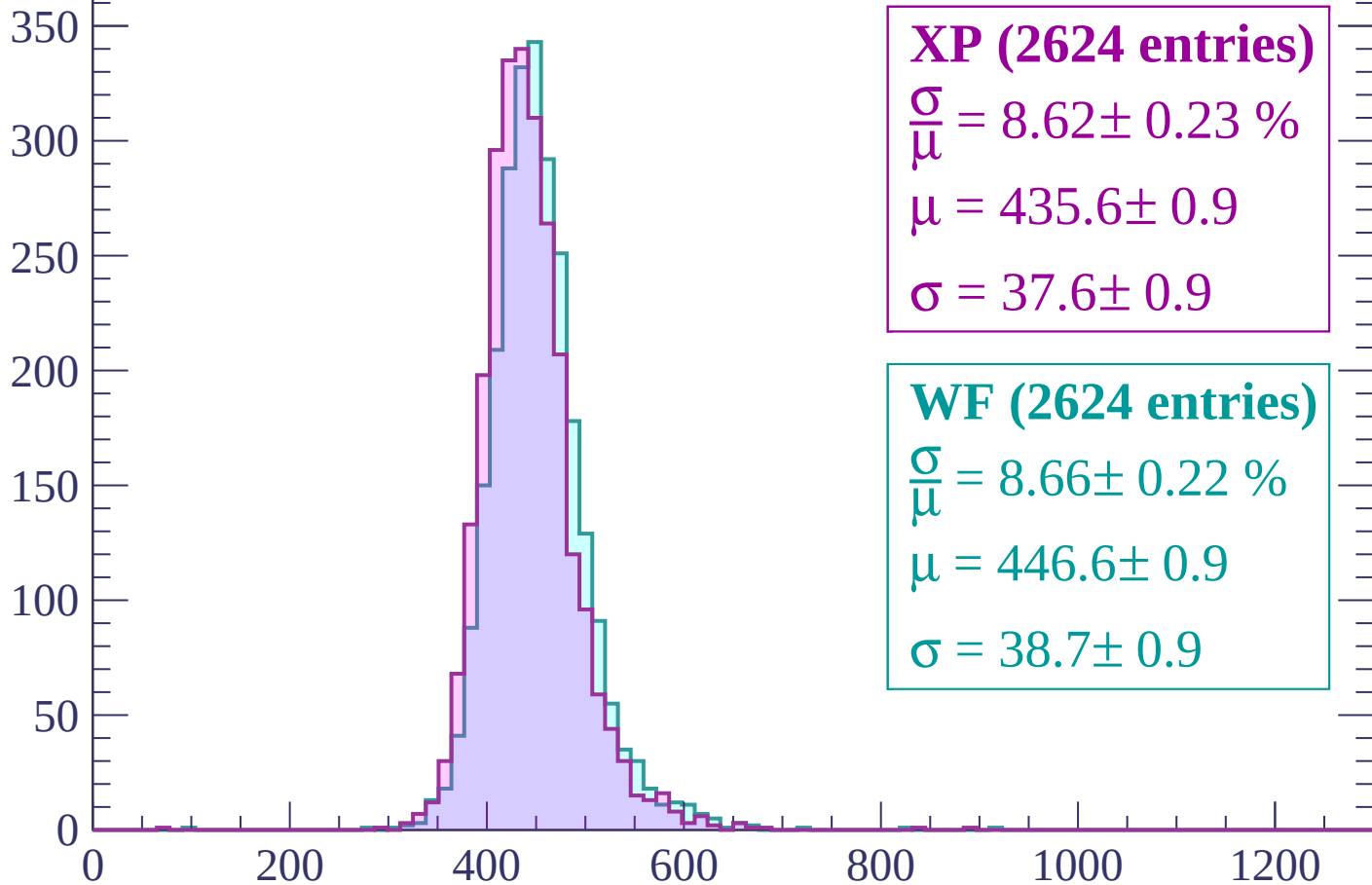
$\mu = 445.1 \pm 0.9$

$\sigma = 39.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (2624 entries)

$\frac{\sigma}{\mu} = 8.62 \pm 0.23 \%$

$\mu = 435.6 \pm 0.9$

$\sigma = 37.6 \pm 0.9$

WF (2624 entries)

$\frac{\sigma}{\mu} = 8.66 \pm 0.22 \%$

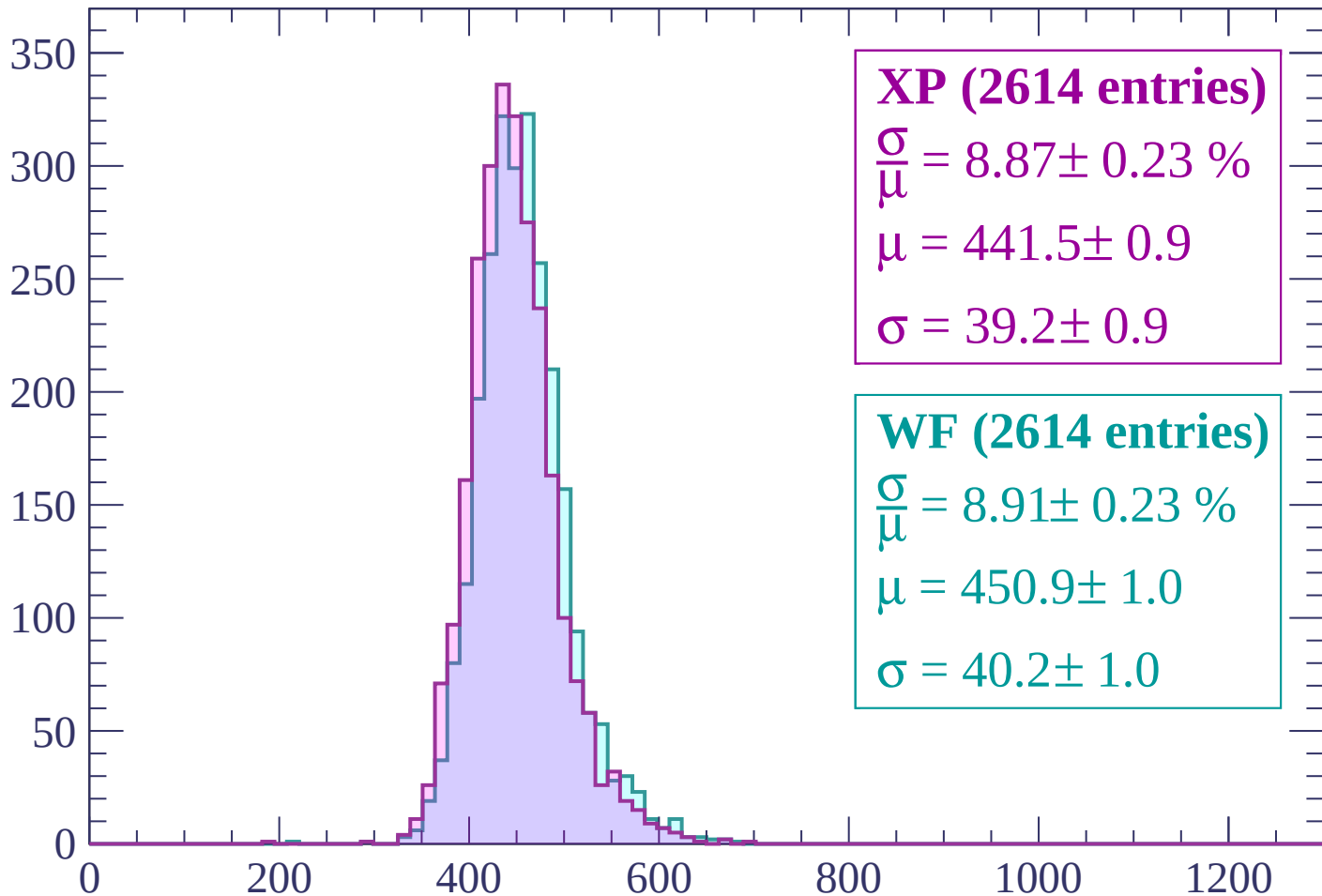
$\mu = 446.6 \pm 0.9$

$\sigma = 38.7 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (2614 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.23 \%$$

$$\mu = 441.5 \pm 0.9$$

$$\sigma = 39.2 \pm 0.9$$

WF (2614 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.23 \%$$

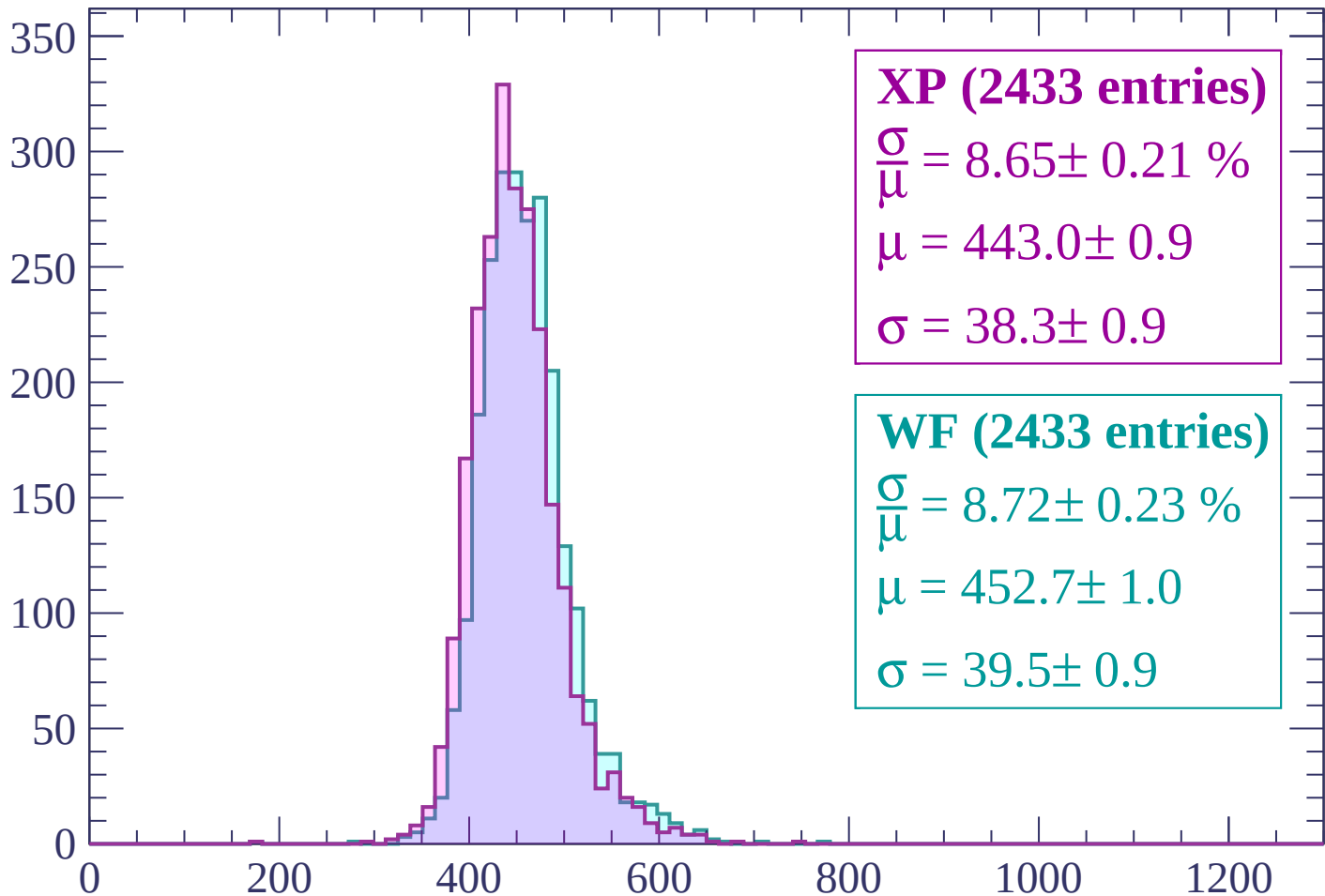
$$\mu = 450.9 \pm 1.0$$

$$\sigma = 40.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (2433 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.21 \%$$

$$\mu = 443.0 \pm 0.9$$

$$\sigma = 38.3 \pm 0.9$$

WF (2433 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.23 \%$$

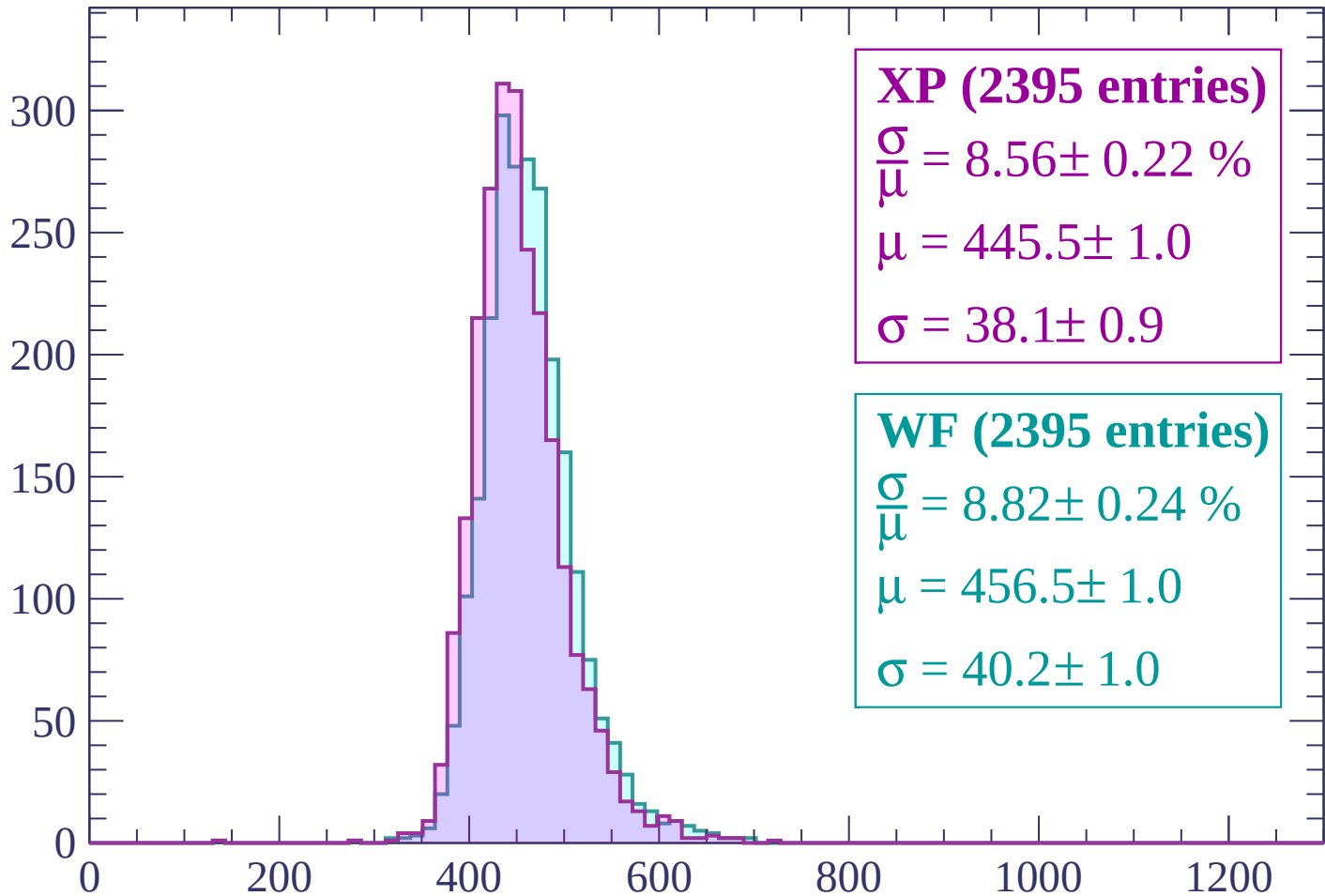
$$\mu = 452.7 \pm 1.0$$

$$\sigma = 39.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (2395 entries)

$$\frac{\sigma}{\mu} = 8.56 \pm 0.22 \%$$

$$\mu = 445.5 \pm 1.0$$

$$\sigma = 38.1 \pm 0.9$$

WF (2395 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.24 \%$$

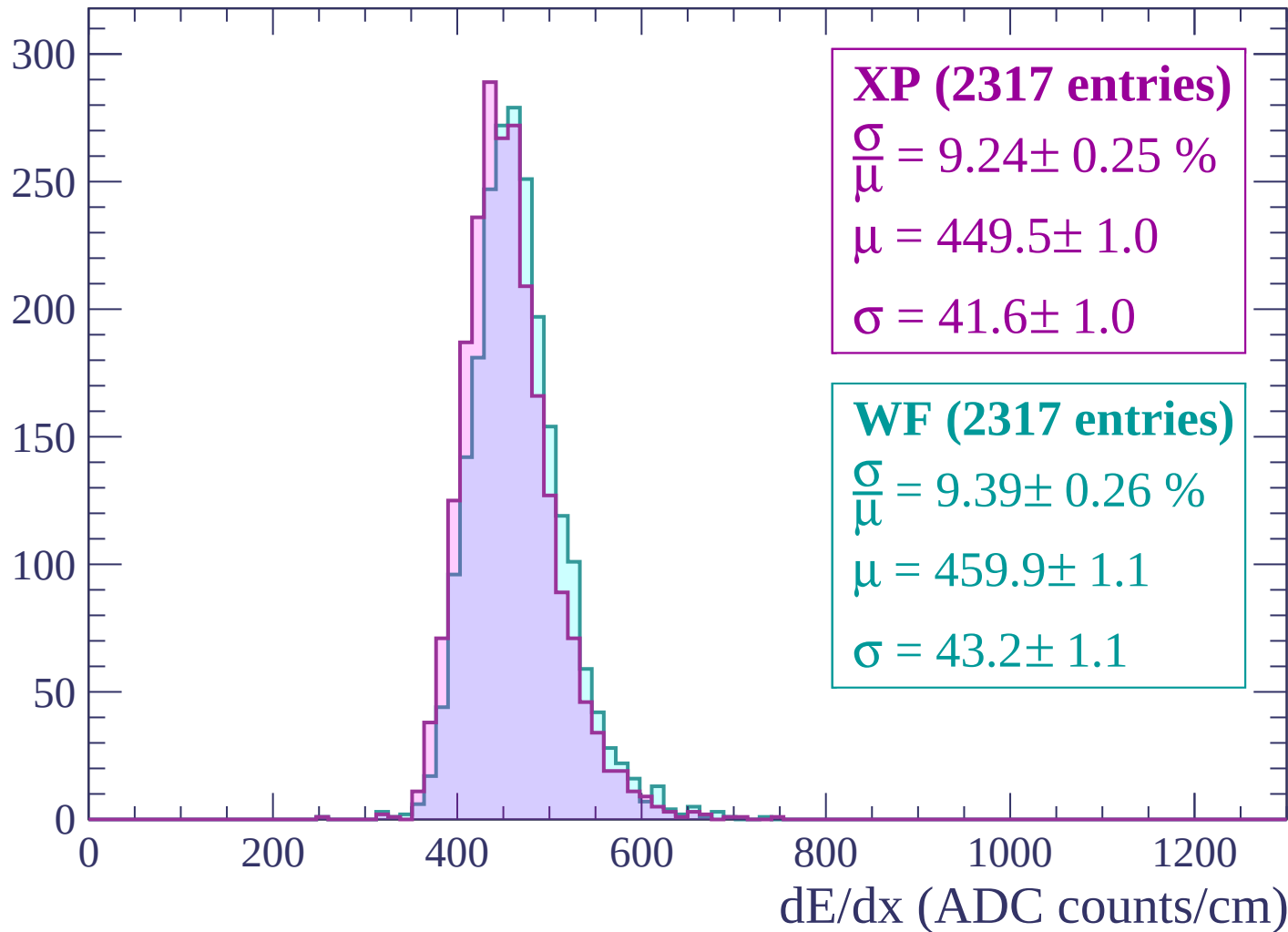
$$\mu = 456.5 \pm 1.0$$

$$\sigma = 40.2 \pm 1.0$$

dE/dx (ADC counts/cm)

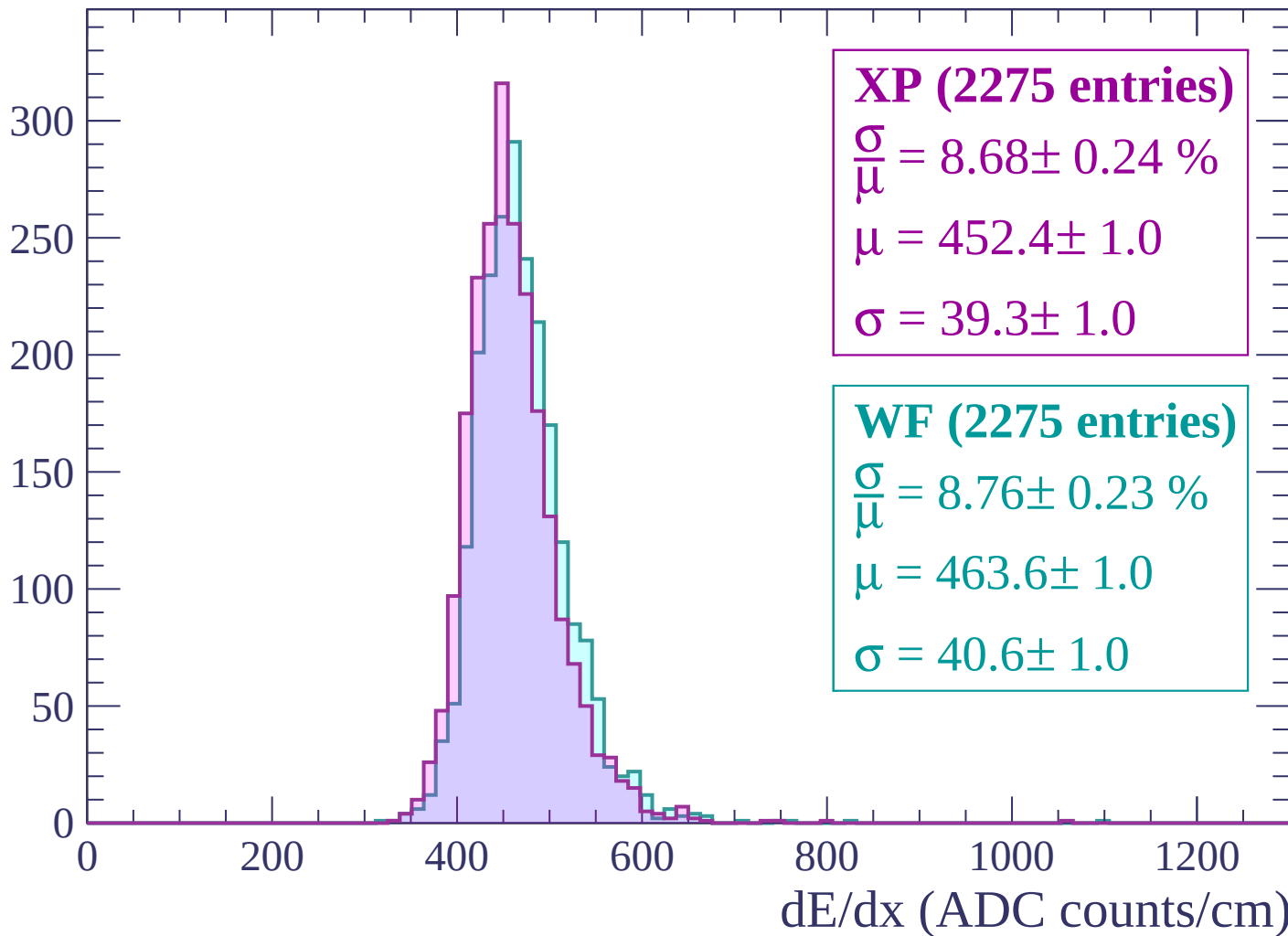
Energy loss | $1380 < p < 1440$

Count



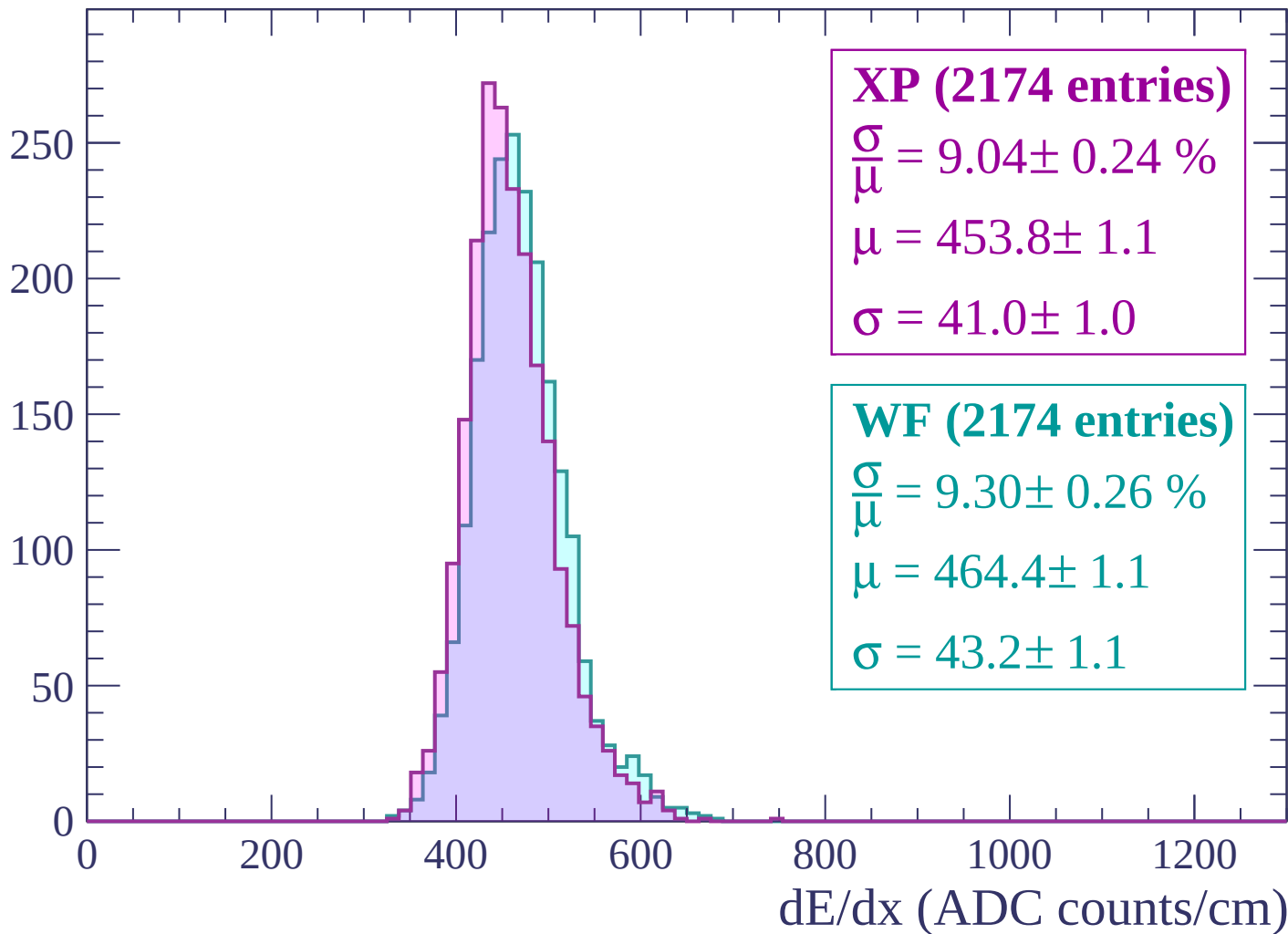
Energy loss | $1440 < p < 1500$

Count



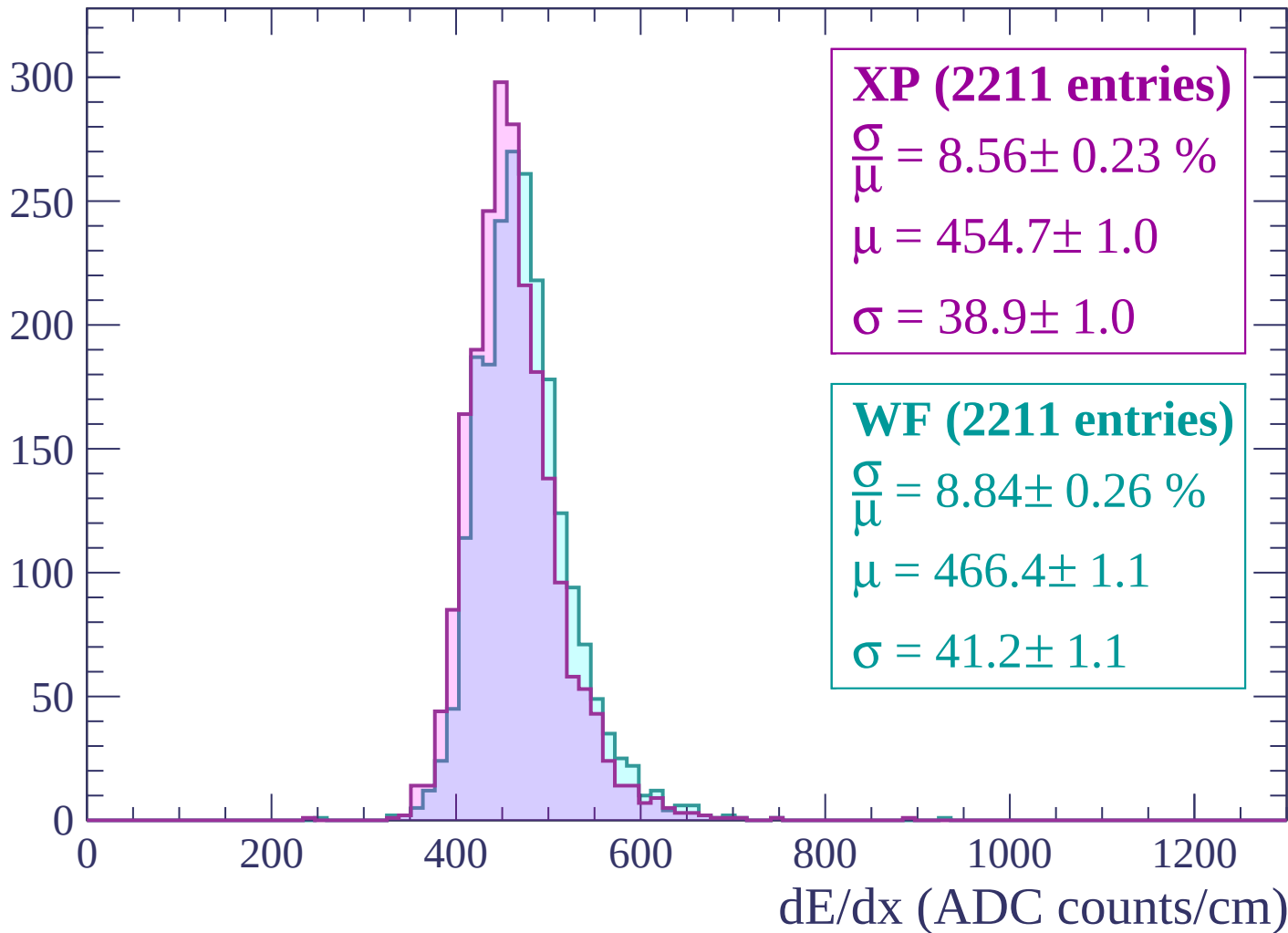
Energy loss | $1500 < p < 1560$

Count



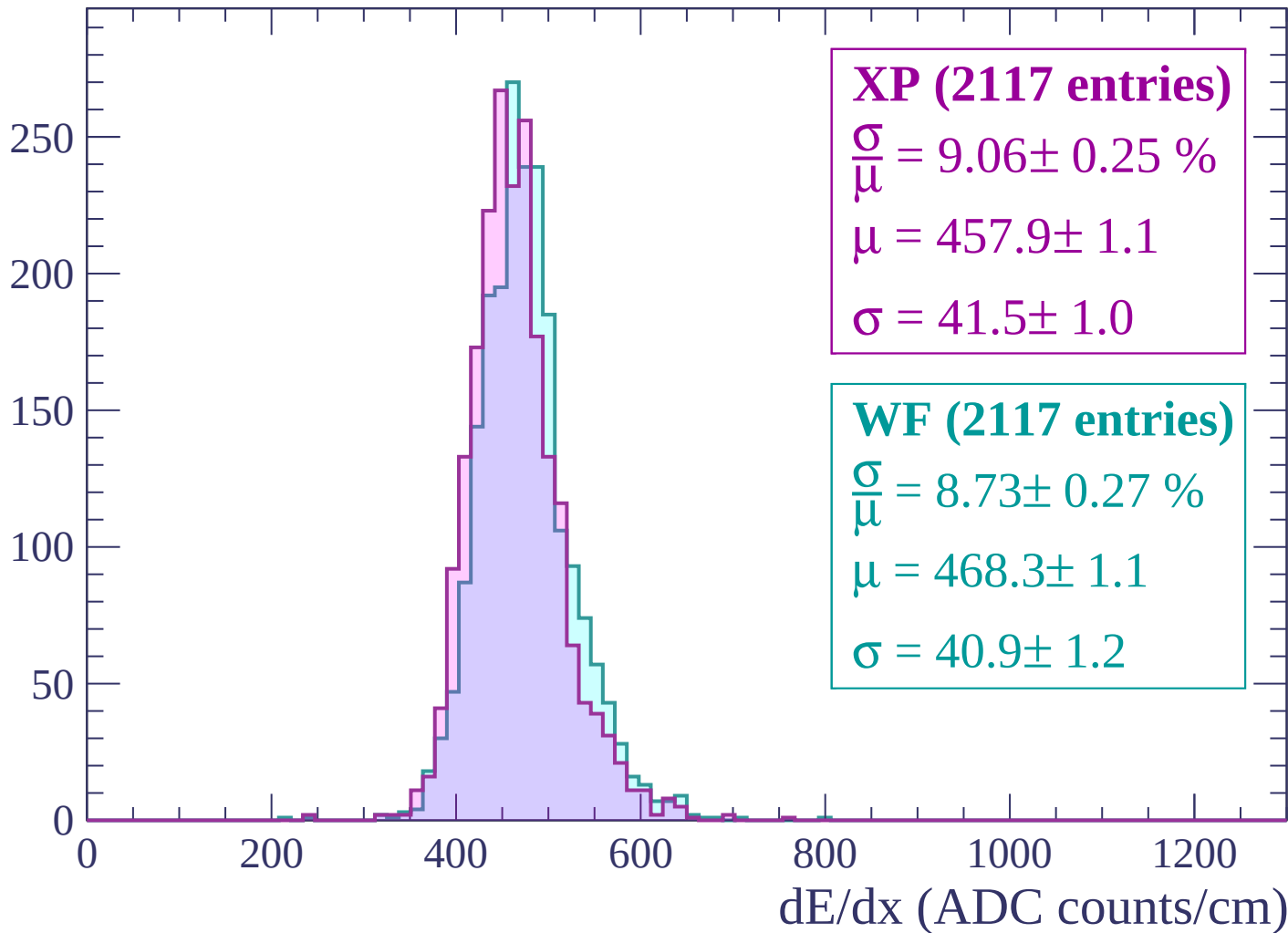
Energy loss | $1560 < p < 1620$

Count



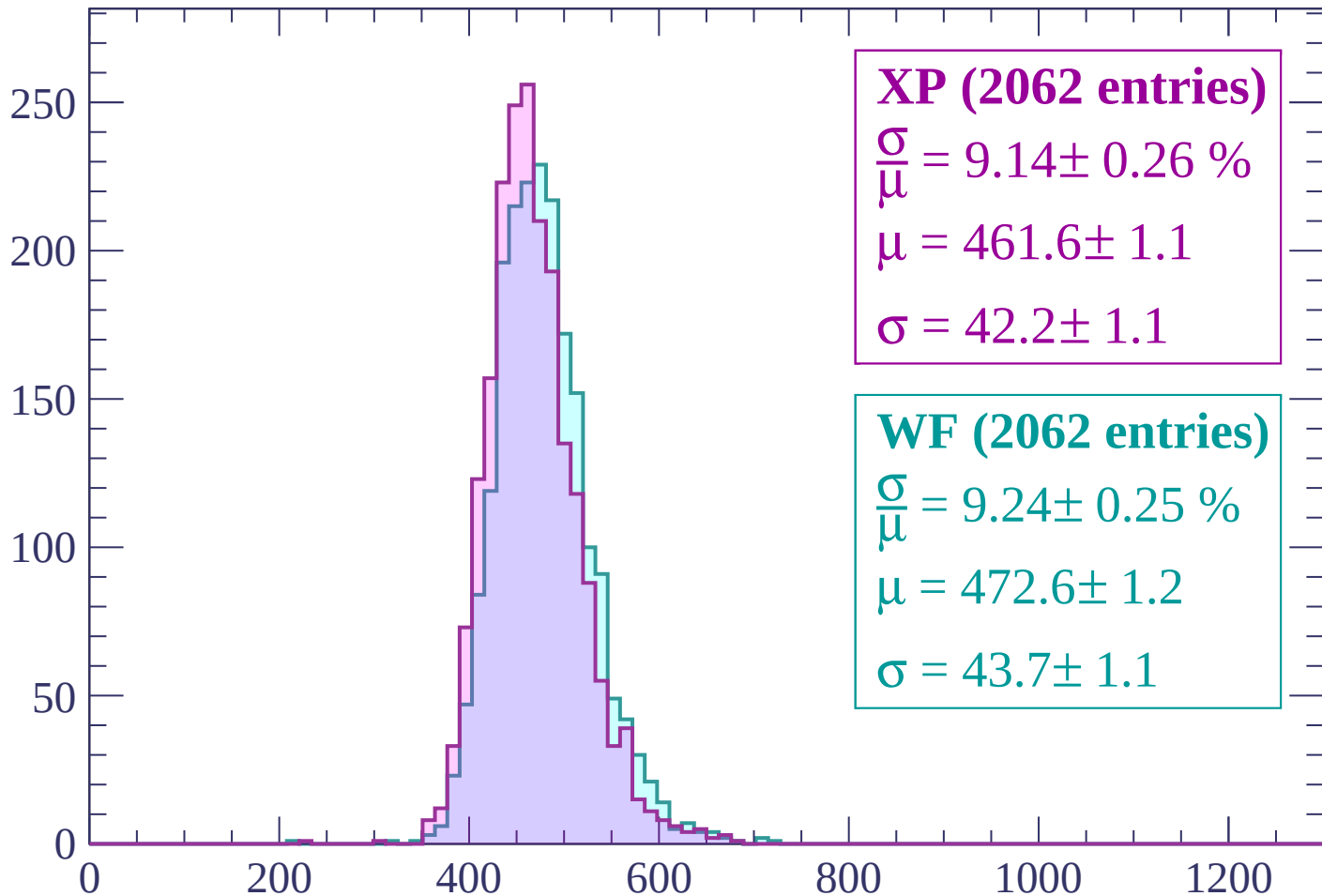
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP (2062 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.26 \%$$

$$\mu = 461.6 \pm 1.1$$

$$\sigma = 42.2 \pm 1.1$$

WF (2062 entries)

$$\frac{\sigma}{\mu} = 9.24 \pm 0.25 \%$$

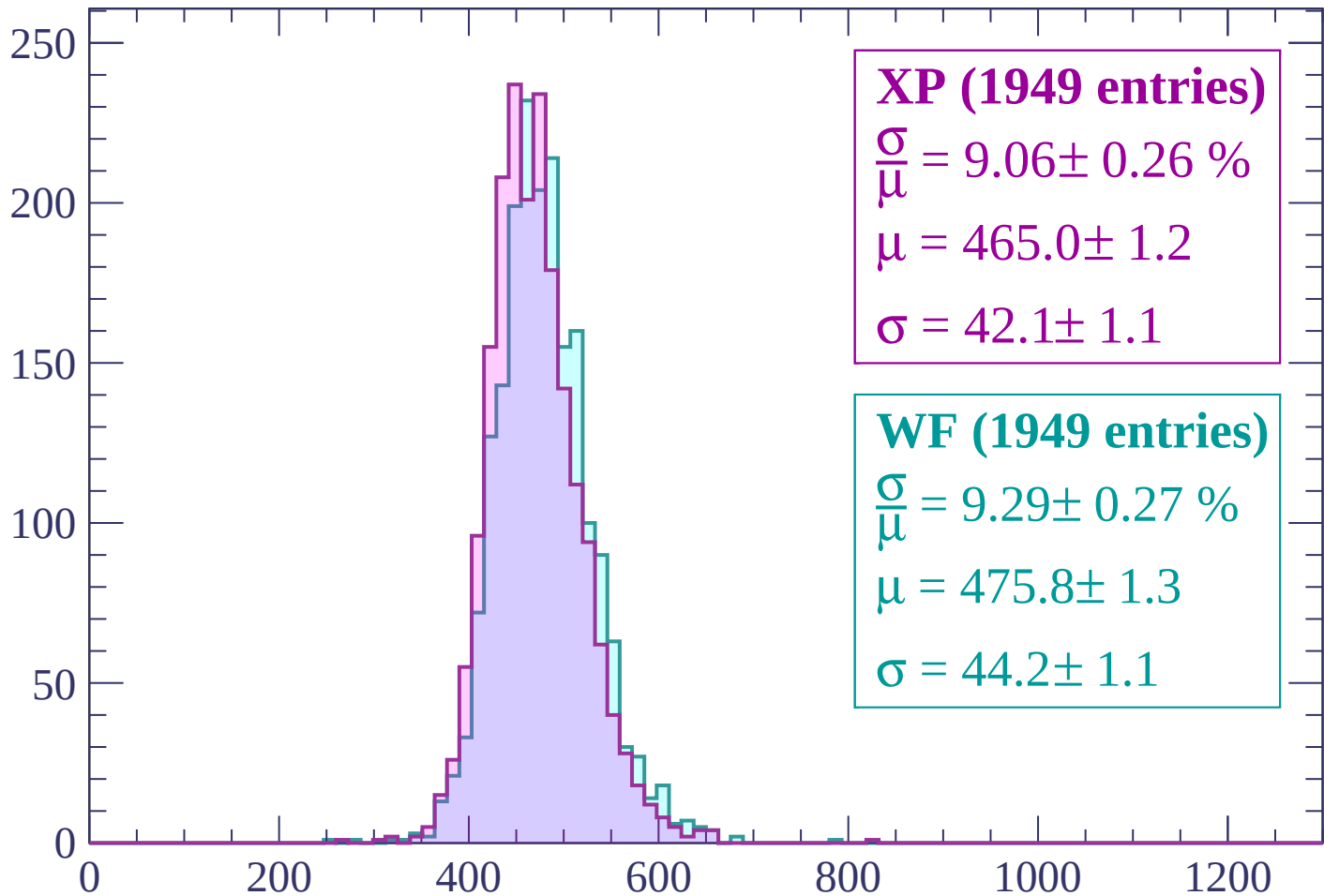
$$\mu = 472.6 \pm 1.2$$

$$\sigma = 43.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (1949 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.26 \%$$

$$\mu = 465.0 \pm 1.2$$

$$\sigma = 42.1 \pm 1.1$$

WF (1949 entries)

$$\frac{\sigma}{\mu} = 9.29 \pm 0.27 \%$$

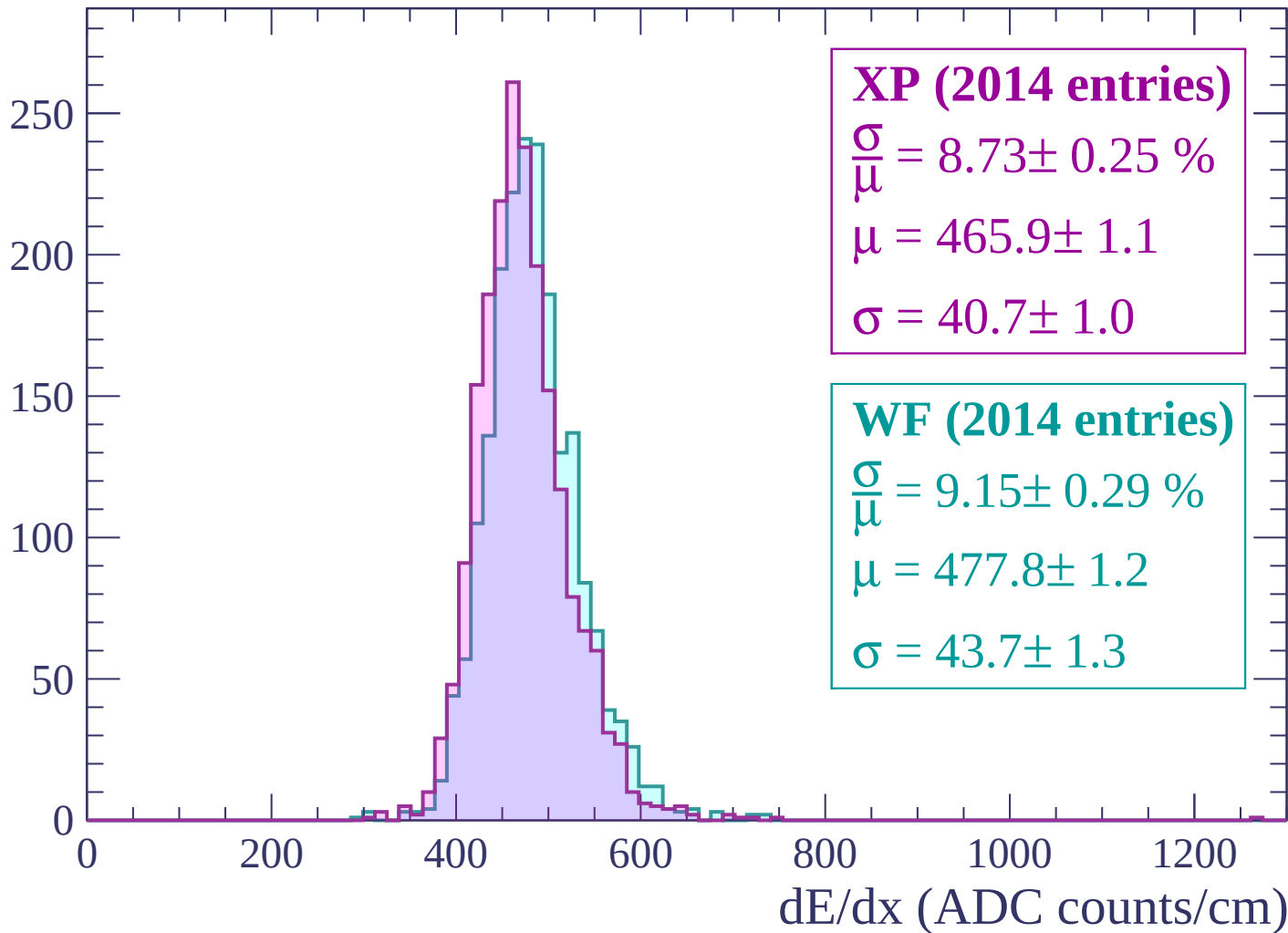
$$\mu = 475.8 \pm 1.3$$

$$\sigma = 44.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1875 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.31 \%$

$\mu = 467.1 \pm 1.2$

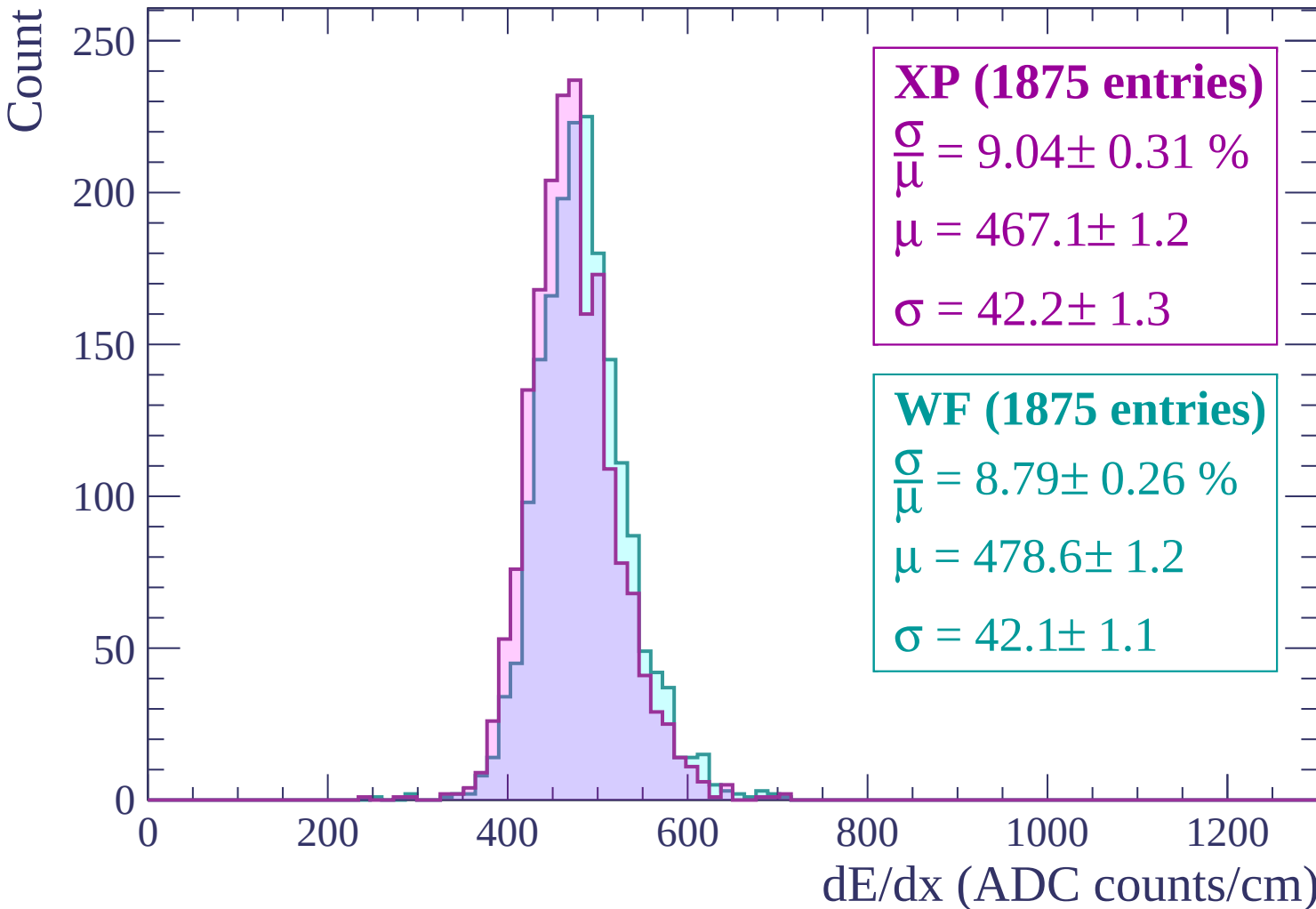
$\sigma = 42.2 \pm 1.3$

WF (1875 entries)

$\frac{\sigma}{\mu} = 8.79 \pm 0.26 \%$

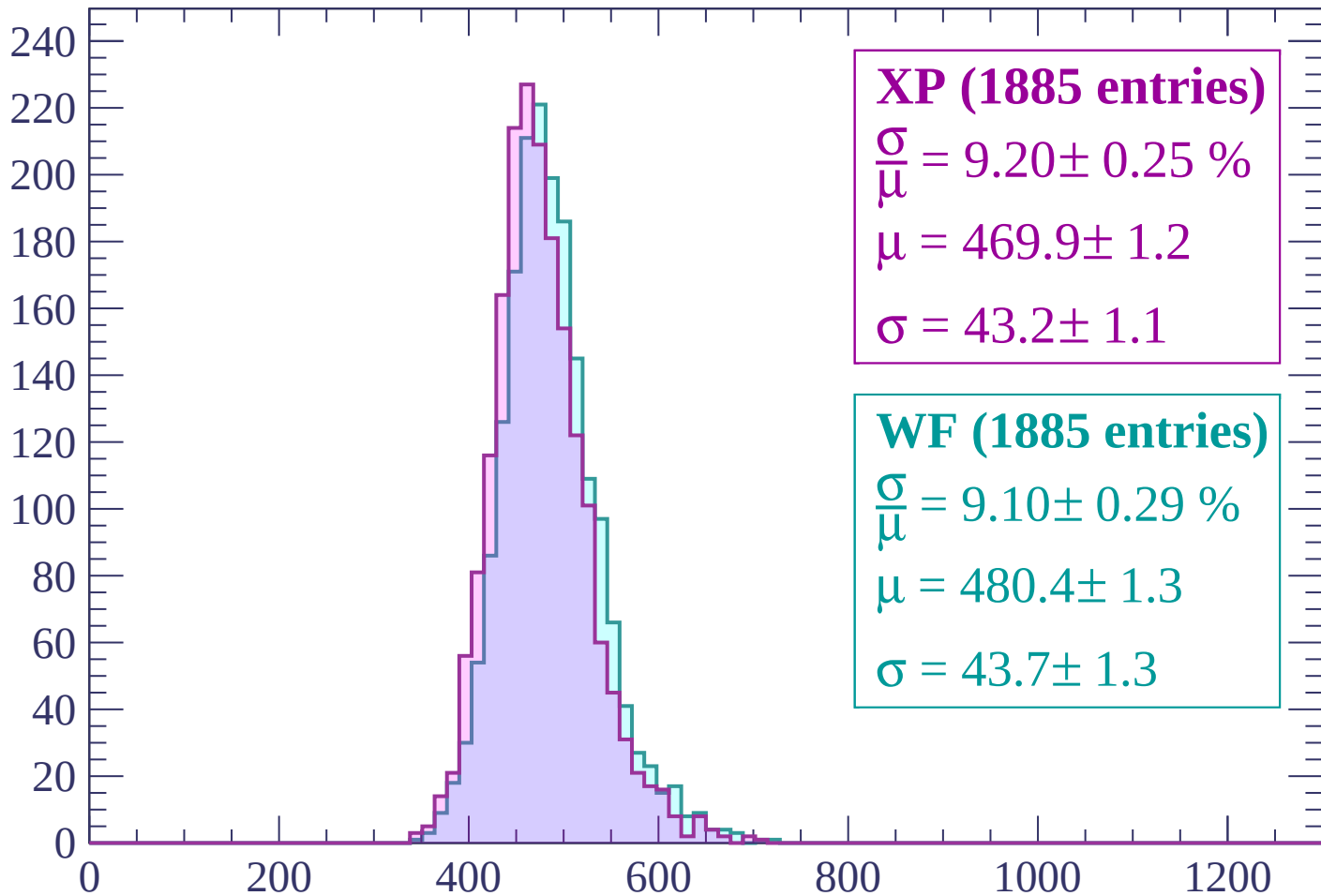
$\mu = 478.6 \pm 1.2$

$\sigma = 42.1 \pm 1.1$



Energy loss | $1920 < p < 1980$

Count



XP (1885 entries)

$$\frac{\sigma}{\mu} = 9.20 \pm 0.25 \%$$

$$\mu = 469.9 \pm 1.2$$

$$\sigma = 43.2 \pm 1.1$$

WF (1885 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.29 \%$$

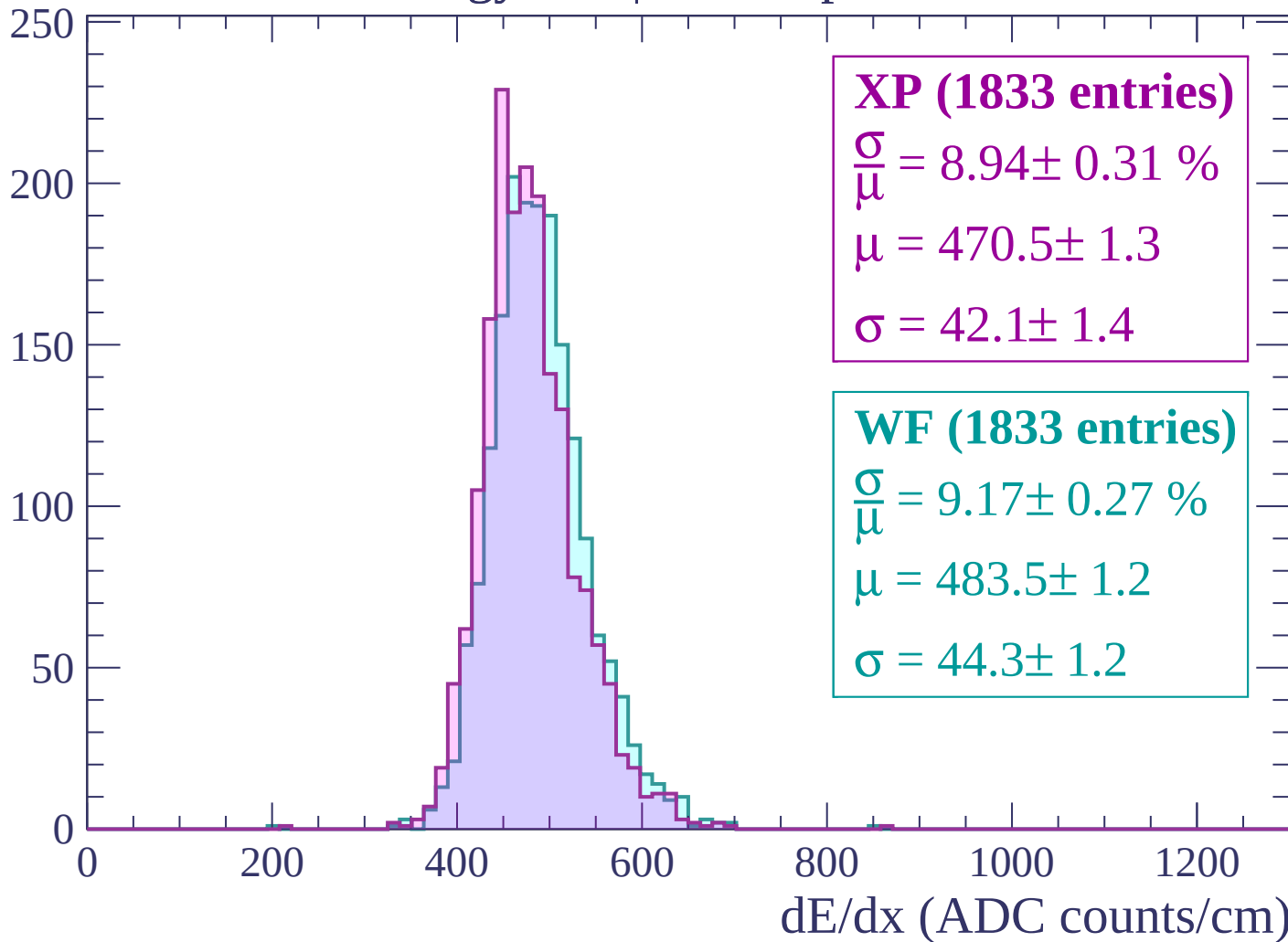
$$\mu = 480.4 \pm 1.3$$

$$\sigma = 43.7 \pm 1.3$$

dE/dx (ADC counts/cm)

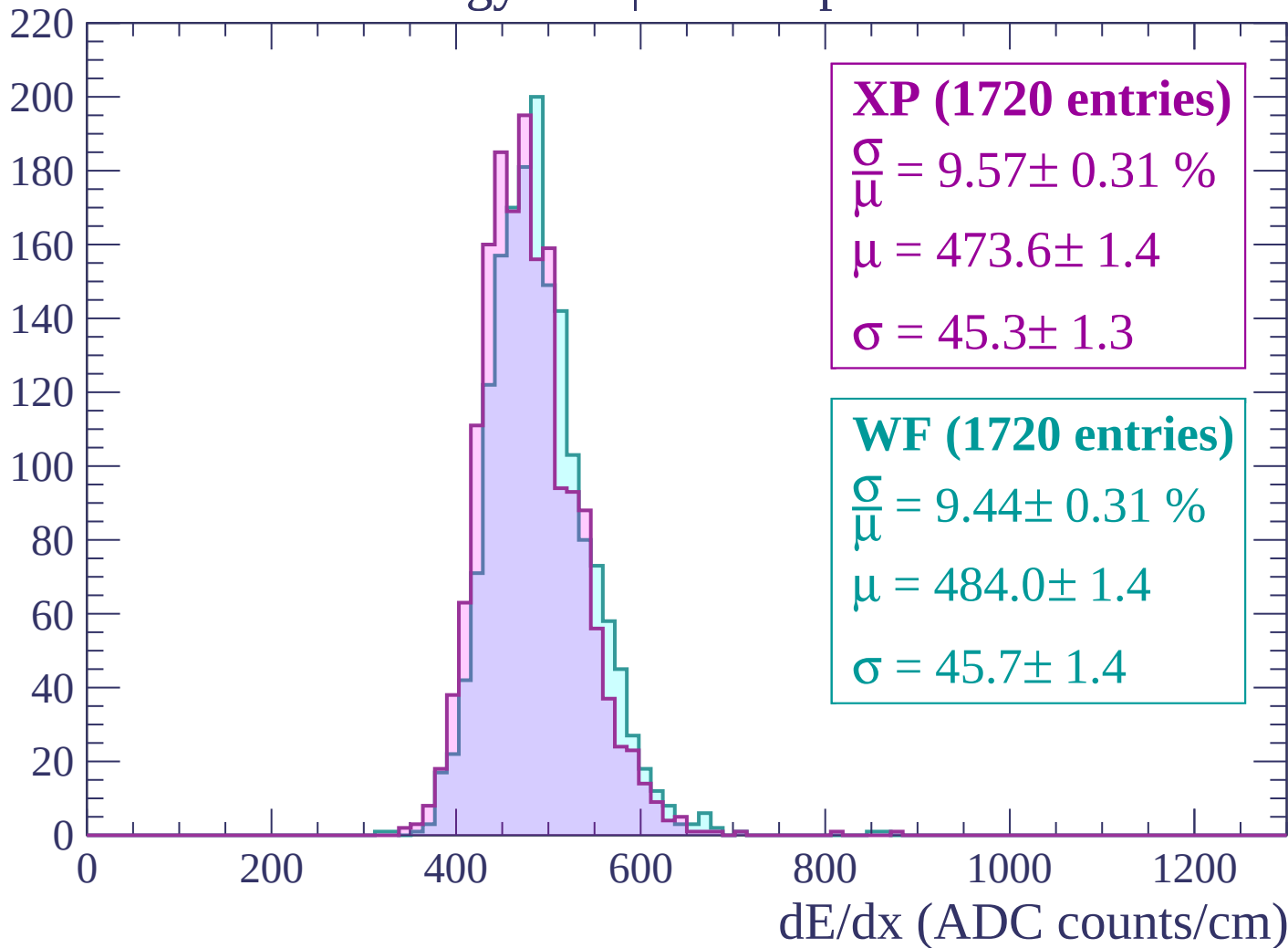
Energy loss | $1980 < p < 2040$

Count



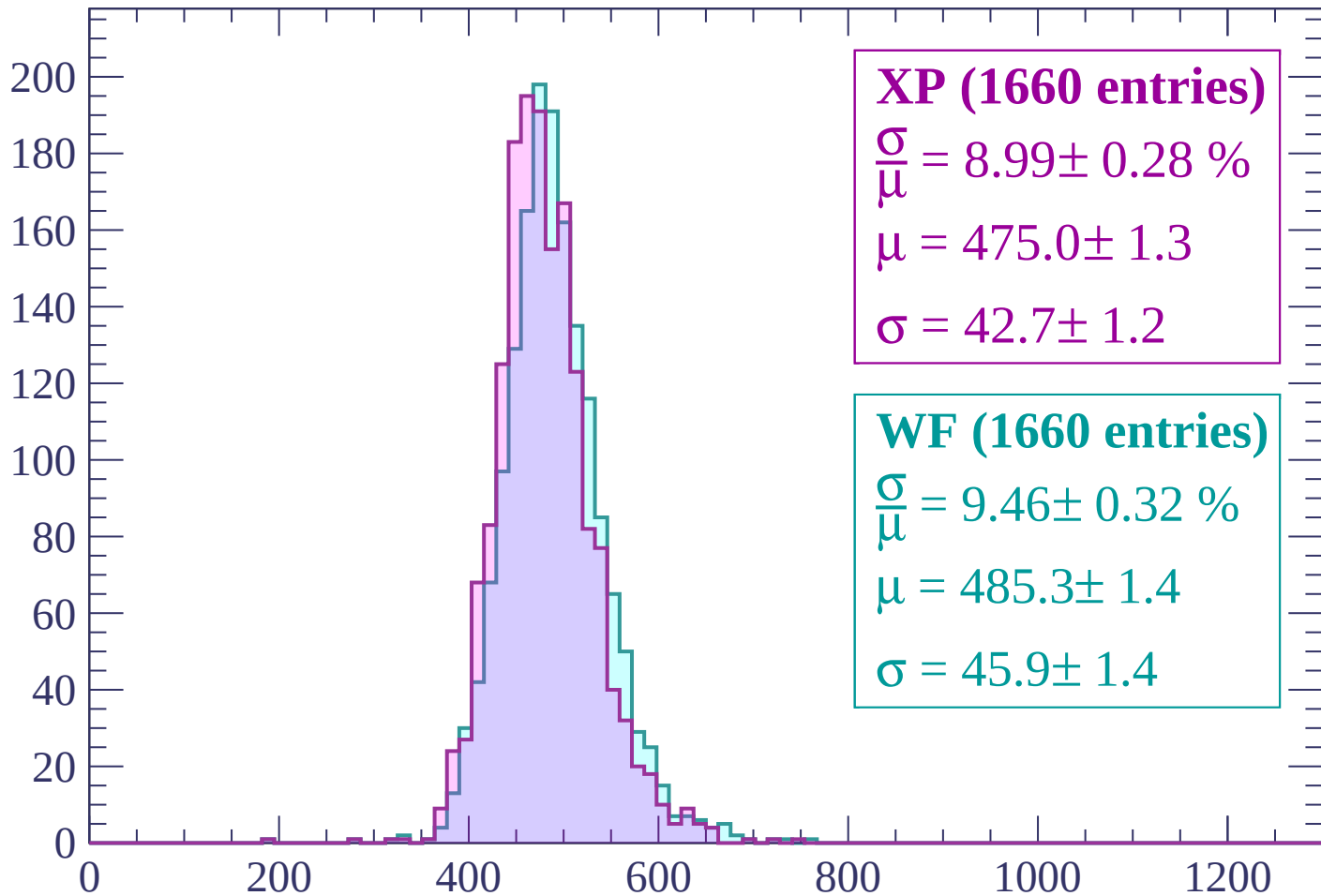
Energy loss | $2040 < p < 2100$

Count



Energy loss | $2100 < p < 2160$

Count



XP (1660 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.28 \%$$

$$\mu = 475.0 \pm 1.3$$

$$\sigma = 42.7 \pm 1.2$$

WF (1660 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.32 \%$$

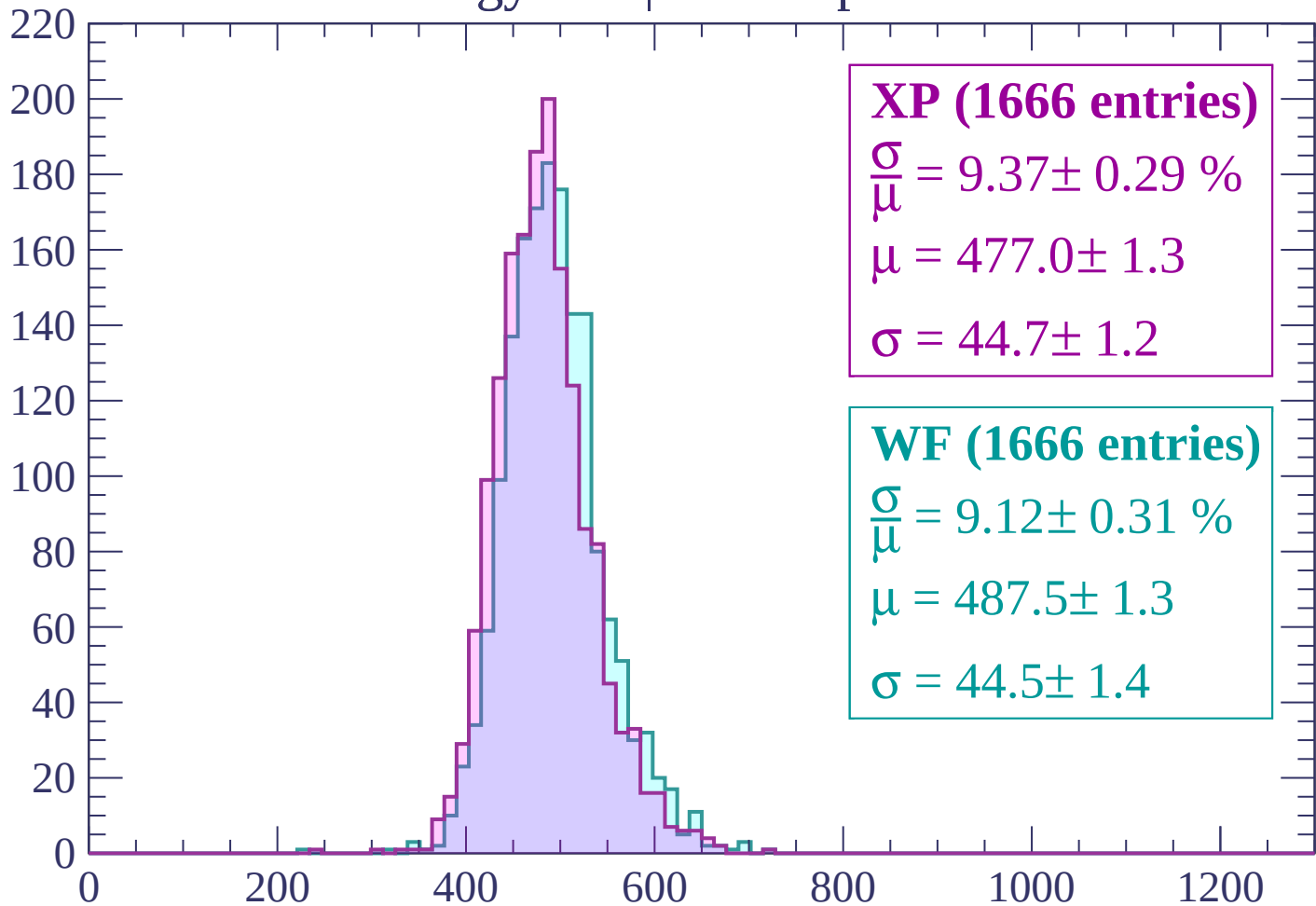
$$\mu = 485.3 \pm 1.4$$

$$\sigma = 45.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (1666 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.29 \%$$

$$\mu = 477.0 \pm 1.3$$

$$\sigma = 44.7 \pm 1.2$$

WF (1666 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 0.31 \%$$

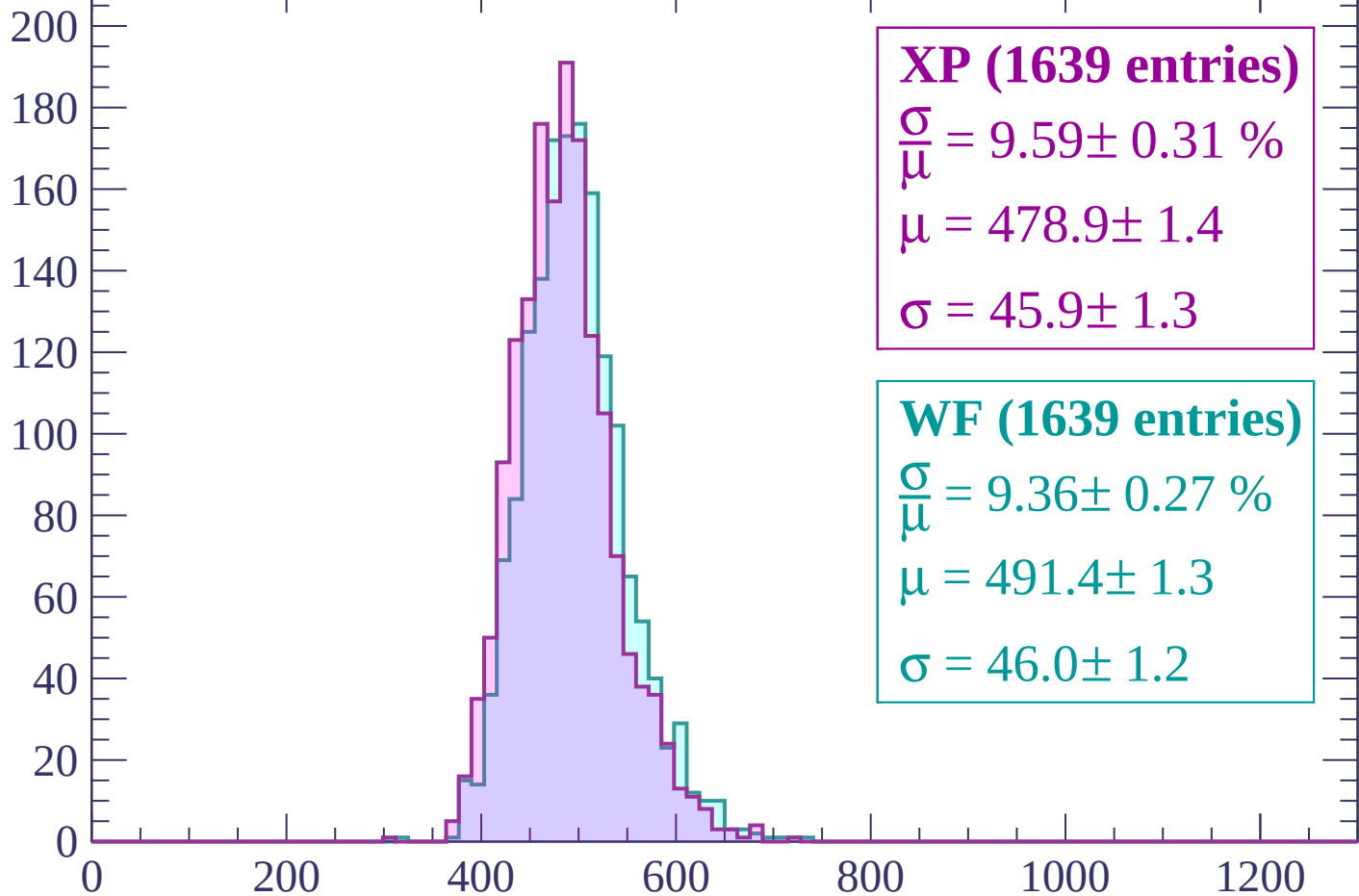
$$\mu = 487.5 \pm 1.3$$

$$\sigma = 44.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (1639 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.31 \%$$

$$\mu = 478.9 \pm 1.4$$

$$\sigma = 45.9 \pm 1.3$$

WF (1639 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.27 \%$$

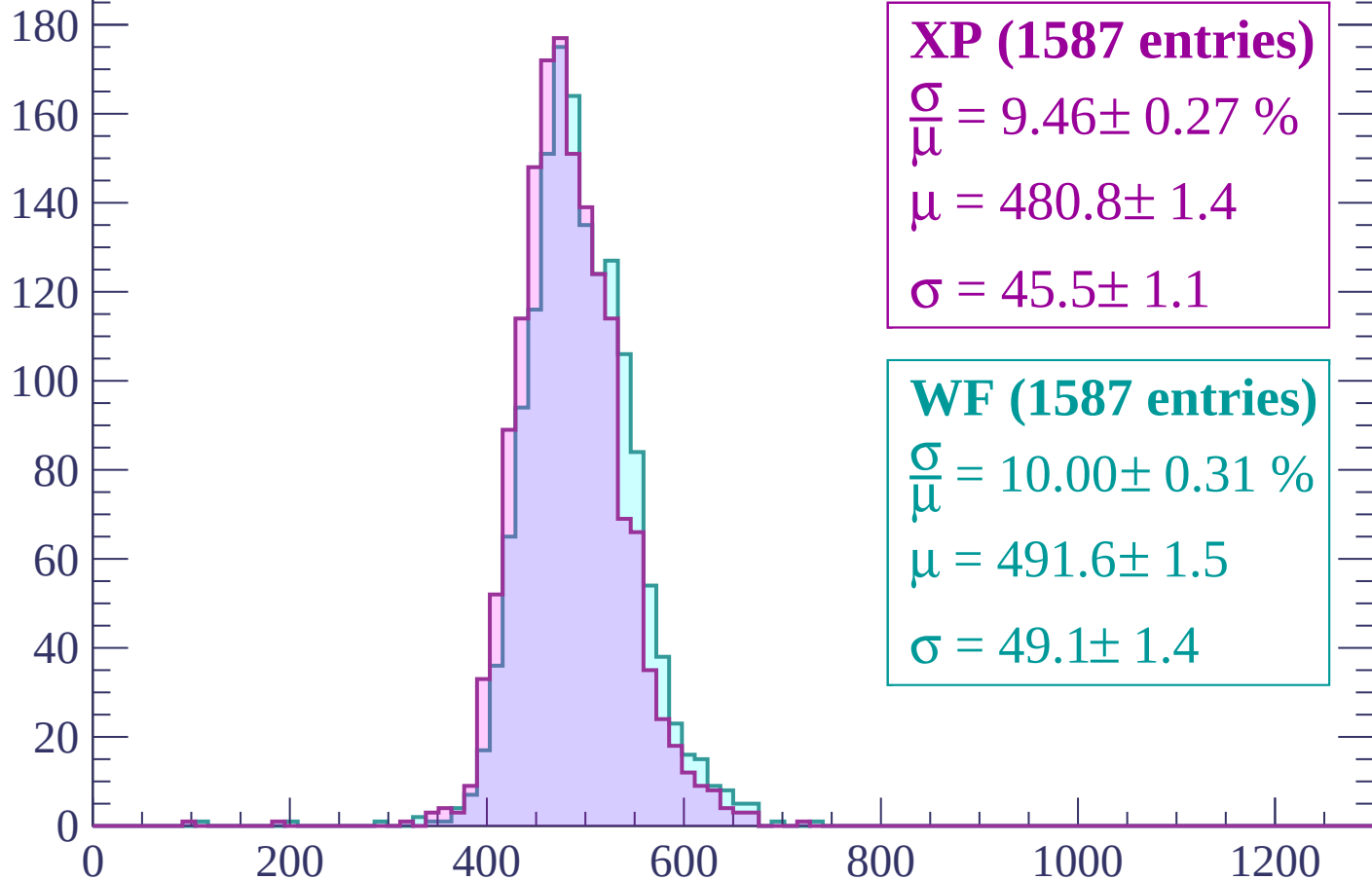
$$\mu = 491.4 \pm 1.3$$

$$\sigma = 46.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (1587 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.27 \%$$

$$\mu = 480.8 \pm 1.4$$

$$\sigma = 45.5 \pm 1.1$$

WF (1587 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.31 \%$$

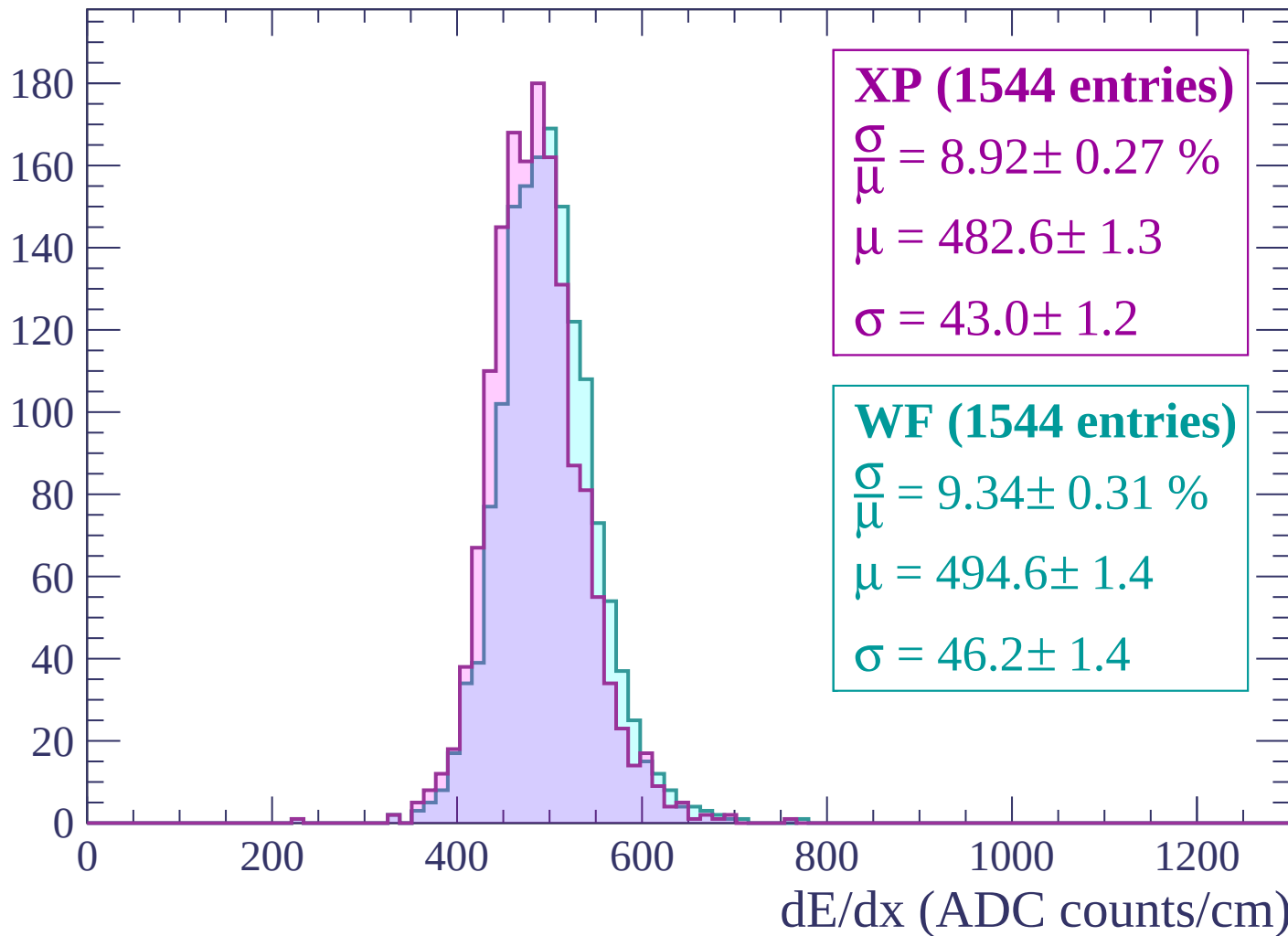
$$\mu = 491.6 \pm 1.5$$

$$\sigma = 49.1 \pm 1.4$$

dE/dx (ADC counts/cm)

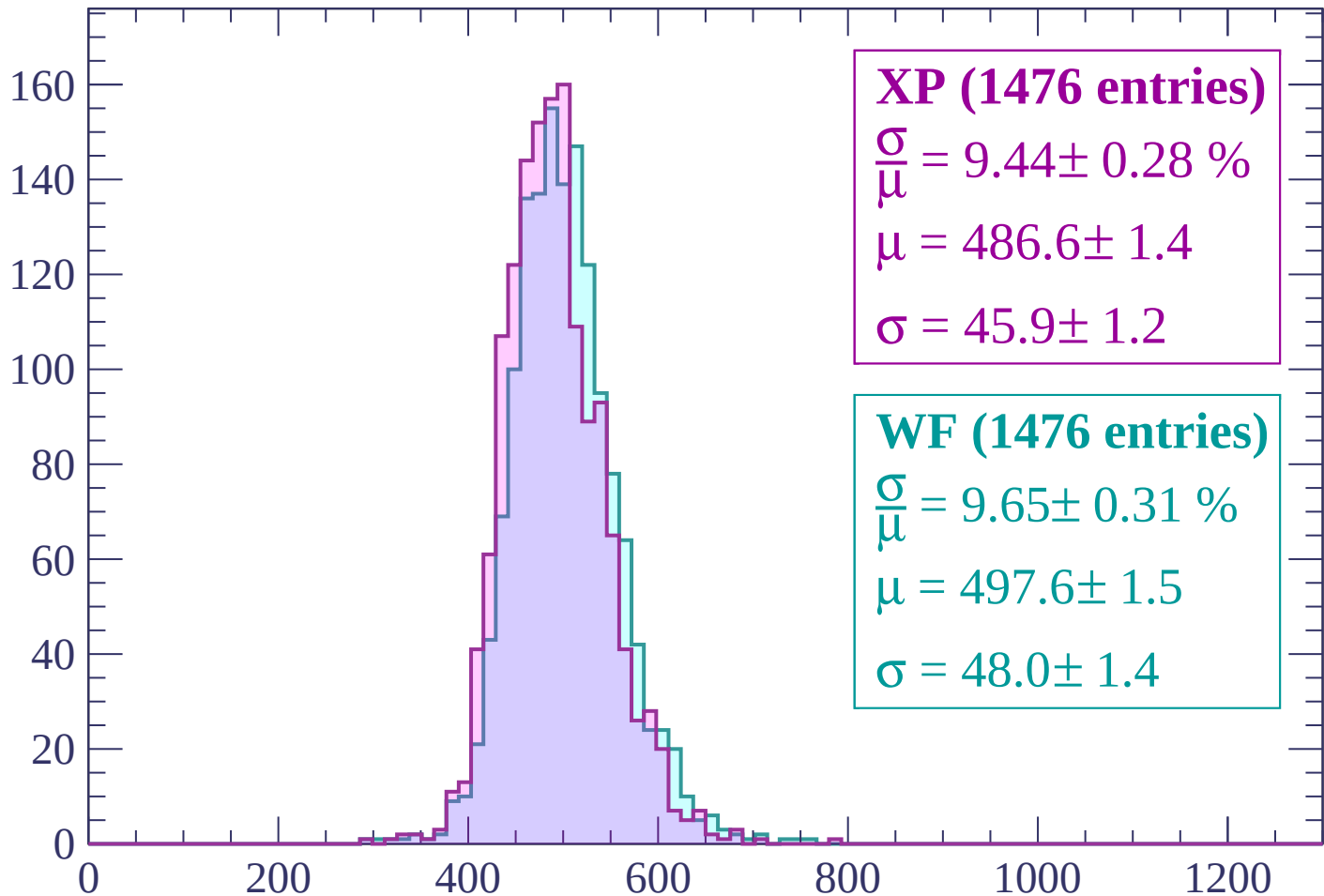
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (1476 entries)

$$\frac{\sigma}{\mu} = 9.44 \pm 0.28 \%$$

$$\mu = 486.6 \pm 1.4$$

$$\sigma = 45.9 \pm 1.2$$

WF (1476 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.31 \%$$

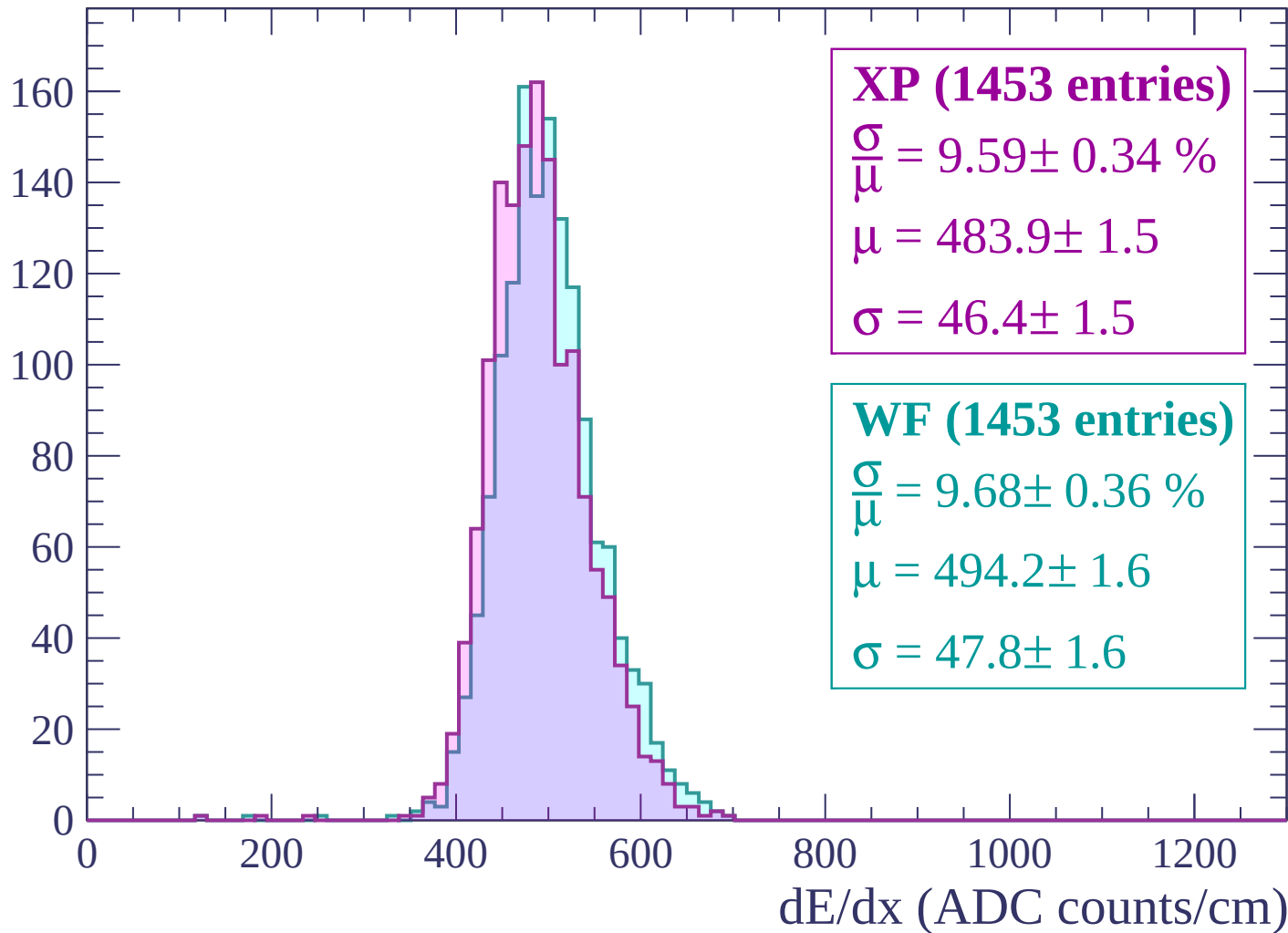
$$\mu = 497.6 \pm 1.5$$

$$\sigma = 48.0 \pm 1.4$$

dE/dx (ADC counts/cm)

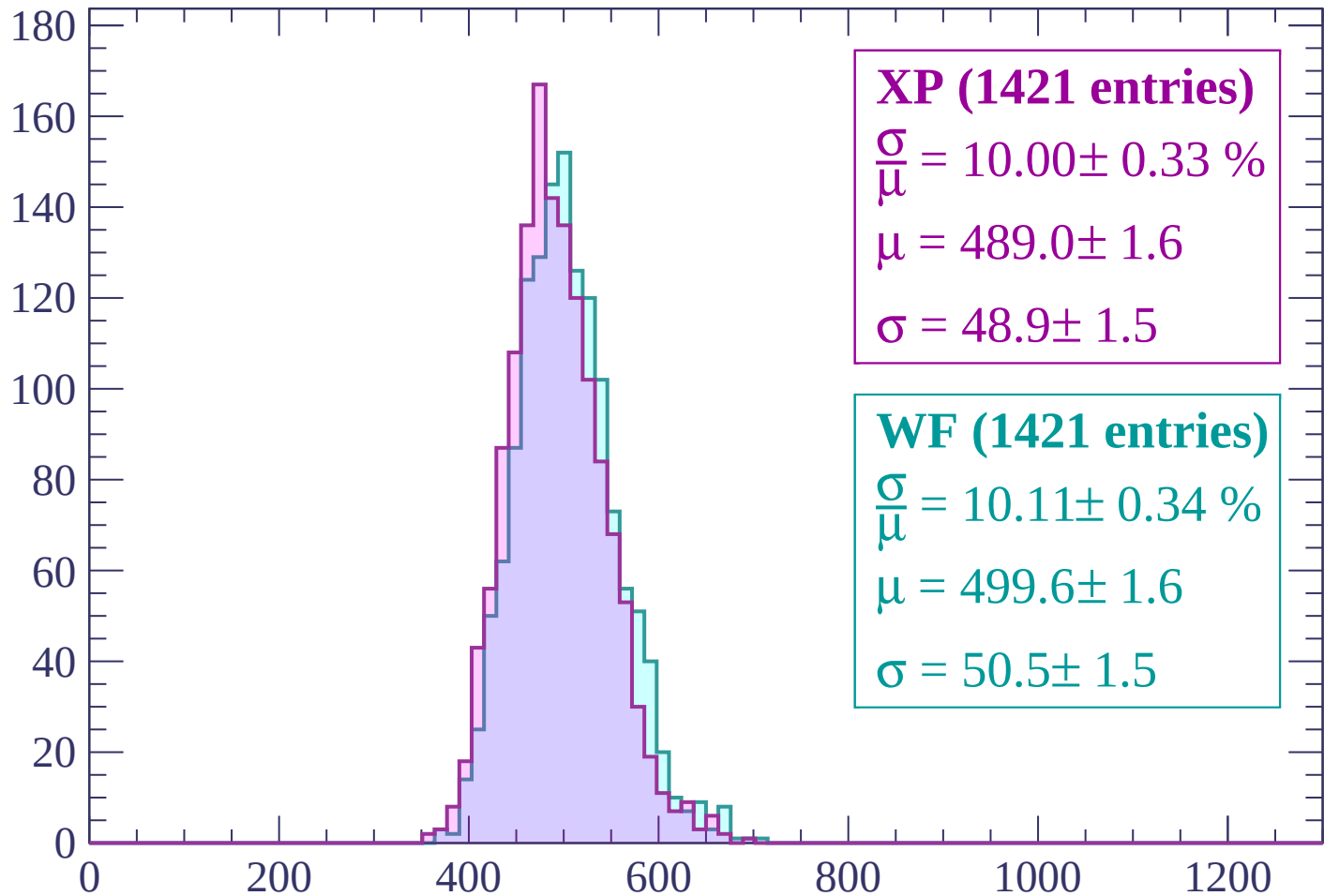
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (1421 entries)

$\frac{\sigma}{\mu} = 10.00 \pm 0.33 \%$

$\mu = 489.0 \pm 1.6$

$\sigma = 48.9 \pm 1.5$

WF (1421 entries)

$\frac{\sigma}{\mu} = 10.11 \pm 0.34 \%$

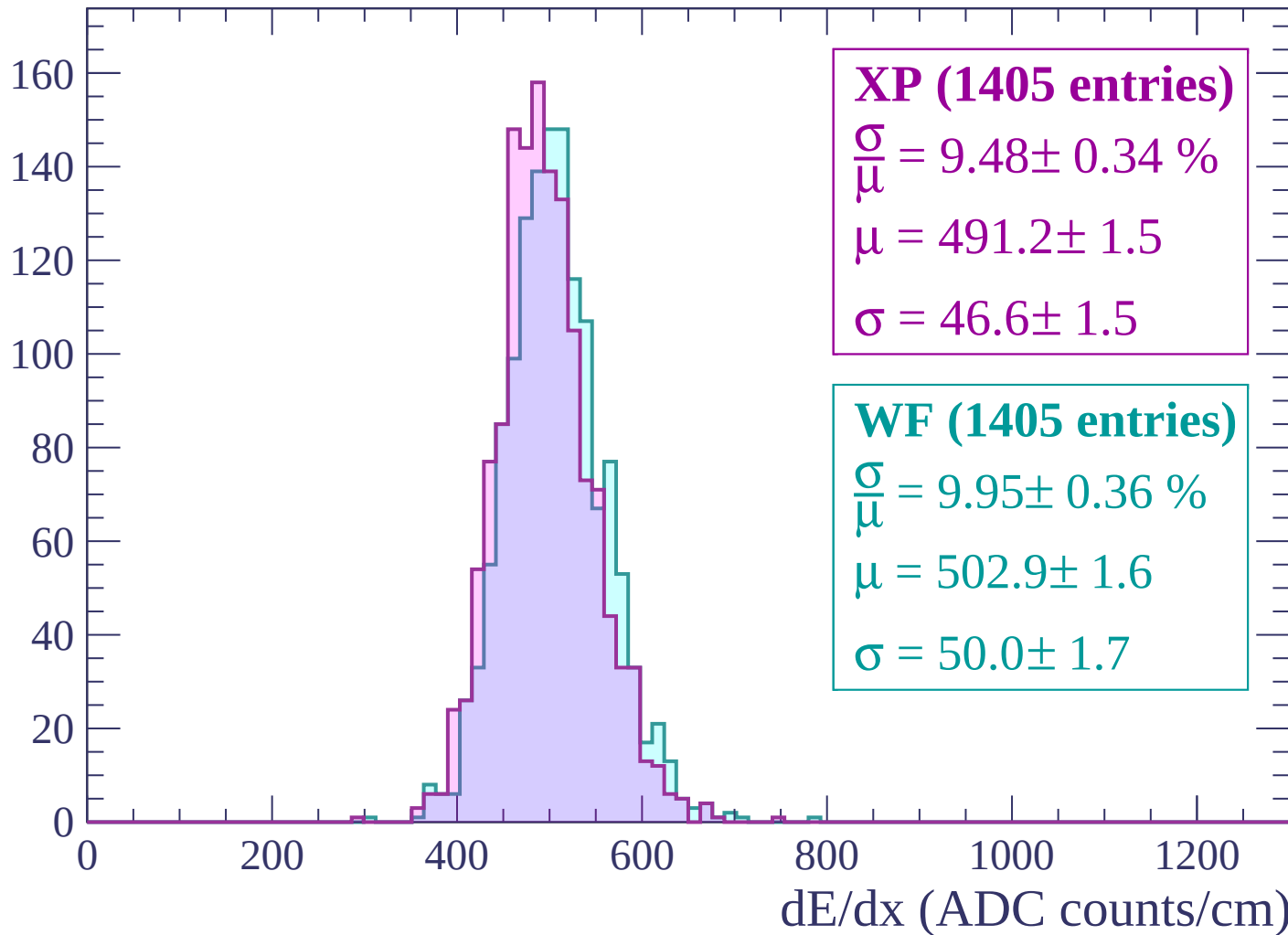
$\mu = 499.6 \pm 1.6$

$\sigma = 50.5 \pm 1.5$

dE/dx (ADC counts/cm)

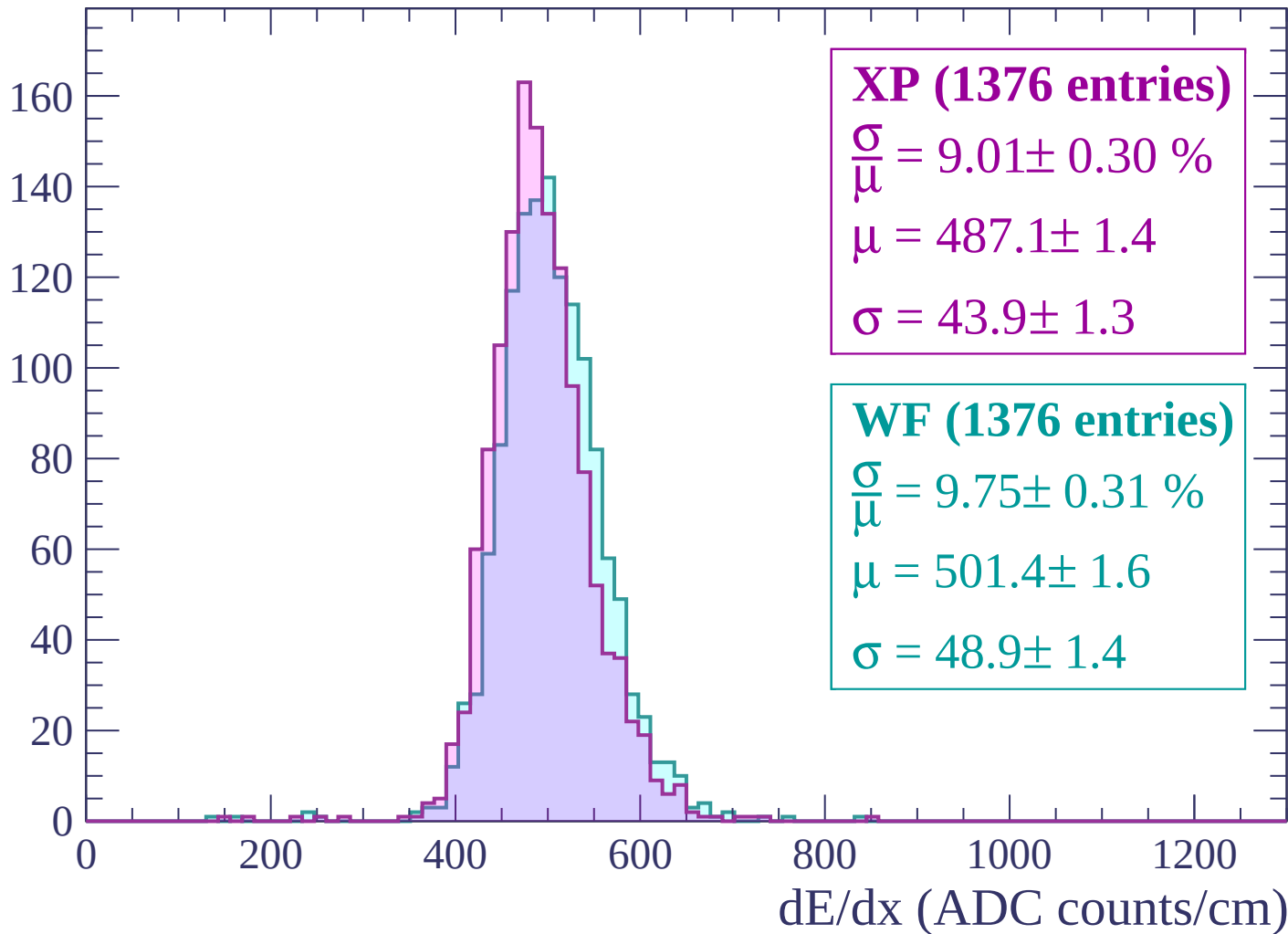
Energy loss | $2580 < p < 2640$

Count



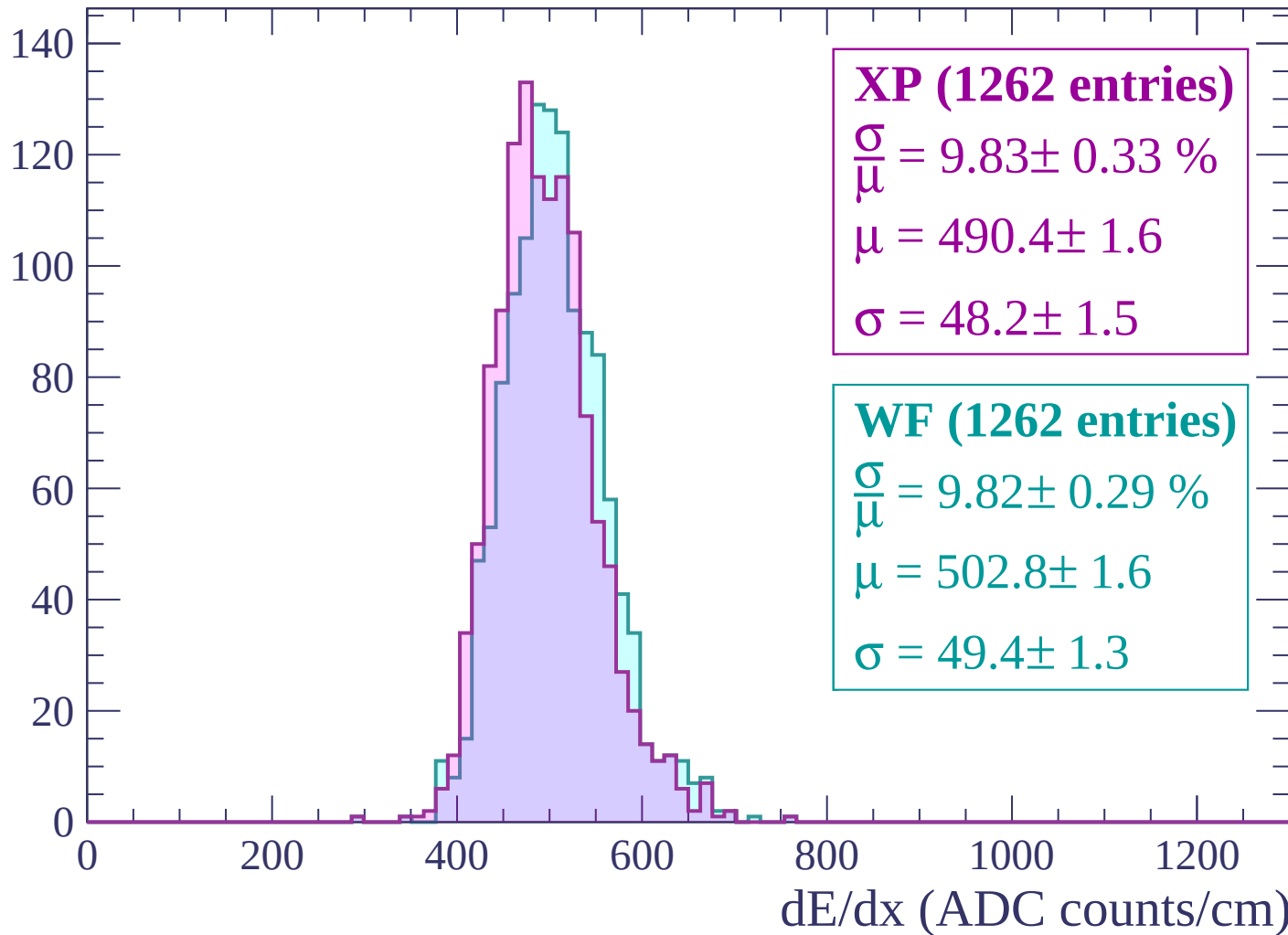
Energy loss | $2640 < p < 2700$

Count



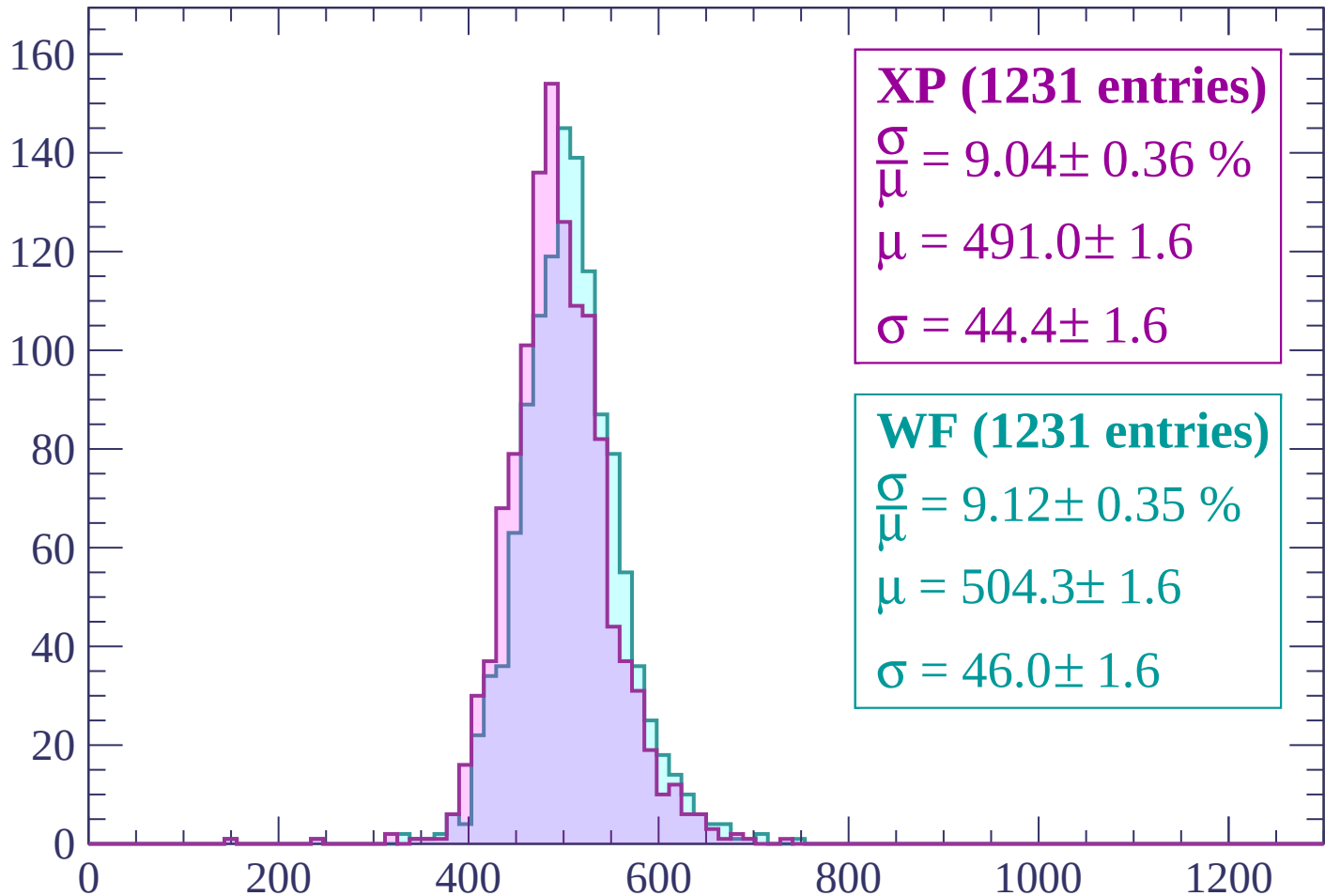
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (1231 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.36 \%$$

$$\mu = 491.0 \pm 1.6$$

$$\sigma = 44.4 \pm 1.6$$

WF (1231 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 0.35 \%$$

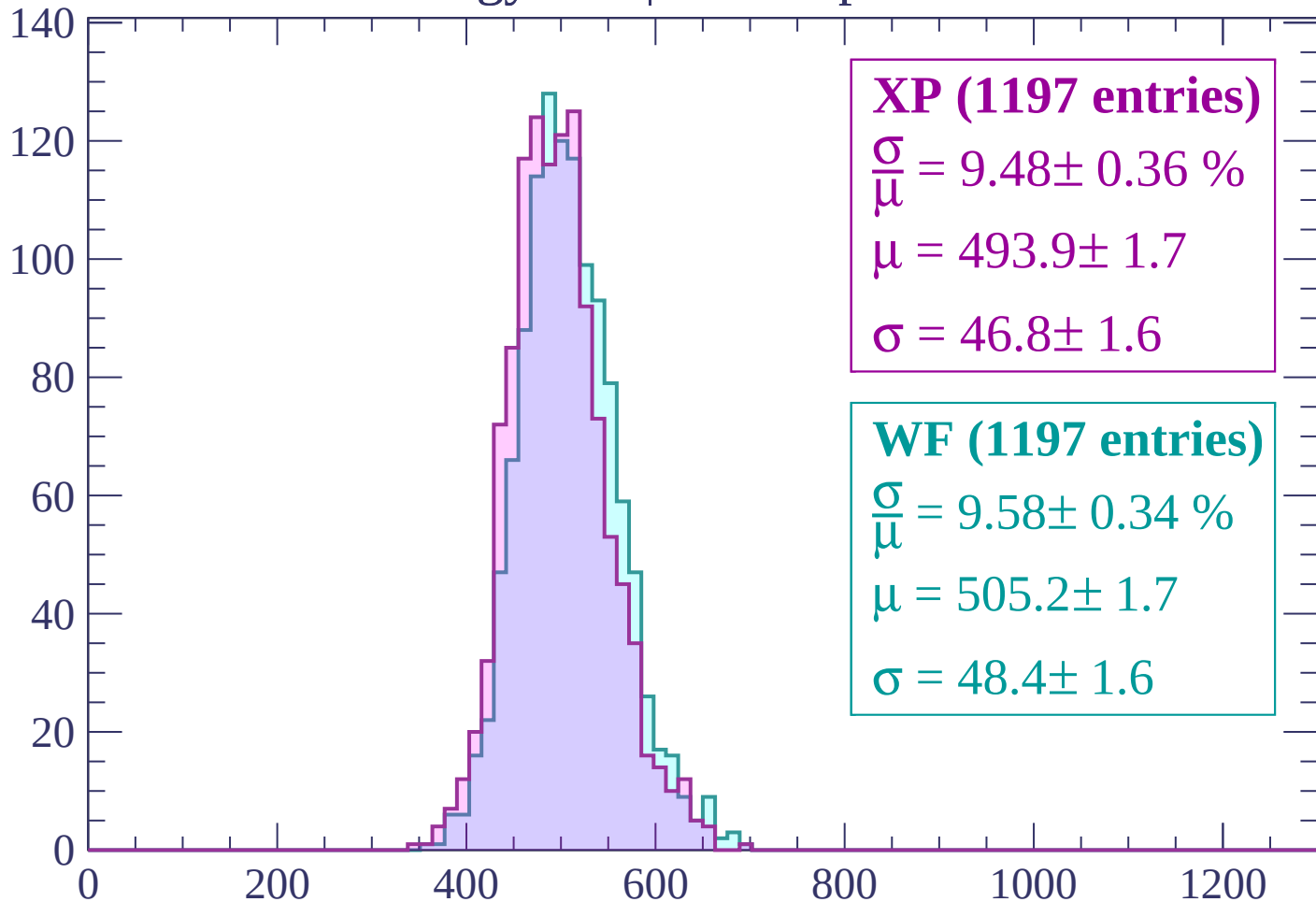
$$\mu = 504.3 \pm 1.6$$

$$\sigma = 46.0 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP (1197 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.36 \%$$

$$\mu = 493.9 \pm 1.7$$

$$\sigma = 46.8 \pm 1.6$$

WF (1197 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.34 \%$$

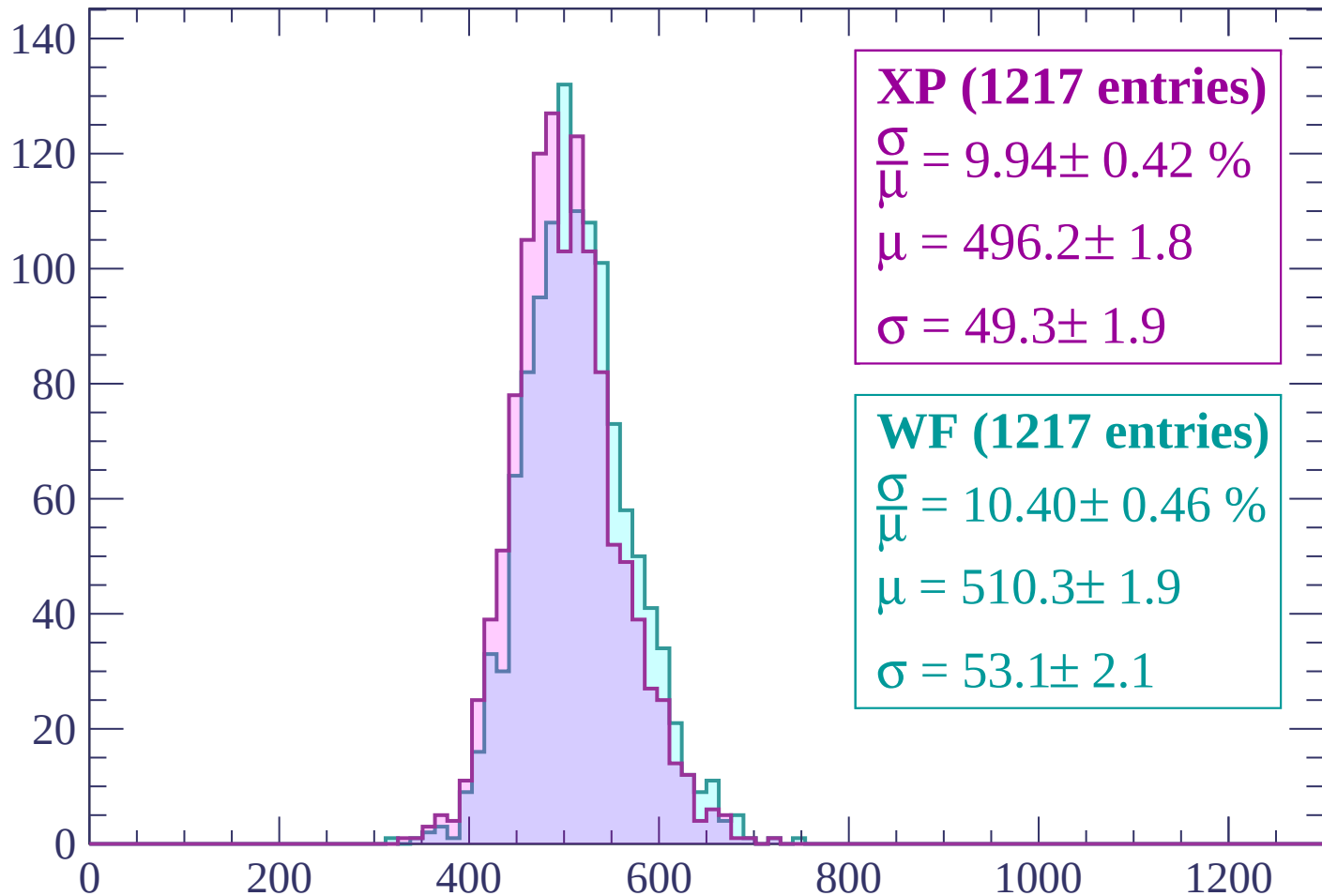
$$\mu = 505.2 \pm 1.7$$

$$\sigma = 48.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (1217 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.42 \%$

$\mu = 496.2 \pm 1.8$

$\sigma = 49.3 \pm 1.9$

WF (1217 entries)

$\frac{\sigma}{\mu} = 10.40 \pm 0.46 \%$

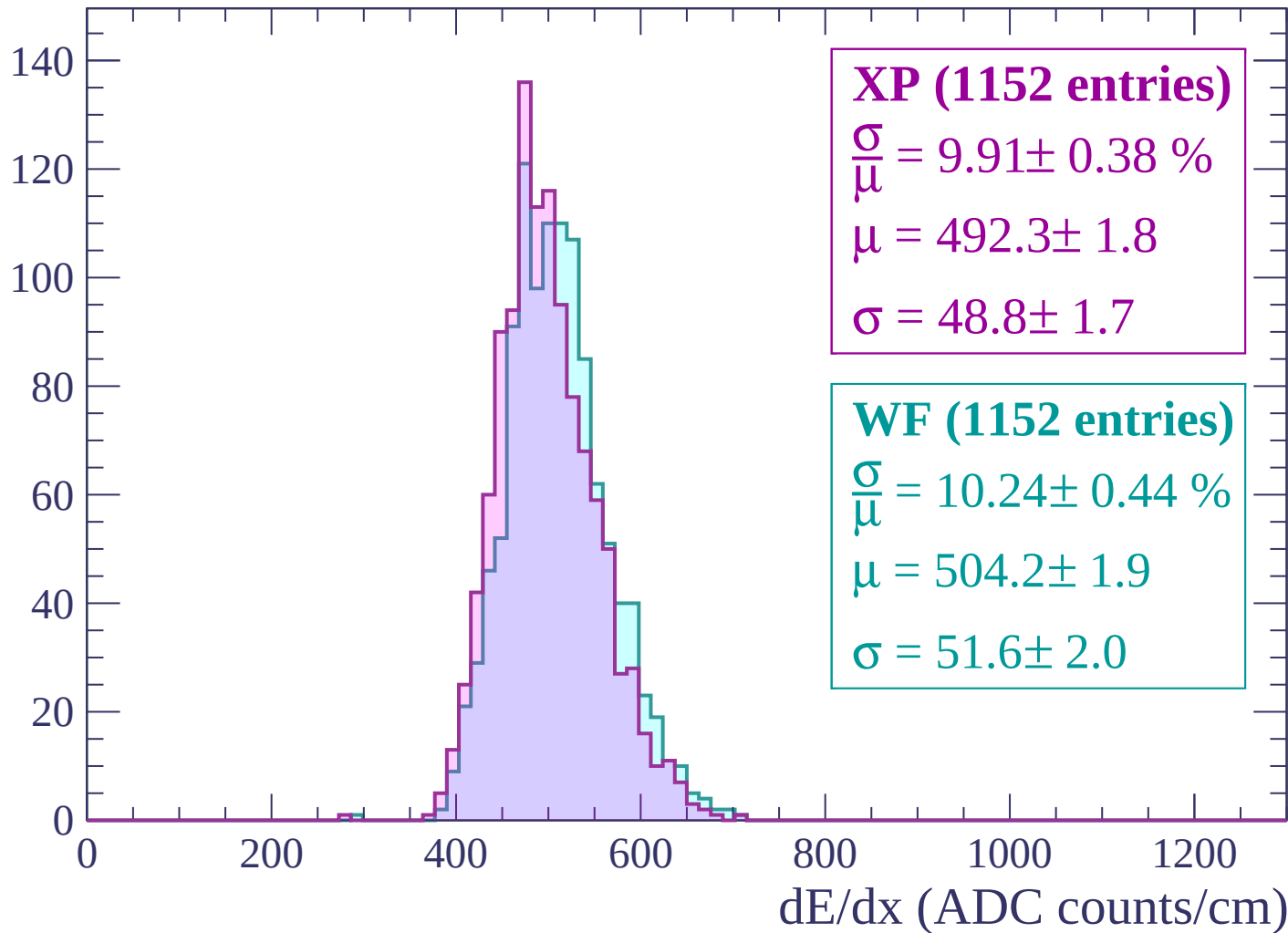
$\mu = 510.3 \pm 1.9$

$\sigma = 53.1 \pm 2.1$

dE/dx (ADC counts/cm)

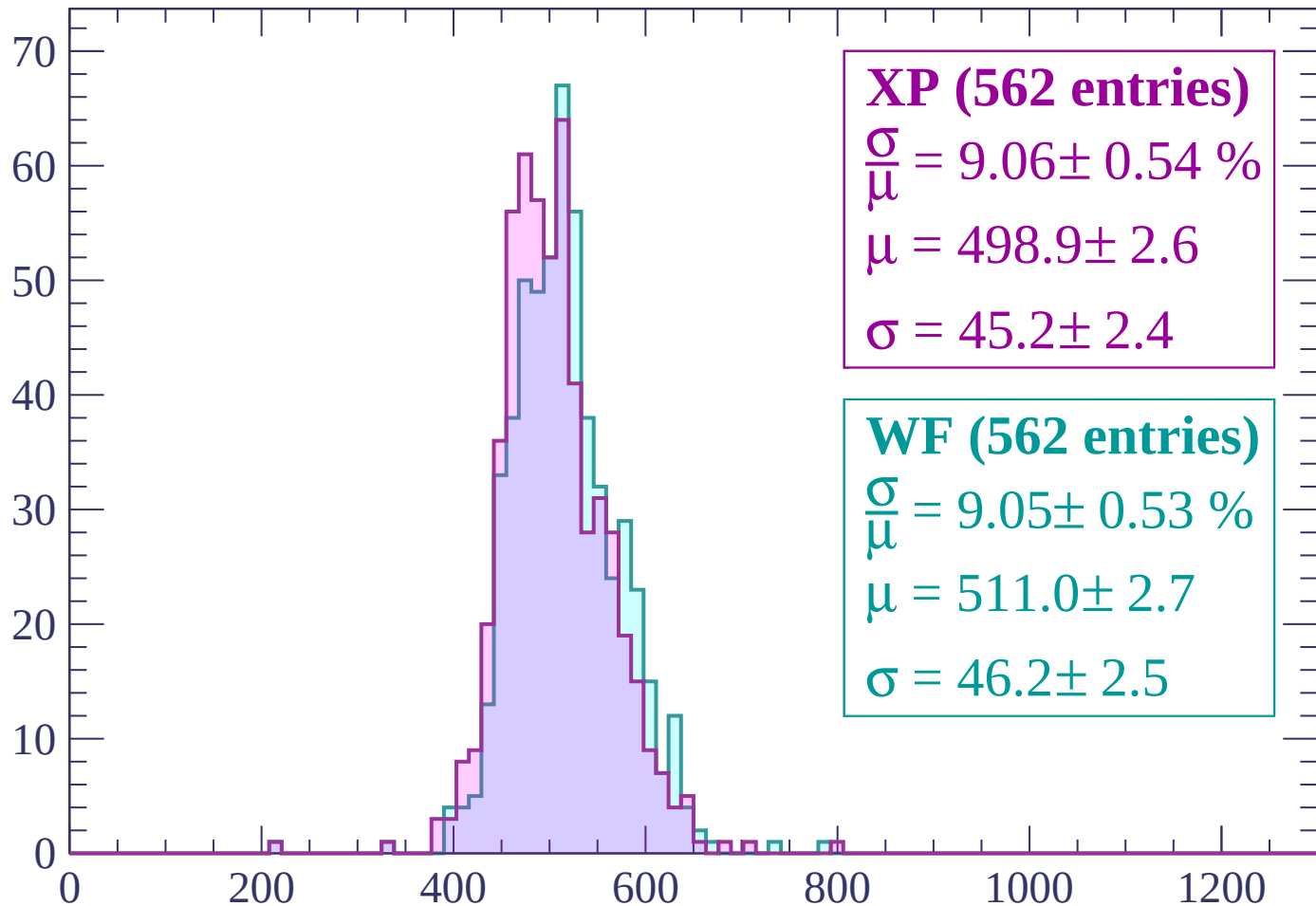
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP (562 entries)

$\frac{\sigma}{\mu} = 9.06 \pm 0.54 \%$

$\mu = 498.9 \pm 2.6$

$\sigma = 45.2 \pm 2.4$

WF (562 entries)

$\frac{\sigma}{\mu} = 9.05 \pm 0.53 \%$

$\mu = 511.0 \pm 2.7$

$\sigma = 46.2 \pm 2.5$

dE/dx (ADC counts/cm)

